Reporte: Core vs Divergent
Metabolismo + Clínica + Supervivencia

Pacientes Core: 1127 | Divergentes: 39

Variables metabólicas analizadas: 248 | Significativas (p_adj<0.05): 93

Soluciones FBA: FBA, pFBA, L1w, L2, L2w

Estructura: Clínica −> TNBC −> Metabolismo −> Integración −> Supervivencia

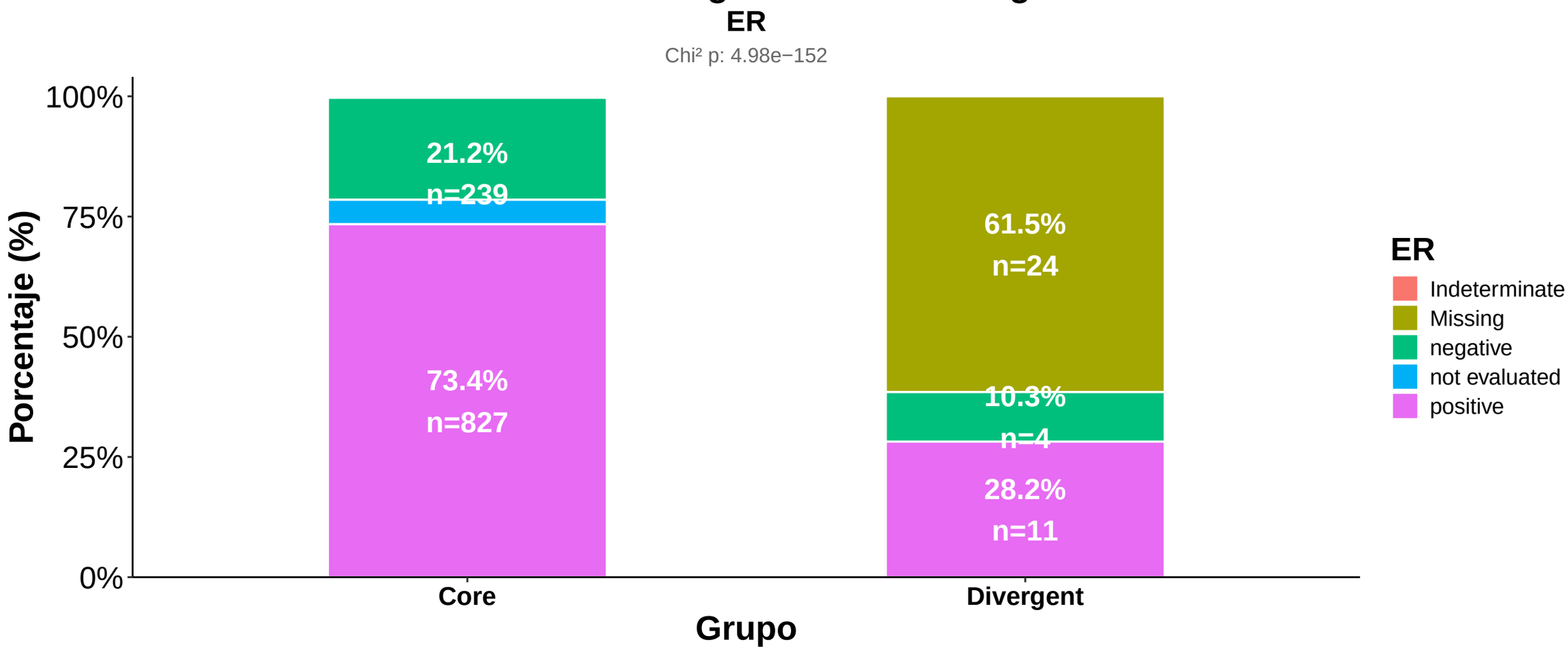
Tamaño de efecto: Cliff's . | >=0.147 pequeño · >=0.330 mediano · >=0.474 grande

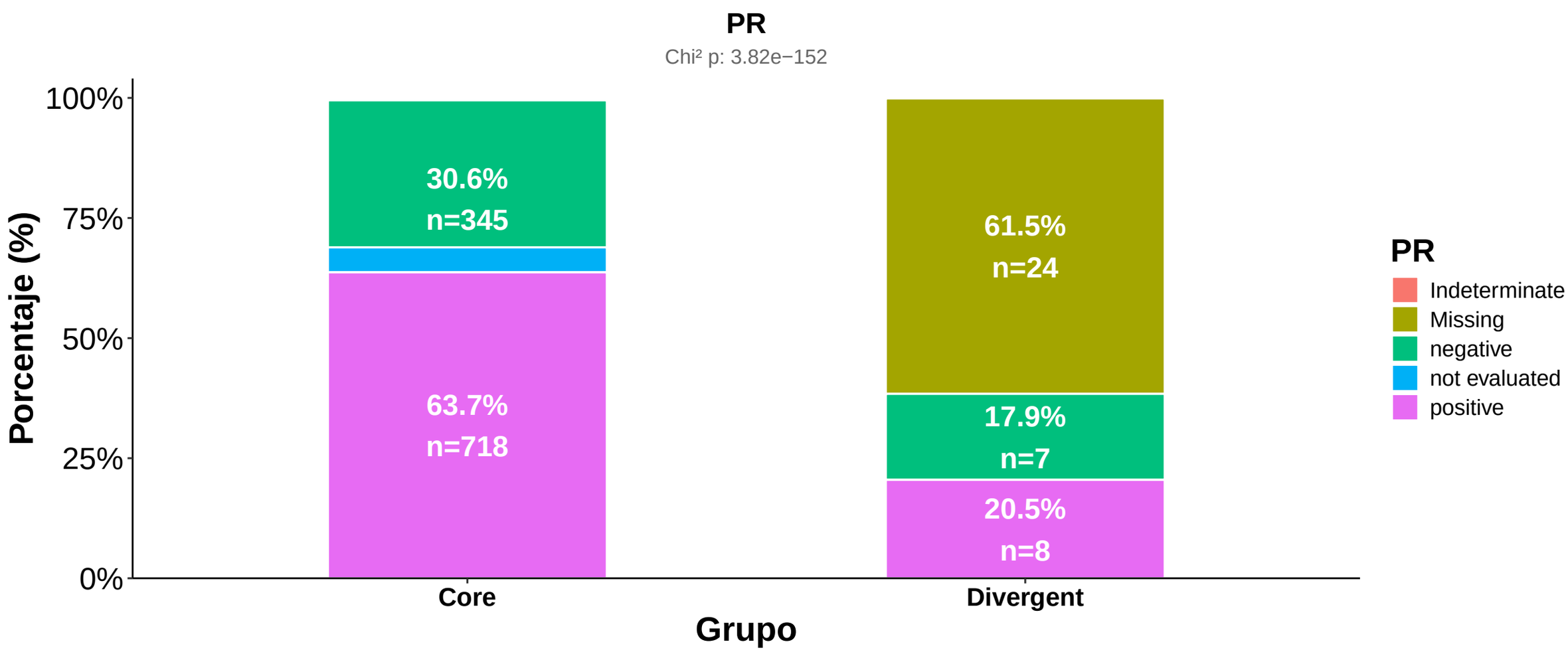
SECCIÓN I

Diferencias Clínicas: Core vs Divergent

Macro−categorías: Biología Tumoral | Estadio | Historia Terapéutica | Factores del Paciente

Clínica Categórica – . A. Biología Tumoral

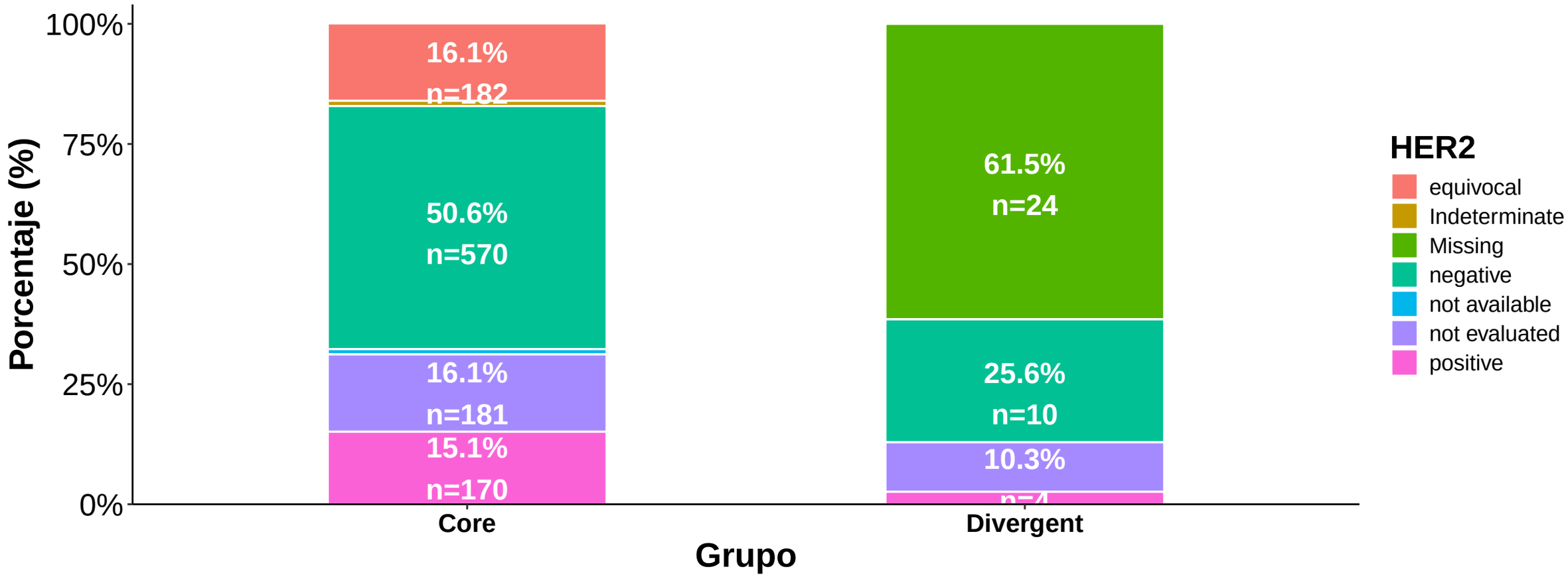




Clínica Categórica − . A. Biología Tumoral

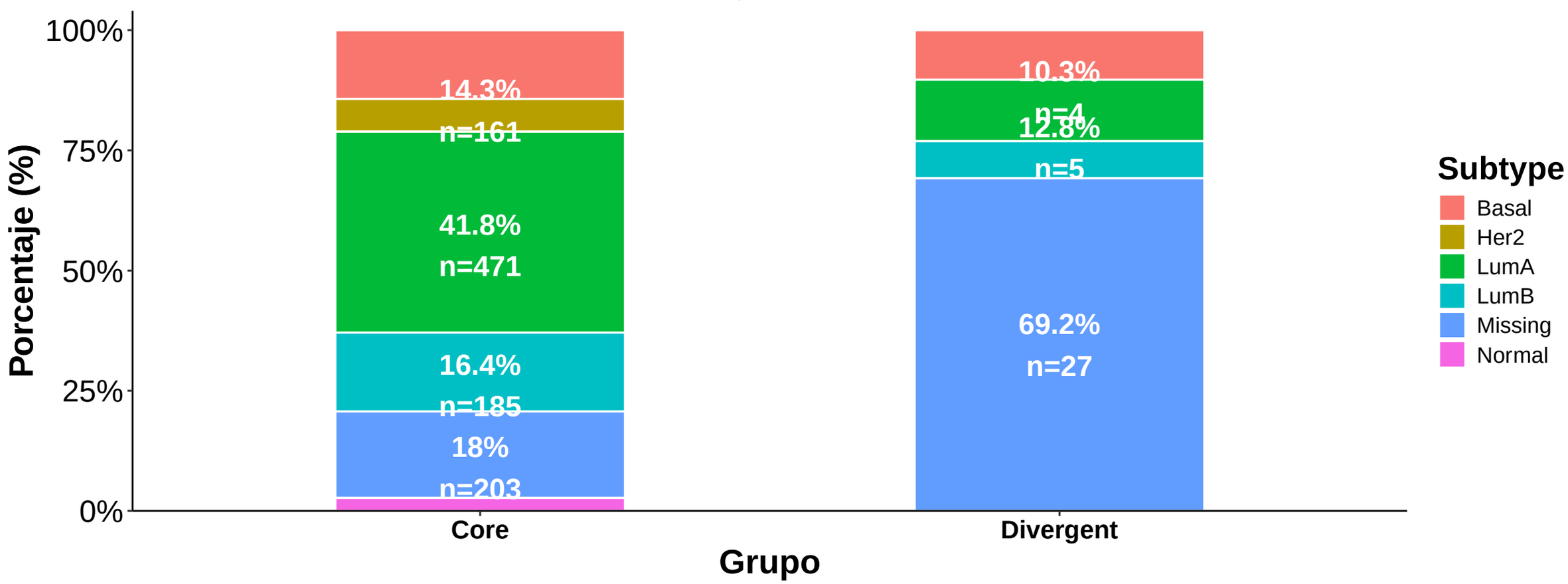
HER2

Chi² p: 3.82e−150

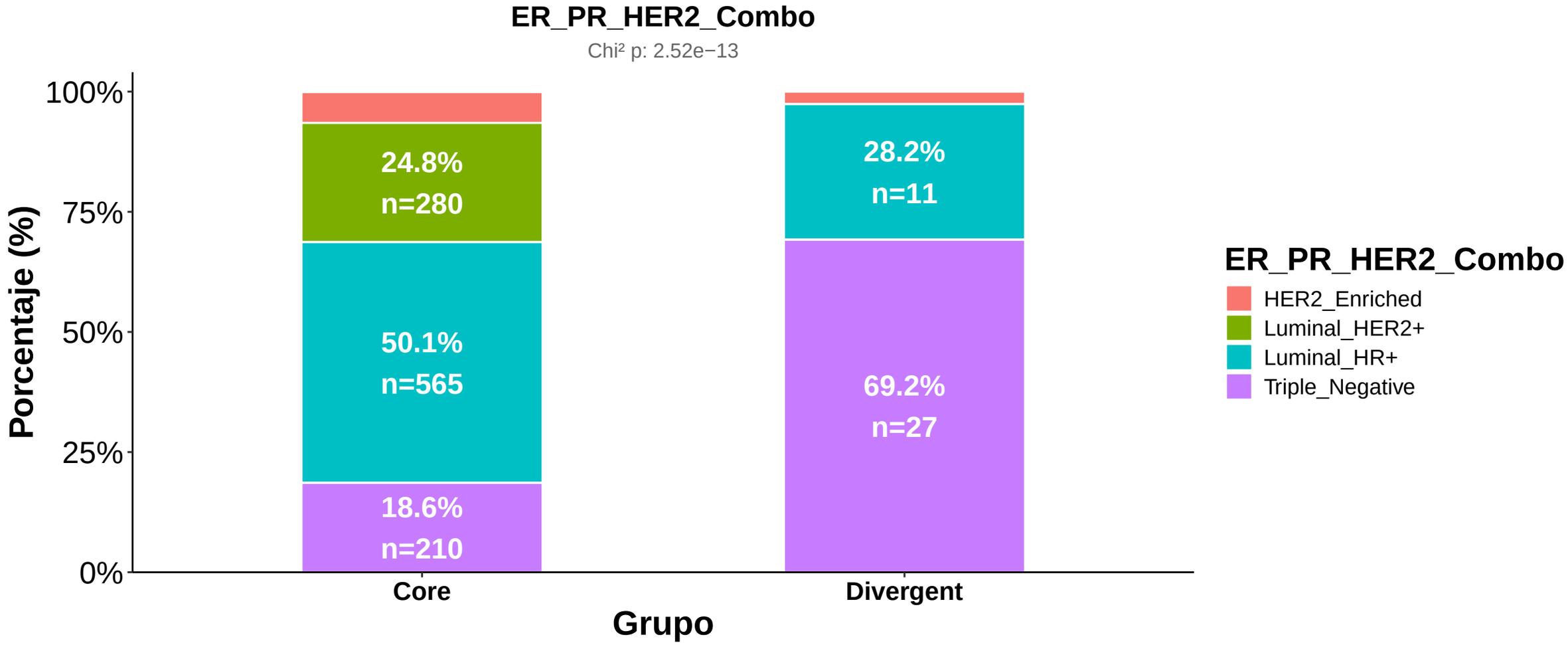


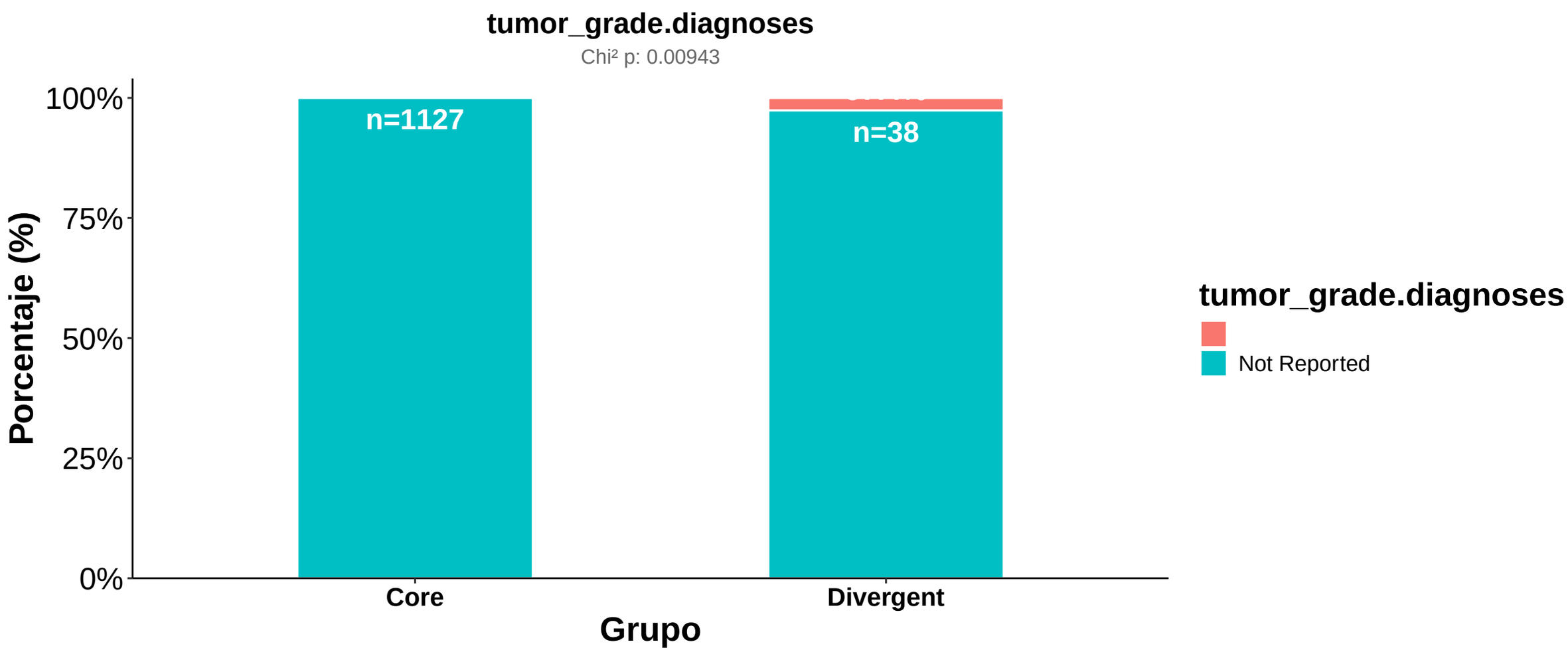
Subtype

Chi² p: 1.99e−12



Clínica Categórica − . A. Biología Tumoral

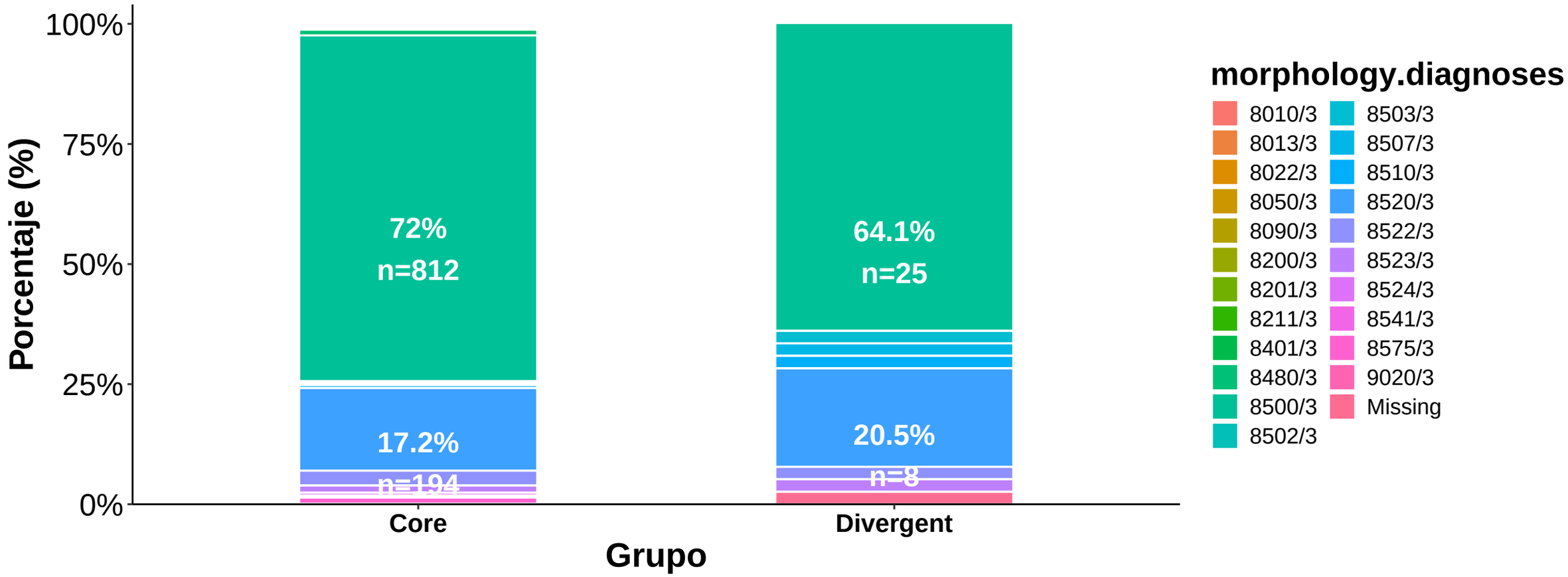




Clínica Categórica – . A. Biología Tumoral

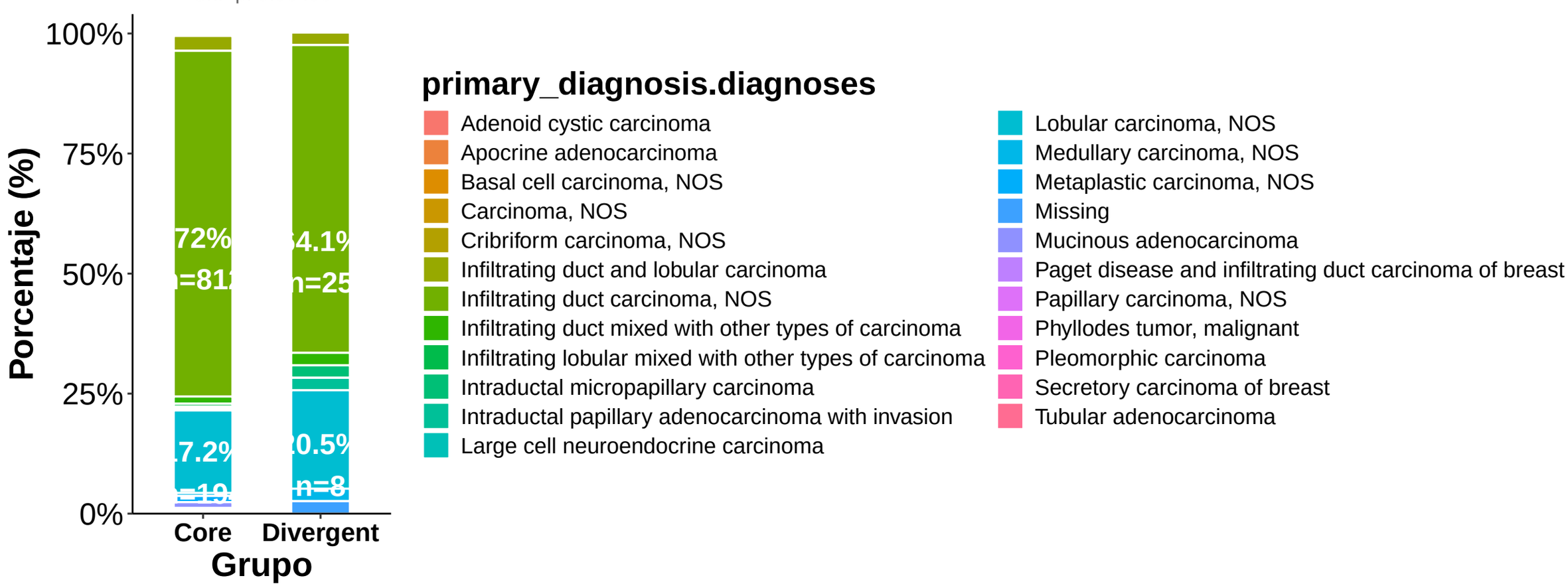
morphology.diagnoses

Chi² p: 0.00498



primary_diagnosis.diagnoses

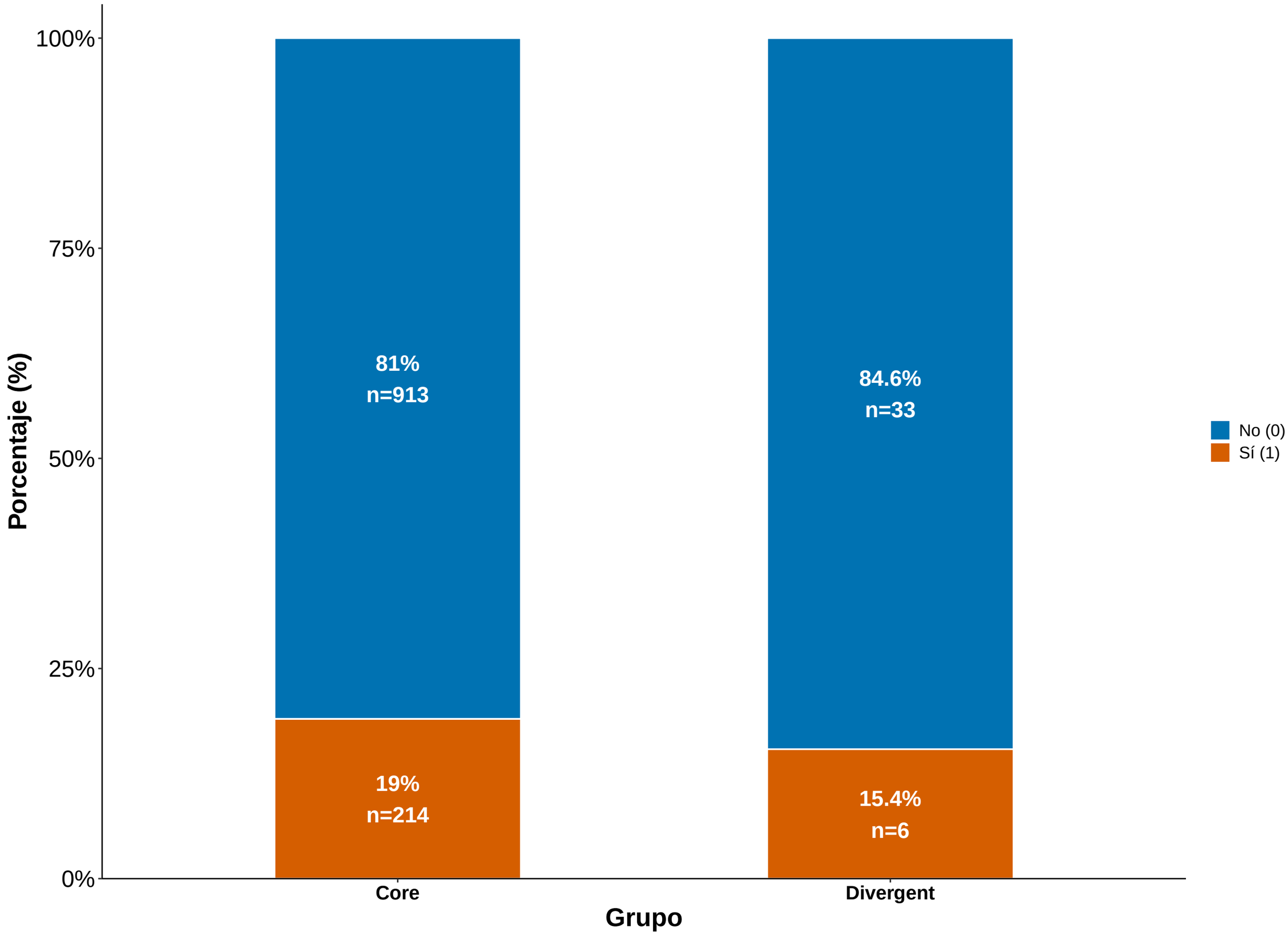
Chi² p: 0.00498



Clínica Numérica − . B. Estadio y Carga Tumoral

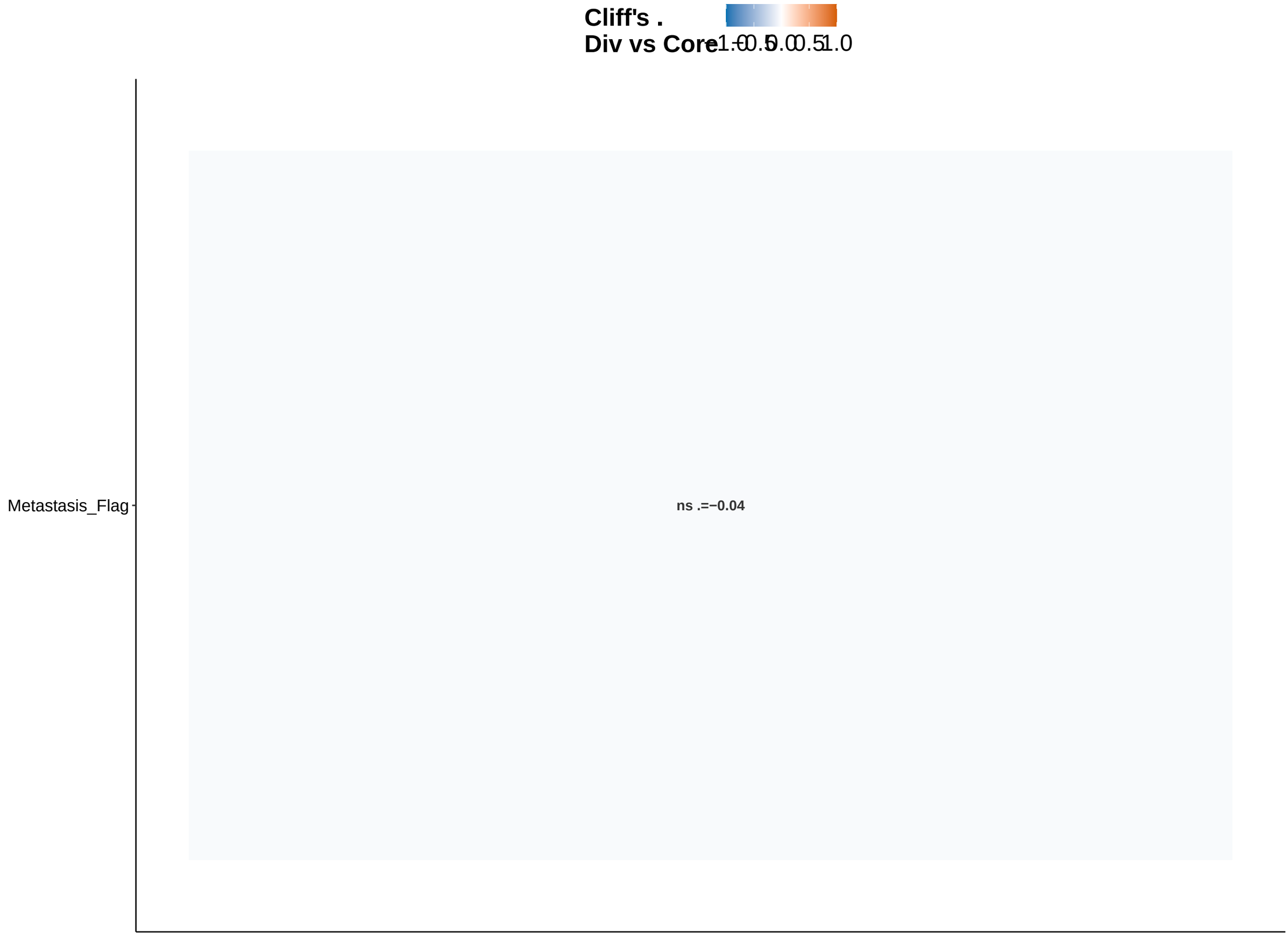
Metastasis_Flag

p−adj: 0.572 | .=−0.04 (Negligible) |



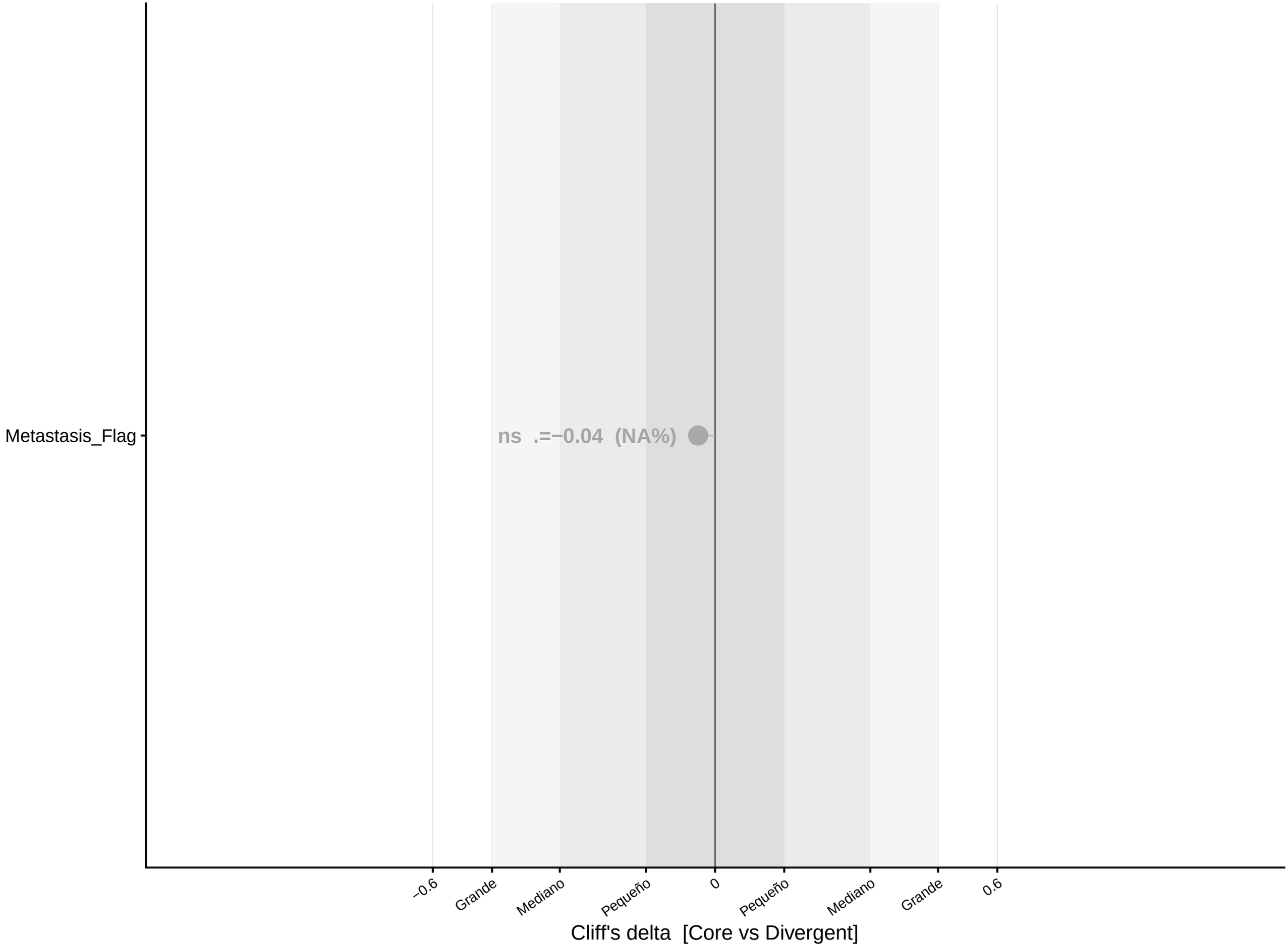
Cliff's .: Clínica: . B. Estadio y Carga Tumoral

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

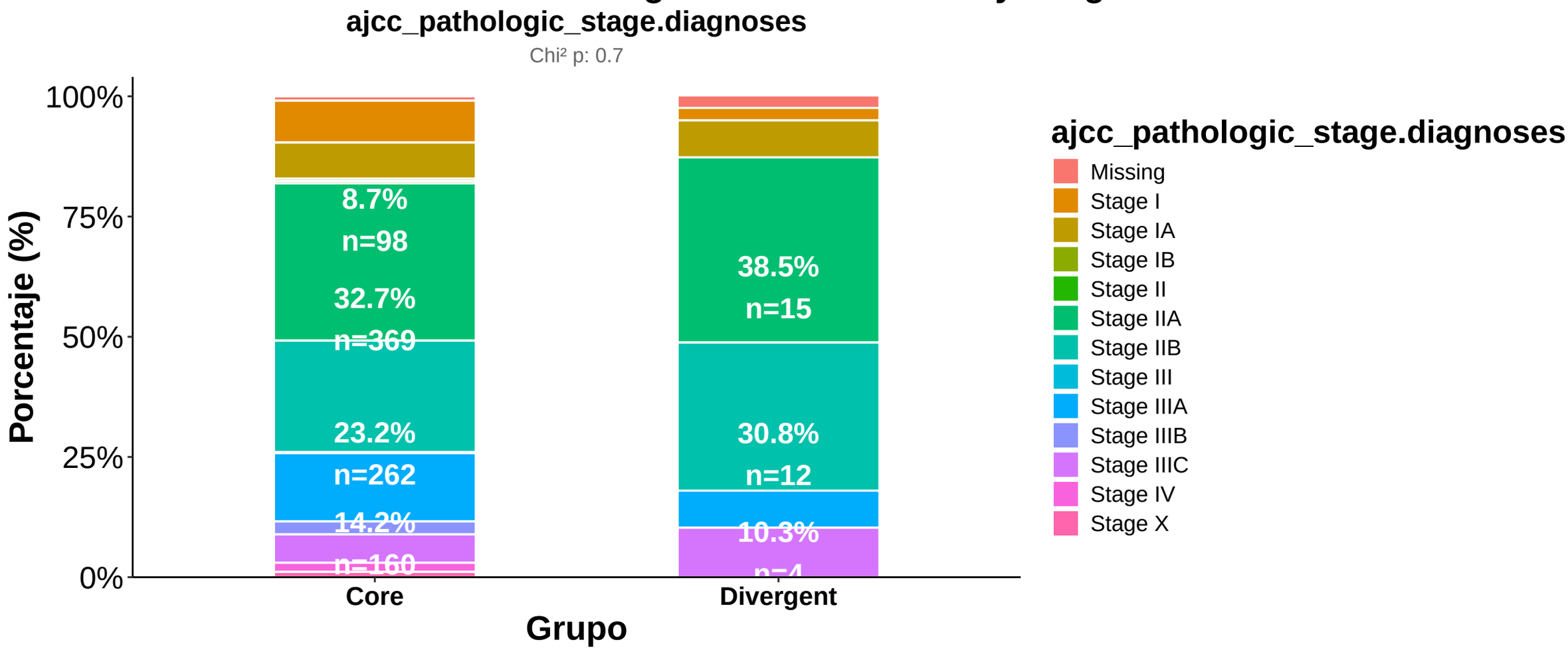


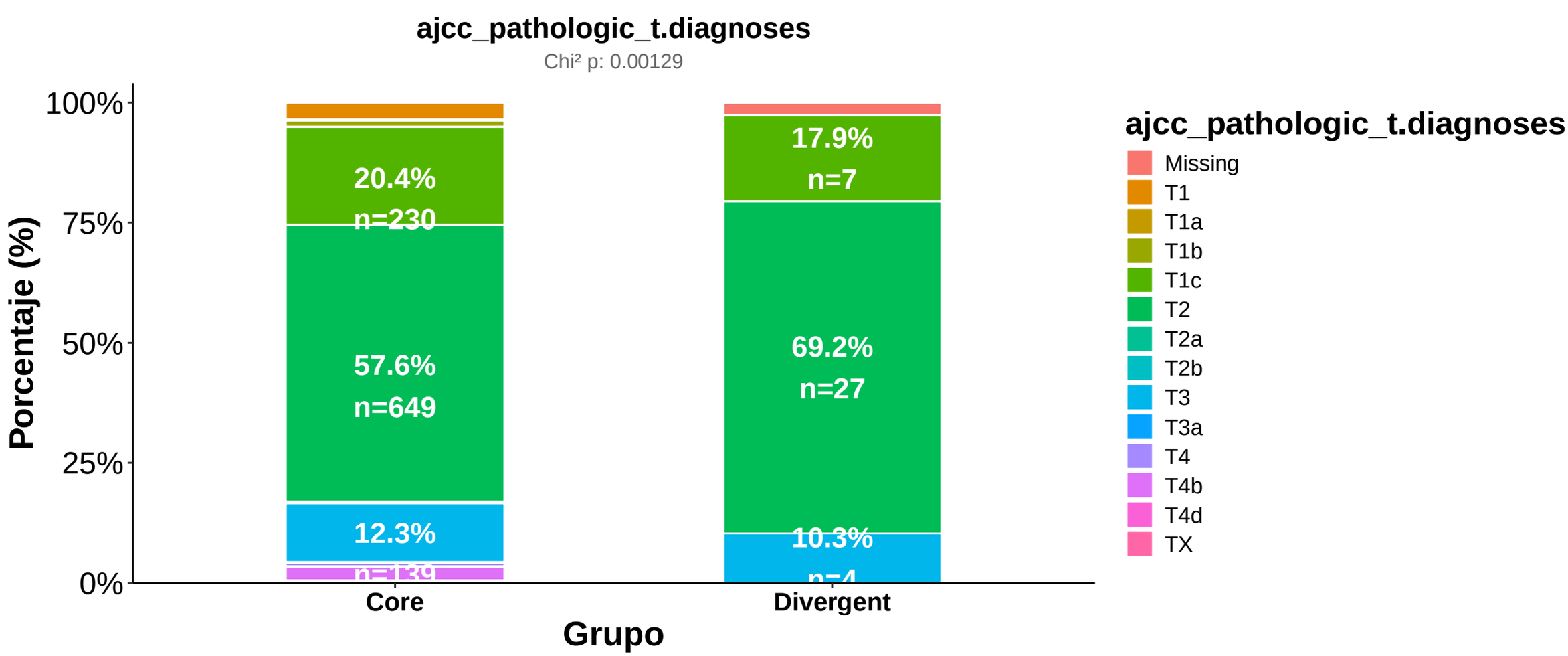
Cliff's . − Clínica: . B. Estadio y Carga Tumoral

Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent

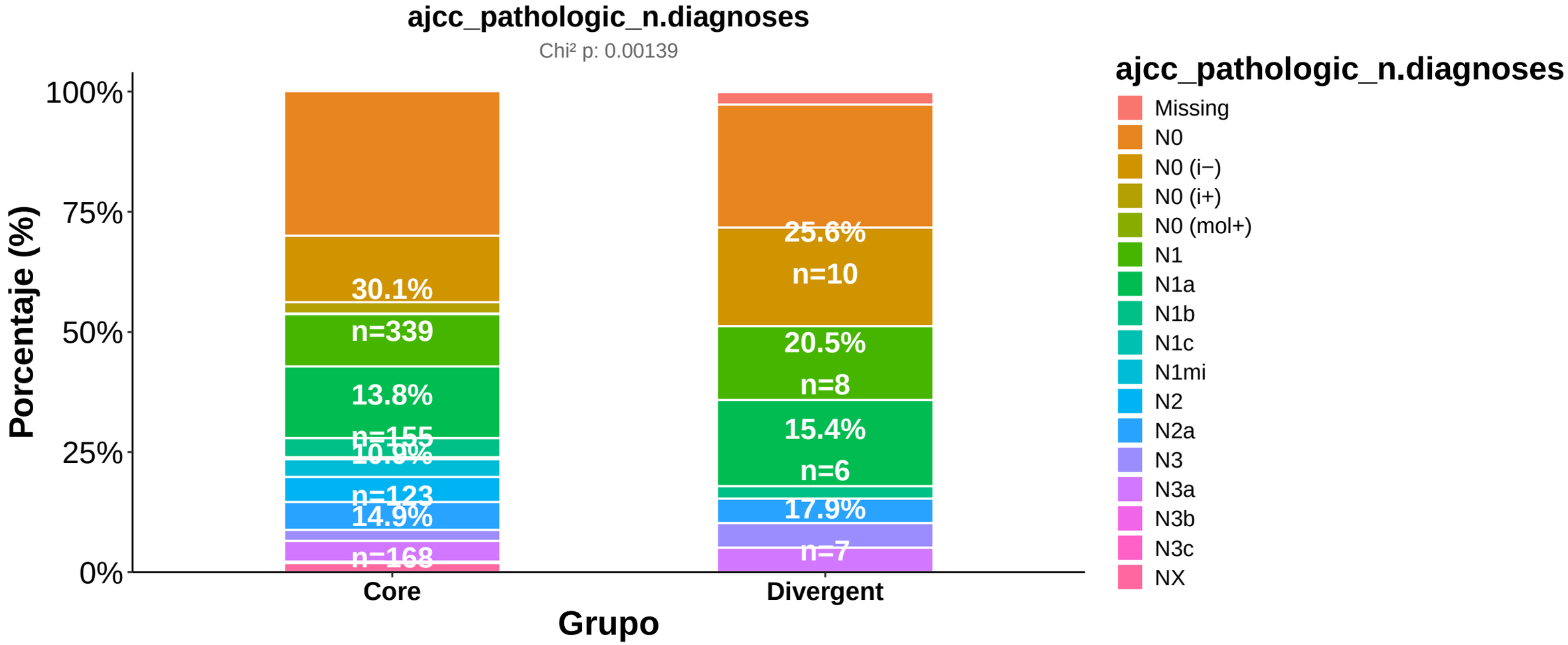


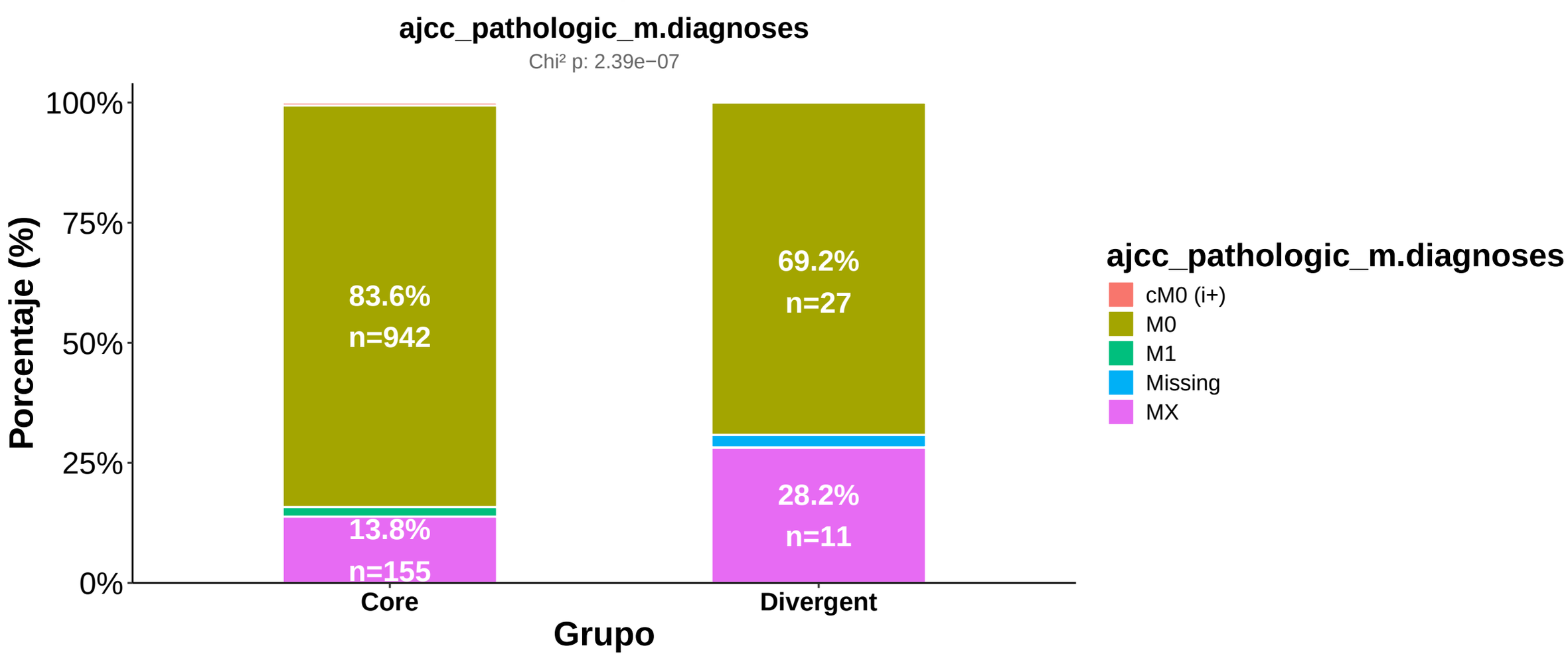
Clínica Categórica – . B. Estadio y Carga Tumoral





Clínica Categórica − . B. Estadio y Carga Tumoral

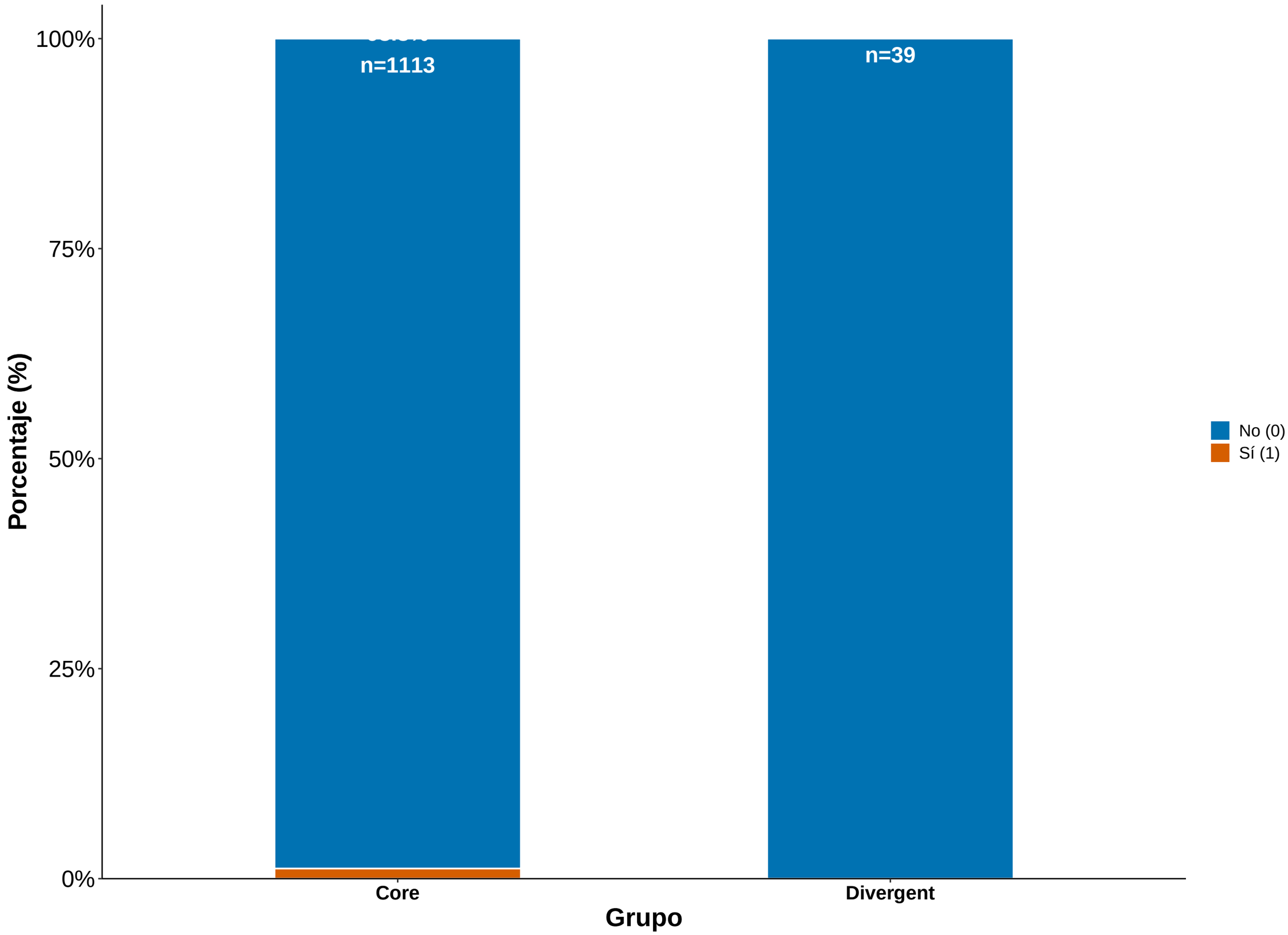




Clínica Numérica − . C. Historia Terapéutica

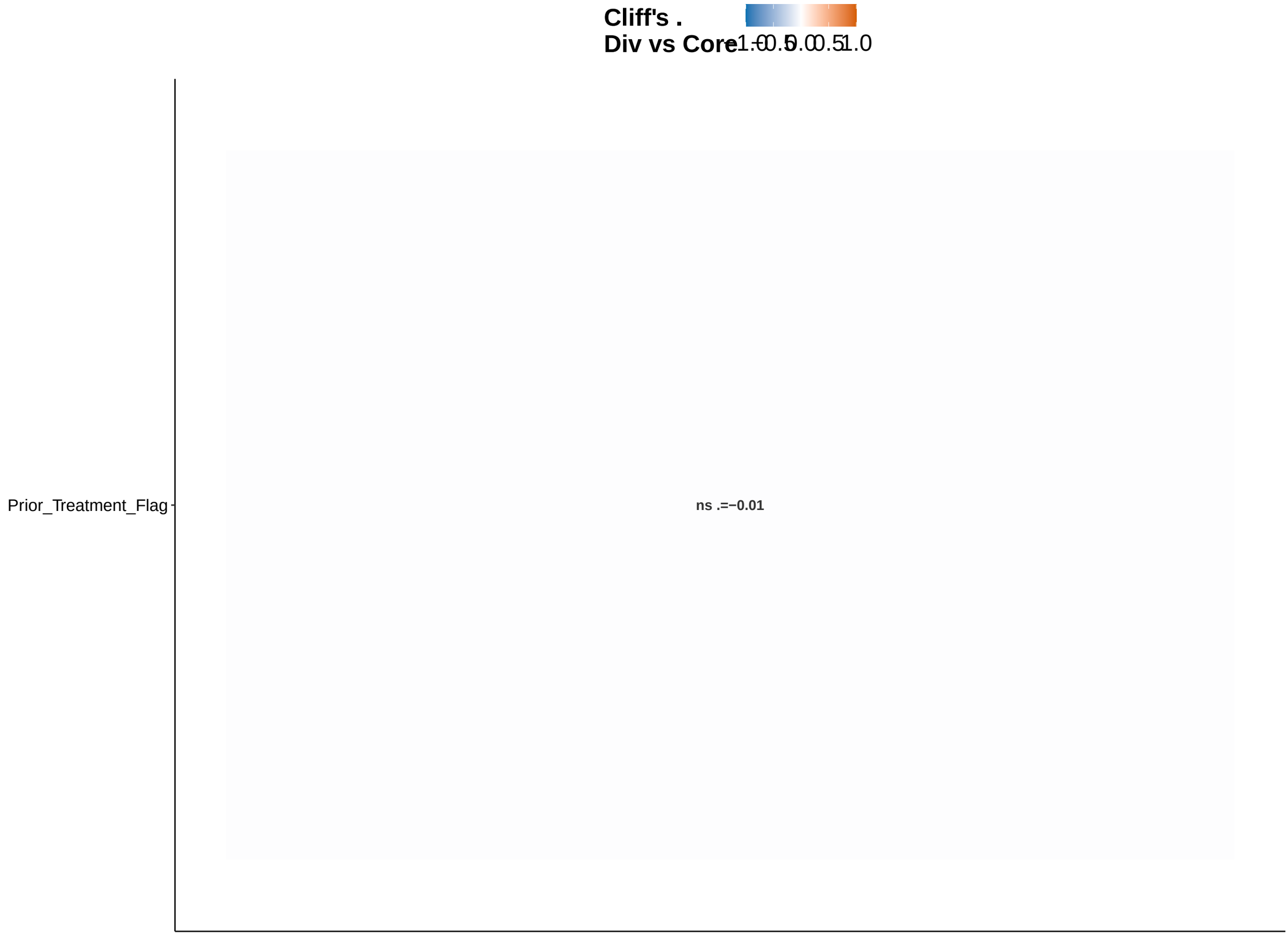
Prior_Treatment_Flag

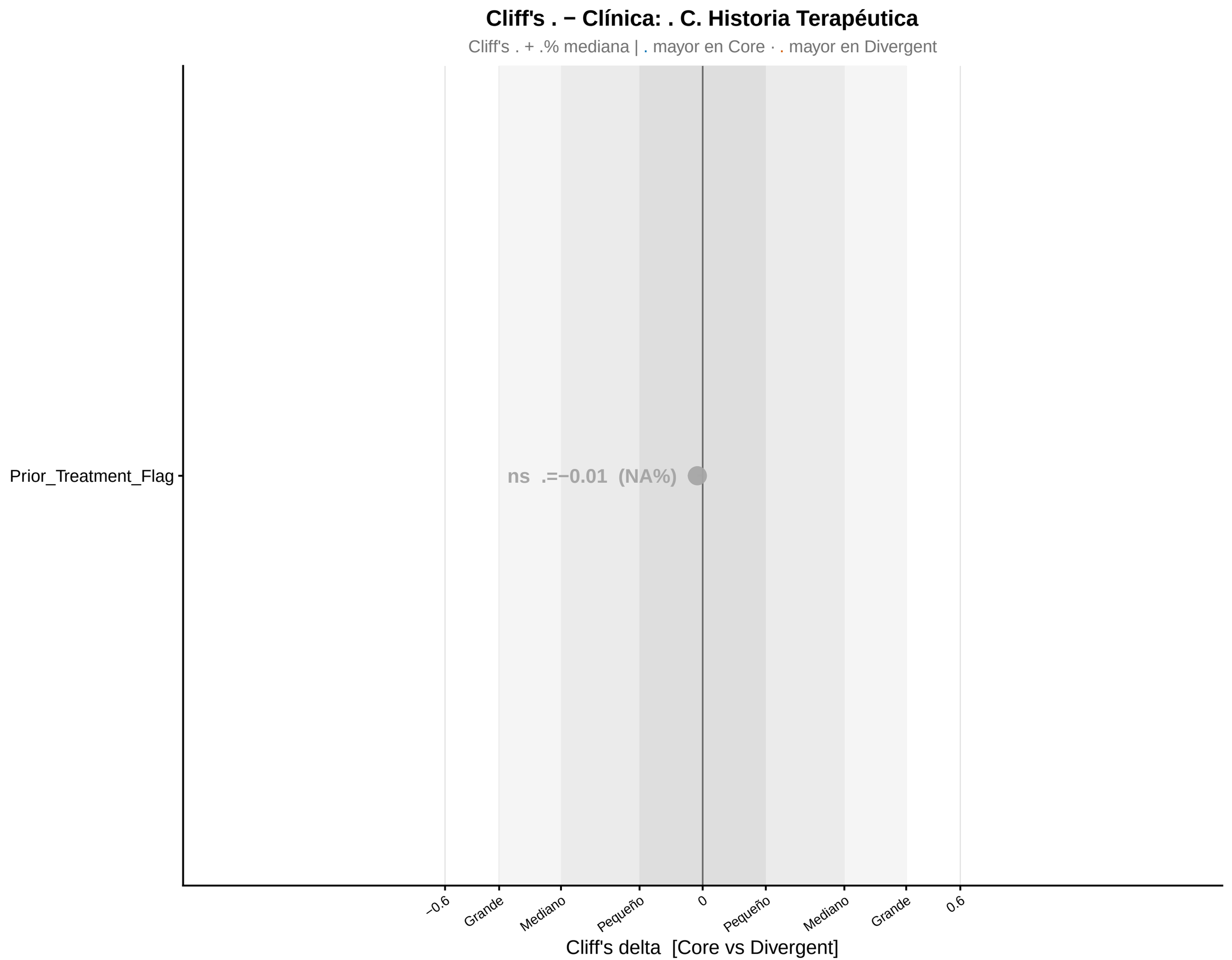
p−adj: 0.572 | .=−0.01 (Negligible) |



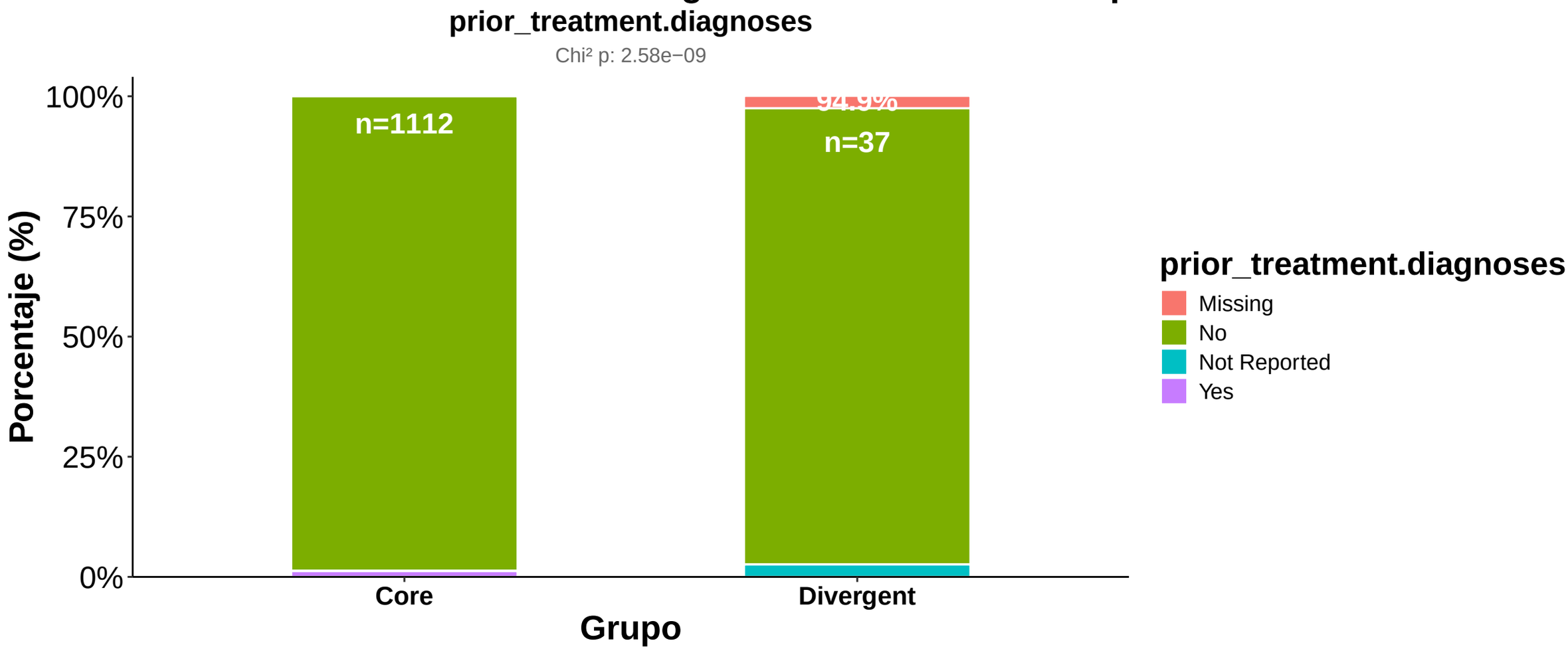
Cliff's .: Clínica: . C. Historia Terapéutica

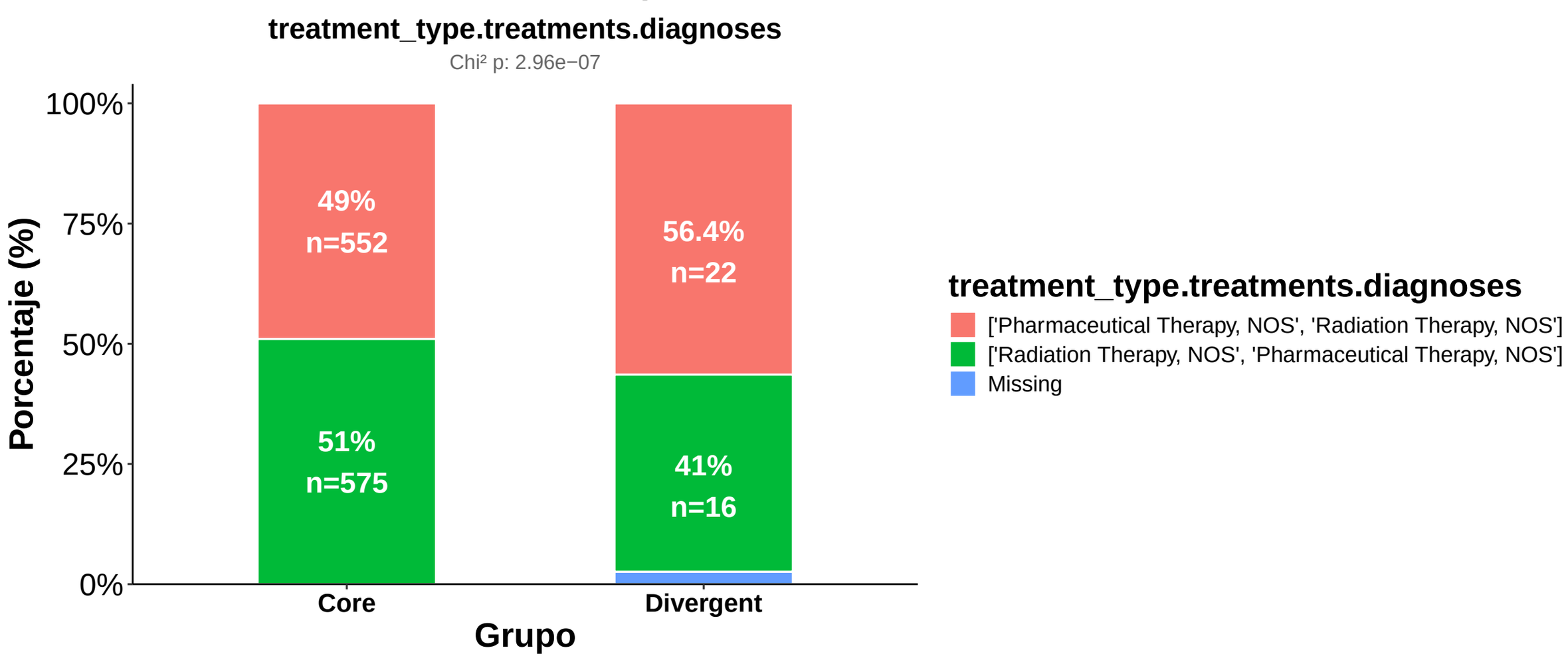
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05



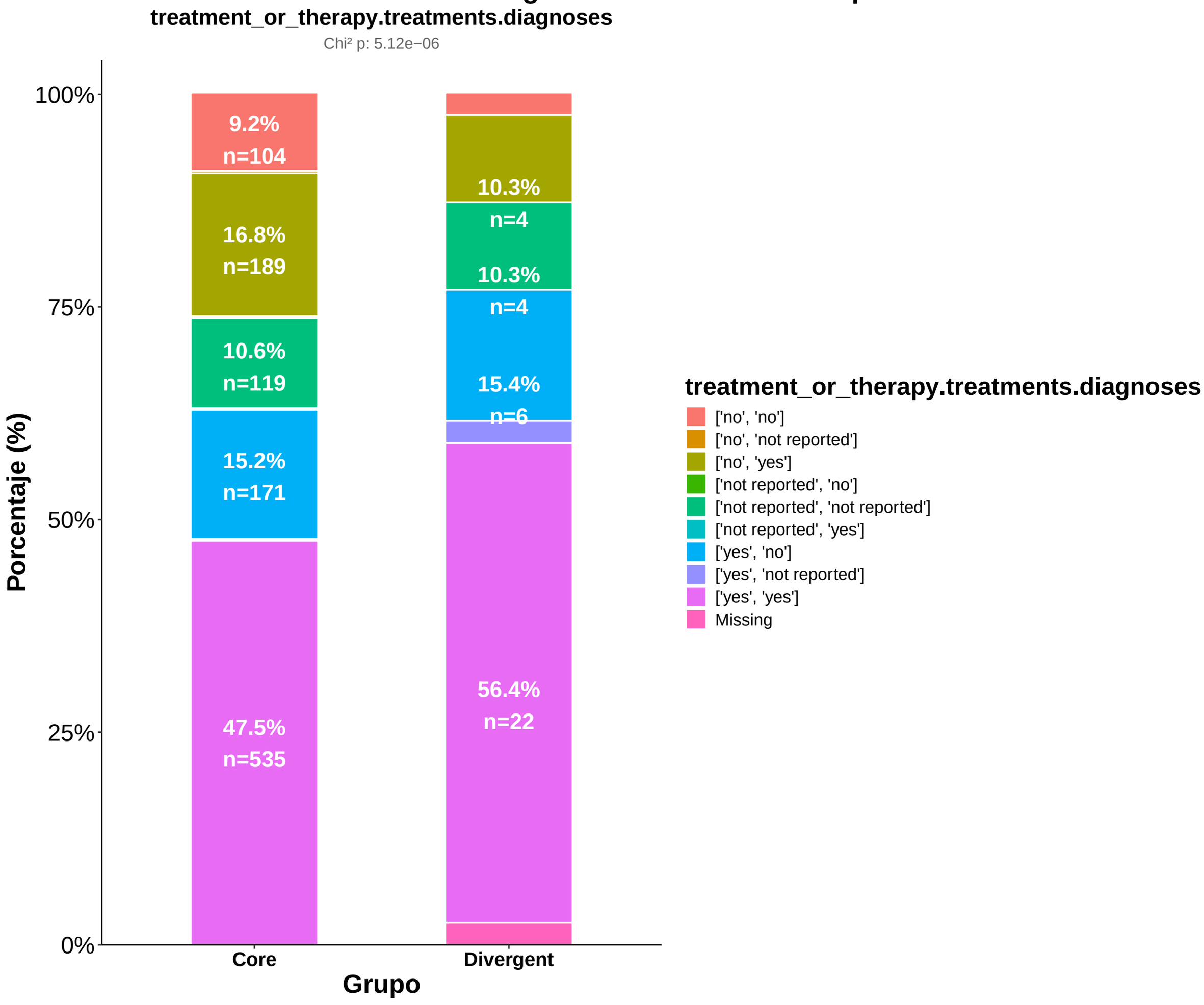

Cliff's . − Clínica: . C. Historia Terapéutica
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
Prior_Treatment_Flag
ns .=−0.01 (NA%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

Clínica Categórica − . C. Historia Terapéutica

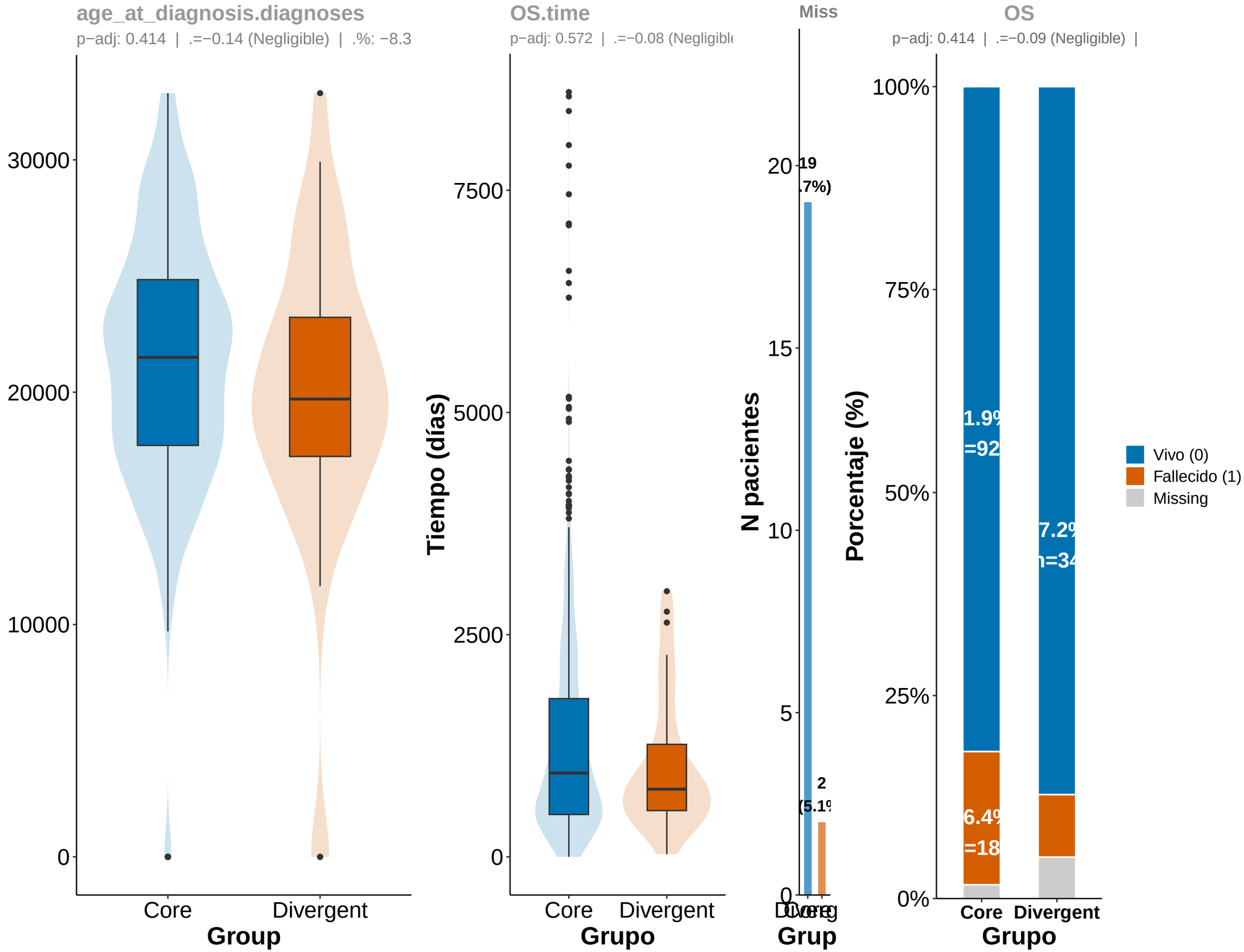




Clínica Categórica − . C. Historia Terapéutica

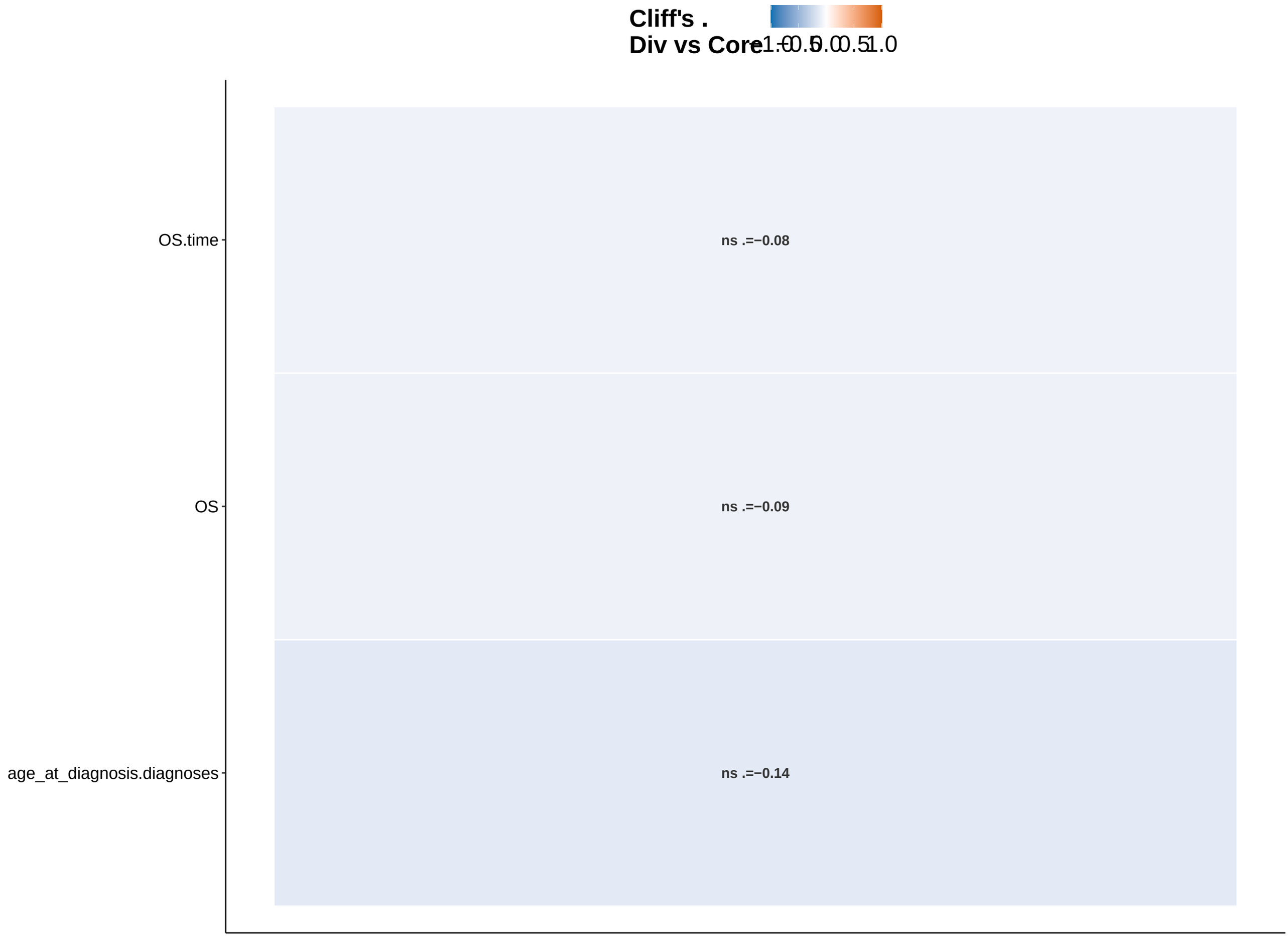


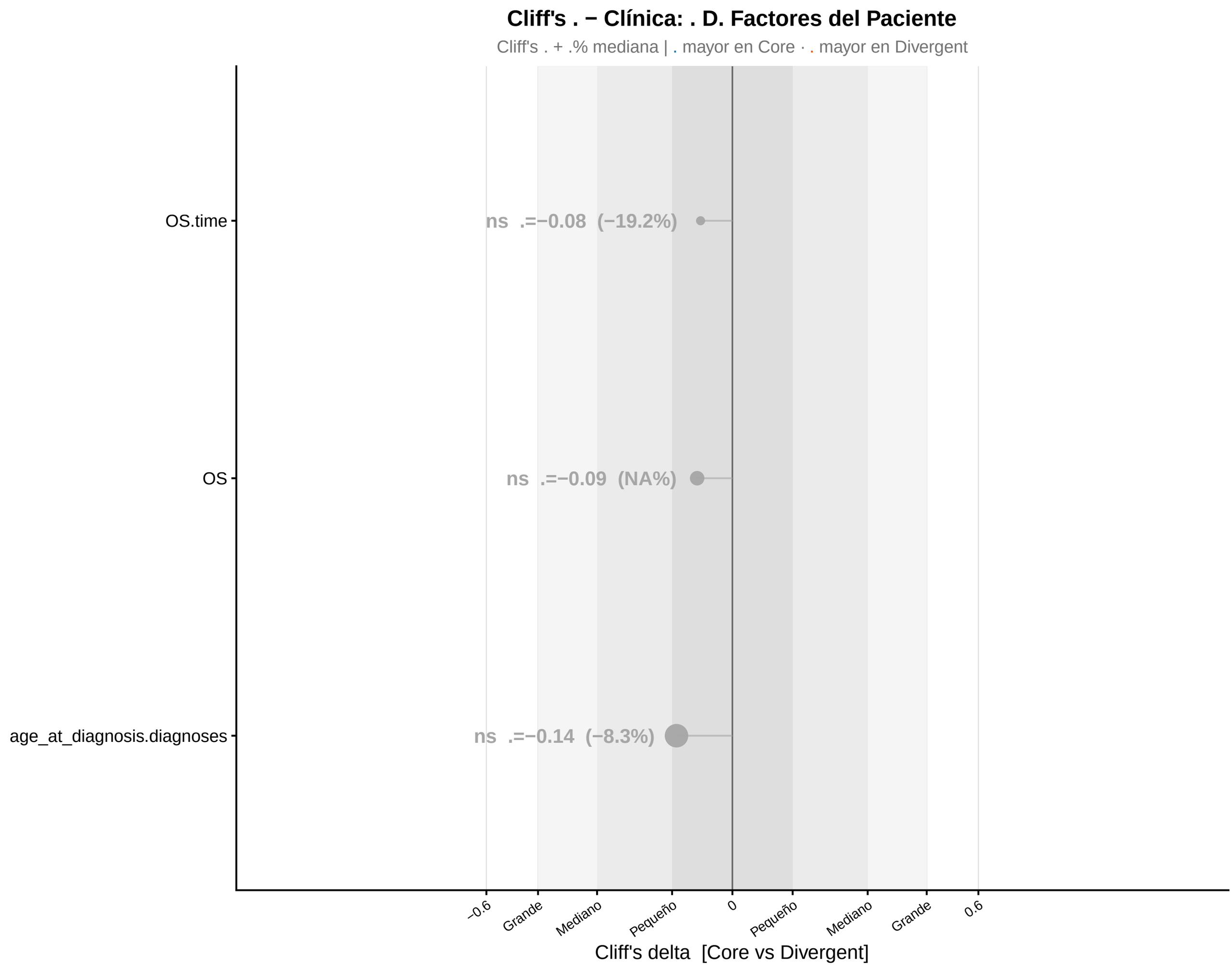
Clínica Numérica − . D. Factores del Paciente



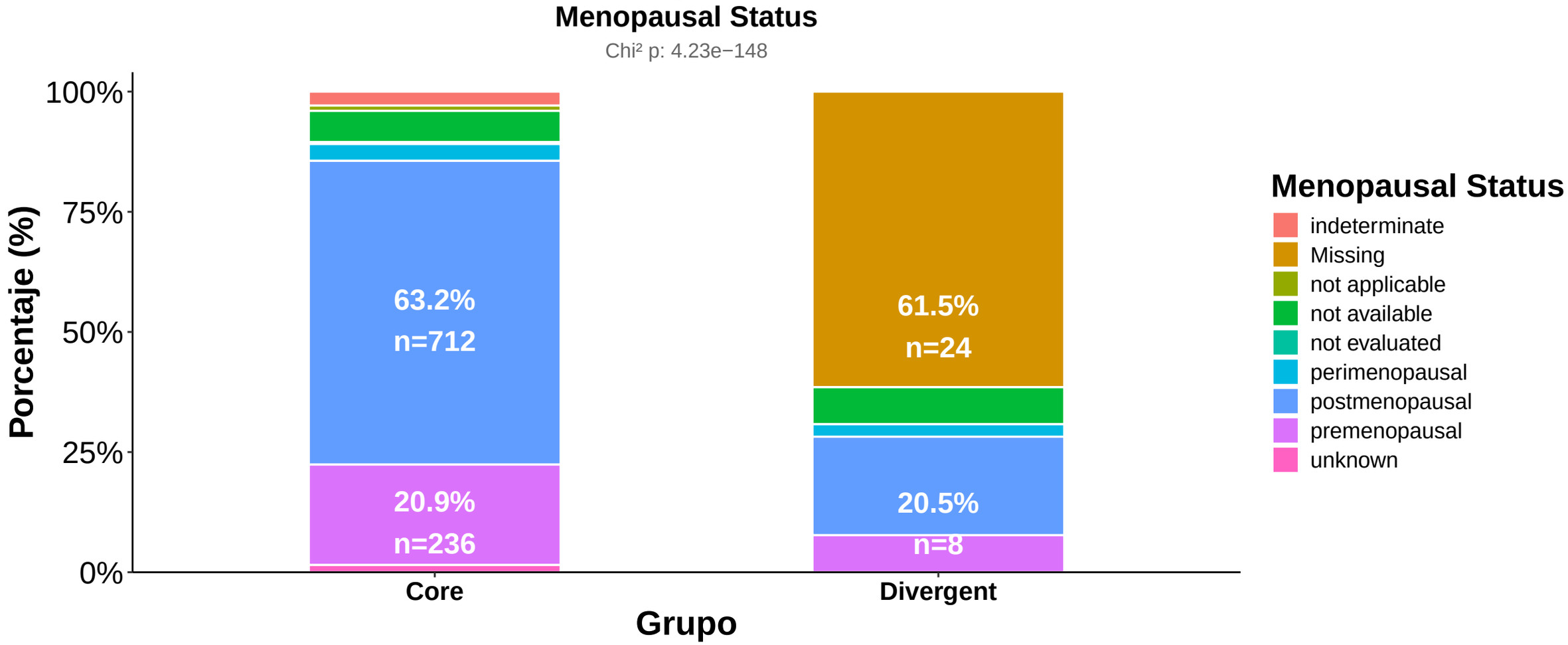
Cliff's .: Clínica: . D. Factores del Paciente

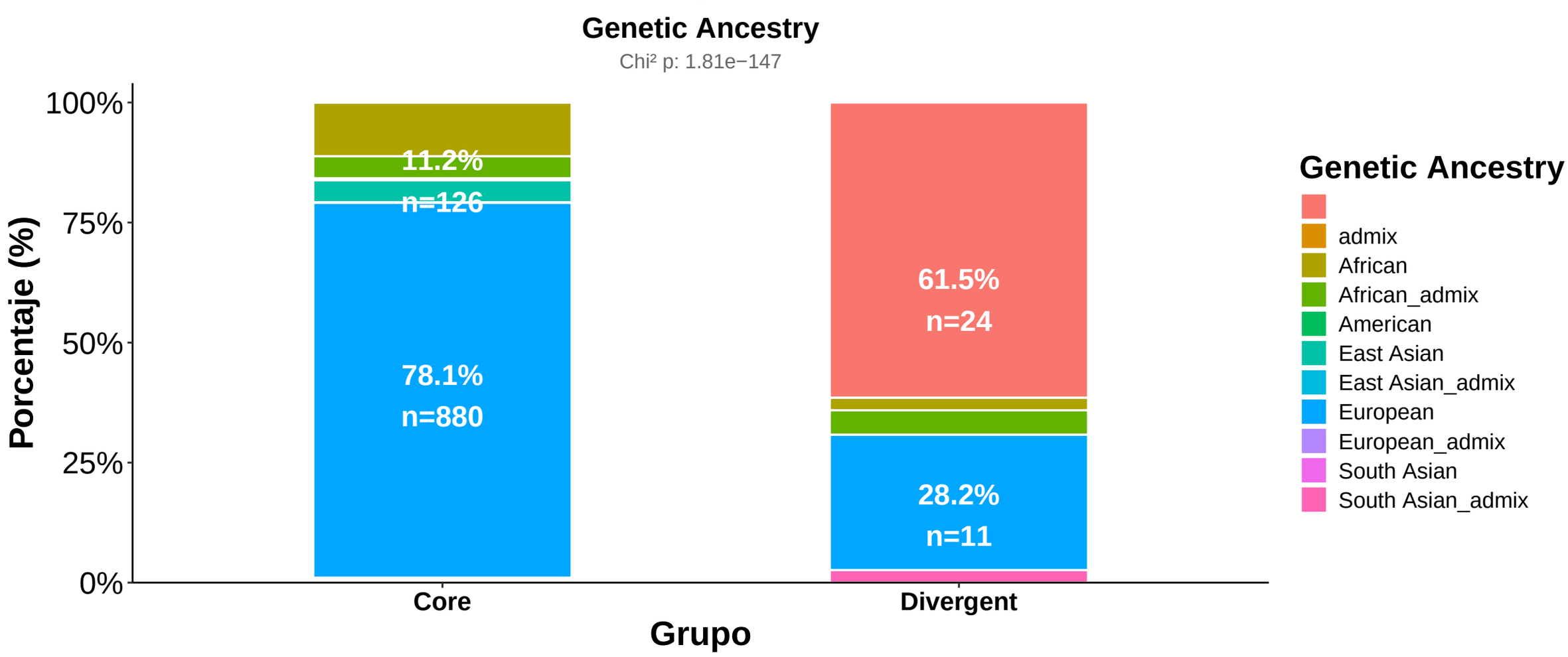
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05



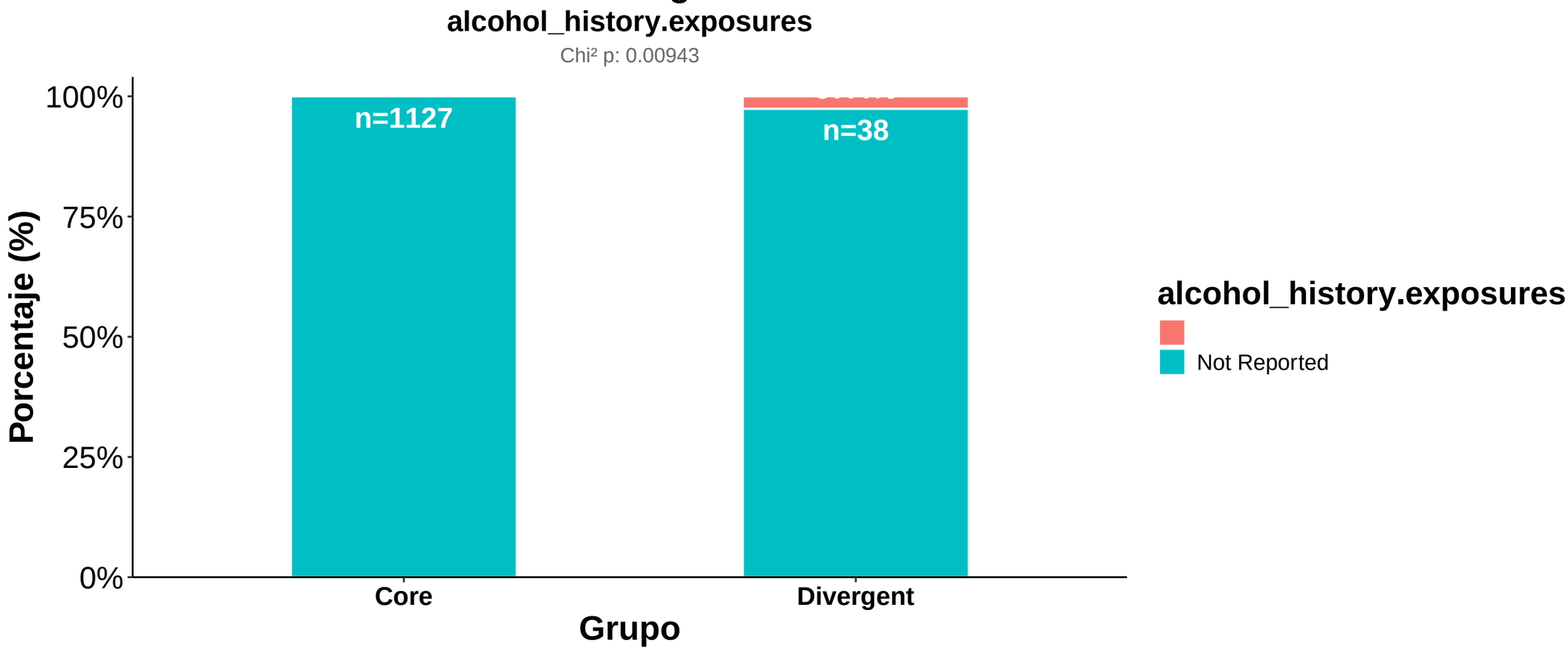

Cliff's . − Clínica: . D. Factores del Paciente
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
OS.time
ns .=−0.08 (−19.2%)
OS
ns .=−0.09 (NA%)
age_at_diagnosis.diagnoses
ns .=−0.14 (−8.3%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

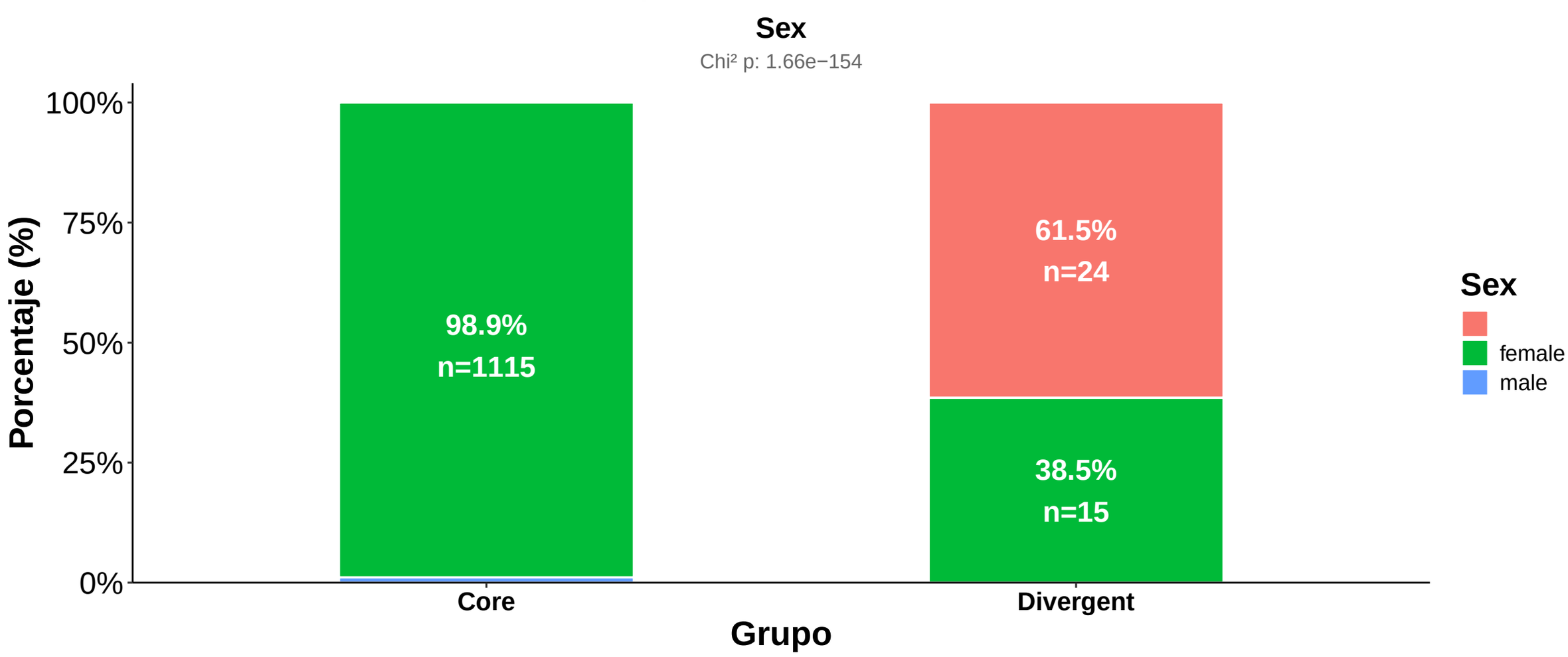
Clínica Categórica – . D. Factores del Paciente



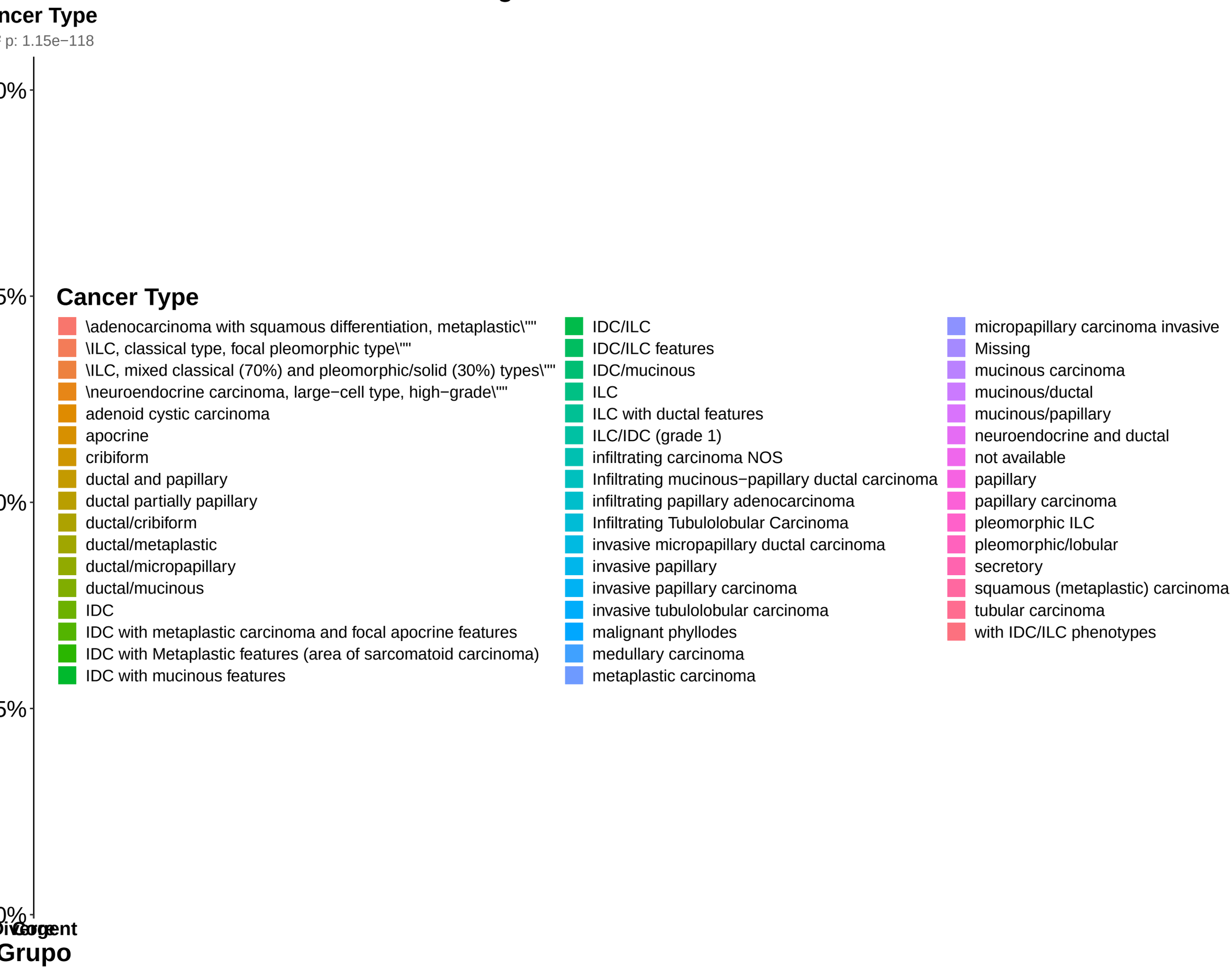


Clínica Categórica – . D. Factores del Paciente





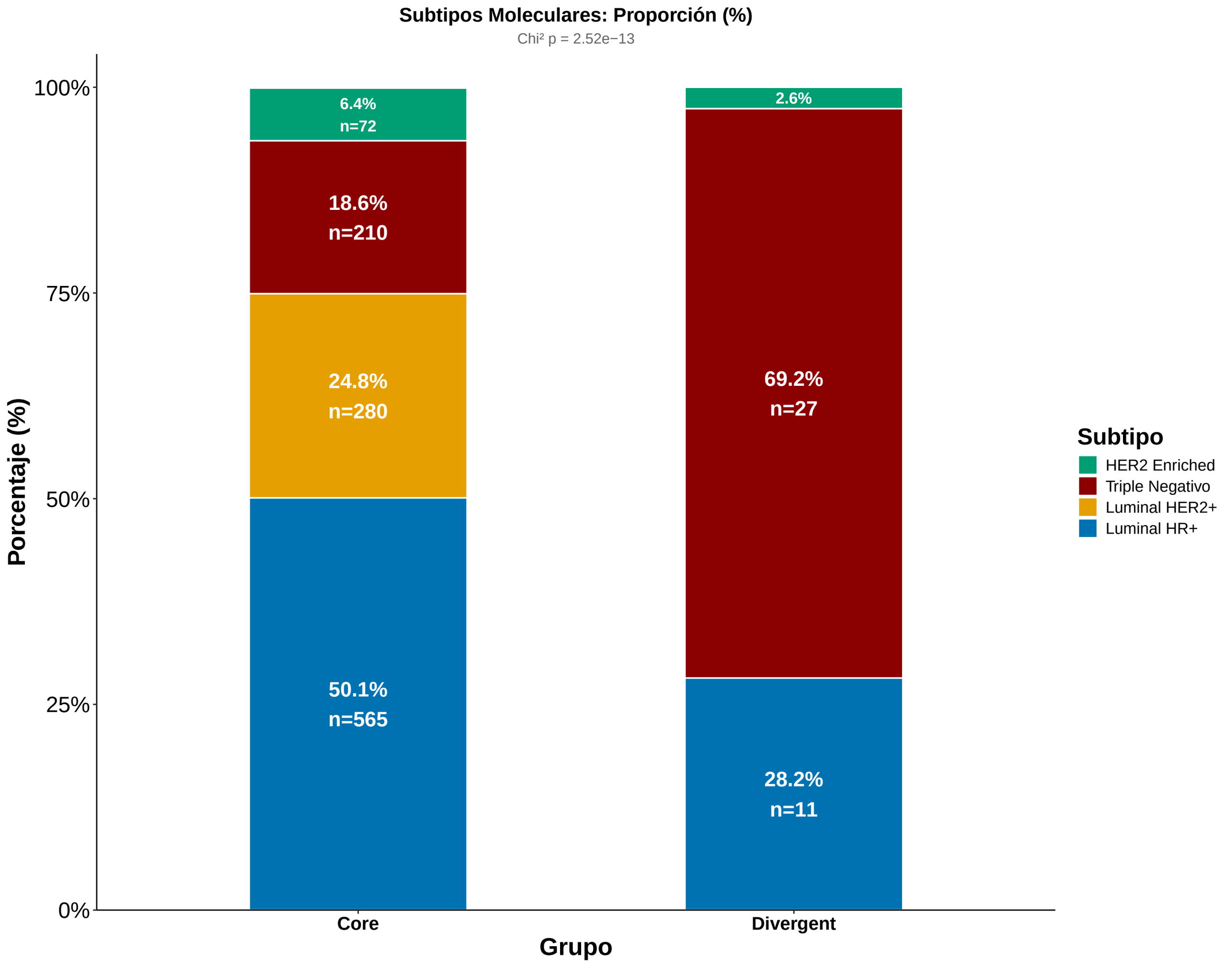
Clínica Categórica − . D. Factores del Paciente

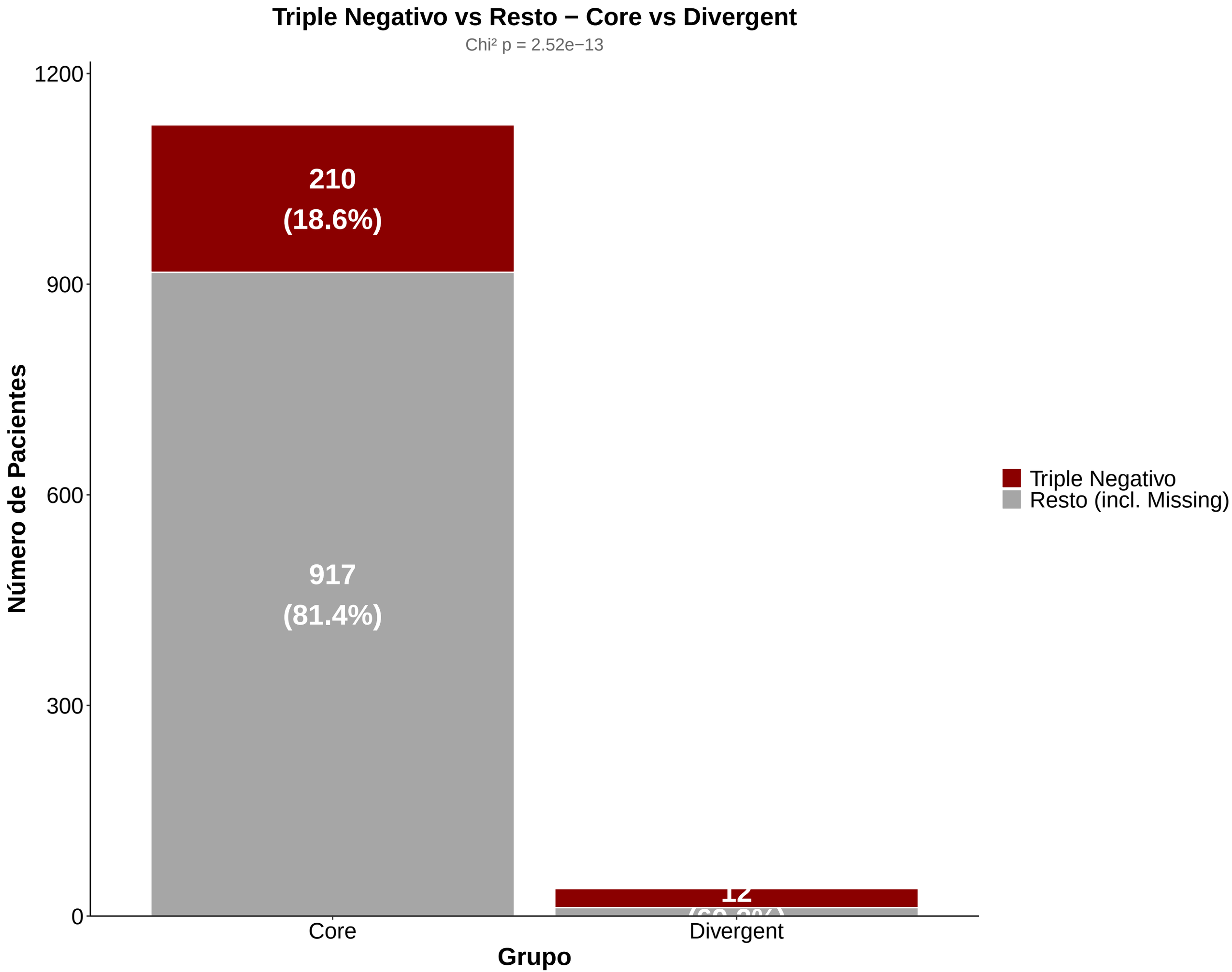


SECCIÓN I−B

Análisis de Subtipos Moleculares: Core vs Divergent

Fuente: ER_PR_HER2_Combo | Triple Negativo · Luminal HR+ · Luminal HER2+ · HER2 Enriched · Missing


Subtipos Moleculares: Proporción (%)
Chi² p = 2.52e−13
100%
75%
50%
25%
0%
Porcentaje (%)
6.4%
n=72
18.6%
n=210
24.8%
n=280
50.1%
n=565
2.6%
69.2%
n=27
28.2%
n=11
Core
Divergent
Grupo
Subtipo
HER2 Enriched
Triple Negativo
Luminal HER2+
Luminal HR+


Triple Negativo vs Resto − Core vs Divergent
Chi² p = 2.52e−13
1200
900
600
300
0
Número de Pacientes
210
(18.6%)
917
(81.4%)
12
Core
Divergent
Grupo
Triple Negativo
Resto (incl. Missing)

Resumen Triple Negativo (fuente: ER_PR_HER2_Combo):

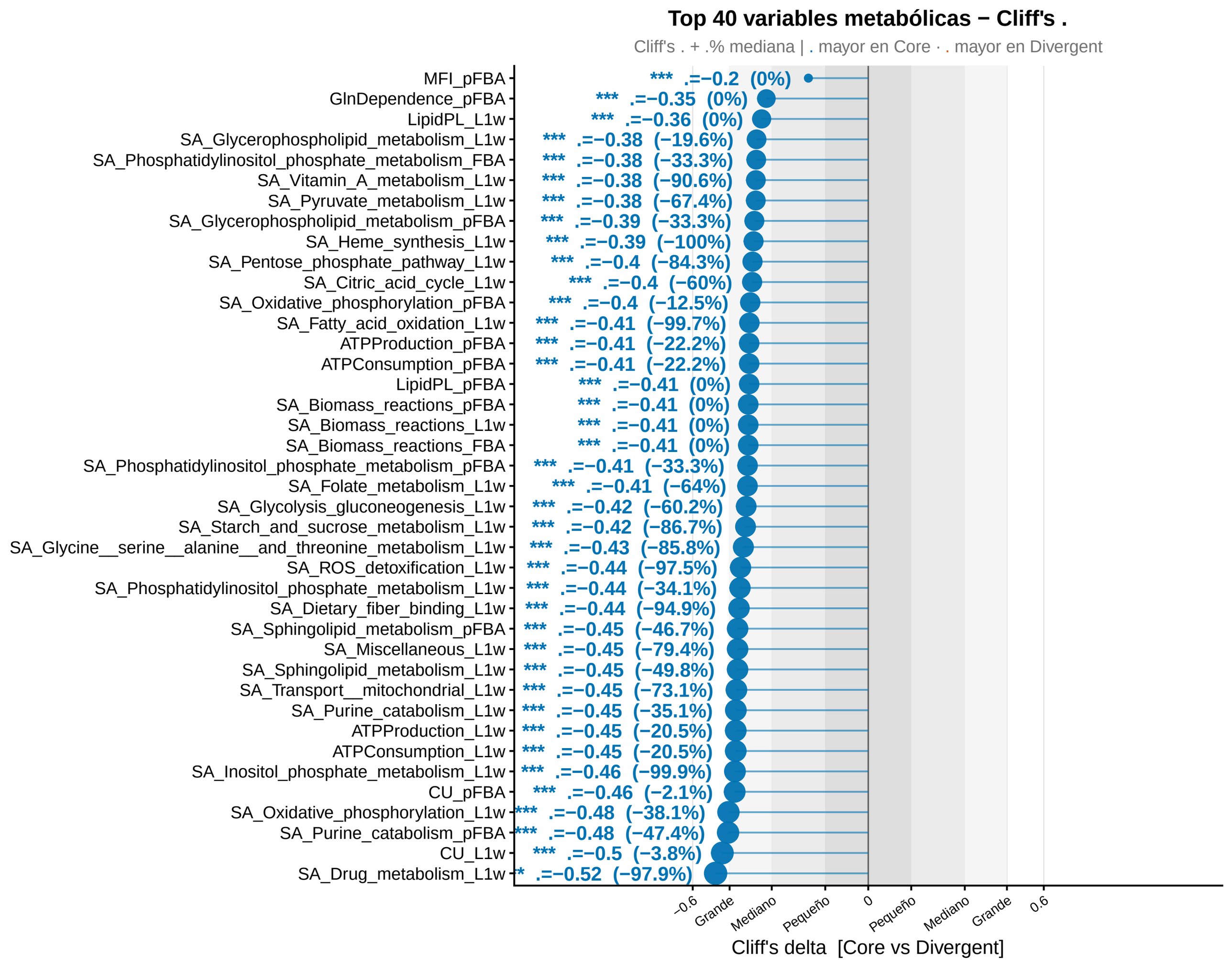
Core: 210 TNBC de 1127 totales (18.6%)
Divergent: 27 TNBC de 39 totales (69.2%)

Chi² test subtipos vs Grupo: Chi² p = 2.52e−13

SECCIÓN II

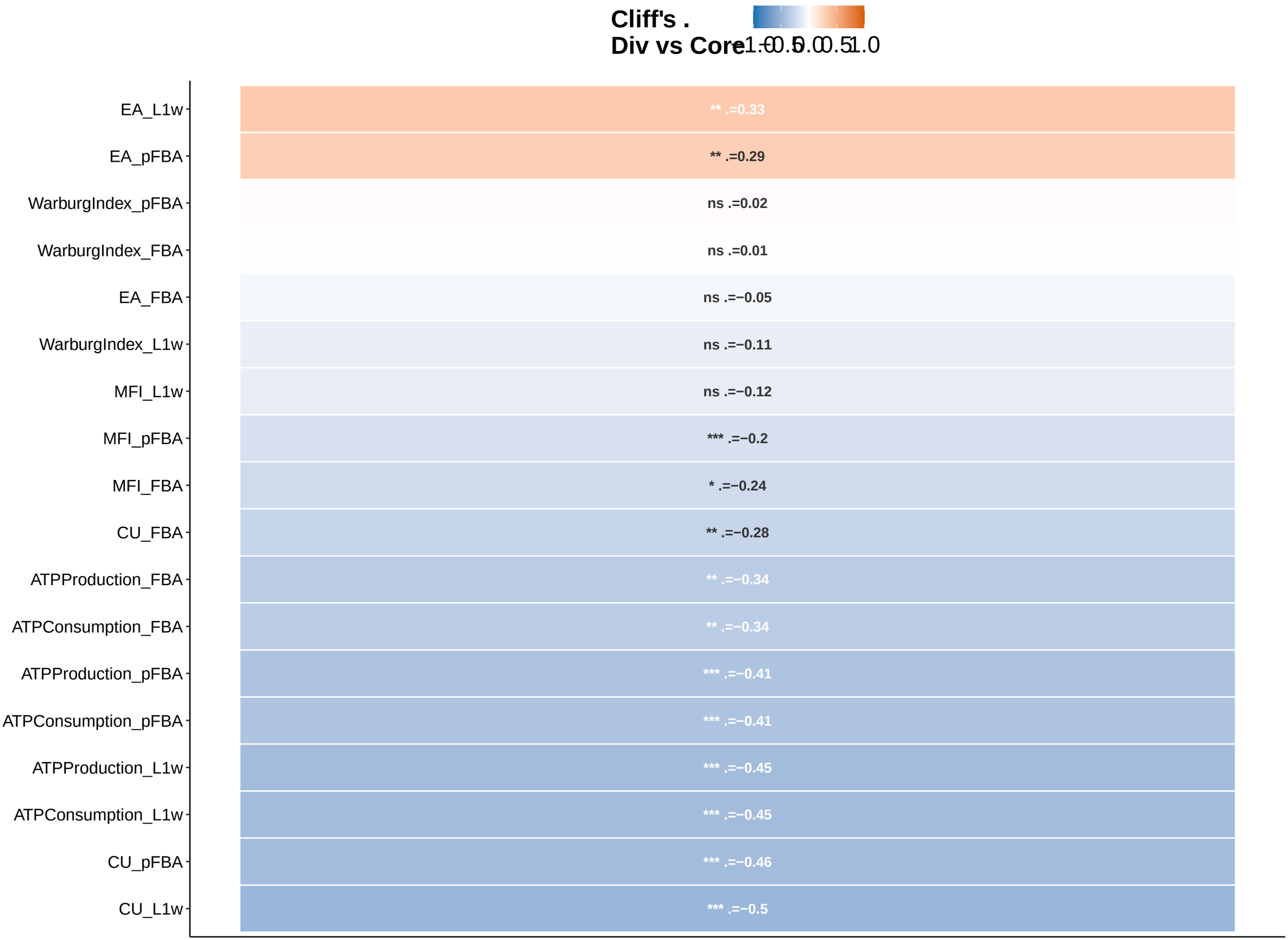
Diferencias Metabólicas: Core vs Divergent

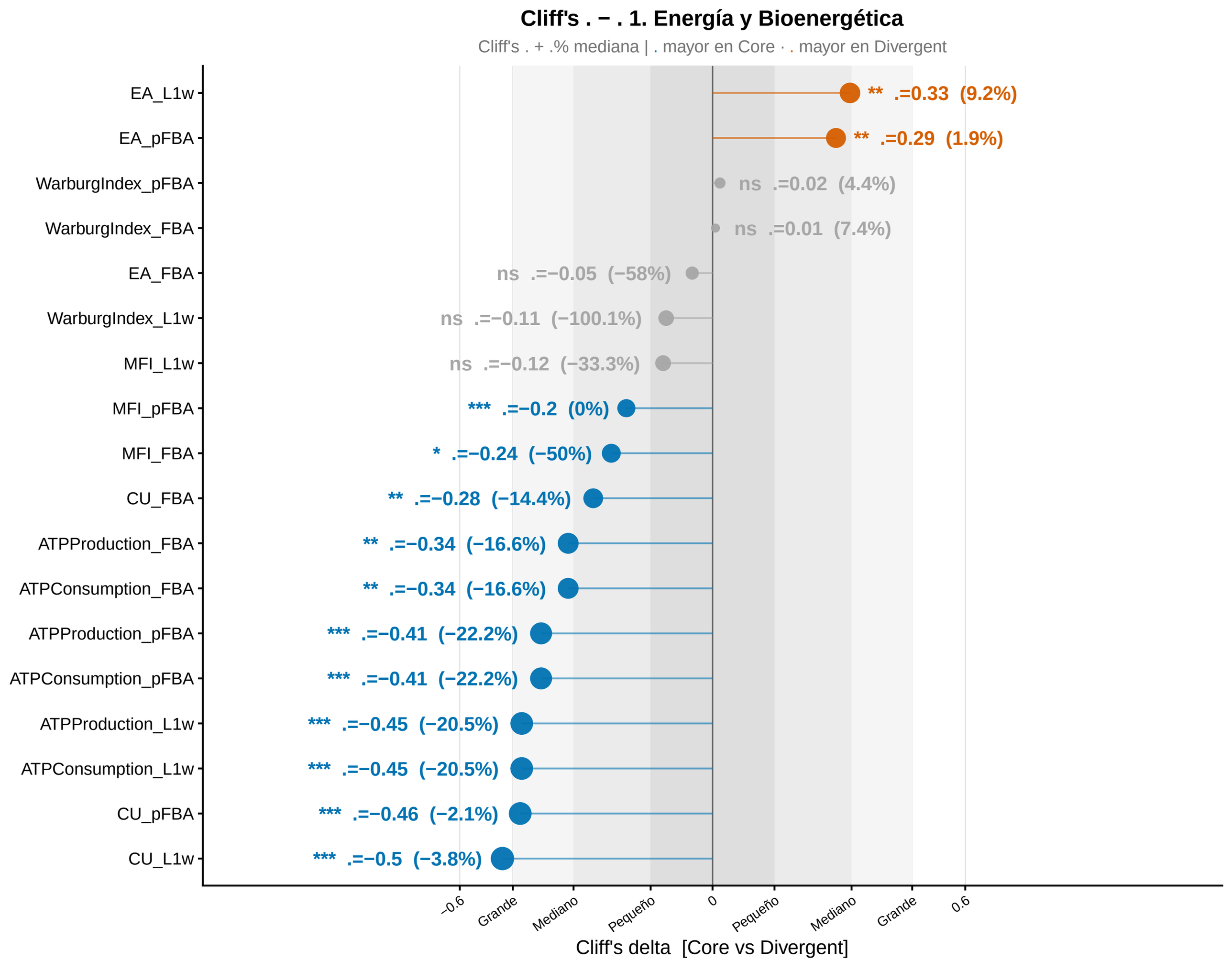
5 bloques funcionales: Energía | Redox | Lípidos | Dependencias | Reprogramación Sistémica


Top 40 variables metabólicas − Cliff's .
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
MFI_pFBA *** .=−0.2 (0%)
GlnDependence_pFBA *** .=−0.35 (0%)
LipidPL_L1w *** .=−0.36 (0%)
SA_Glycerophospholipid_metabolism_L1w *** .=−0.38 (−19.6%)
SA_Phosphatidylinositol_phosphate_metabolism_FBA *** .=−0.38 (−33.3%)
SA_Vitamin_A_metabolism_L1w *** .=−0.38 (−90.6%)
SA_Pyruvate_metabolism_L1w *** .=−0.38 (−67.4%)
SA_Glycerophospholipid_metabolism_pFBA *** .=−0.39 (−33.3%)
SA_Heme_synthesis_L1w *** .=−0.39 (−100%)
SA_Pentose_phosphate_pathway_L1w *** .=−0.4 (−84.3%)
SA_Citric_acid_cycle_L1w *** .=−0.4 (−60%)
SA_Oxidative_phosphorylation_pFBA *** .=−0.4 (−12.5%)
SA_Fatty_acid_oxidation_L1w *** .=−0.41 (−99.7%)
ATPProduction_pFBA *** .=−0.41 (−22.2%)
ATPConsumption_pFBA *** .=−0.41 (−22.2%)
LipidPL_pFBA *** .=−0.41 (0%)
SA_Biomass_reactions_pFBA *** .=−0.41 (0%)
SA_Biomass_reactions_L1w *** .=−0.41 (0%)
SA_Biomass_reactions_FBA *** .=−0.41 (0%)
SA_Phosphatidylinositol_phosphate_metabolism_pFBA *** .=−0.41 (−33.3%)
SA_Folate_metabolism_L1w *** .=−0.41 (−64%)
SA_Glycolysis_gluconeogenesis_L1w *** .=−0.42 (−60.2%)
SA_Starch_and_sucrose_metabolism_L1w *** .=−0.42 (−86.7%)
SA_Glycine__serine__alanine__and_threonine_metabolism_L1w *** .=−0.43 (−85.8%)
SA_ROS_detoxification_L1w *** .=−0.44 (−97.5%)
SA_Phosphatidylinositol_phosphate_metabolism_L1w *** .=−0.44 (−34.1%)
SA_Dietary_fiber_binding_L1w *** .=−0.44 (−94.9%)
SA_Sphingolipid_metabolism_pFBA *** .=−0.45 (−46.7%)
SA_Miscellaneous_L1w *** .=−0.45 (−79.4%)
SA_Sphingolipid_metabolism_L1w *** .=−0.45 (−49.8%)
SA_Transport__mitochondrial_L1w *** .=−0.45 (−73.1%)
SA_Purine_catabolism_L1w *** .=−0.45 (−35.1%)
ATPProduction_L1w *** .=−0.45 (−20.5%)
ATPConsumption_L1w *** .=−0.45 (−20.5%)
SA_Inositol_phosphate_metabolism_L1w *** .=−0.46 (−99.9%)
CU_pFBA *** .=−0.46 (−2.1%)
SA_Oxidative_phosphorylation_L1w *** .=−0.48 (−38.1%)
SA_Purine_catabolism_pFBA *** .=−0.48 (−47.4%)
CU_L1w *** .=−0.5 (−3.8%)
SA_Drug_metabolism_L1w * .=−0.52 (−97.9%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

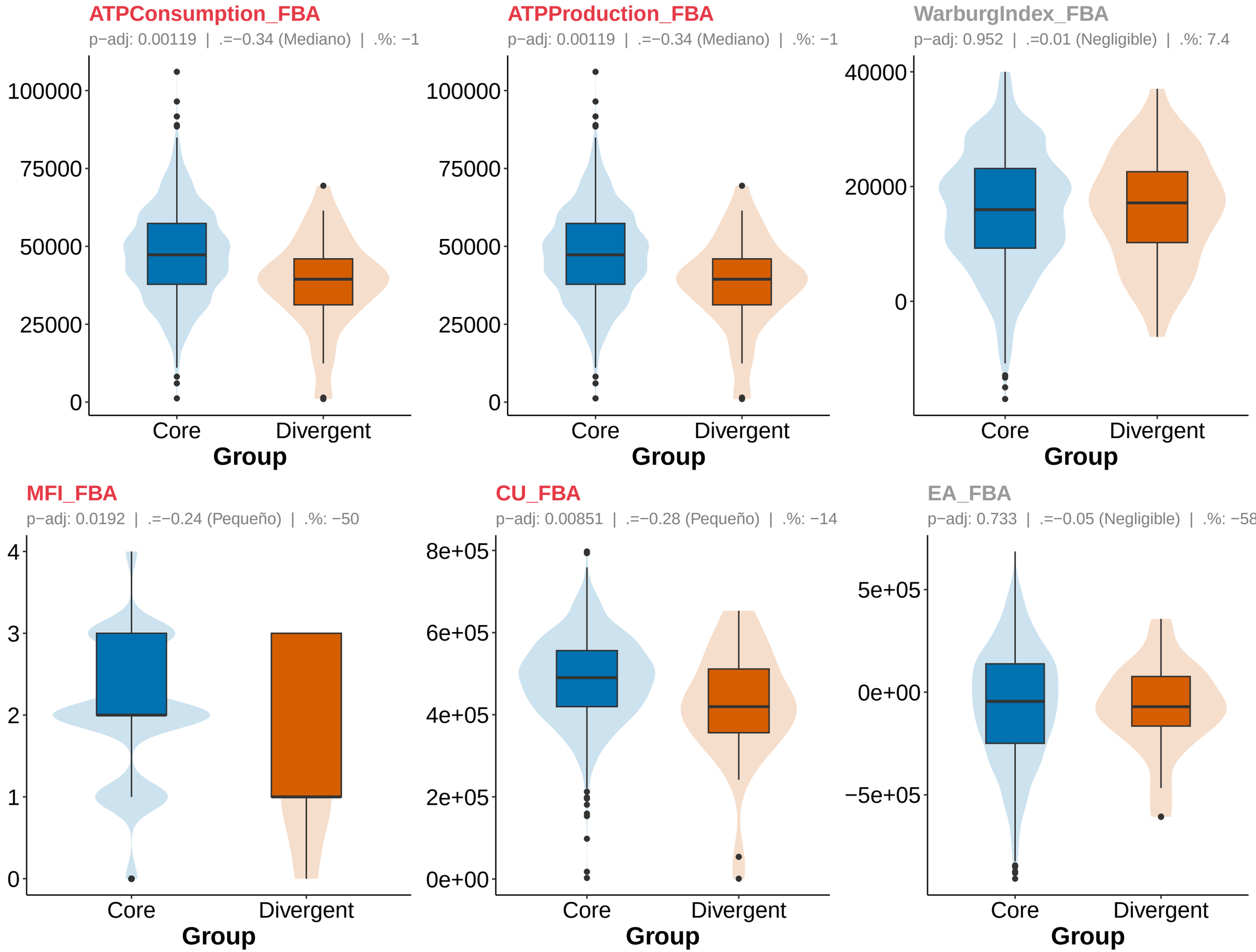
Cliff's .: . 1. Energía y Bioenergética

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

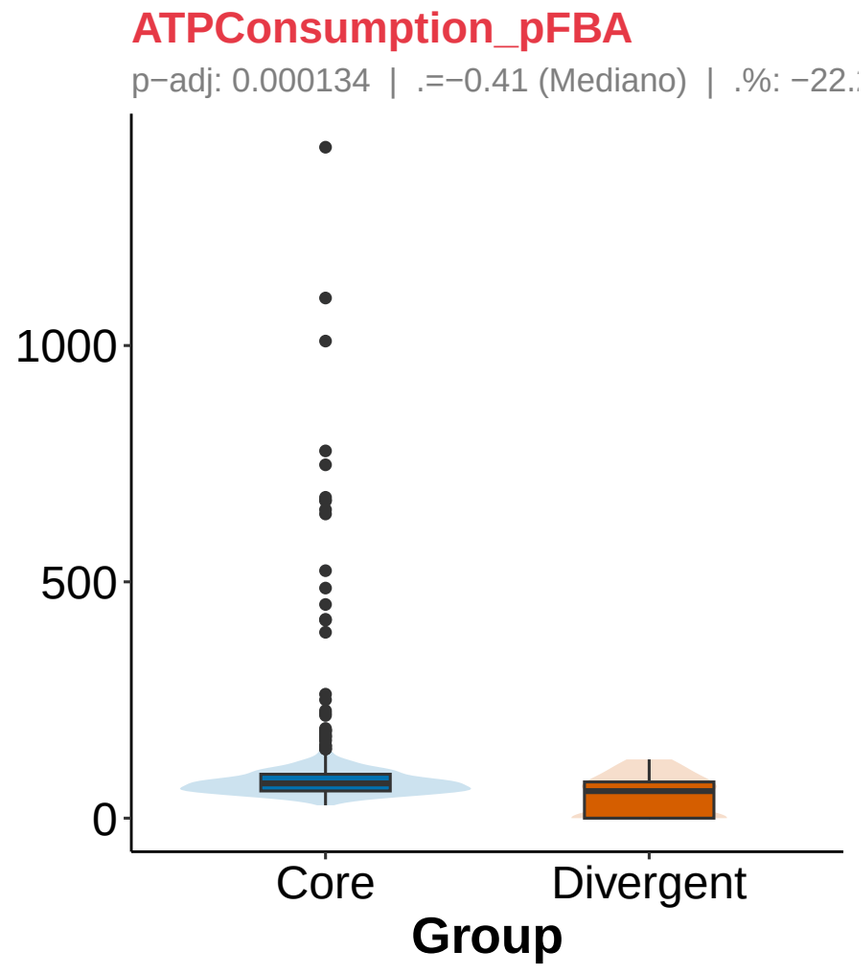


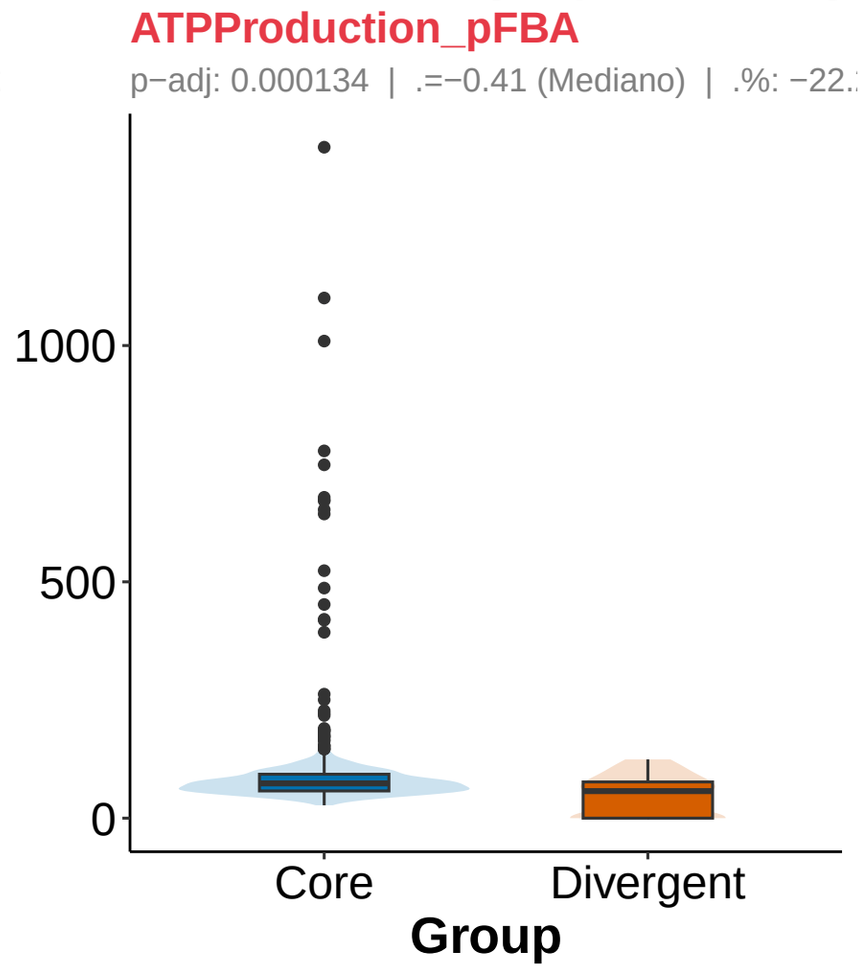

Cliff's . − . 1. Energía y Bioenergética
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
EA_L1w
EA_pFBA
WarburgIndex_pFBA
WarburgIndex_FBA
EA_FBA
WarburgIndex_L1w
MFI_L1w
MFI_pFBA
MFI_FBA
CU_FBA
ATPProduction_FBA
ATPConsumption_FBA
ATPProduction_pFBA
ATPConsumption_pFBA
ATPProduction_L1w
ATPConsumption_L1w
CU_pFBA
CU_L1w
** .=0.33 (9.2%)
** .=0.29 (1.9%)
ns .=0.02 (4.4%)
ns .=0.01 (7.4%)
ns .=−0.05 (−58%)
ns .=−0.11 (−100.1%)
ns .=−0.12 (−33.3%)
*** .=−0.2 (0%)
* .=−0.24 (−50%)
** .=−0.28 (−14.4%)
** .=−0.34 (−16.6%)
** .=−0.34 (−16.6%)
*** .=−0.41 (−22.2%)
*** .=−0.41 (−22.2%)
*** .=−0.45 (−20.5%)
*** .=−0.45 (−20.5%)
*** .=−0.46 (−2.1%)
*** .=−0.5 (−3.8%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

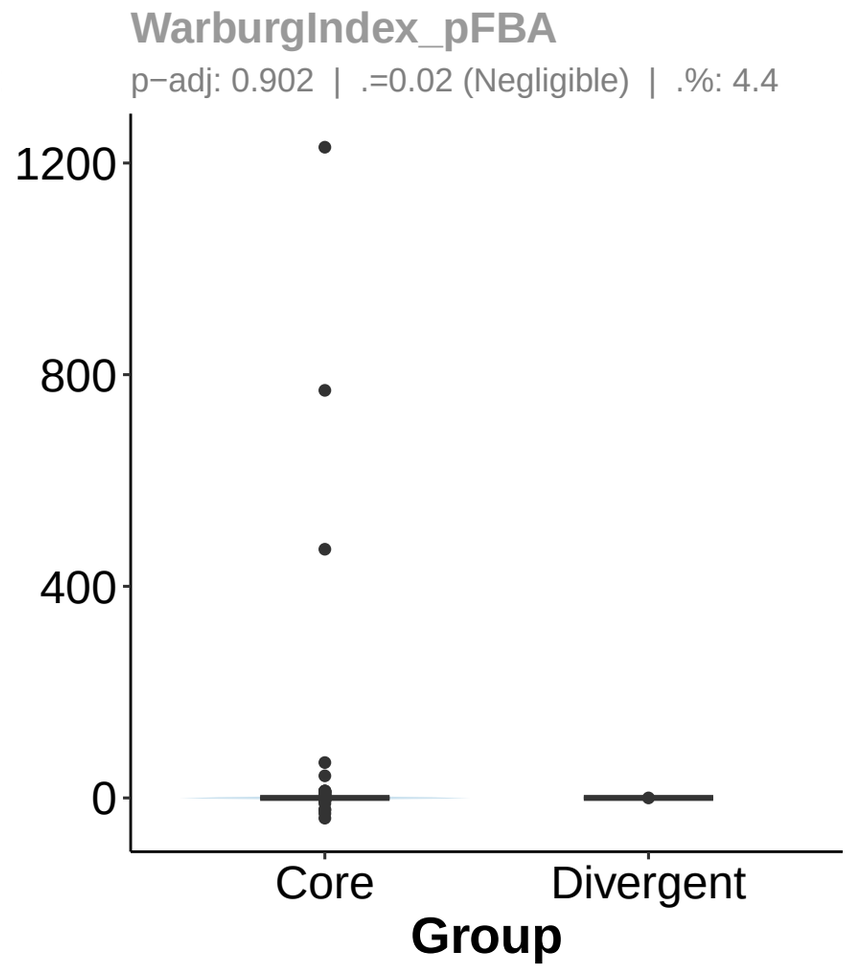
Metabolismo − . 1. Energía y Bioenergética

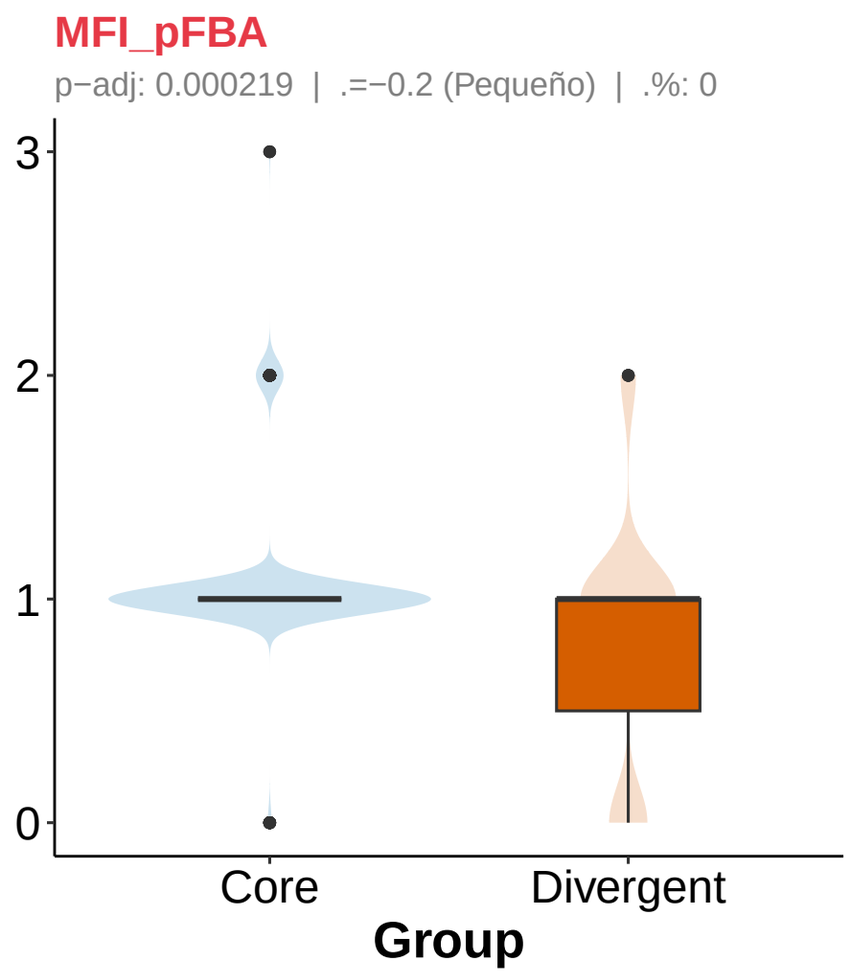


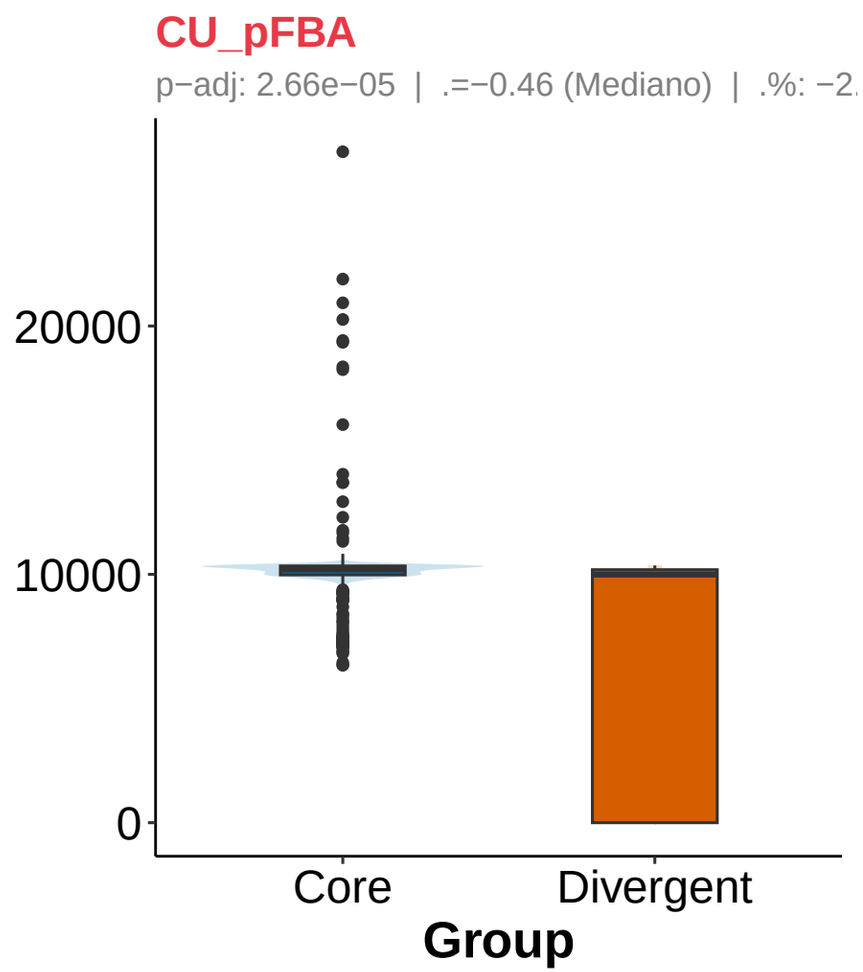
Metabolismo − . 1. Energía y Bioenergética

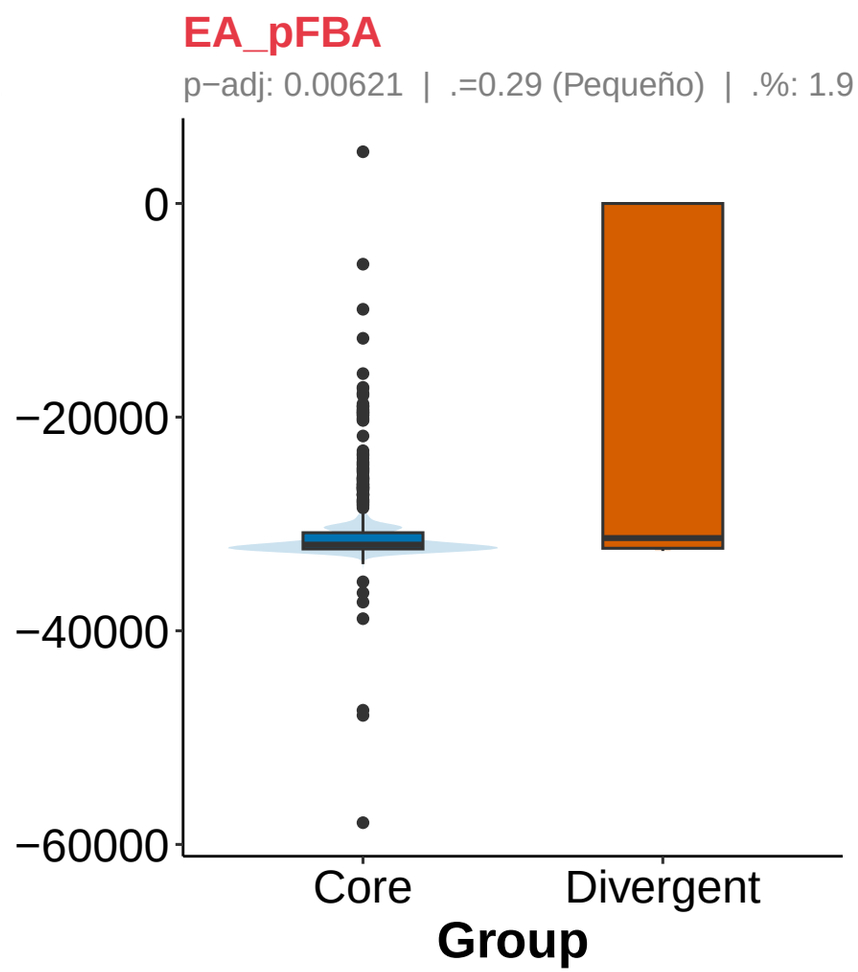




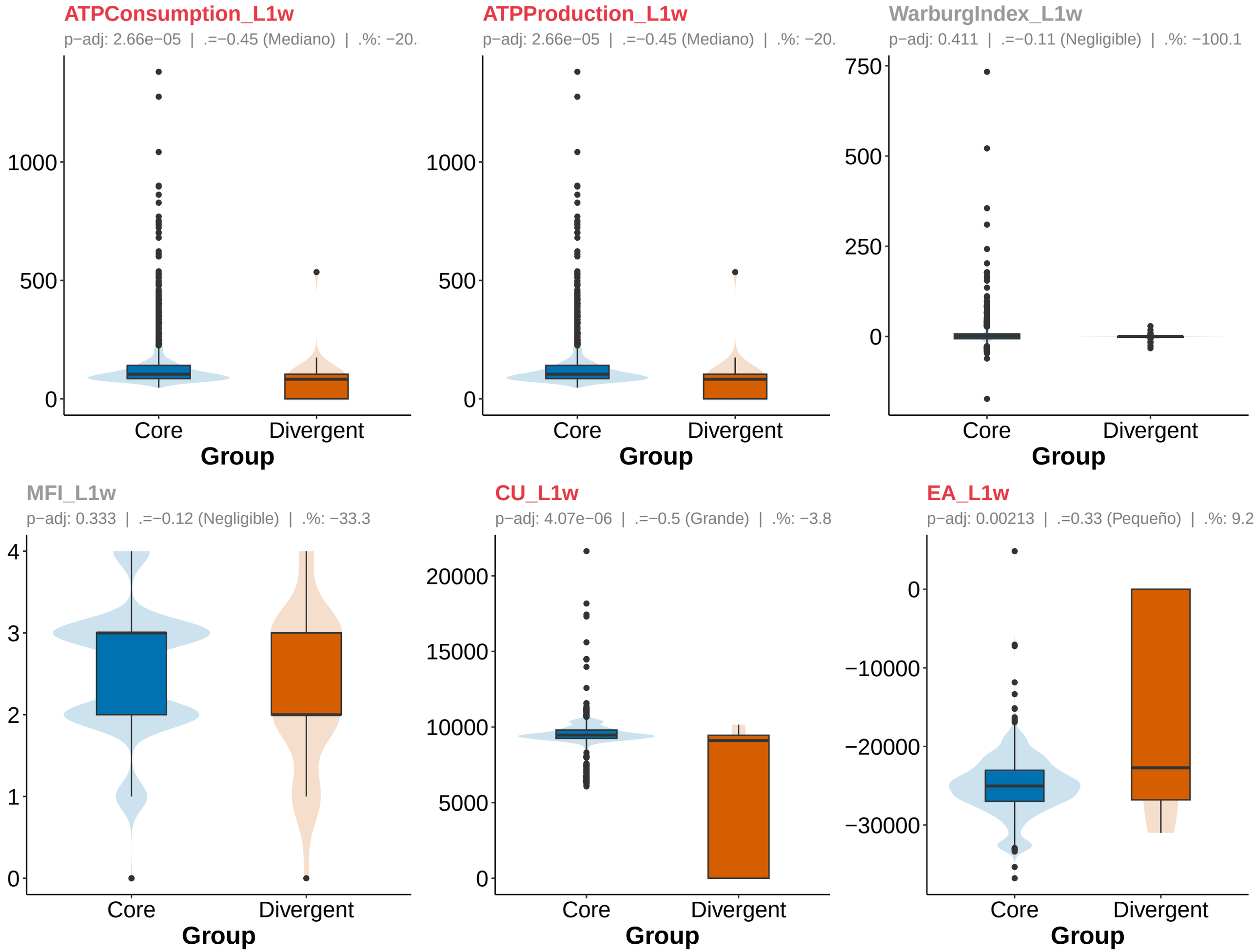






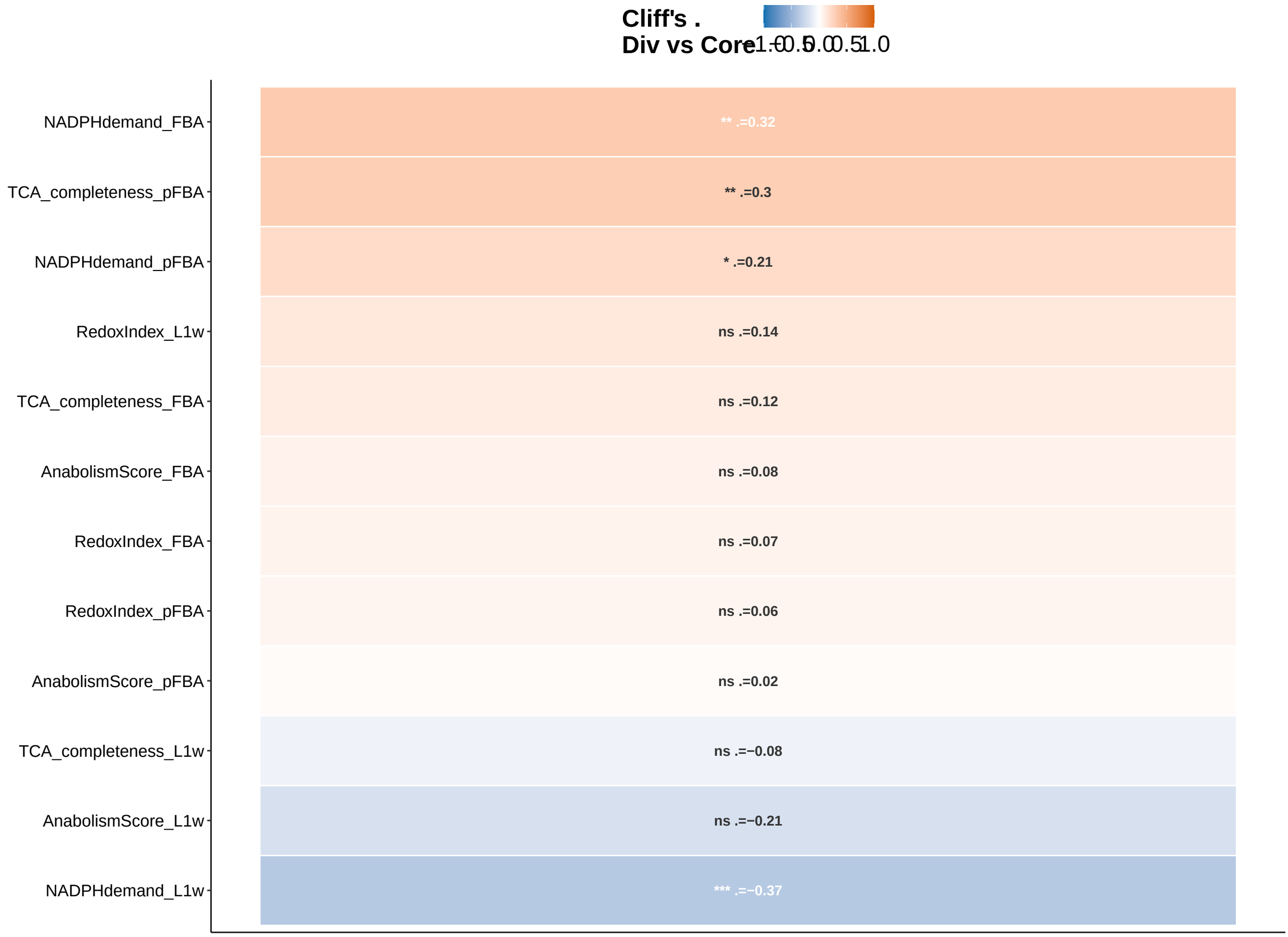


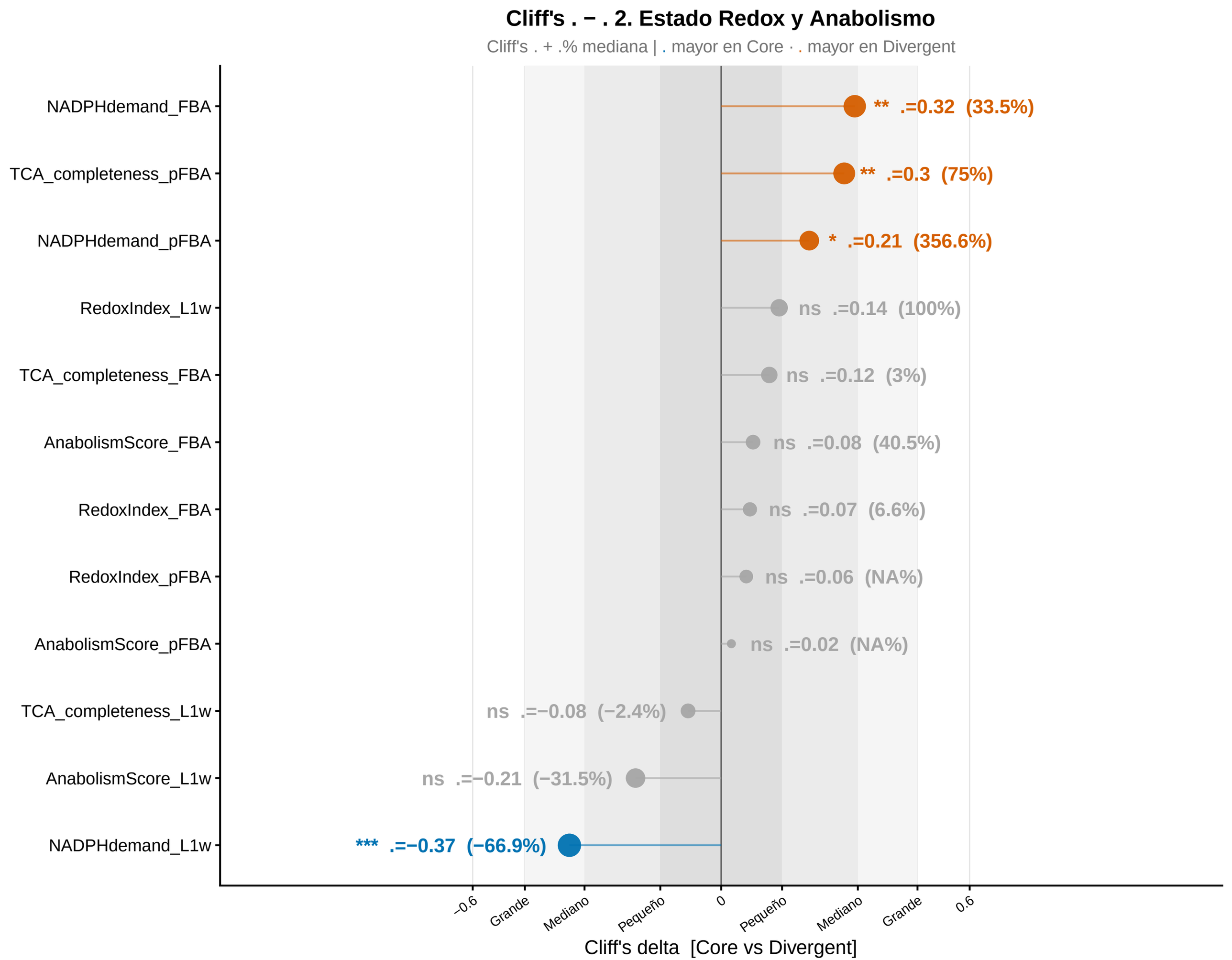
Metabolismo − . 1. Energía y Bioenergética



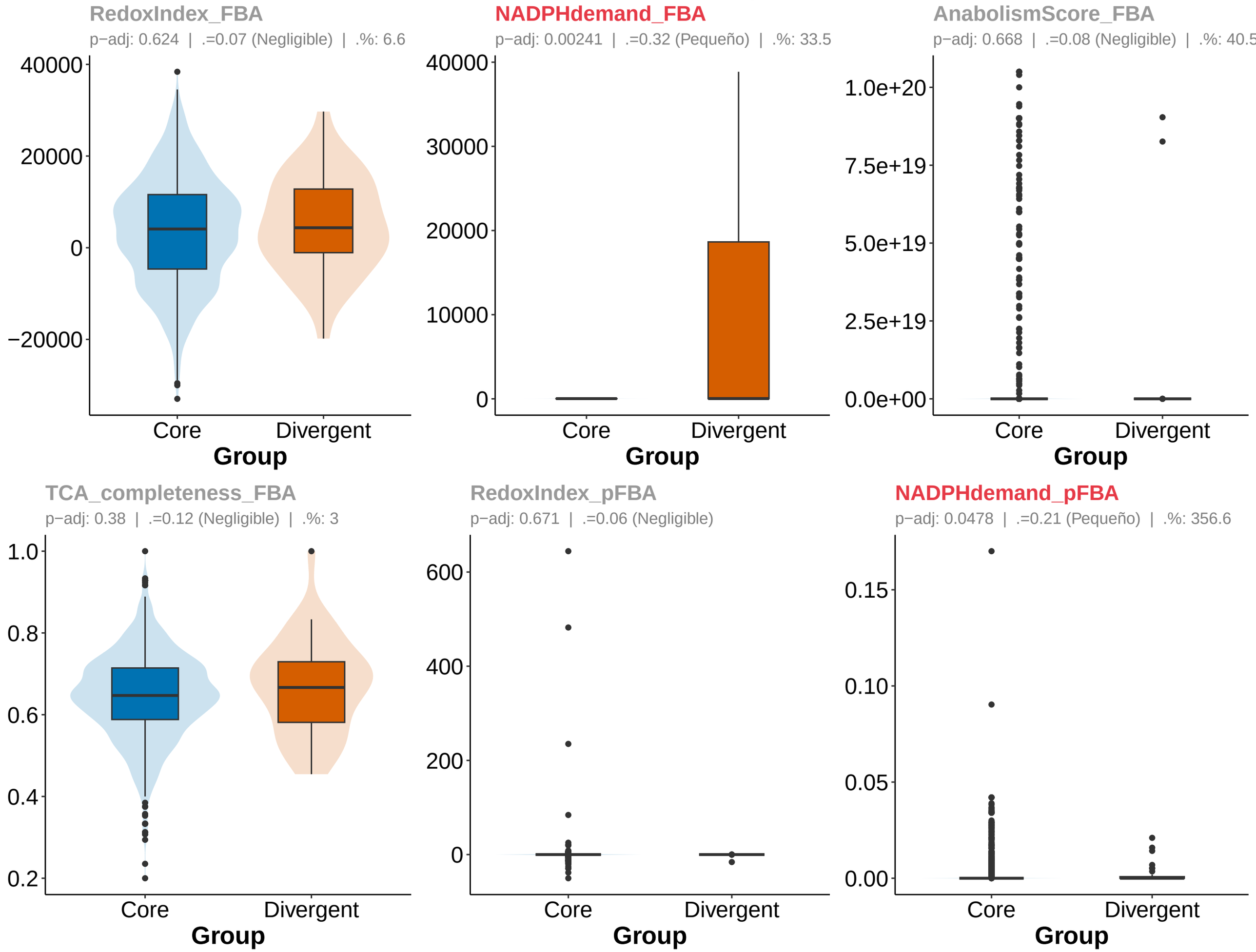
Cliff's .: . 2. Estado Redox y Anabolismo

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

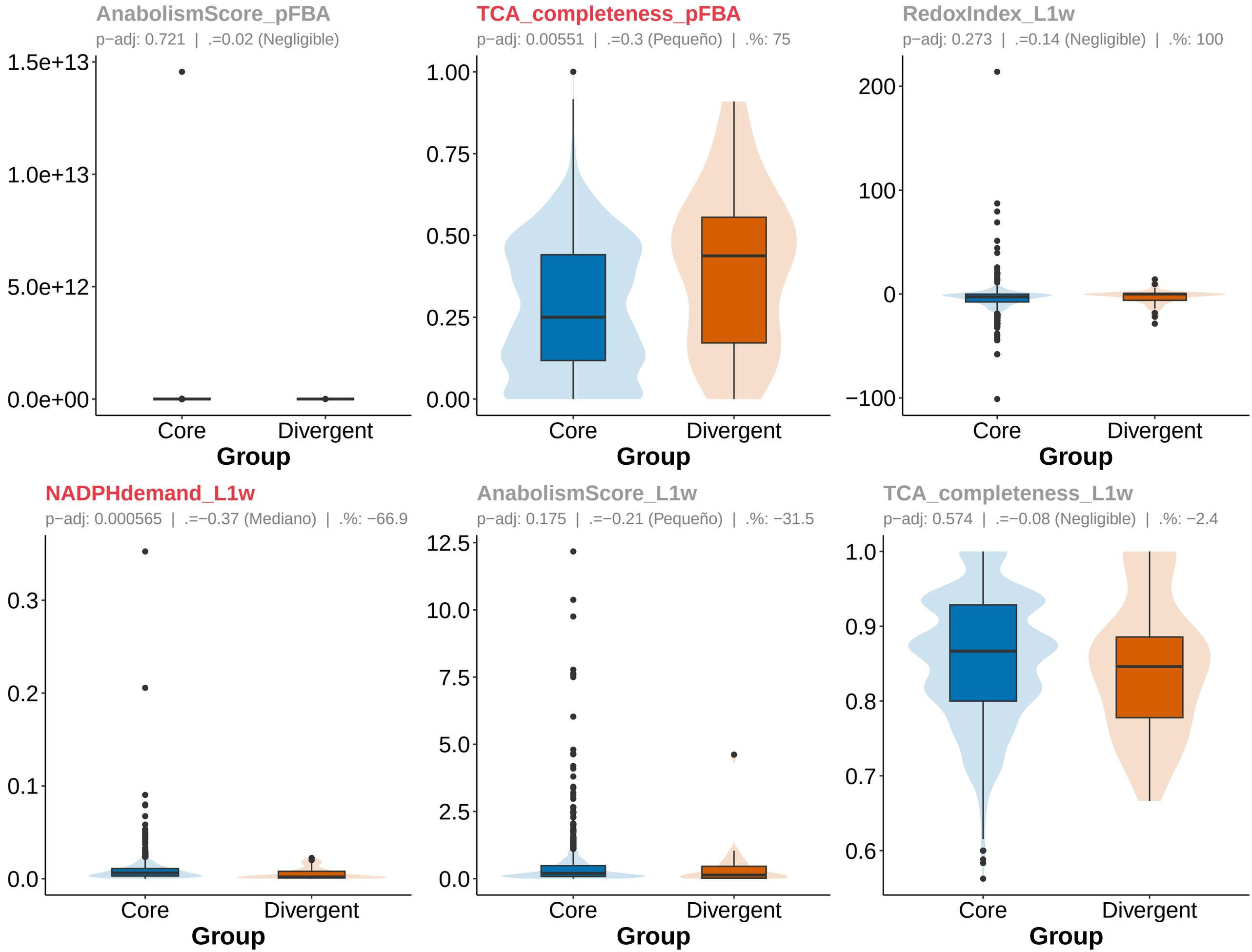



Cliff's . − . 2. Estado Redox y Anabolismo
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
NADPHdemand_FBA
** .=0.32 (33.5%)
TCA_completeness_pFBA
** .=0.3 (75%)
NADPHdemand_pFBA
* .=0.21 (356.6%)
RedoxIndex_L1w
ns .=0.14 (100%)
TCA_completeness_FBA
ns .=0.12 (3%)
AnabolismScore_FBA
ns .=0.08 (40.5%)
RedoxIndex_FBA
ns .=0.07 (6.6%)
RedoxIndex_pFBA
ns .=0.06 (NA%)
AnabolismScore_pFBA
ns .=0.02 (NA%)
TCA_completeness_L1w
ns .=−0.08 (−2.4%)
AnabolismScore_L1w
ns .=−0.21 (−31.5%)
NADPHdemand_L1w
*** .=−0.37 (−66.9%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

Metabolismo − . 2. Estado Redox y Anabolismo

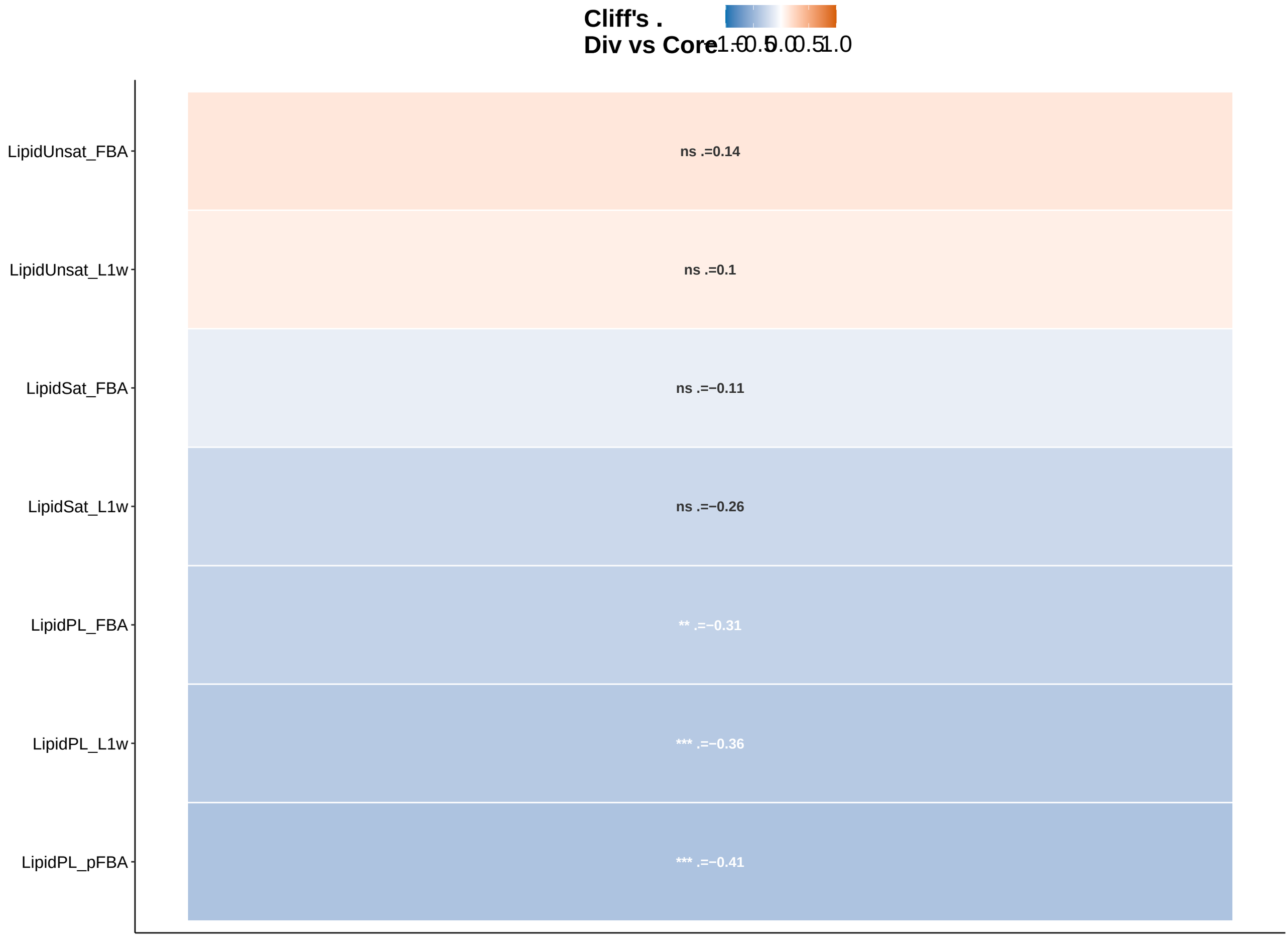


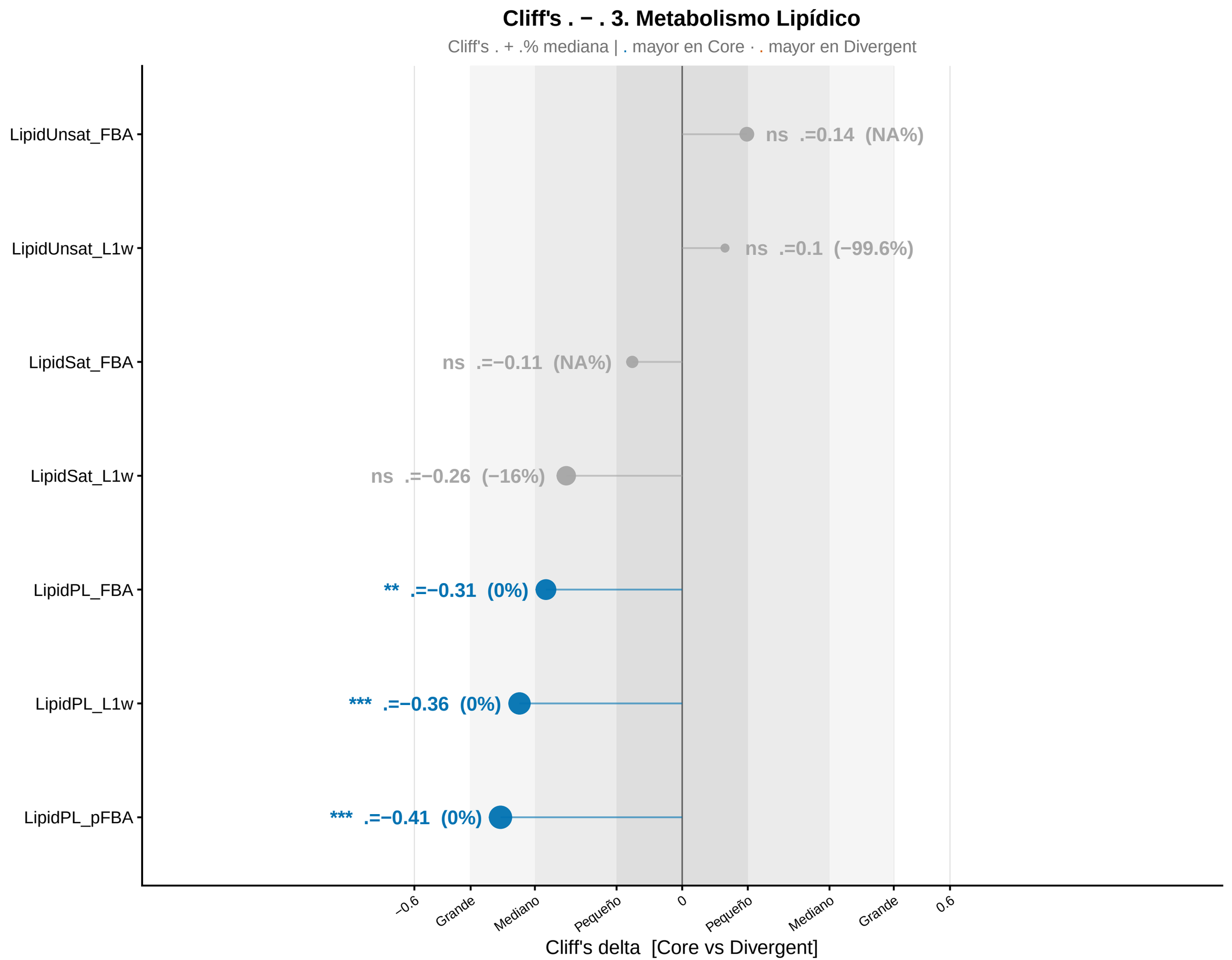
Metabolismo − . 2. Estado Redox y Anabolismo



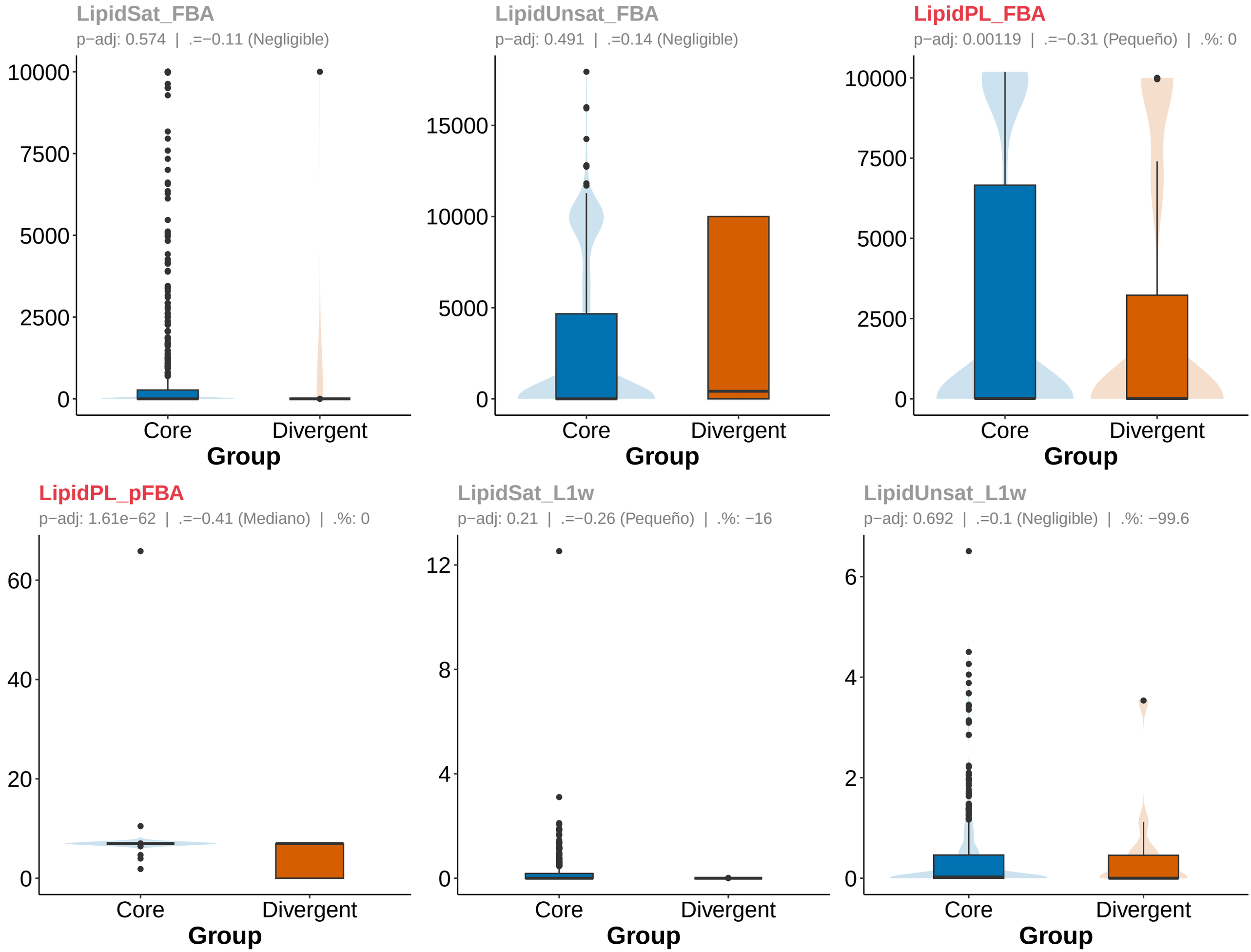
Cliff's .: . 3. Metabolismo Lipídico

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

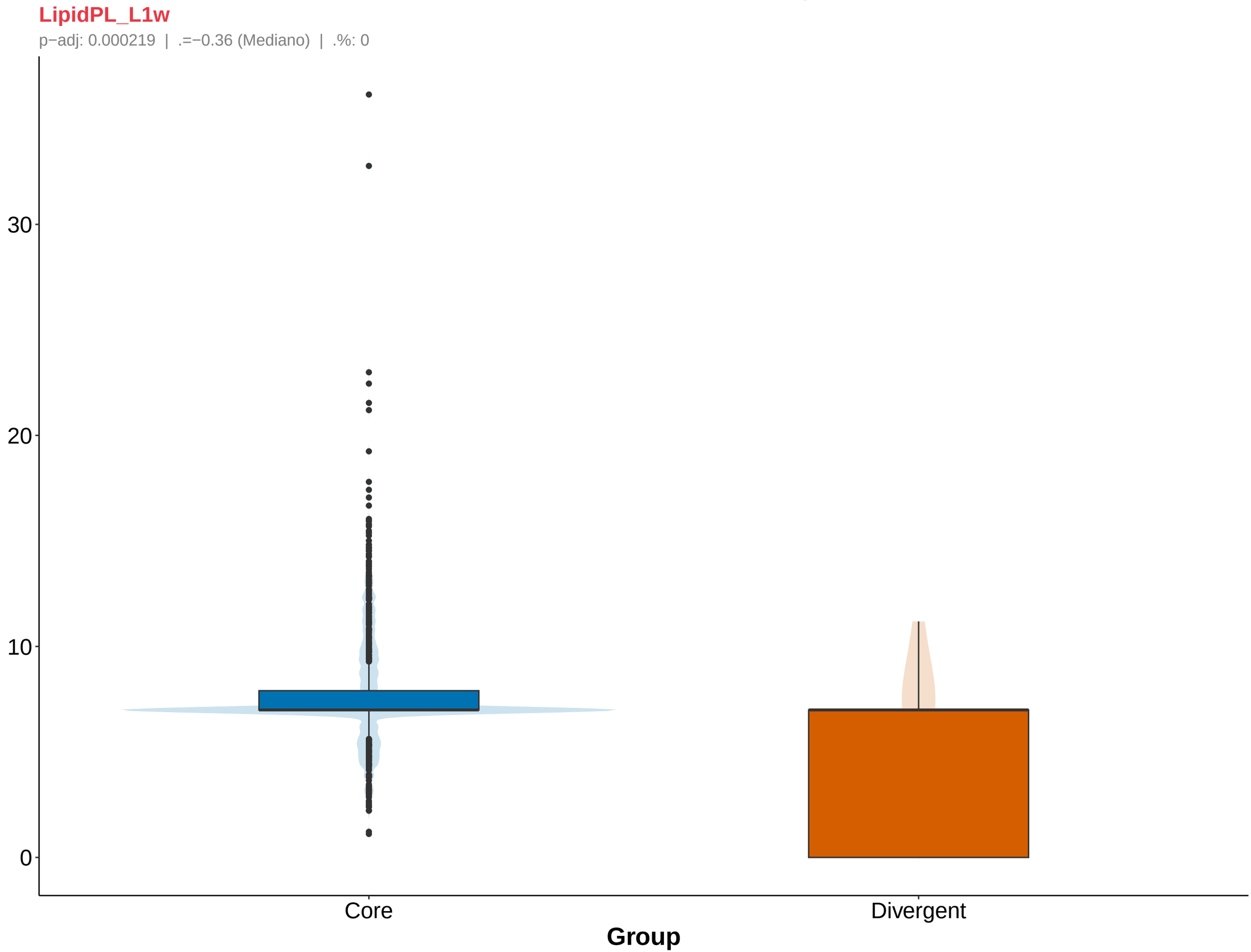



Cliff's . − . 3. Metabolismo Lipídico
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
LipidUnsat_FBA
ns .=0.14 (NA%)
LipidUnsat_L1w
ns .=0.1 (−99.6%)
LipidSat_FBA
ns .=−0.11 (NA%)
LipidSat_L1w
ns .=−0.26 (−16%)
LipidPL_FBA
** .=−0.31 (0%)
LipidPL_L1w
*** .=−0.36 (0%)
LipidPL_pFBA
*** .=−0.41 (0%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

Metabolismo − . 3. Metabolismo Lipídico

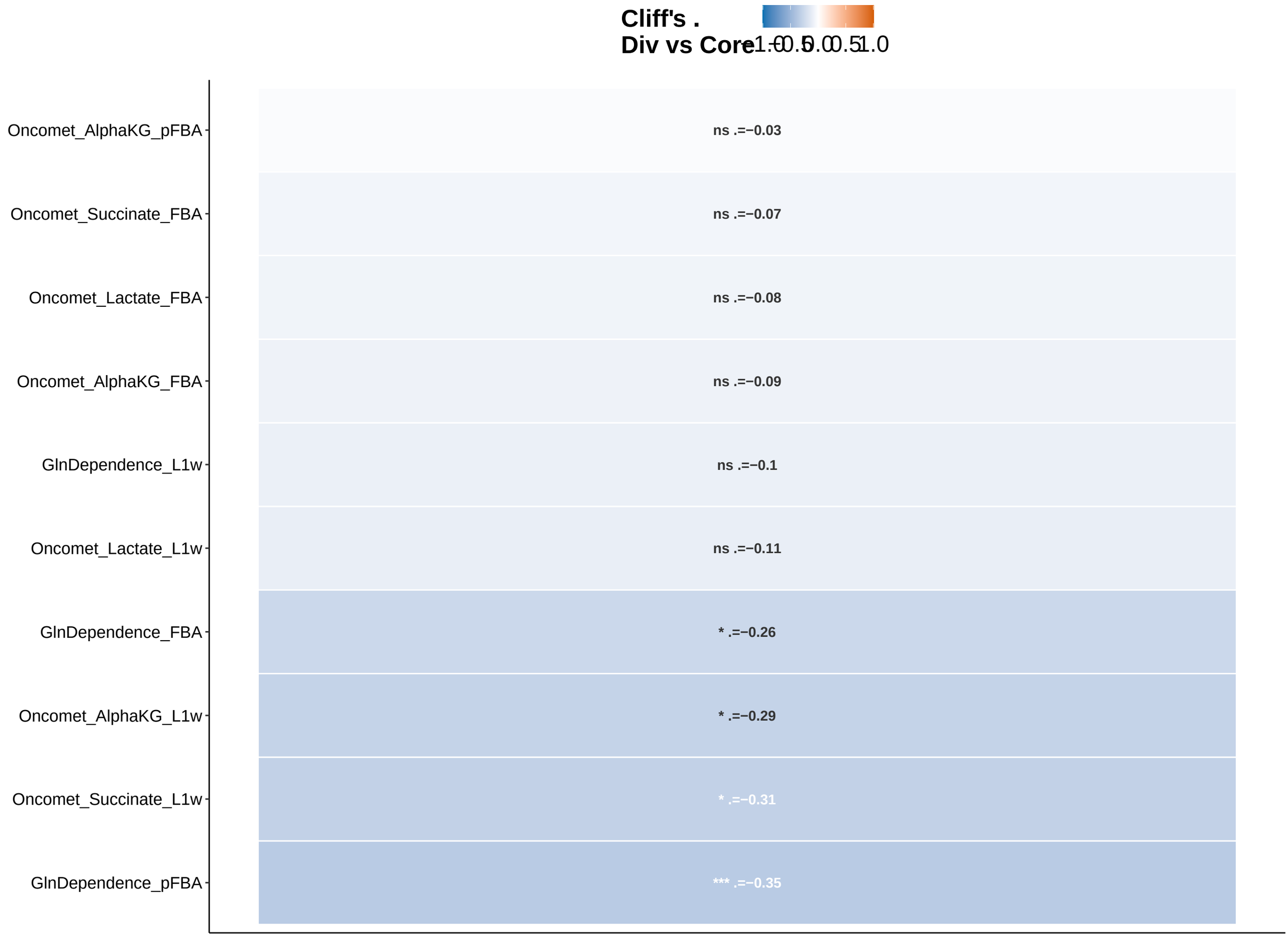


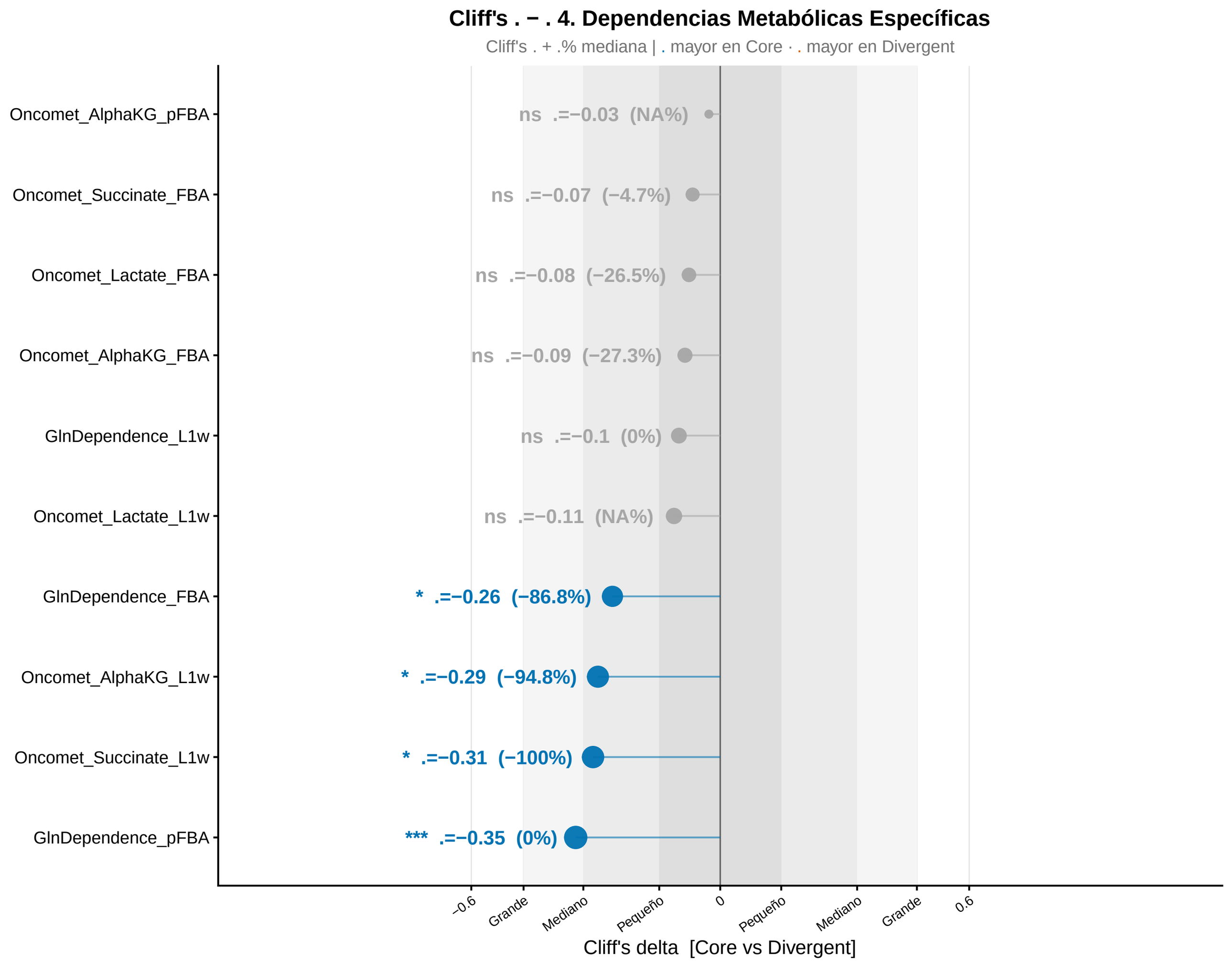
Metabolismo − . 3. Metabolismo Lipídico



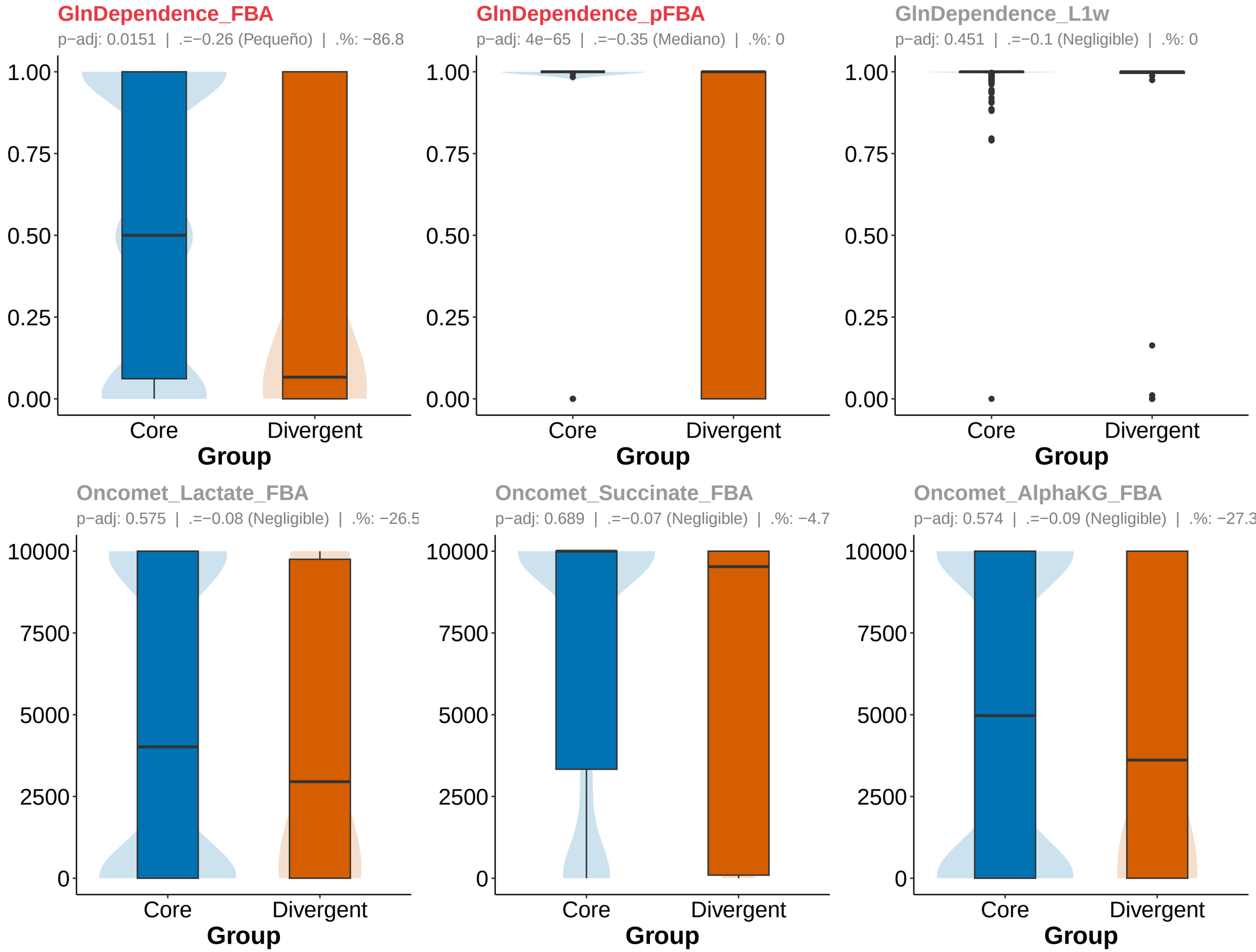
Cliff's .: . 4. Dependencias Metabólicas Específicas

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

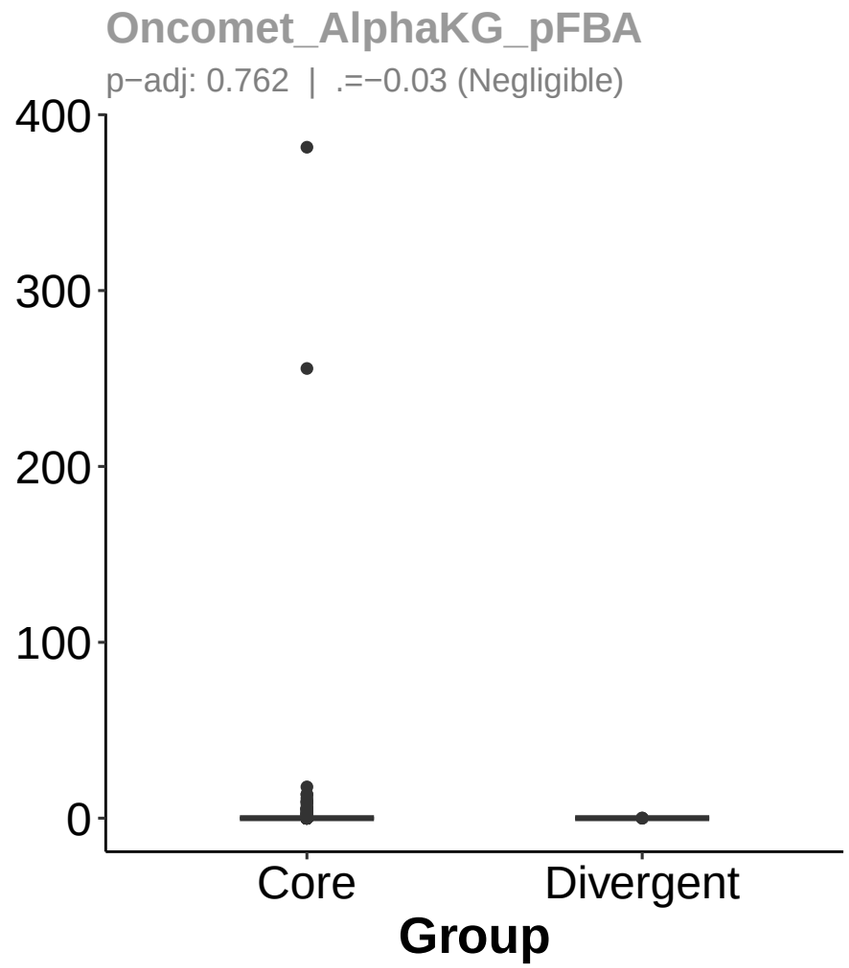


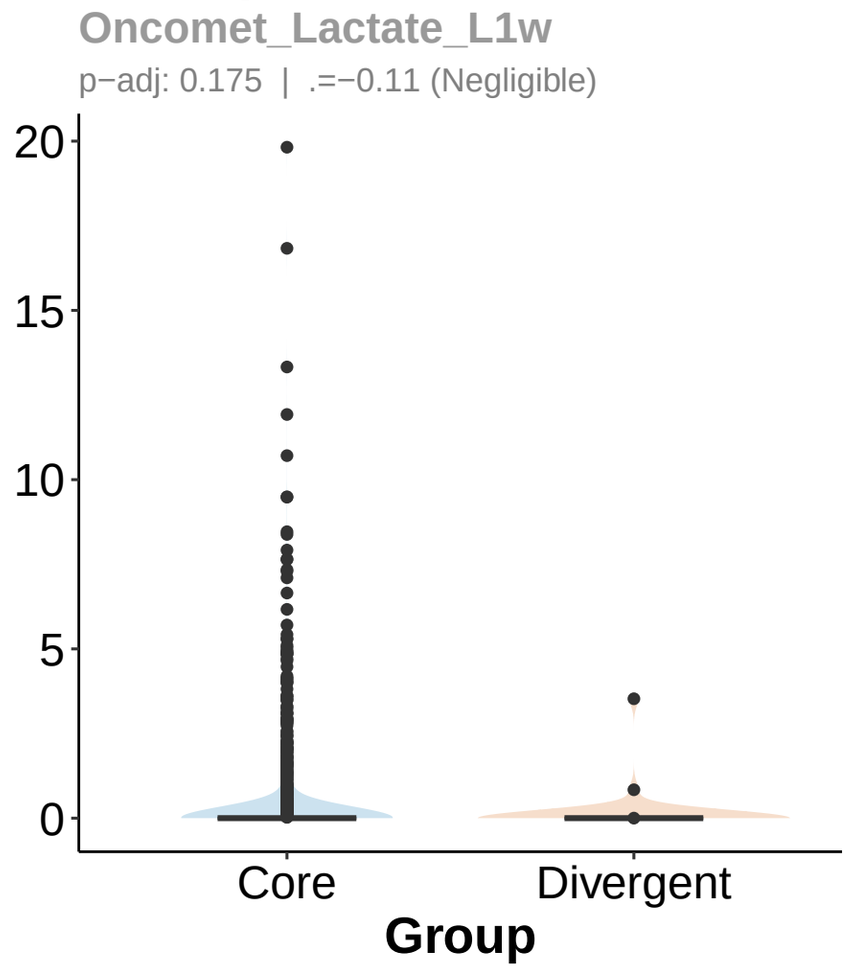

Cliff's . − . 4. Dependencias Metabólicas Específicas
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
Oncomet_AlphaKG_pFBA
ns .=−0.03 (NA%)
Oncomet_Succinate_FBA
ns .=−0.07 (−4.7%)
Oncomet_Lactate_FBA
ns .=−0.08 (−26.5%)
Oncomet_AlphaKG_FBA
ns .=−0.09 (−27.3%)
GlnDependence_L1w
ns .=−0.1 (0%)
Oncomet_Lactate_L1w
ns .=−0.11 (NA%)
GlnDependence_FBA
* .=−0.26 (−86.8%)
Oncomet_AlphaKG_L1w
* .=−0.29 (−94.8%)
Oncomet_Succinate_L1w
* .=−0.31 (−100%)
GlnDependence_pFBA
*** .=−0.35 (0%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

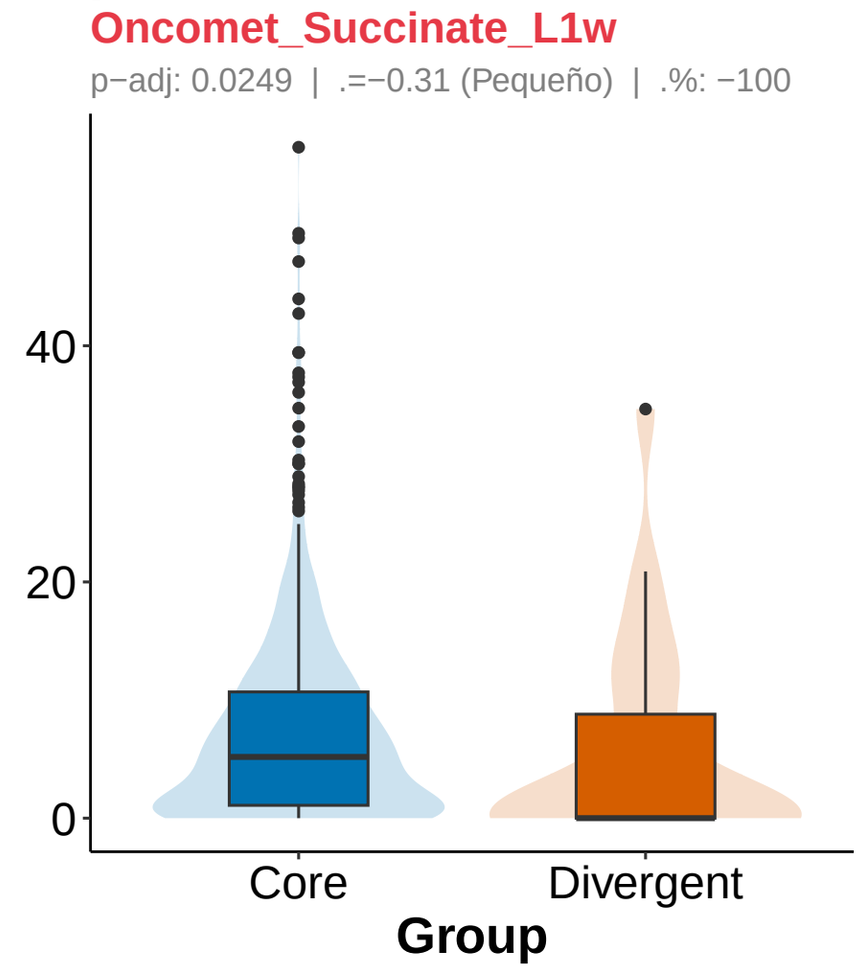
Metabolismo − . 4. Dependencias Metabólicas Específicas

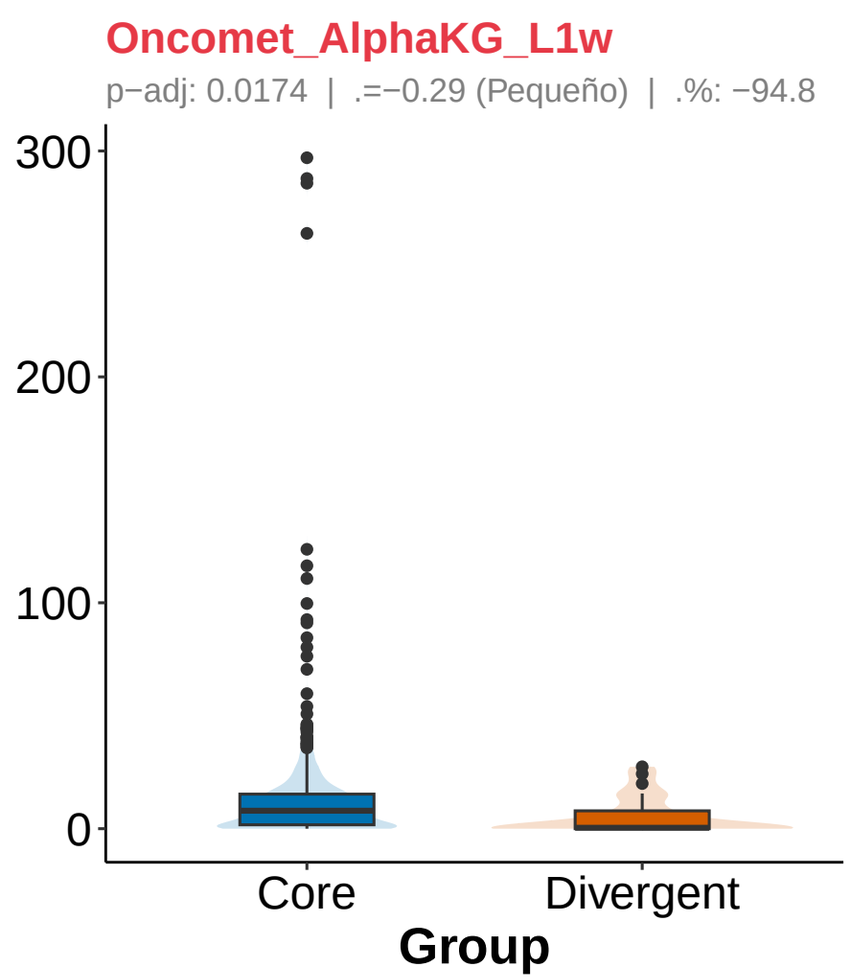


Metabolismo − . 4. Dependencias Metabólicas Específicas



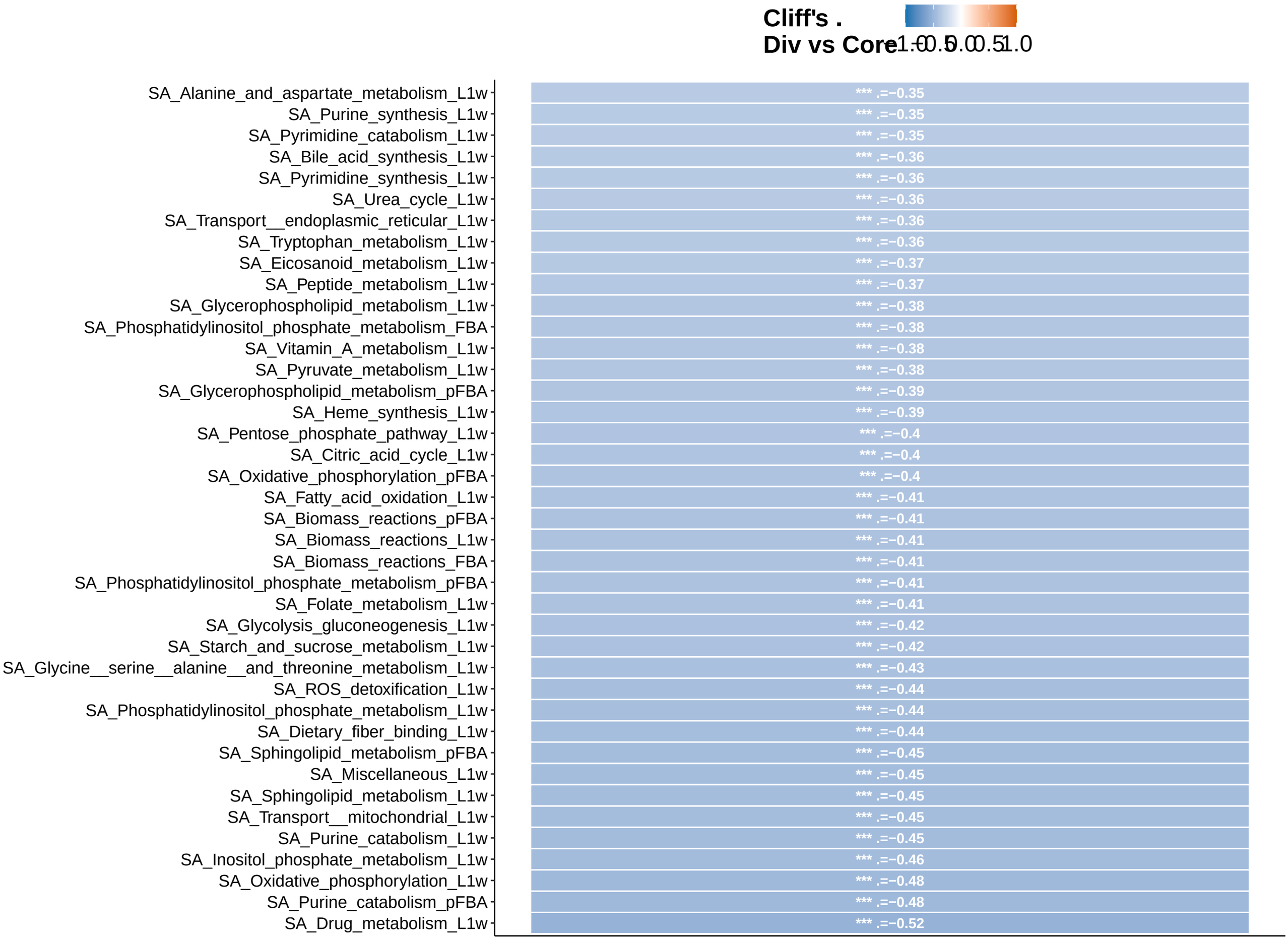


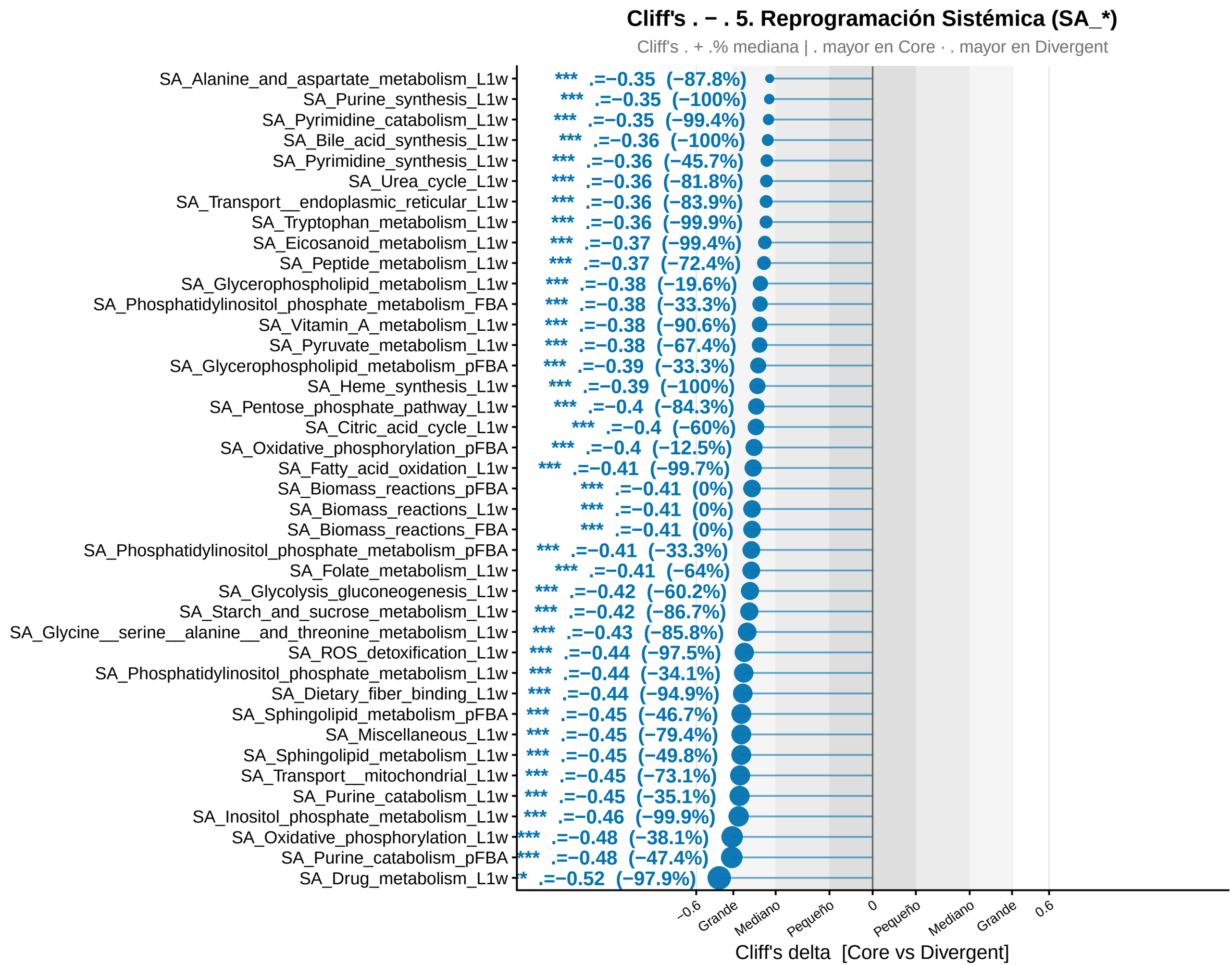




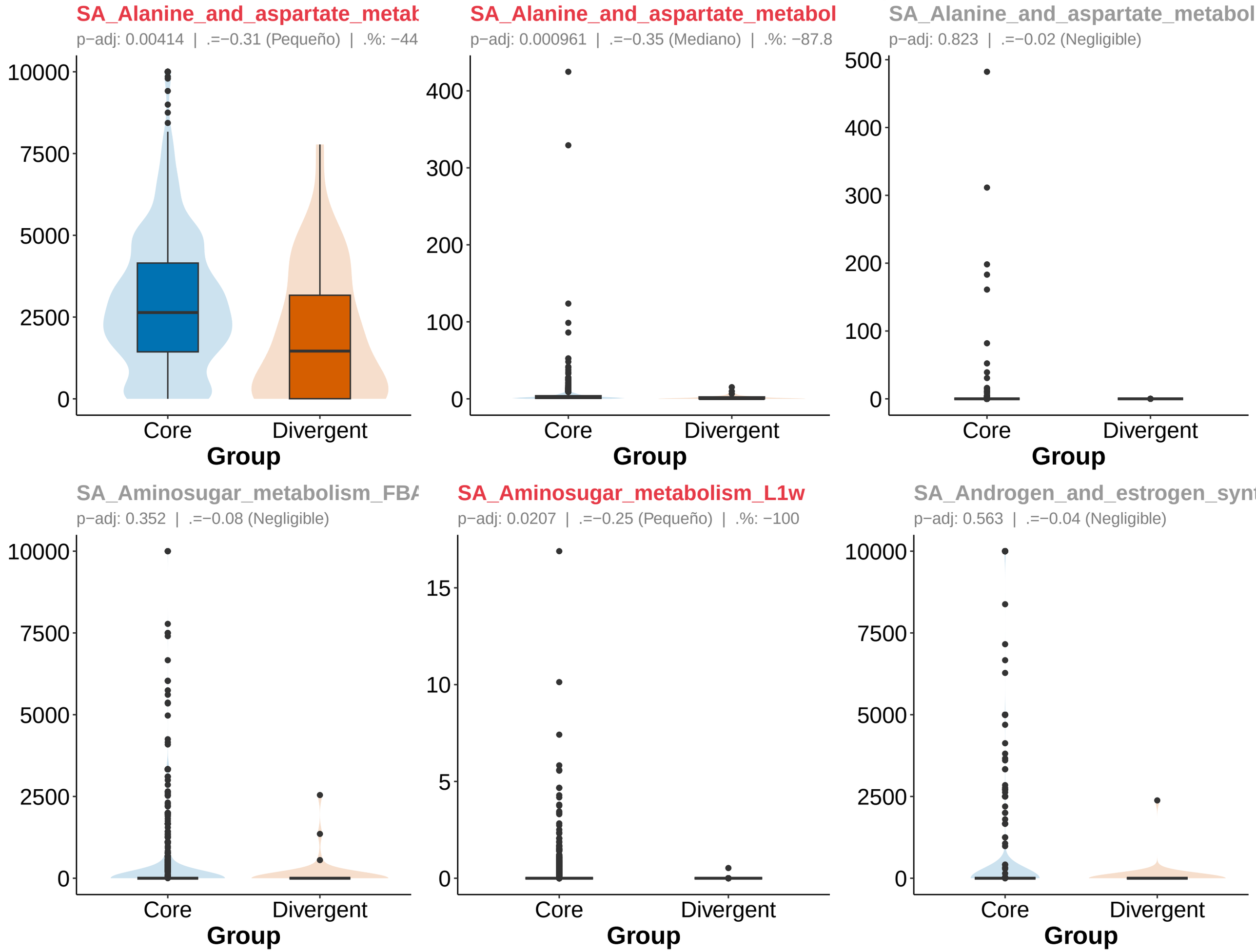
Cliff's .: . 5. Reprogramación Sistémica (SA_*)

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

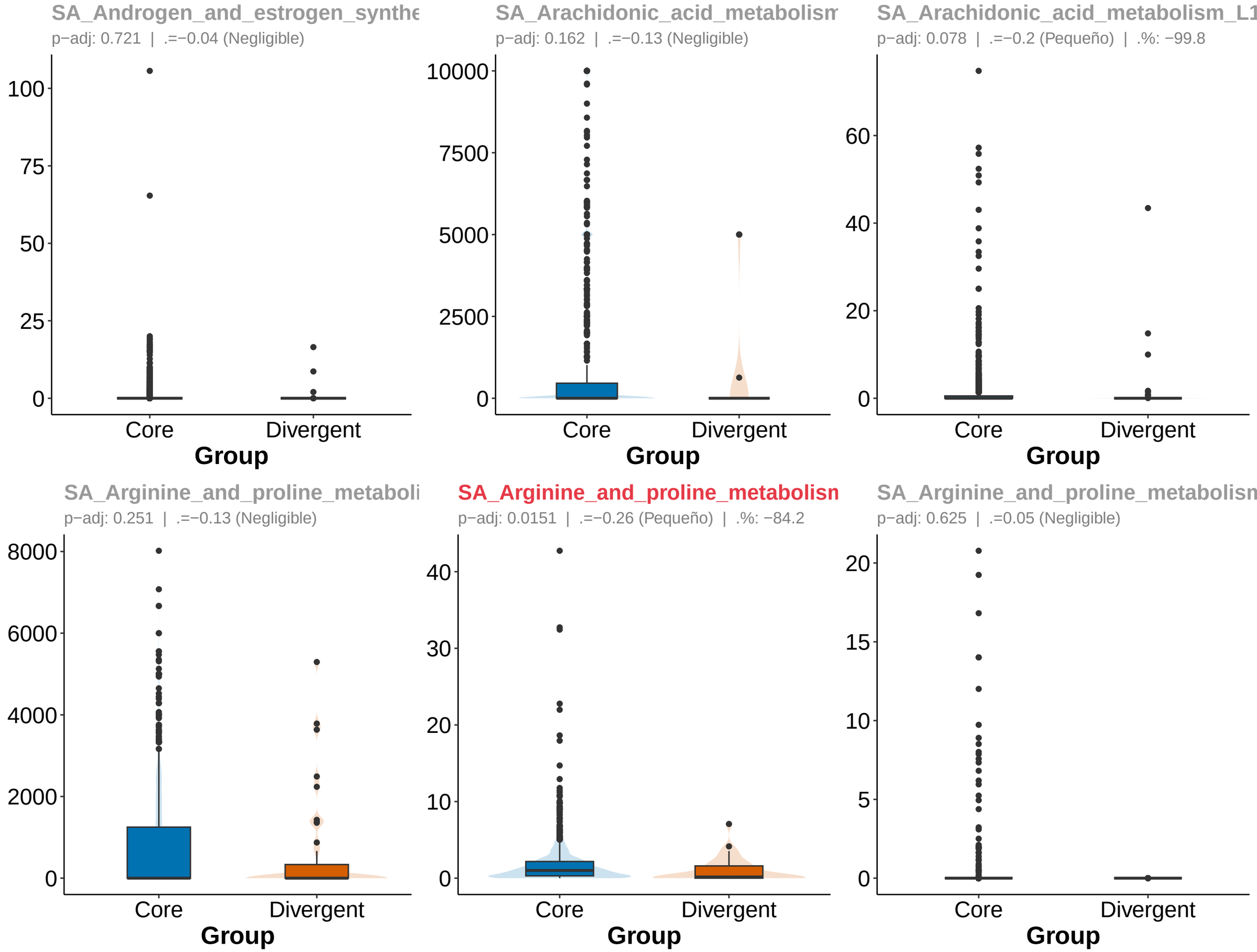



Cliff's . − . 5. Reprogramación Sistémica (SA_*)
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
SA_Alanine_and_aspartate_metabolism_L1w *** .=−0.35 (−87.8%)
SA_Purine_synthesis_L1w *** .=−0.35 (−100%)
SA_Pyrimidine_catabolism_L1w *** .=−0.35 (−99.4%)
SA_Bile_acid_synthesis_L1w *** .=−0.36 (−100%)
SA_Pyrimidine_synthesis_L1w *** .=−0.36 (−45.7%)
SA_Urea_cycle_L1w *** .=−0.36 (−81.8%)
SA_Transport__endoplasmic_reticular_L1w *** .=−0.36 (−83.9%)
SA_Tryptophan_metabolism_L1w *** .=−0.36 (−99.9%)
SA_Eicosanoid_metabolism_L1w *** .=−0.37 (−99.4%)
SA_Peptide_metabolism_L1w *** .=−0.37 (−72.4%)
SA_Glycerophospholipid_metabolism_L1w *** .=−0.38 (−19.6%)
SA_Phosphatidylinositol_phosphate_metabolism_FBA *** .=−0.38 (−33.3%)
SA_Vitamin_A_metabolism_L1w *** .=−0.38 (−90.6%)
SA_Pyruvate_metabolism_L1w *** .=−0.38 (−67.4%)
SA_Glycerophospholipid_metabolism_pFBA *** .=−0.39 (−33.3%)
SA_Heme_synthesis_L1w *** .=−0.39 (−100%)
SA_Pentose_phosphate_pathway_L1w *** .=−0.4 (−84.3%)
SA_Citric_acid_cycle_L1w *** .=−0.4 (−60%)
SA_Oxidative_phosphorylation_pFBA *** .=−0.4 (−12.5%)
SA_Fatty_acid_oxidation_L1w *** .=−0.41 (−99.7%)
SA_Biomass_reactions_pFBA *** .=−0.41 (0%)
SA_Biomass_reactions_L1w *** .=−0.41 (0%)
SA_Biomass_reactions_FBA *** .=−0.41 (0%)
SA_Phosphatidylinositol_phosphate_metabolism_pFBA *** .=−0.41 (−33.3%)
SA_Folate_metabolism_L1w *** .=−0.41 (−64%)
SA_Glycolysis_gluconeogenesis_L1w *** .=−0.42 (−60.2%)
SA_Starch_and_sucrose_metabolism_L1w *** .=−0.42 (−86.7%)
SA_Glycine__serine__alanine__and_threonine_metabolism_L1w *** .=−0.43 (−85.8%)
SA_ROS_detoxification_L1w *** .=−0.44 (−97.5%)
SA_Phosphatidylinositol_phosphate_metabolism_L1w *** .=−0.44 (−34.1%)
SA_Dietary_fiber_binding_L1w *** .=−0.44 (−94.9%)
SA_Sphingolipid_metabolism_pFBA *** .=−0.45 (−46.7%)
SA_Miscellaneous_L1w *** .=−0.45 (−79.4%)
SA_Sphingolipid_metabolism_L1w *** .=−0.45 (−49.8%)
SA_Transport__mitochondrial_L1w *** .=−0.45 (−73.1%)
SA_Purine_catabolism_L1w *** .=−0.45 (−35.1%)
SA_Inositol_phosphate_metabolism_L1w *** .=−0.46 (−99.9%)
SA_Oxidative_phosphorylation_L1w *** .=−0.48 (−38.1%)
SA_Purine_catabolism_pFBA *** .=−0.48 (−47.4%)
SA_Drug_metabolism_L1w * .=−0.52 (−97.9%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

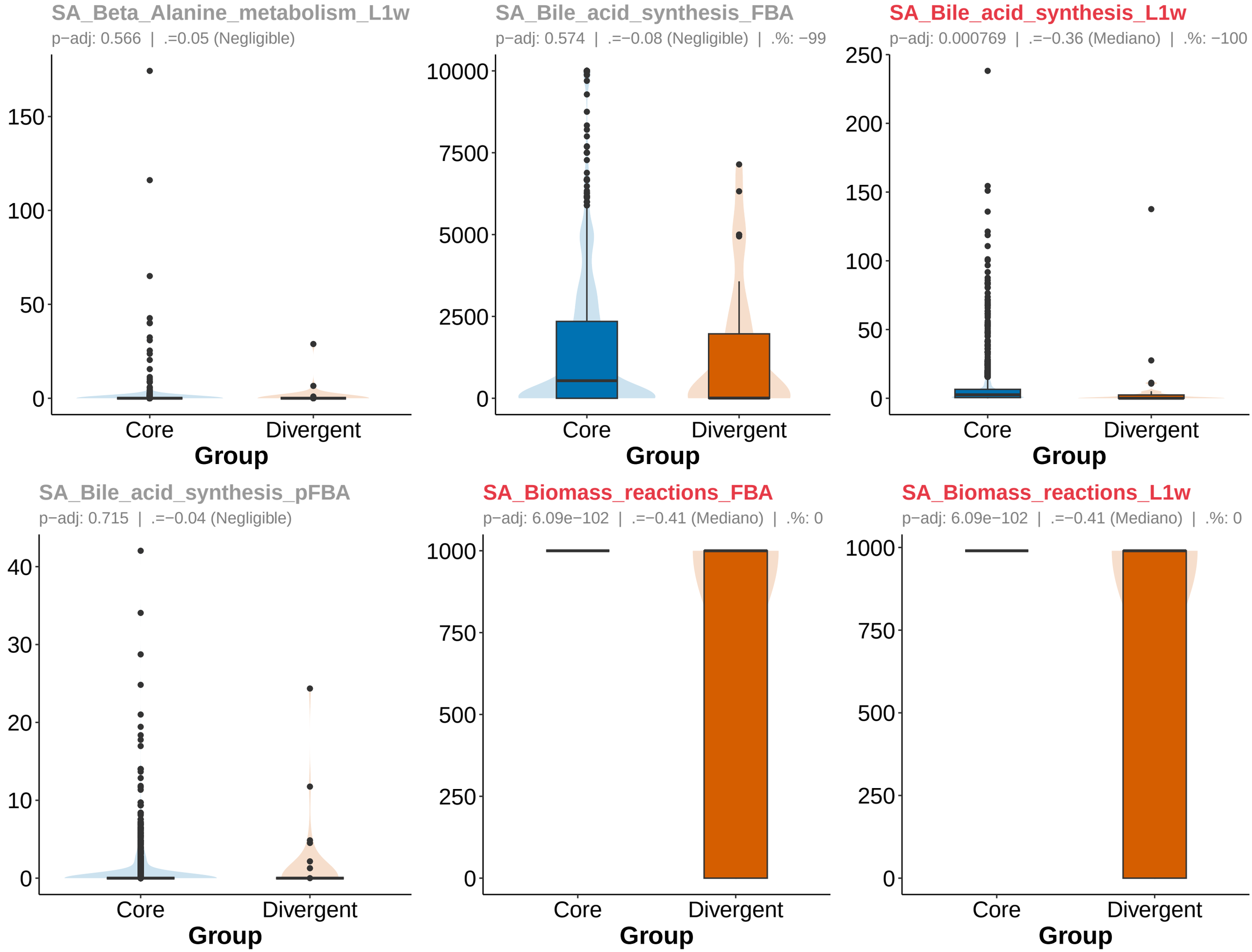
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



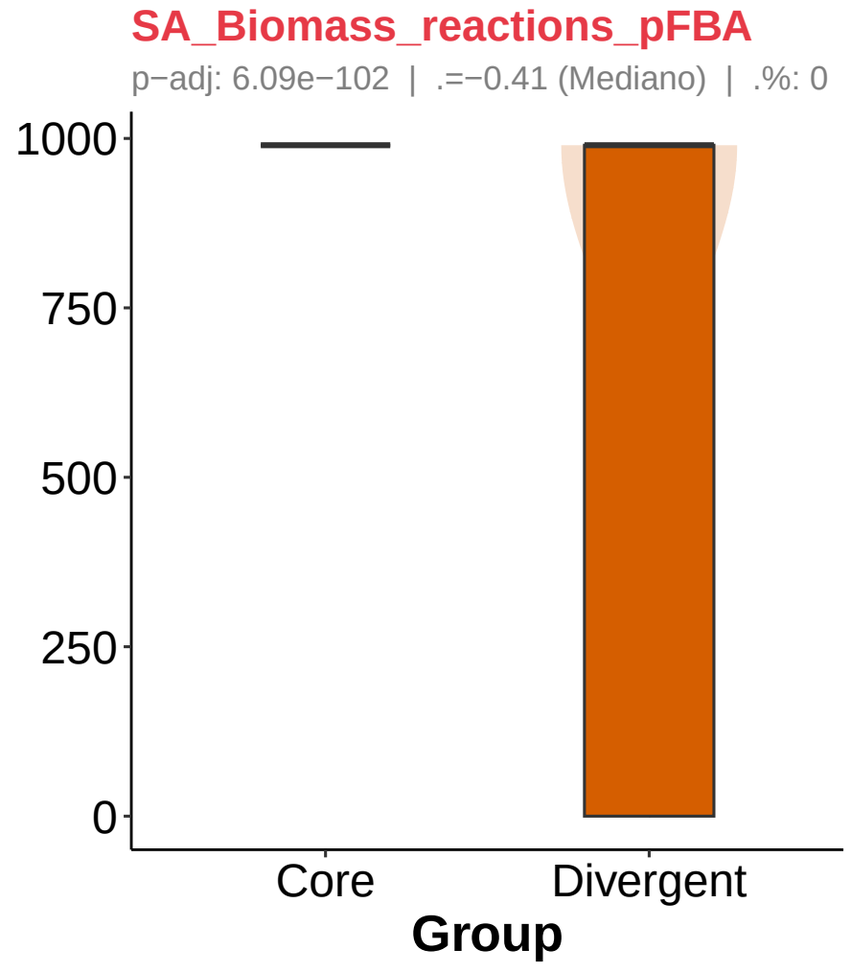
Metabolismo – . 5. Reprogramación Sistémica (SA_*)

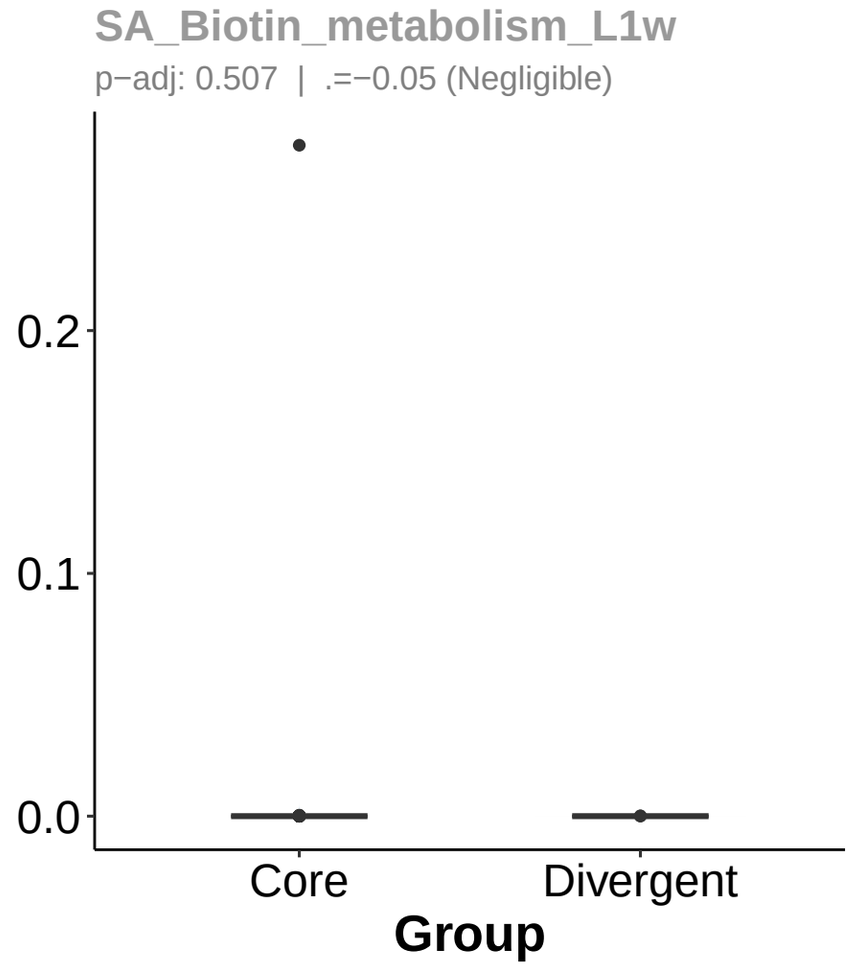


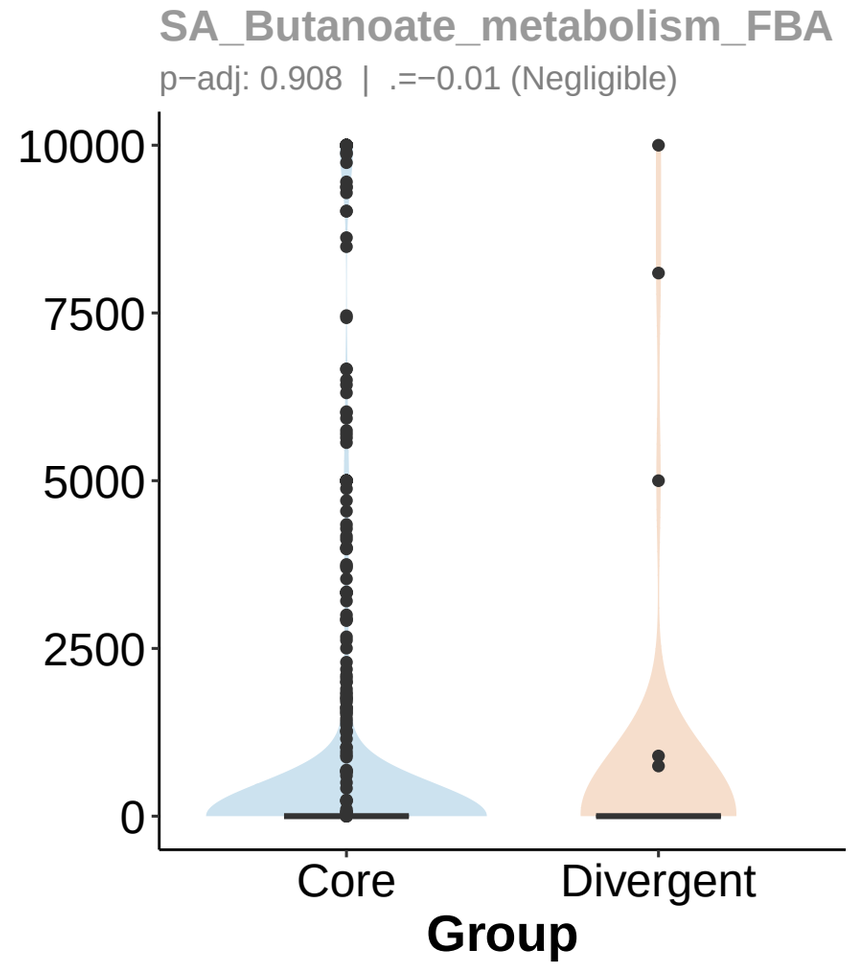
Metabolismo − . 5. Reprogramación Sistémica (SA_*)

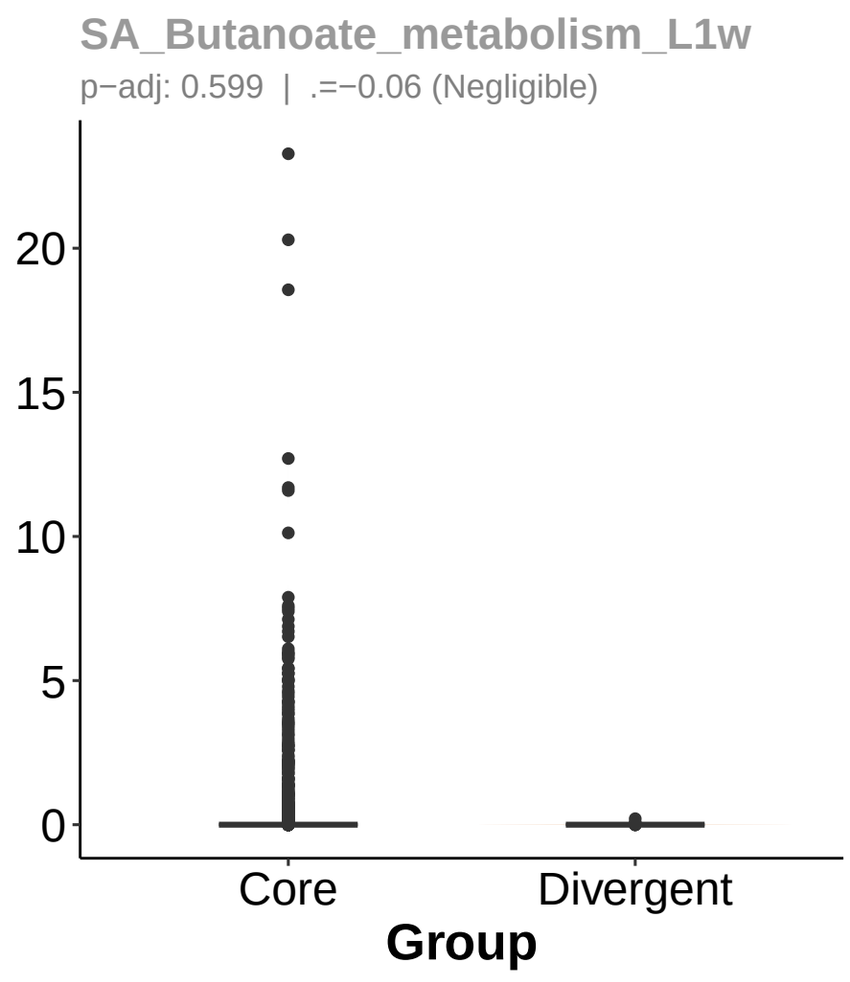


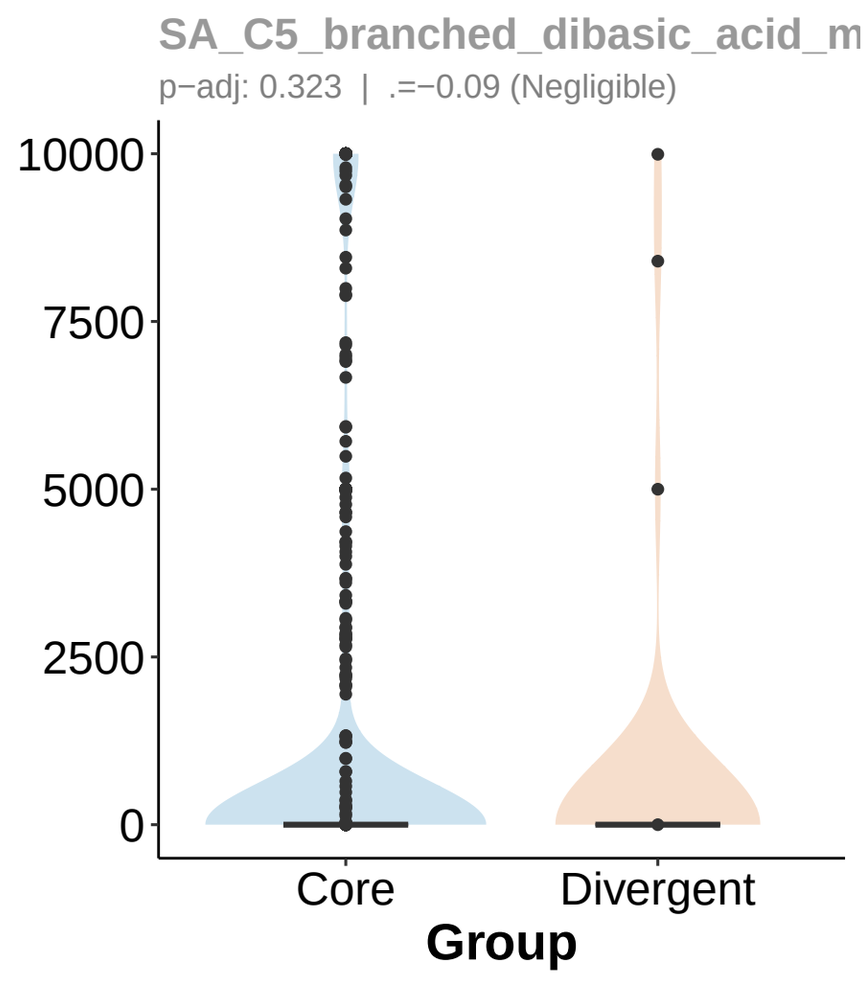
Metabolismo − . 5. Reprogramación Sistémica (SA_*)

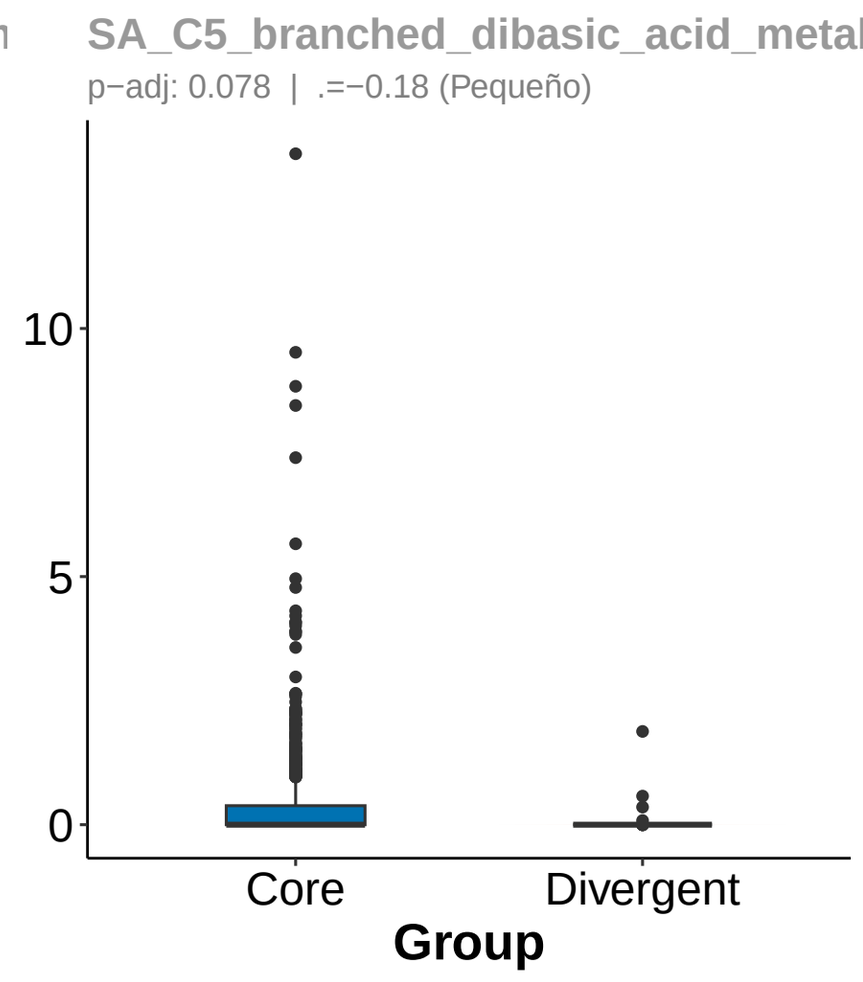




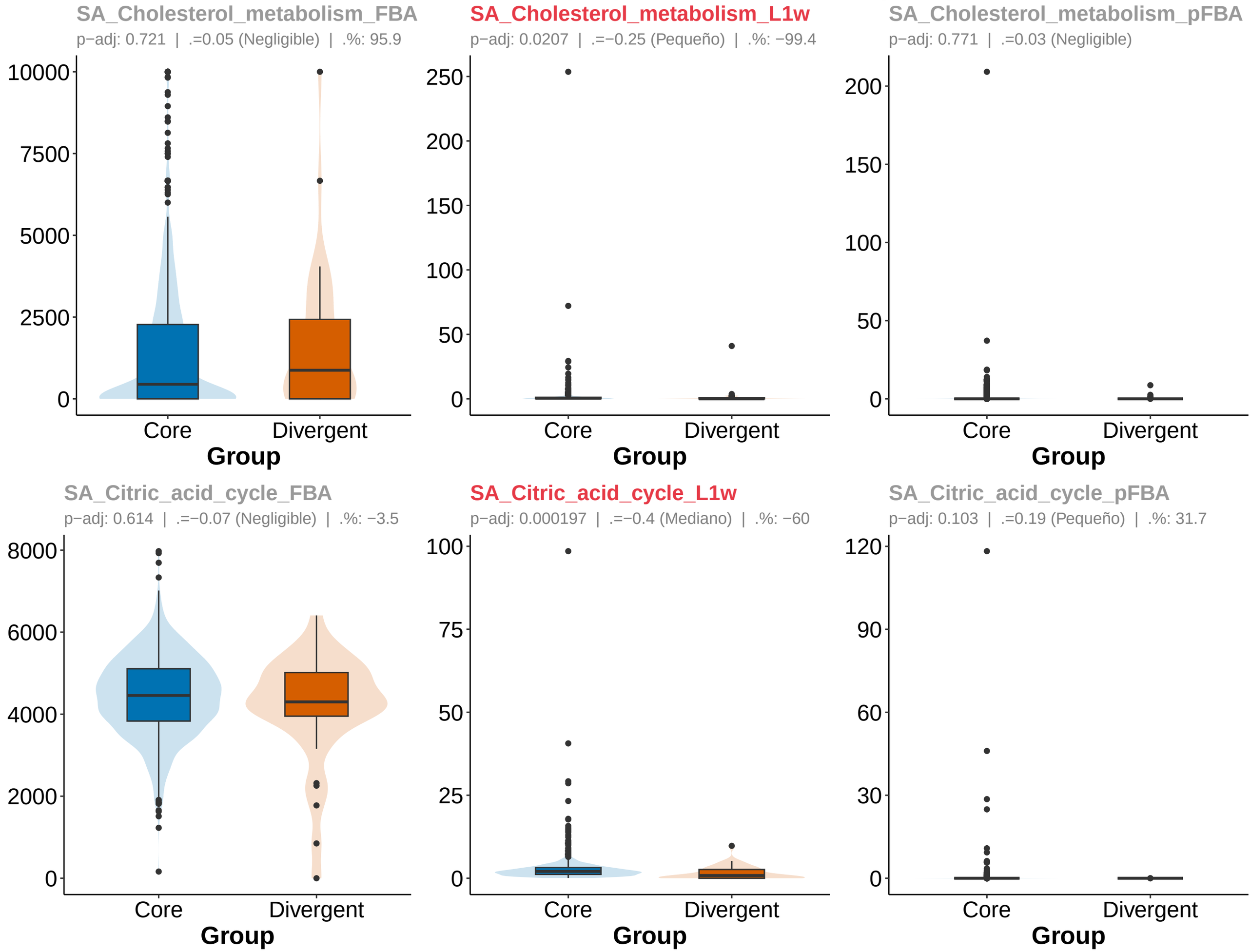




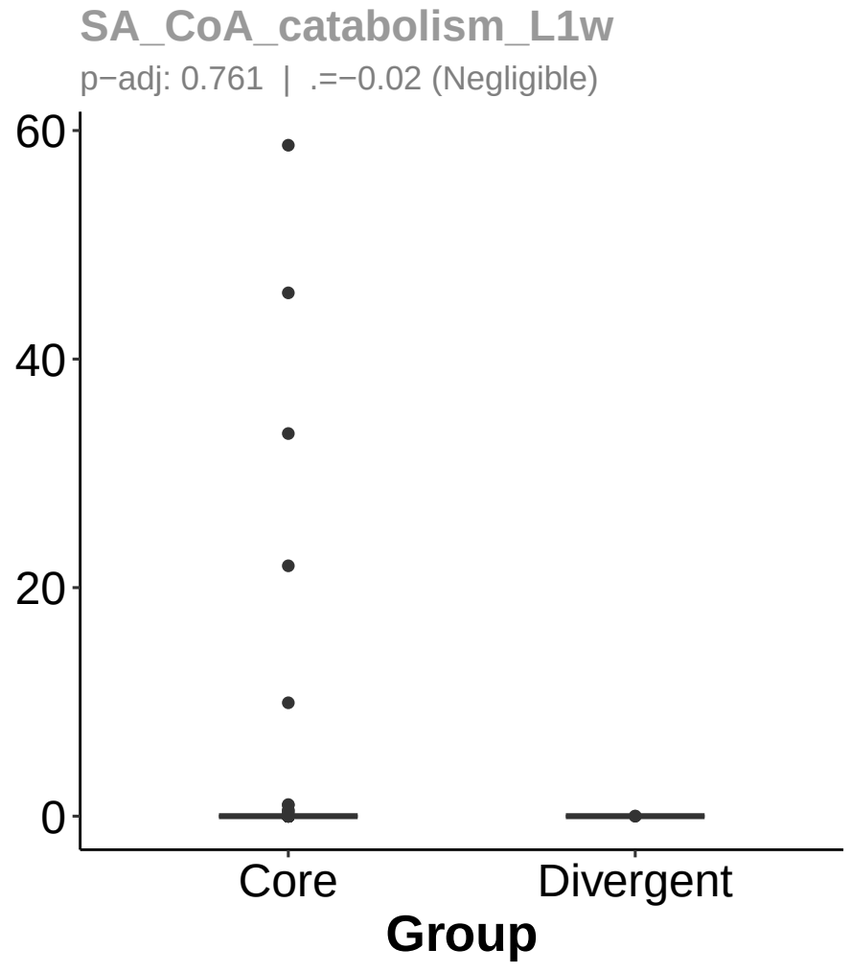


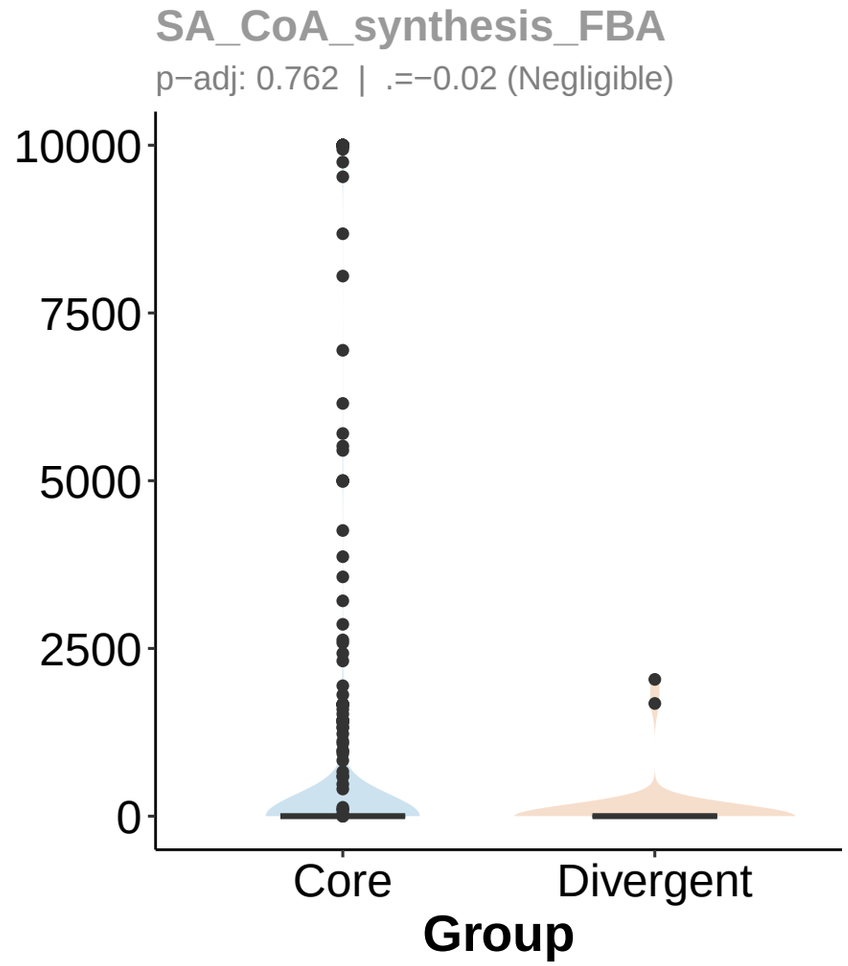


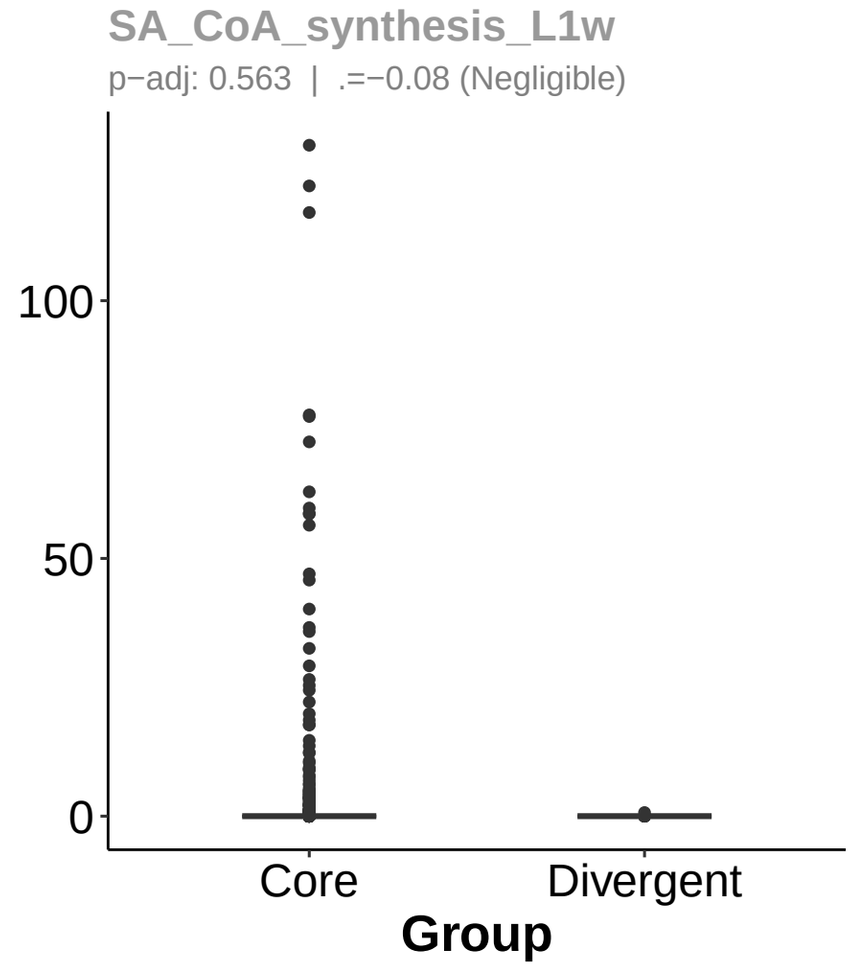
Metabolismo − . 5. Reprogramación Sistémica (SA_*)

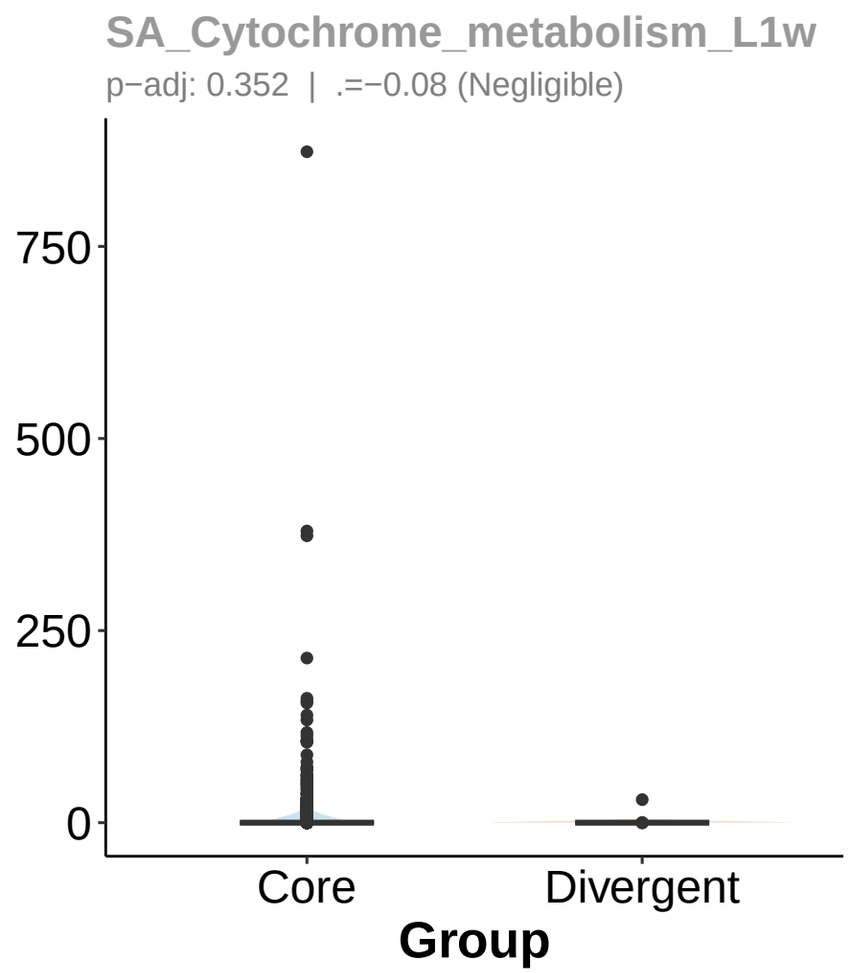


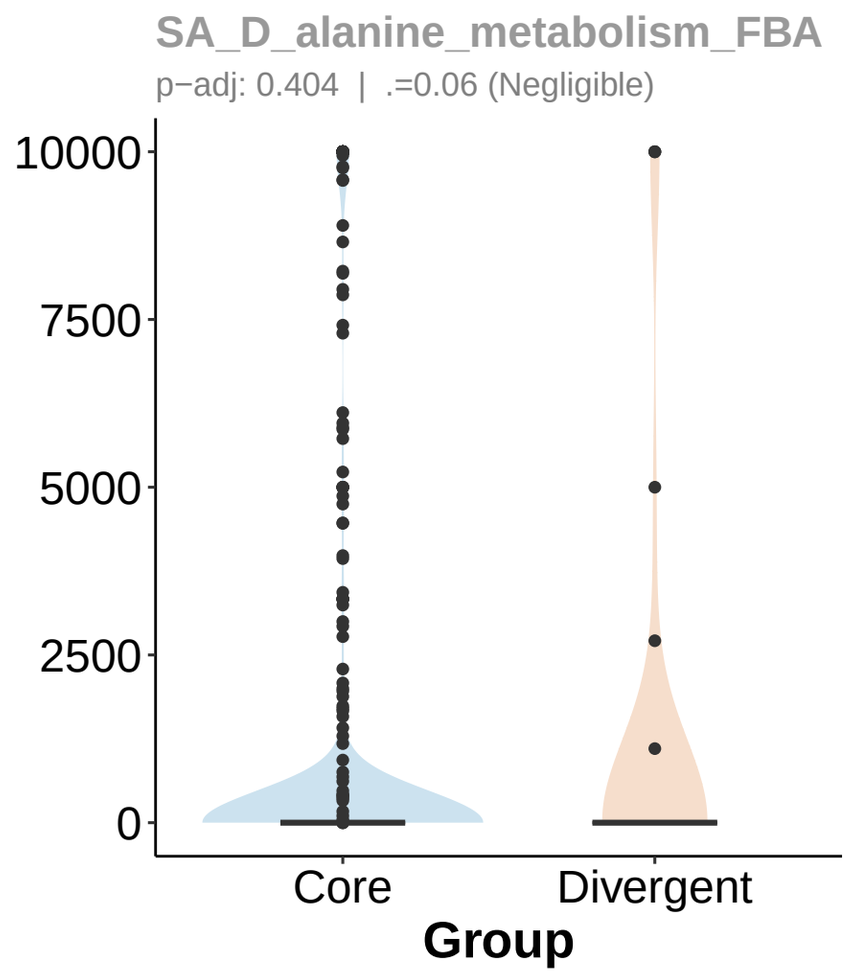
Metabolismo − . 5. Reprogramación Sistémica (SA_*)

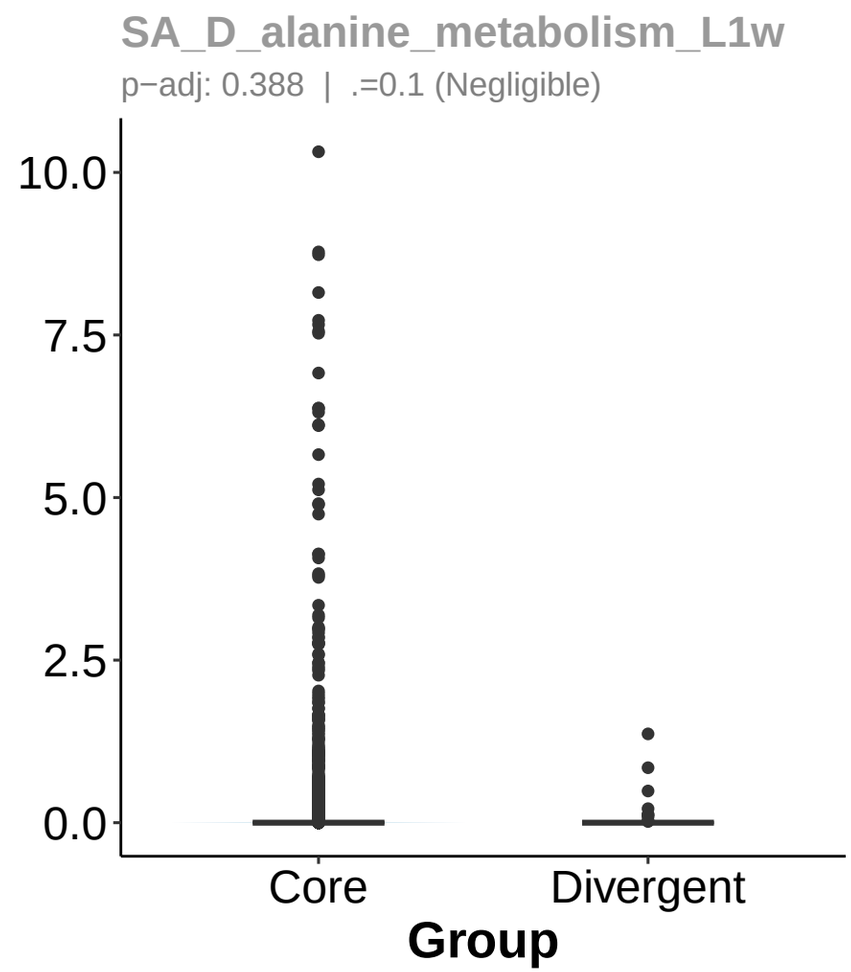




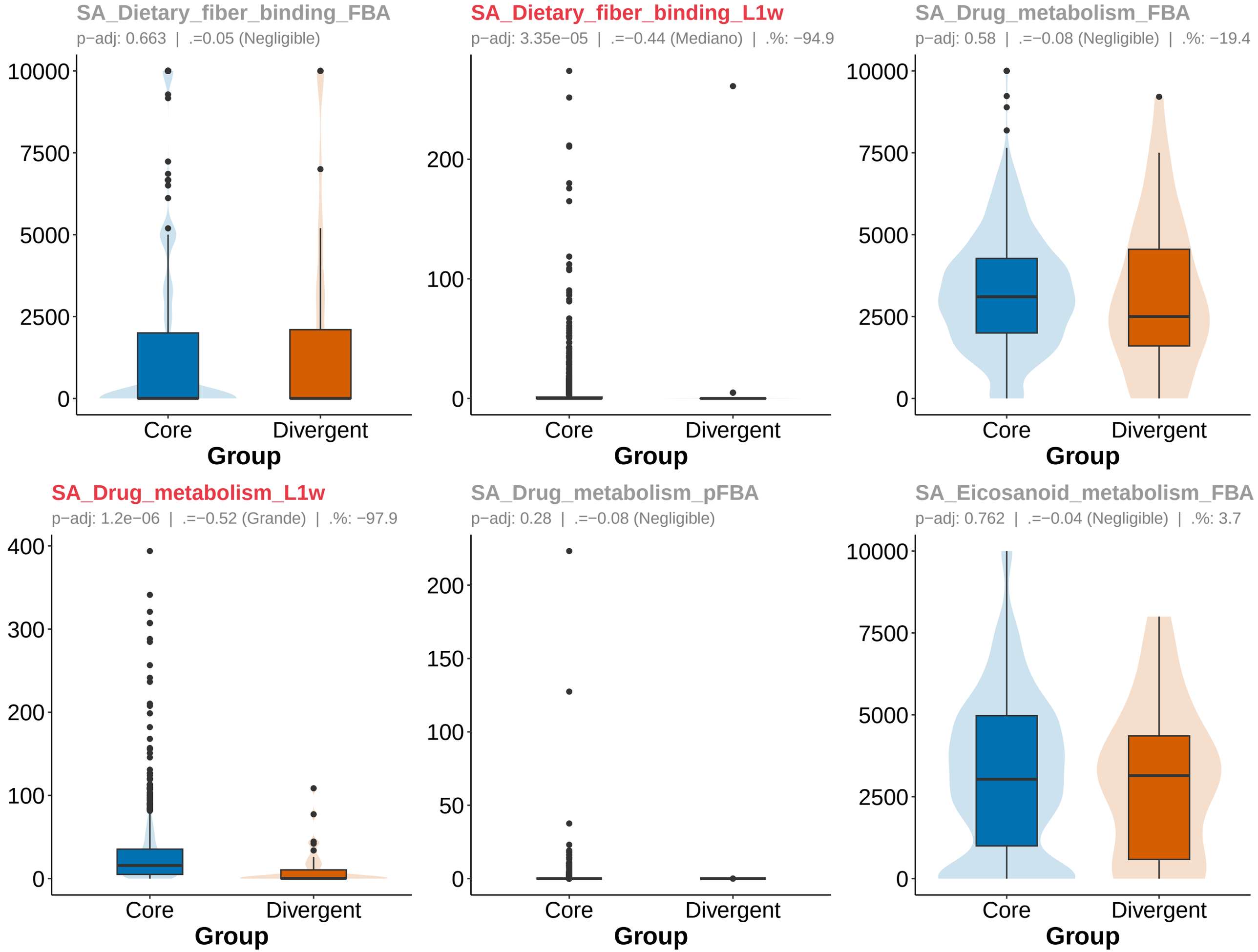








Metabolismo − . 5. Reprogramación Sistémica (SA_*)



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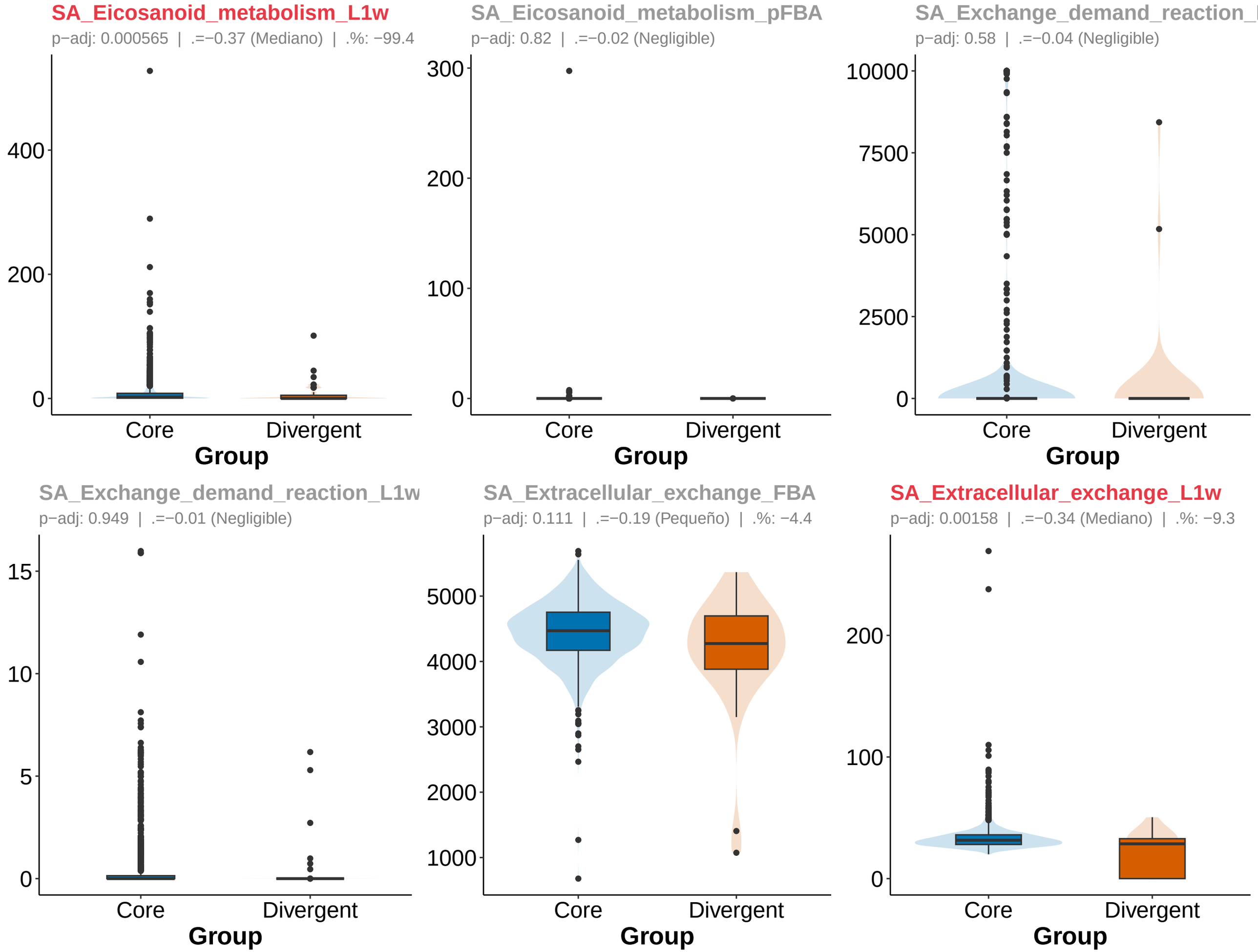

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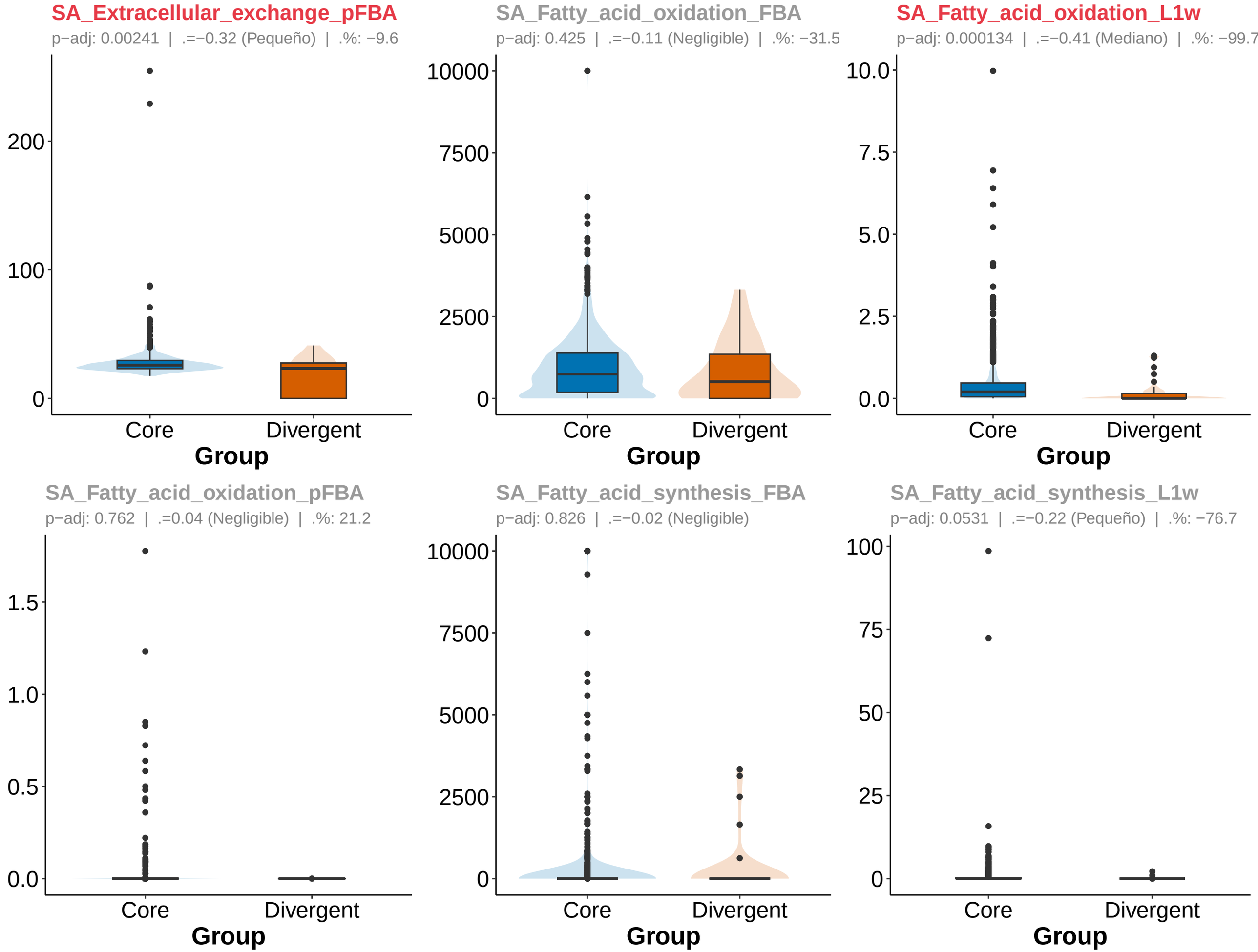

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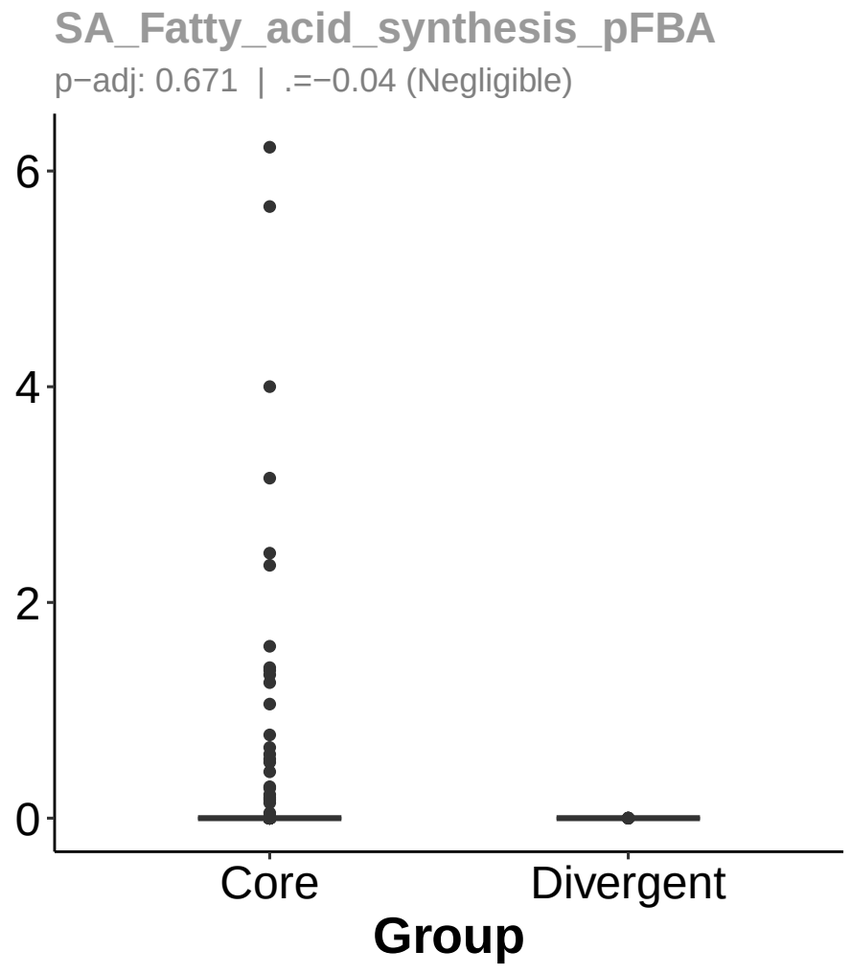

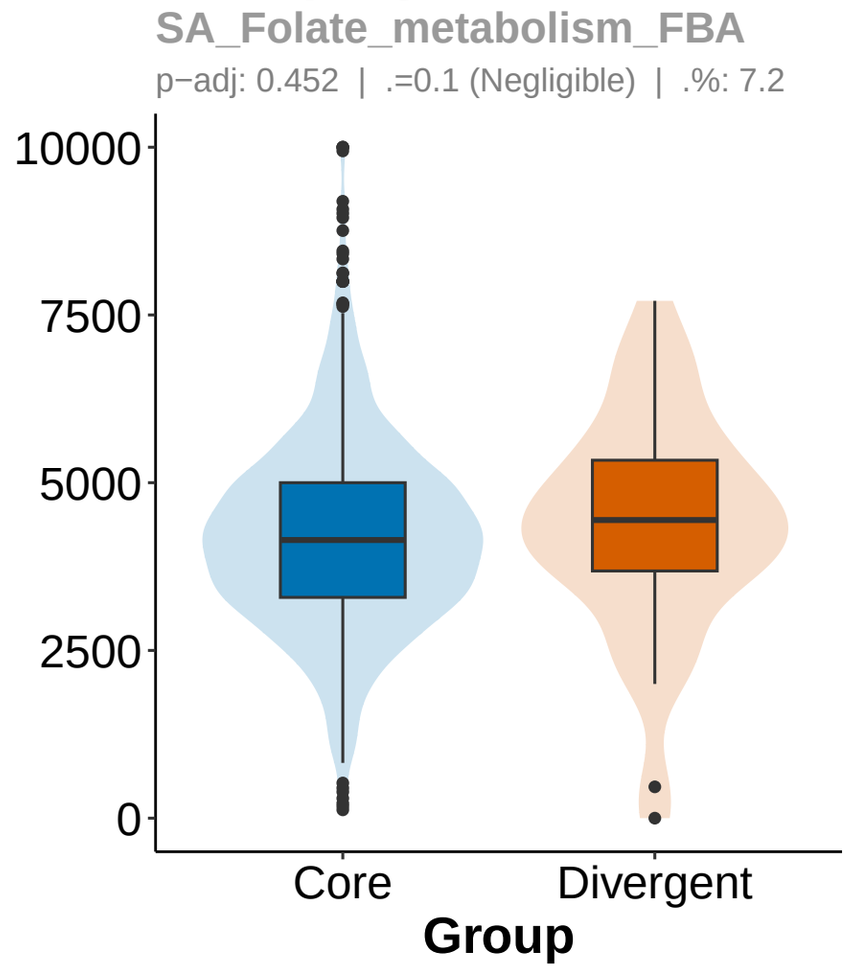

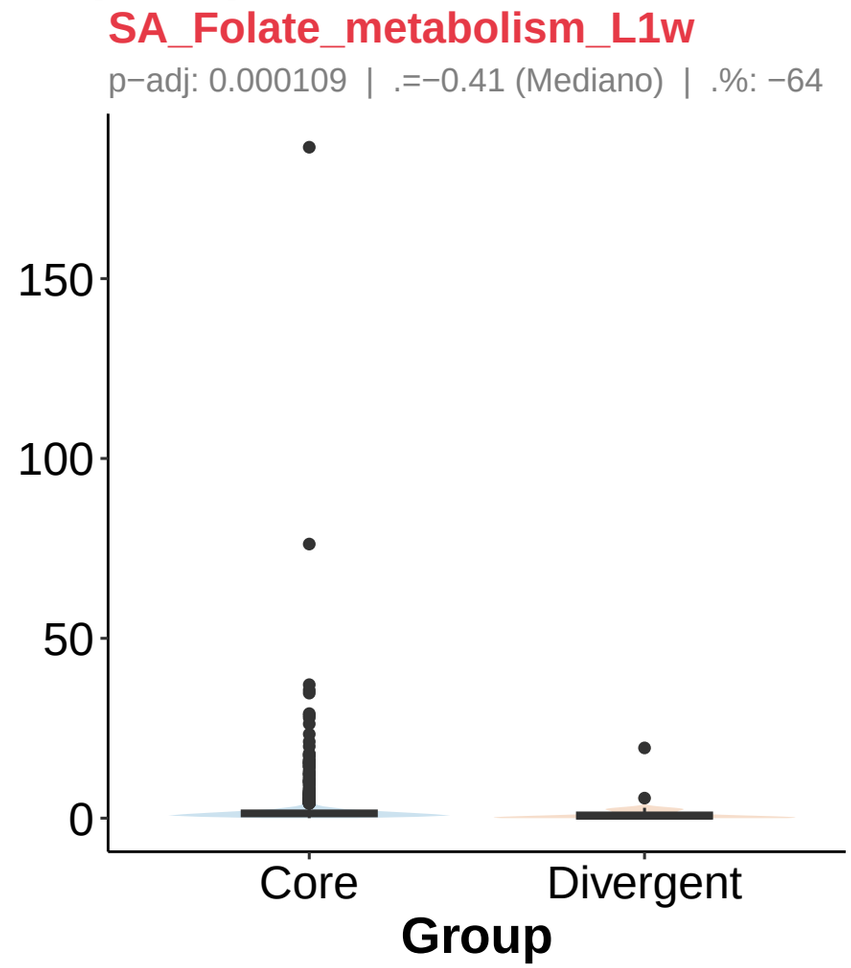

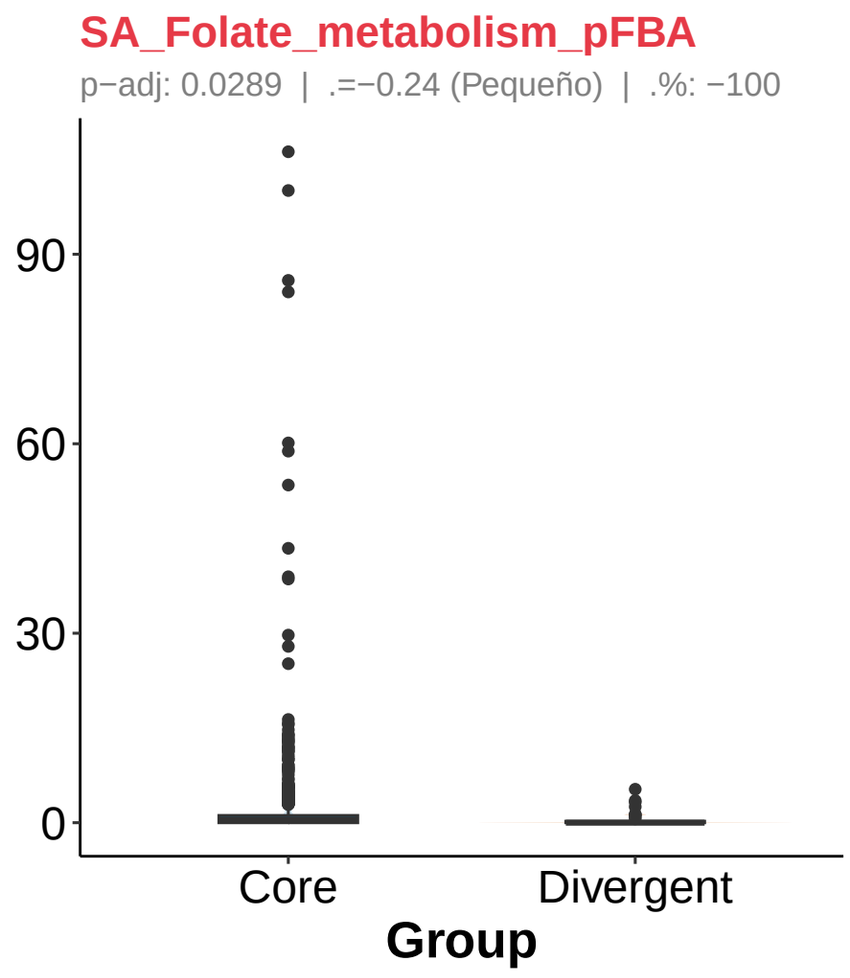

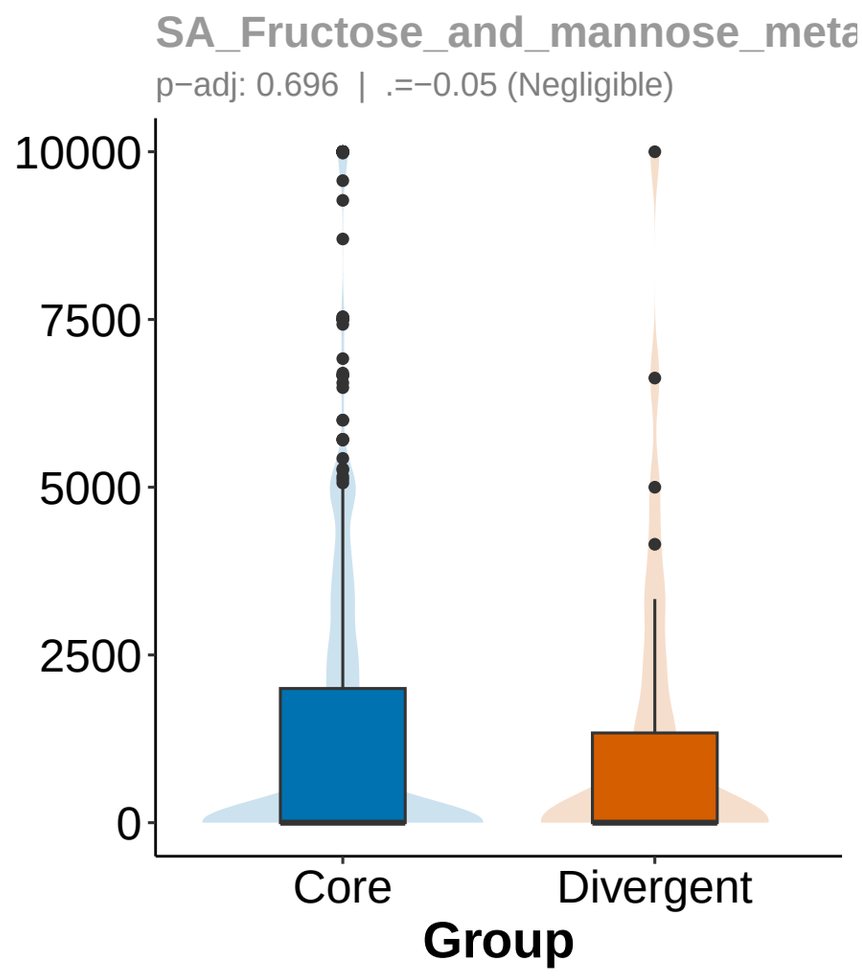

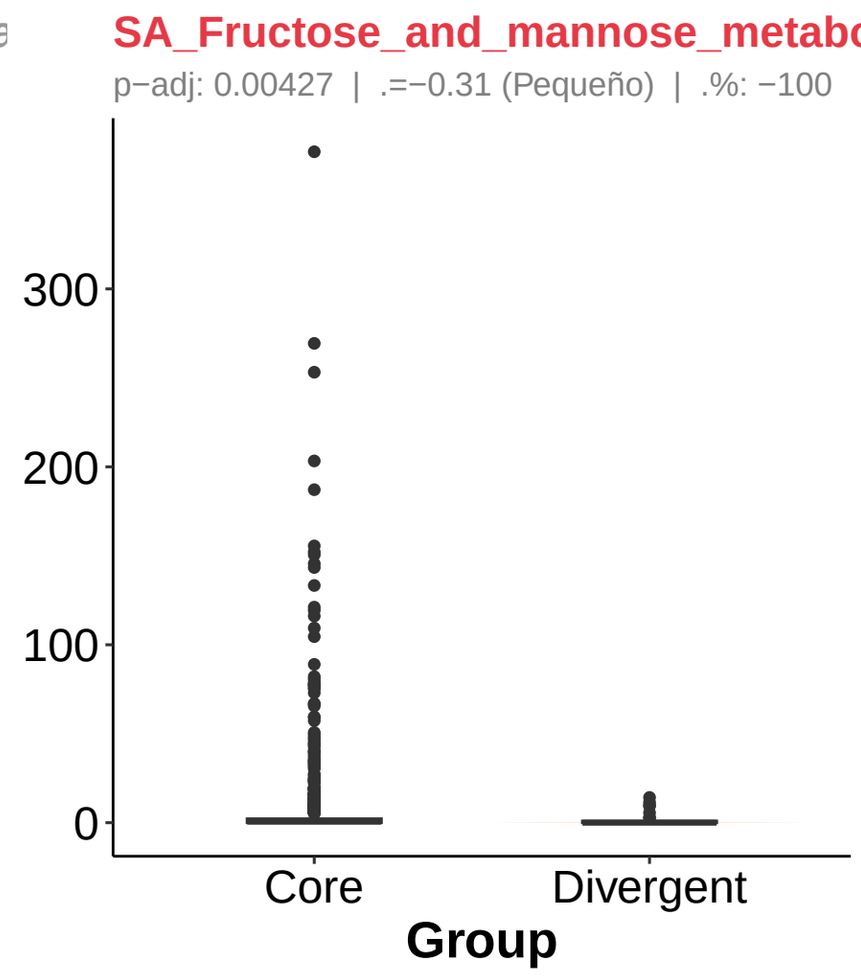

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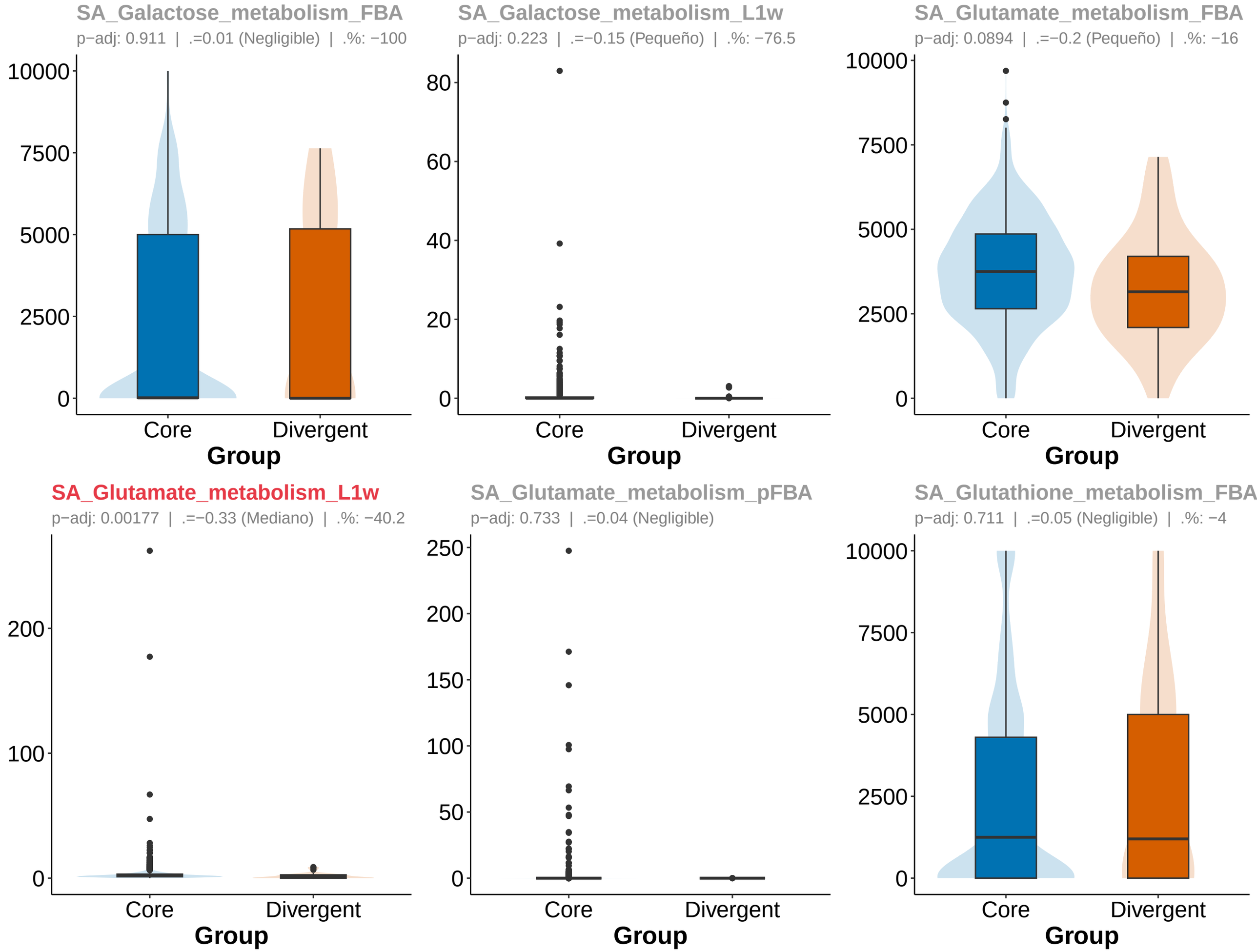

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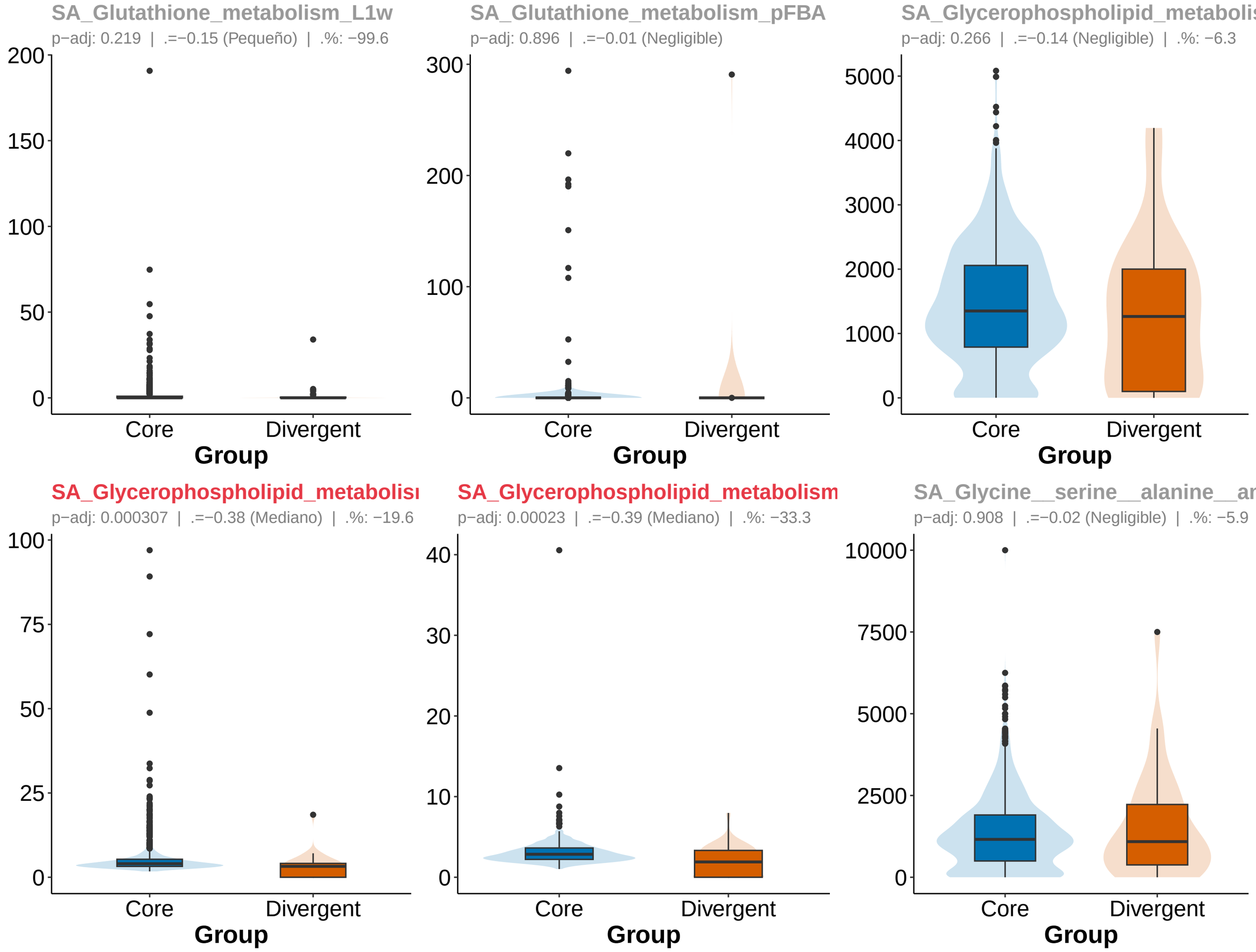

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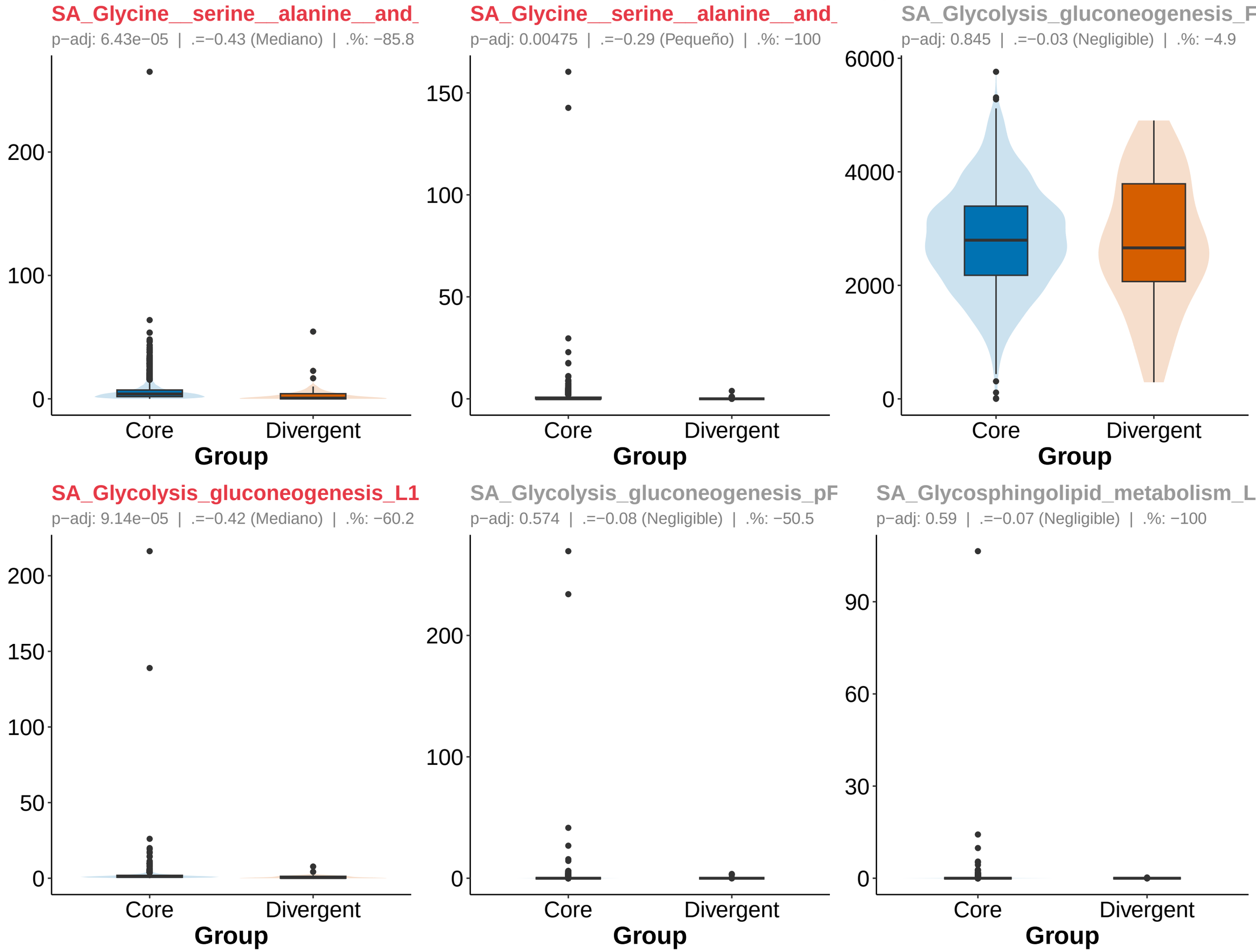

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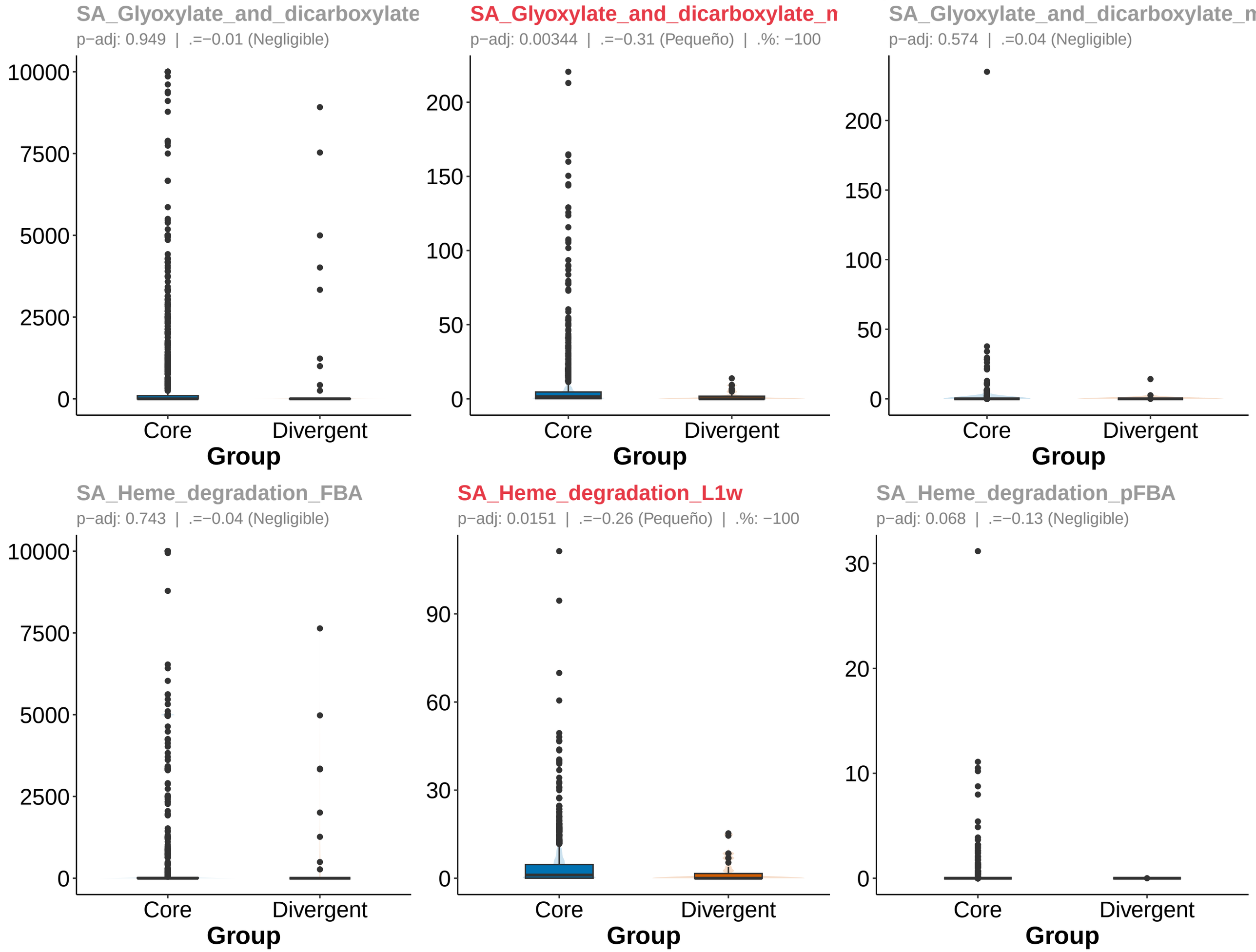

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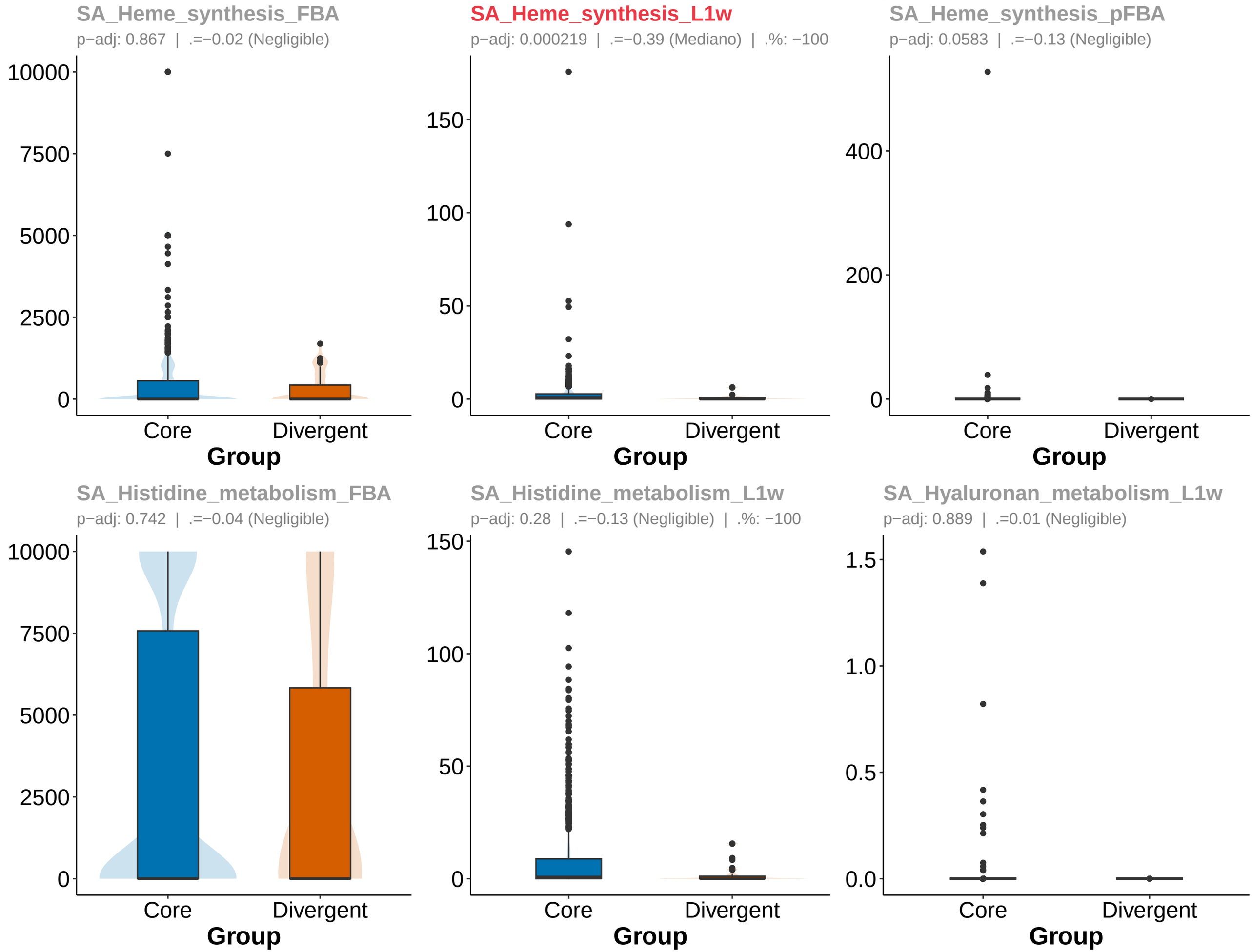

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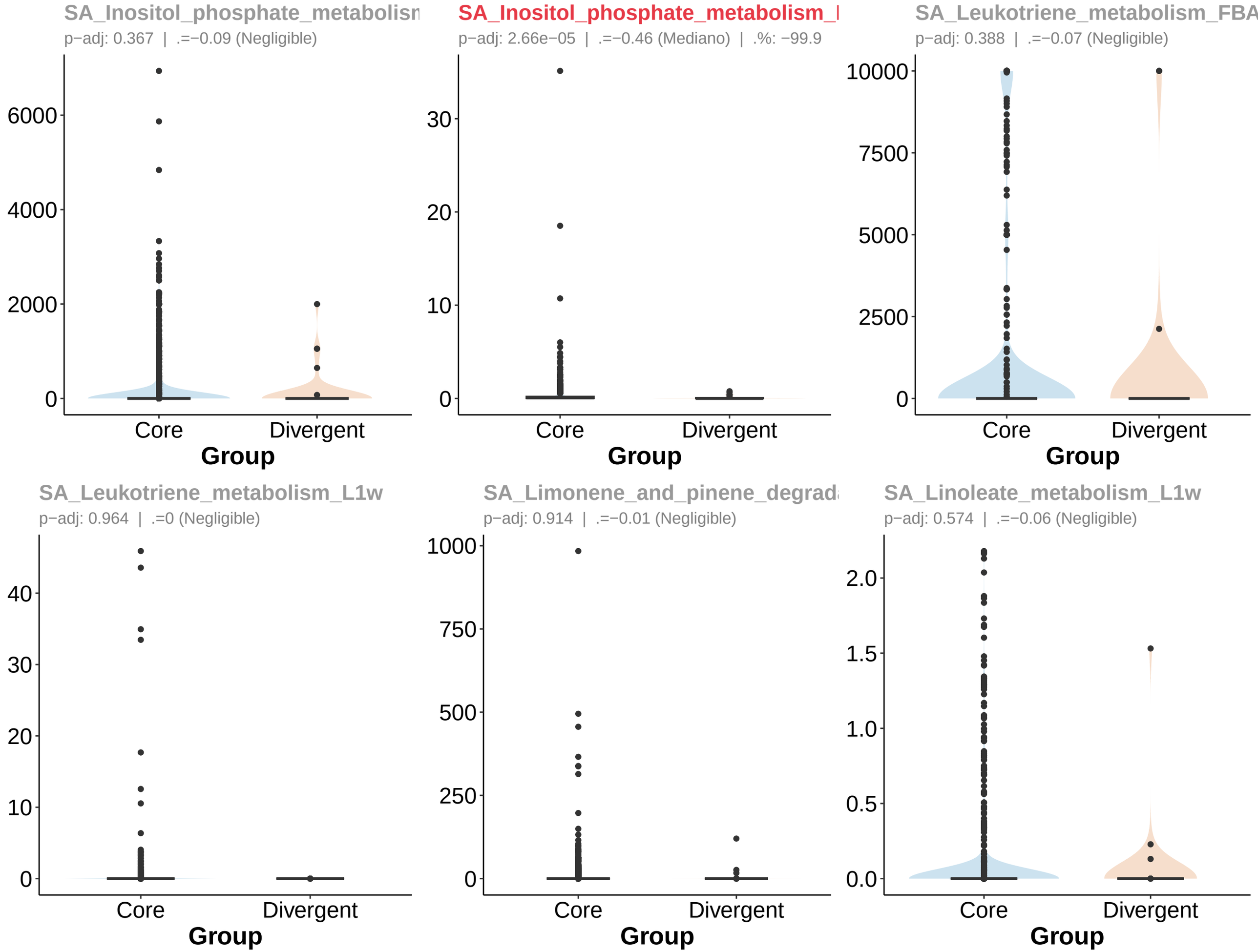

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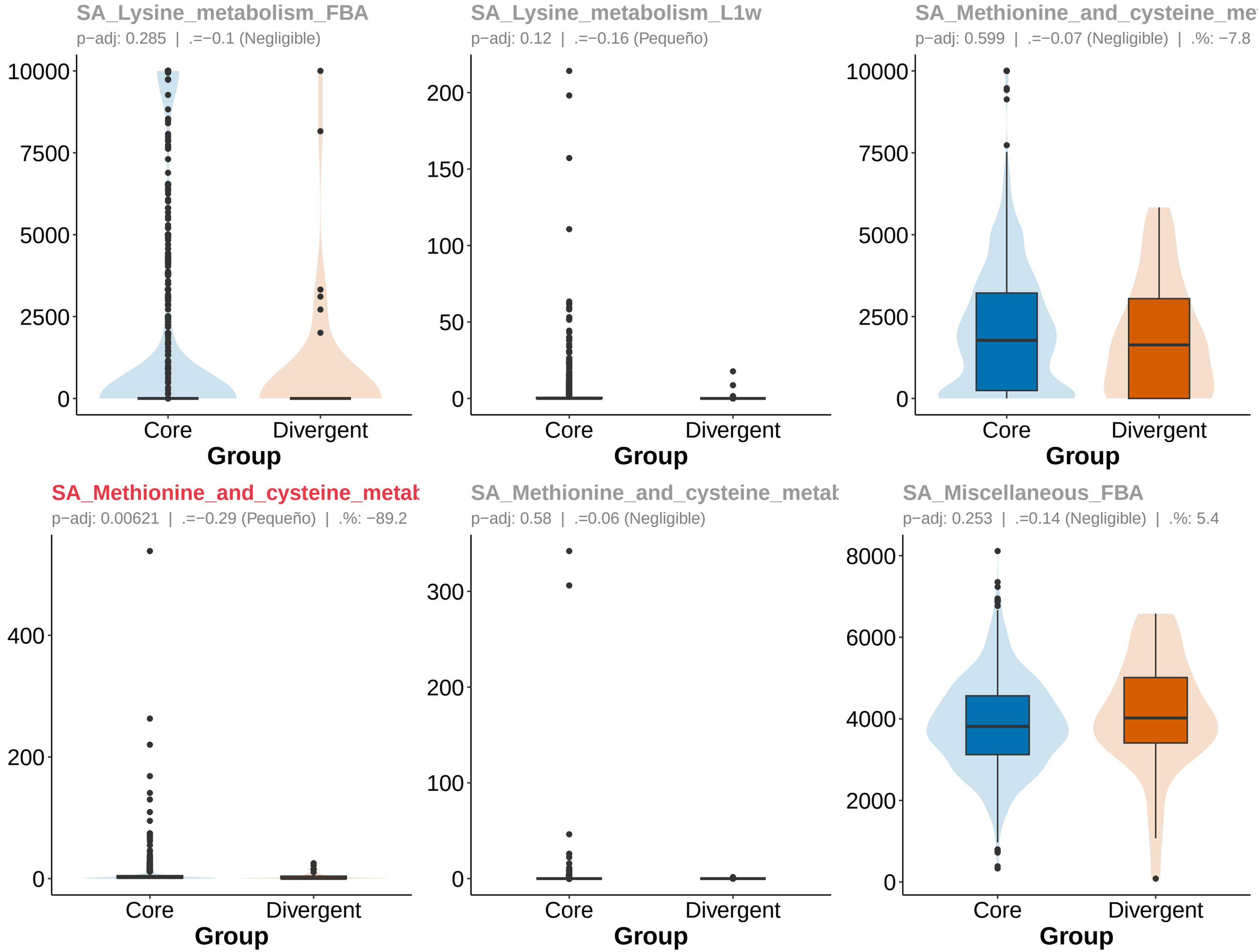

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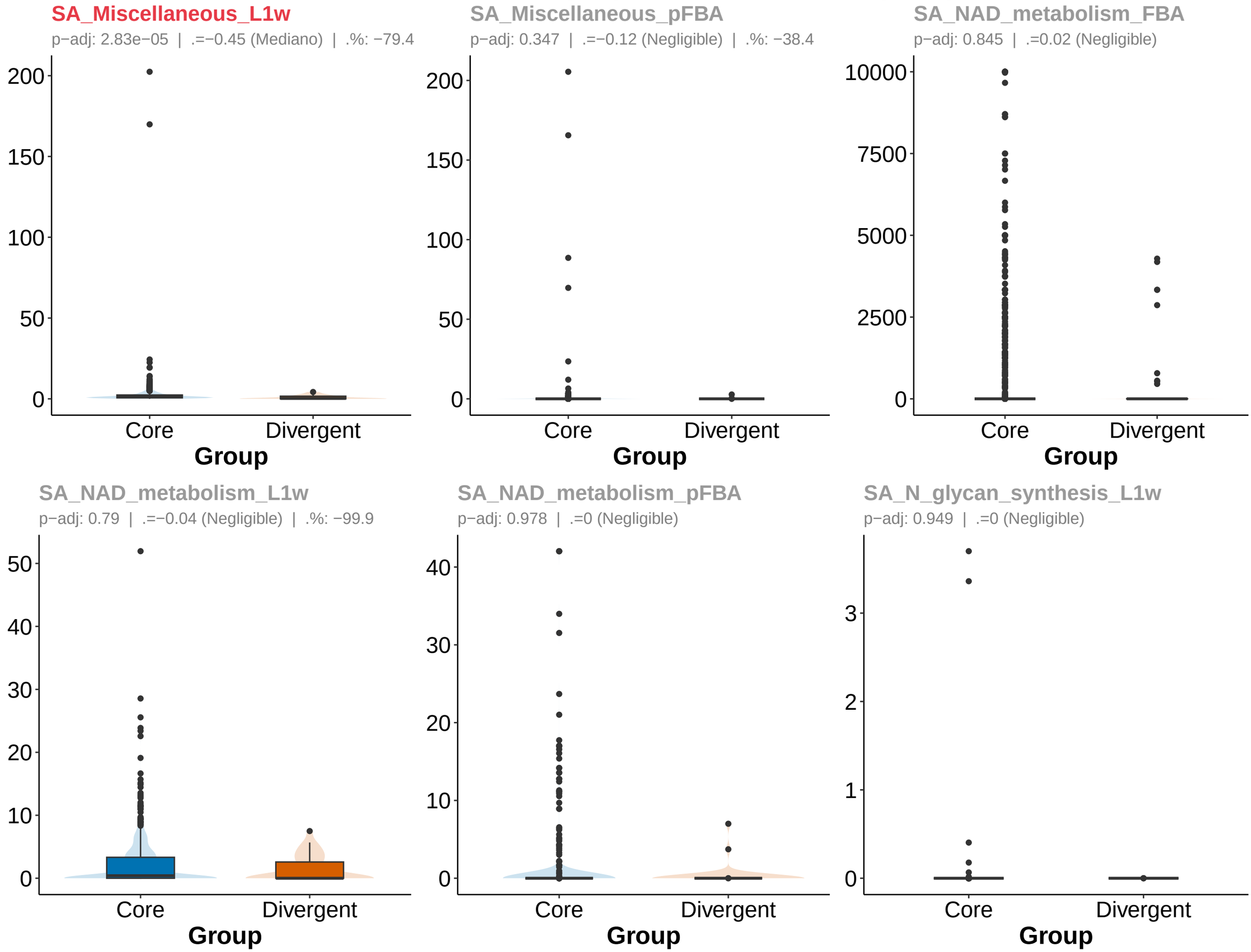

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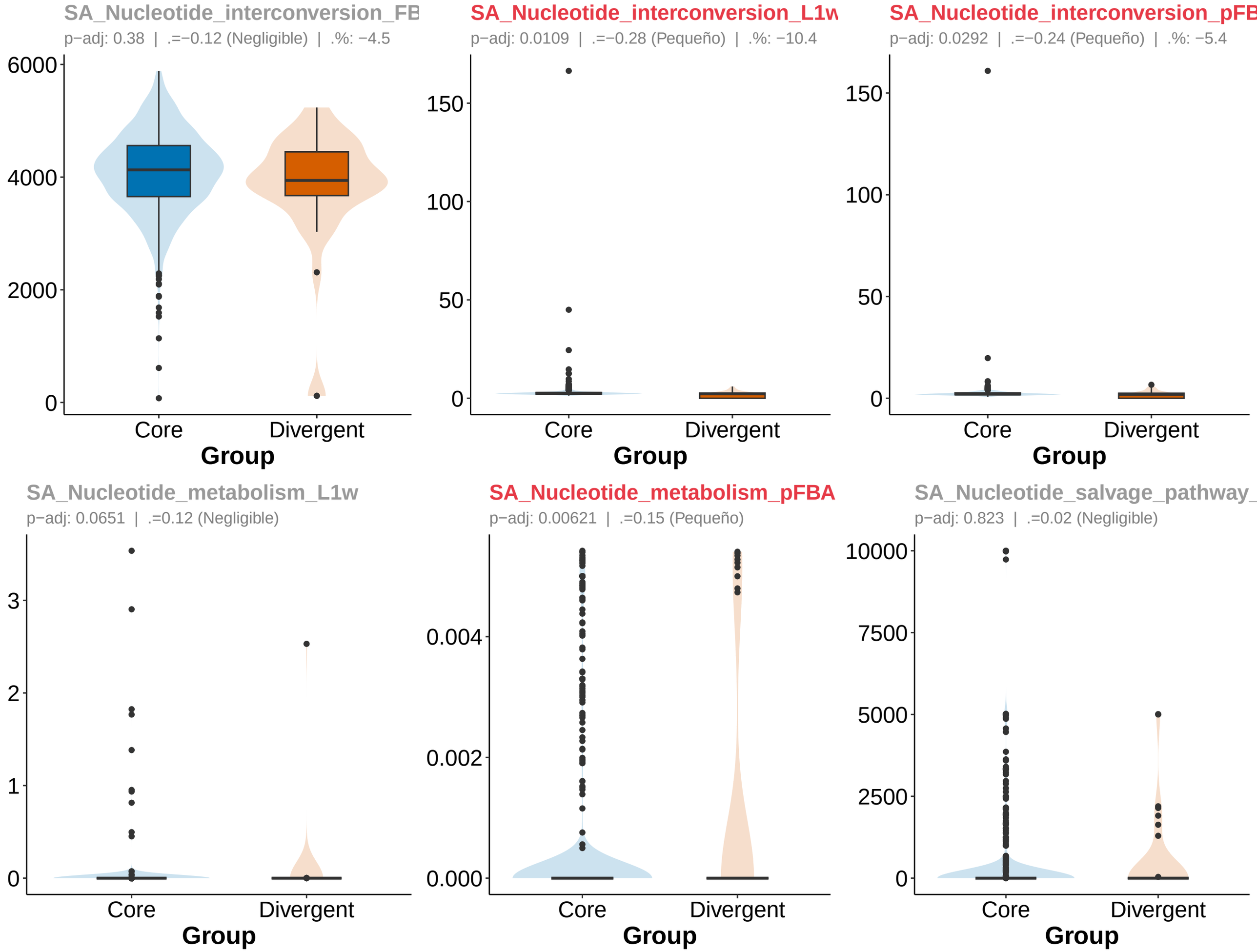

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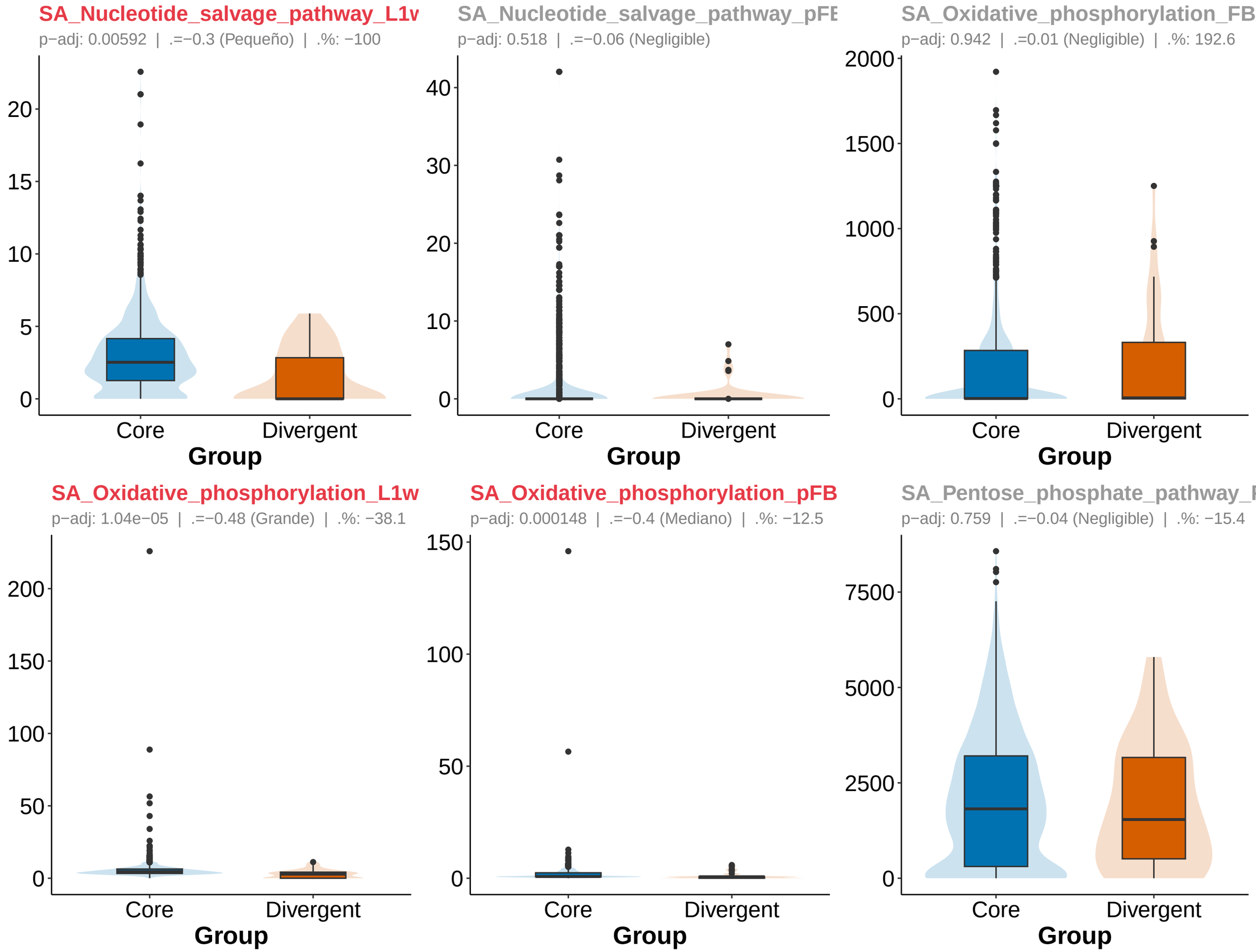

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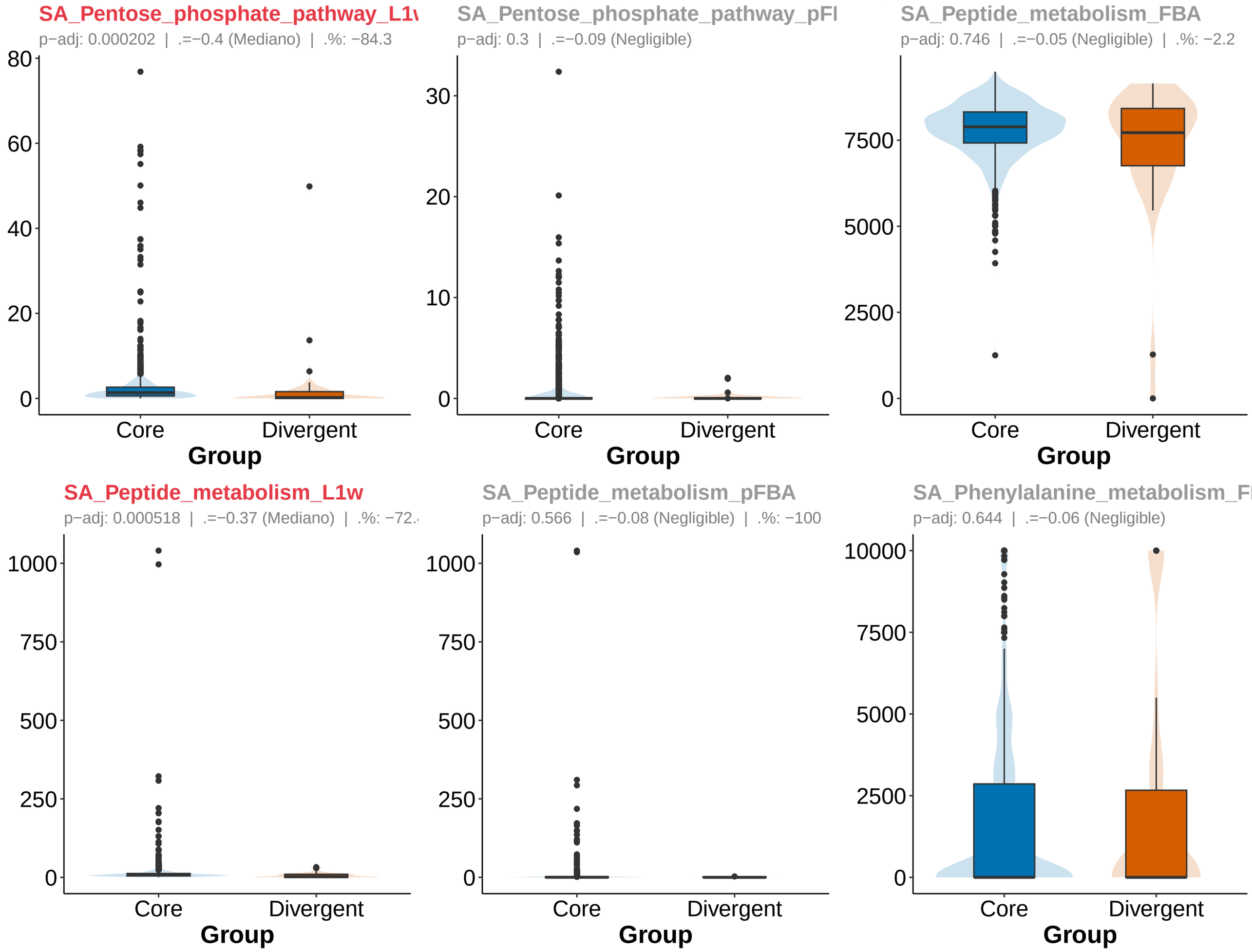

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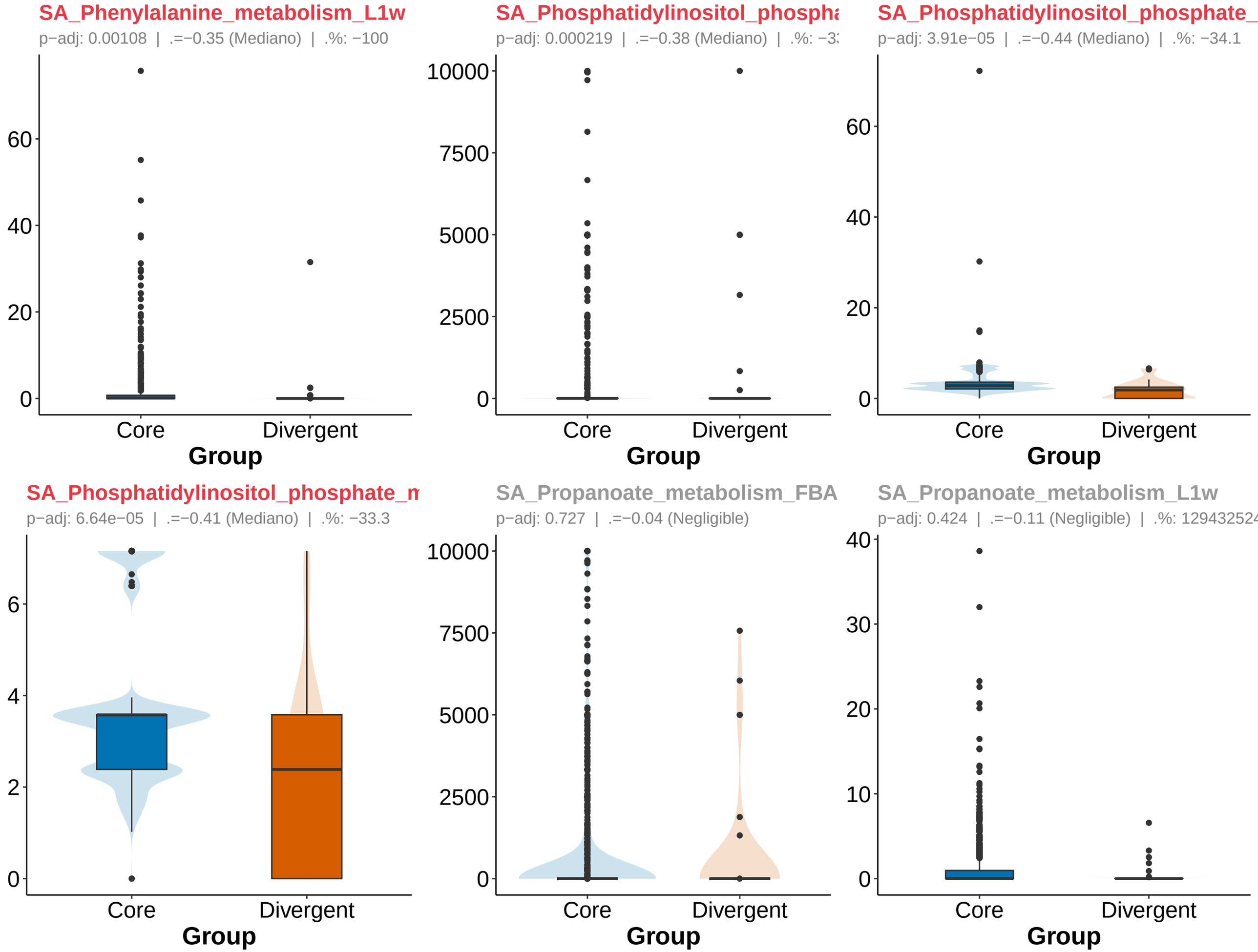

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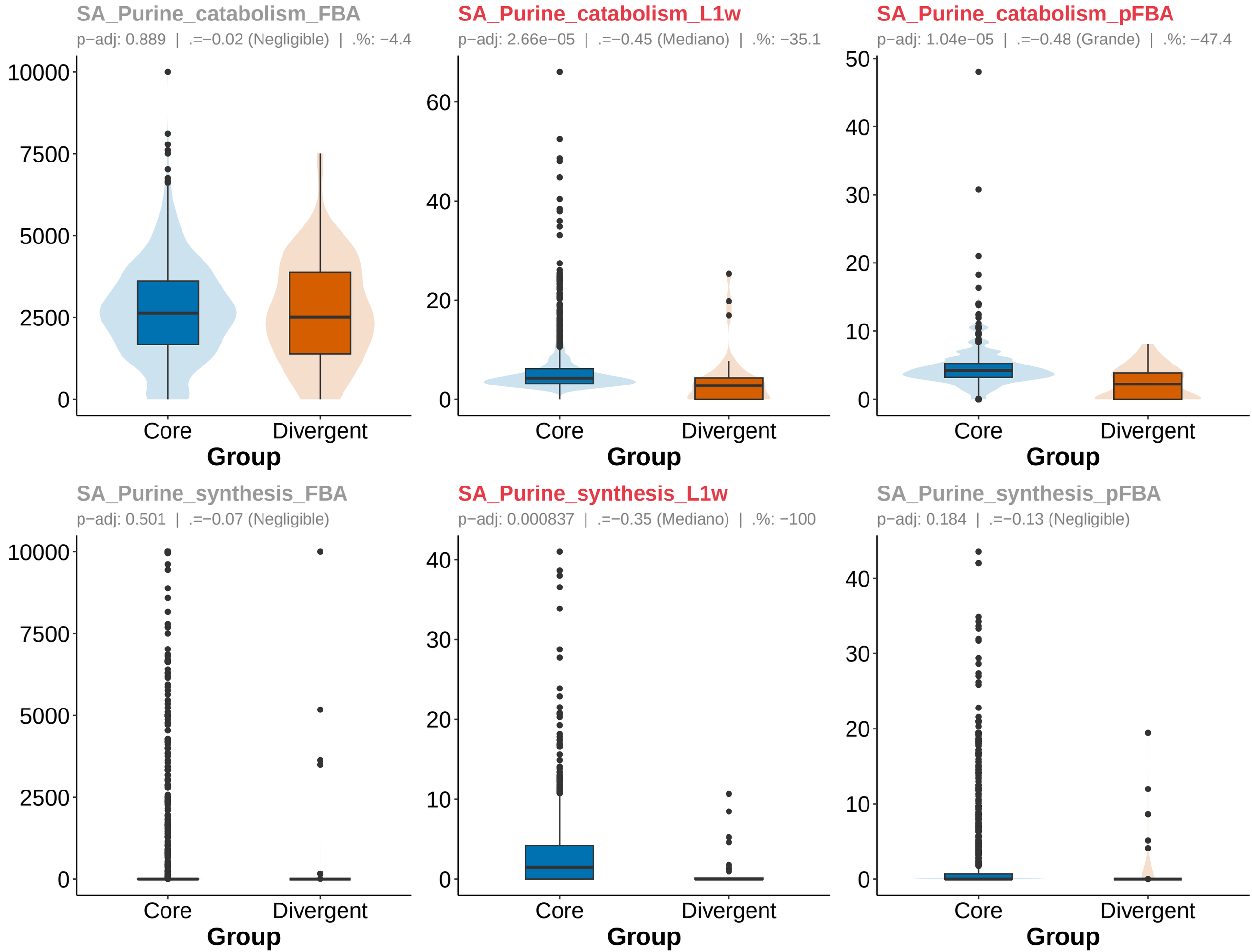

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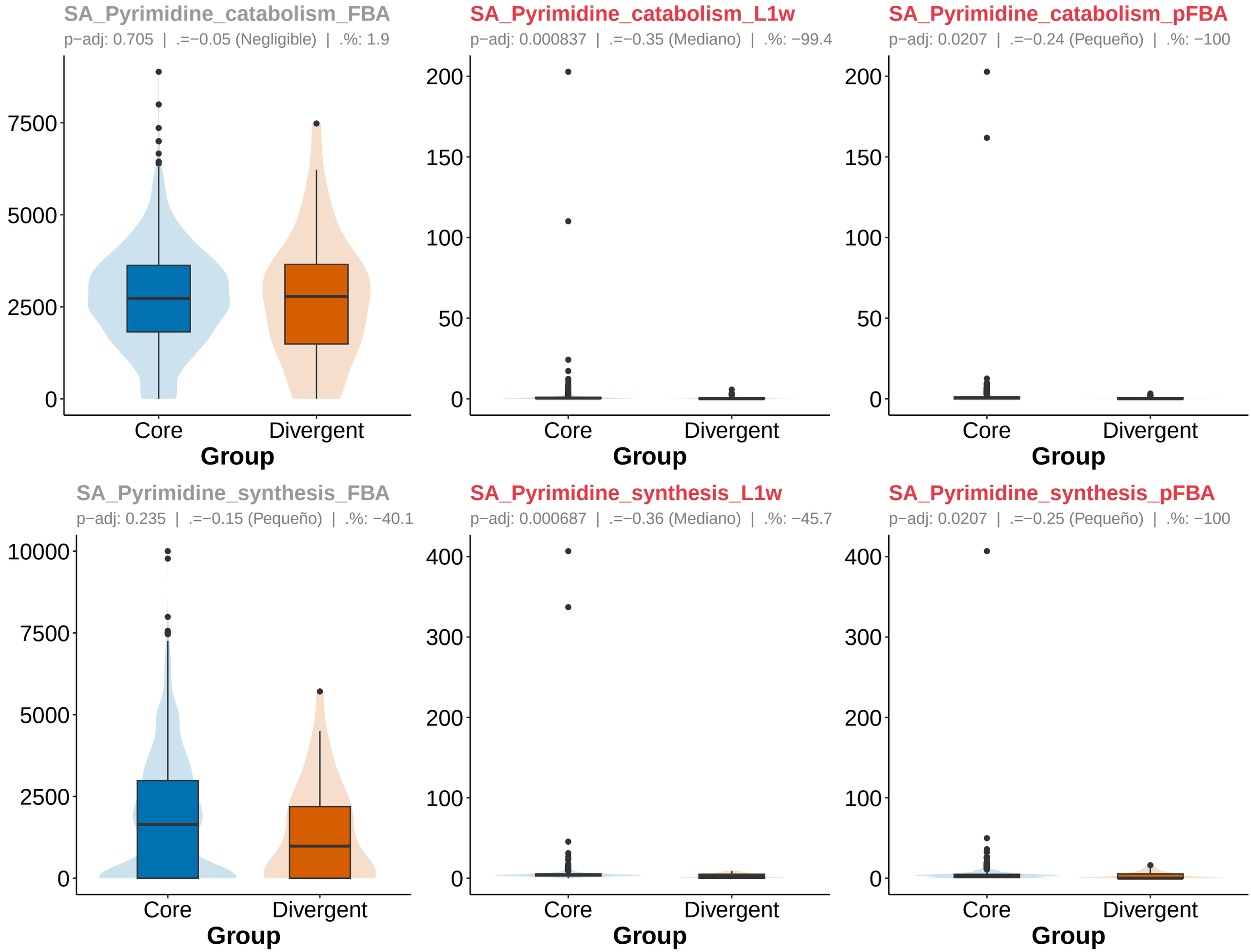

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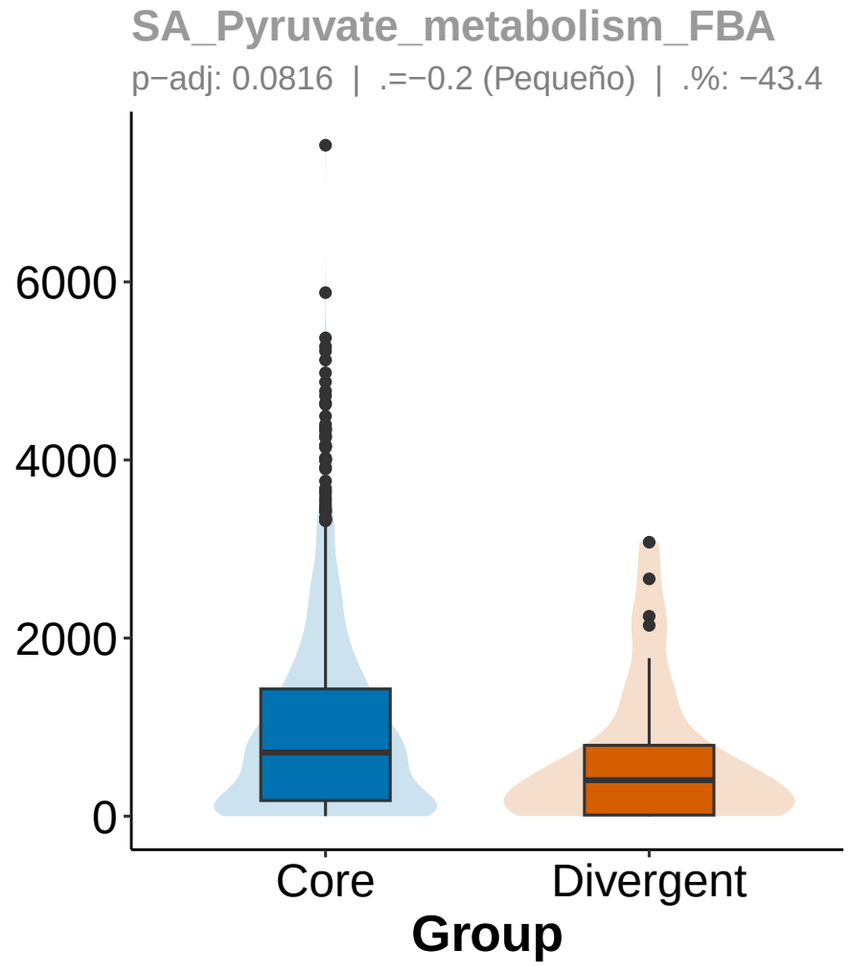

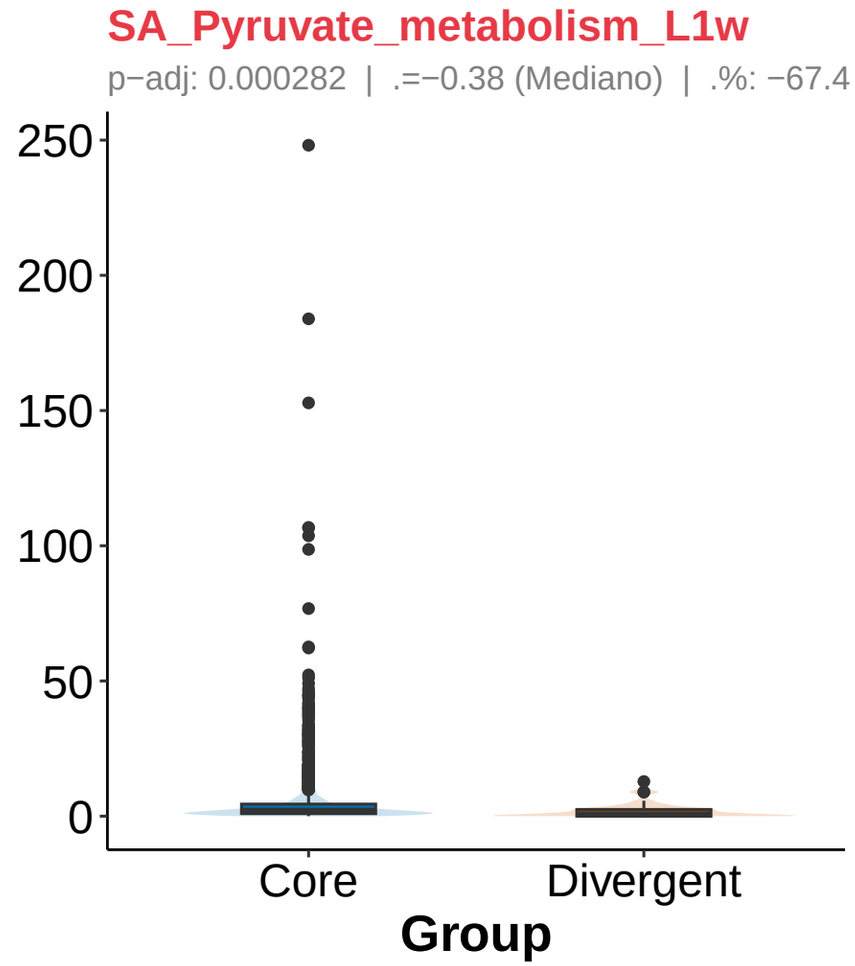

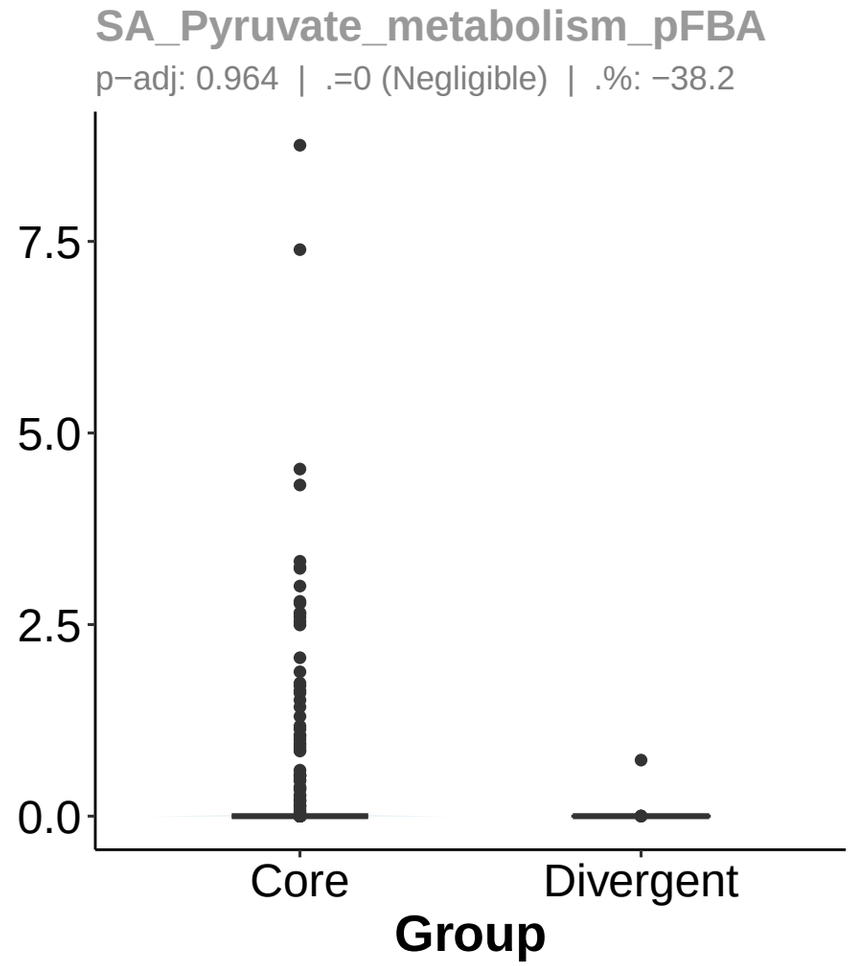

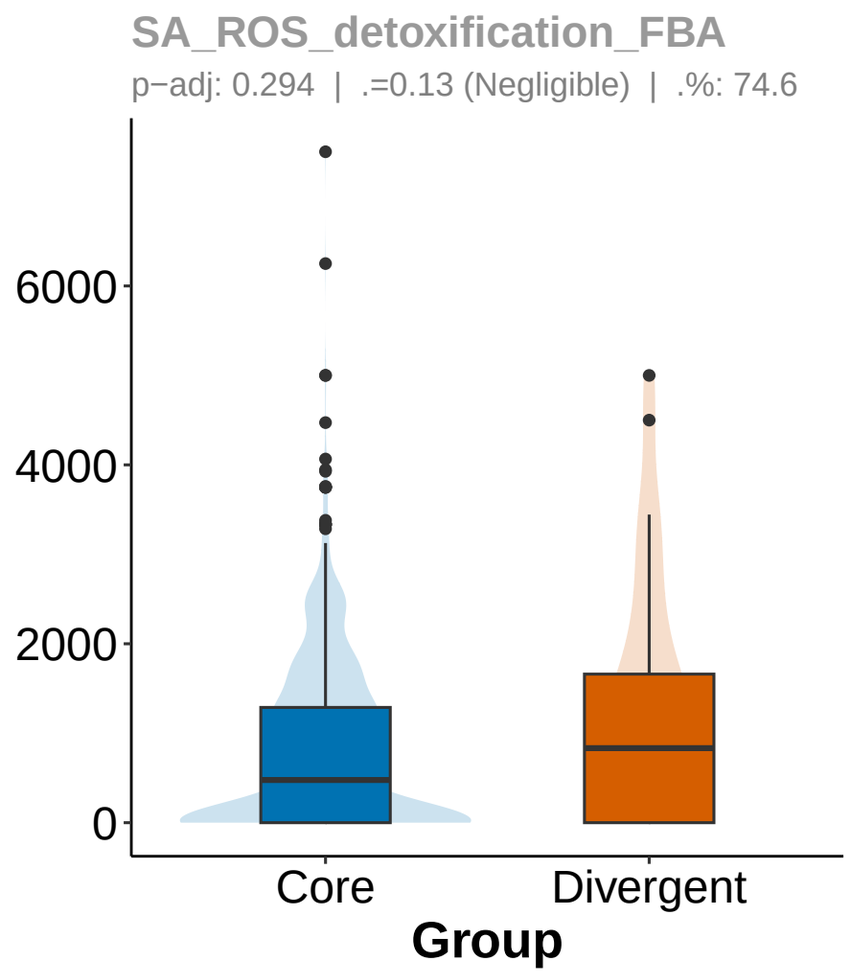

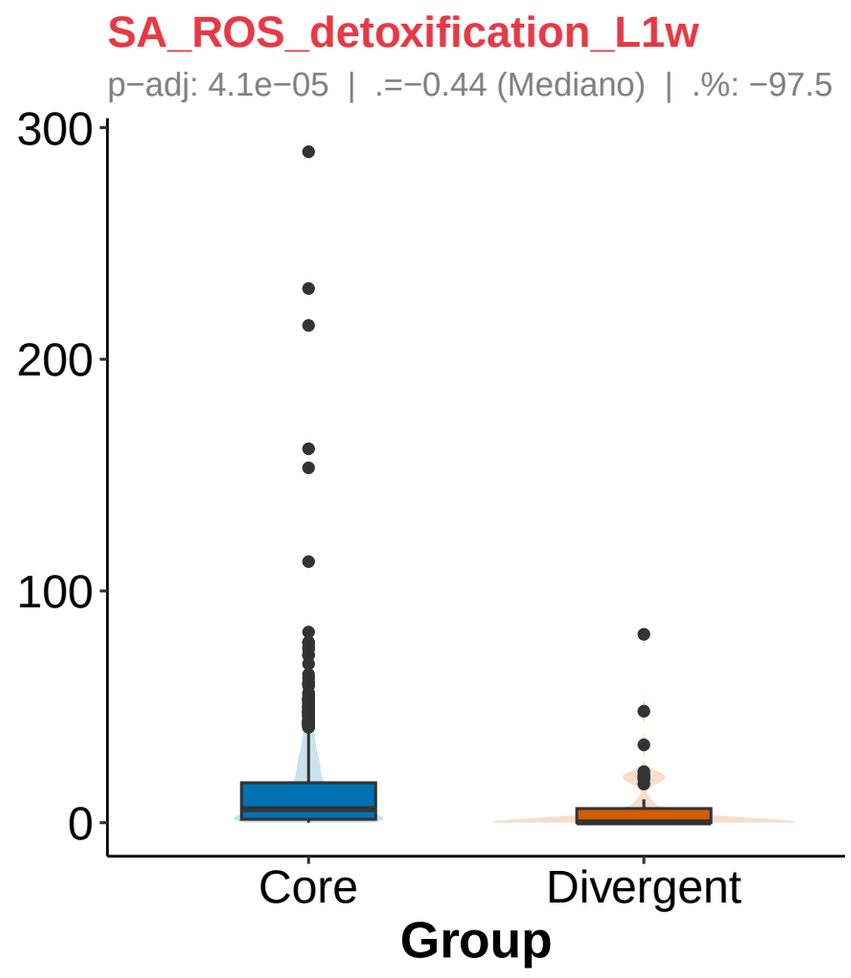

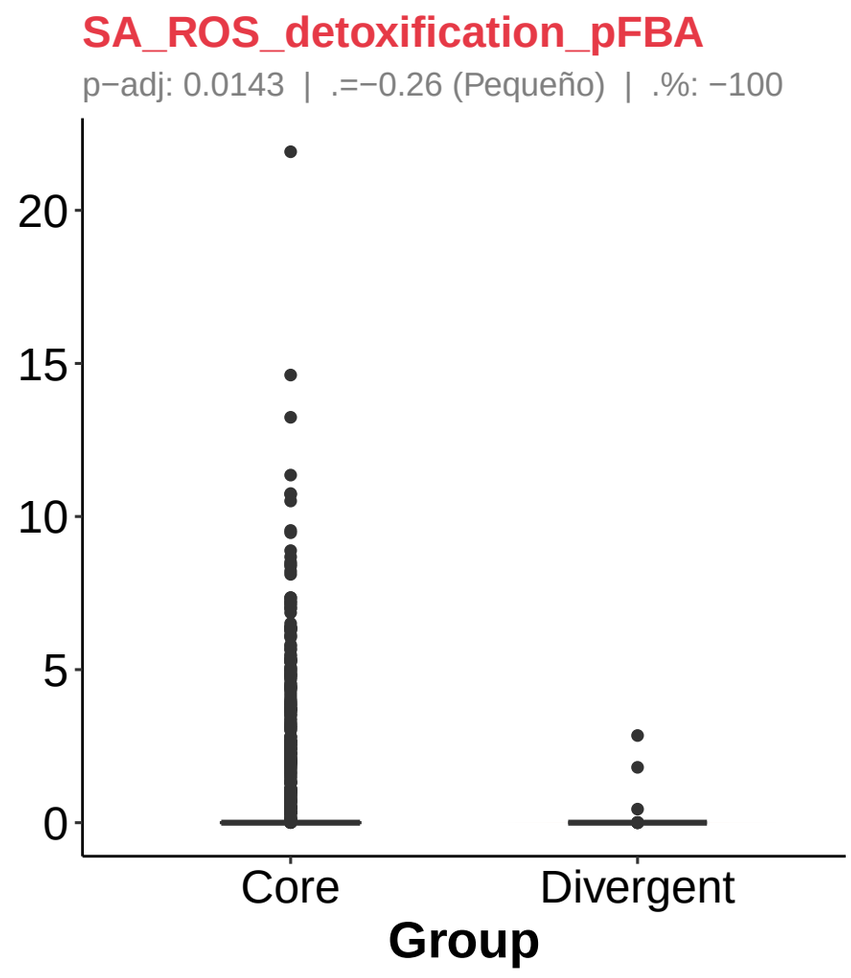

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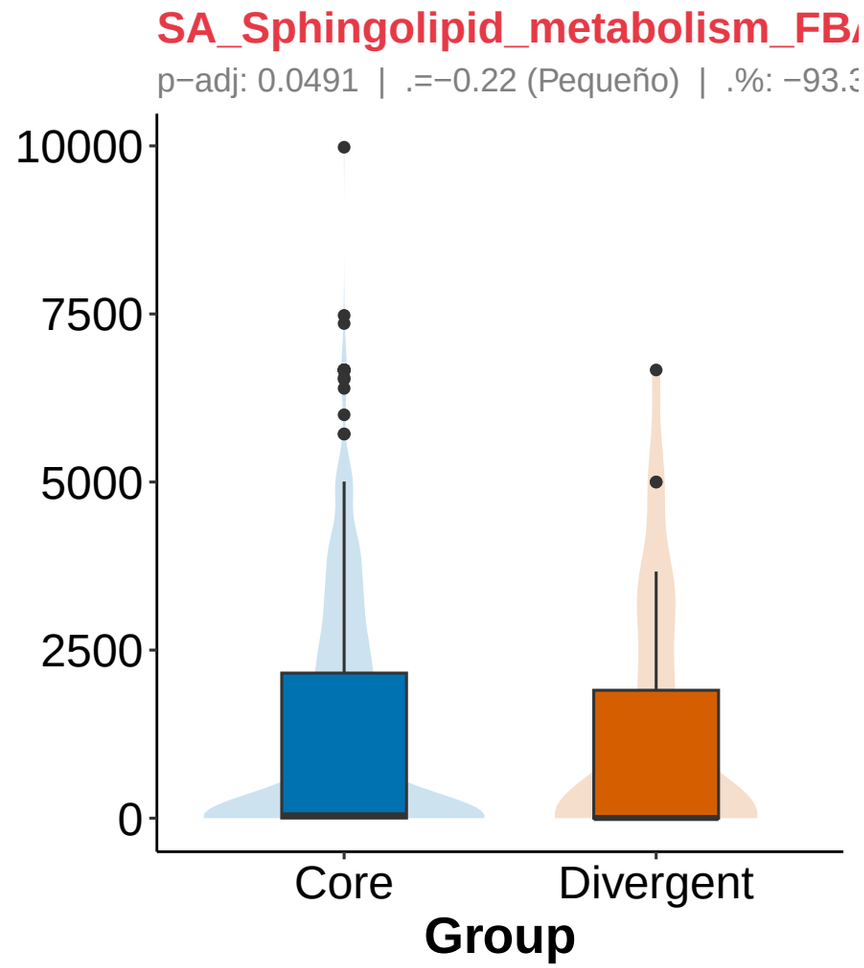

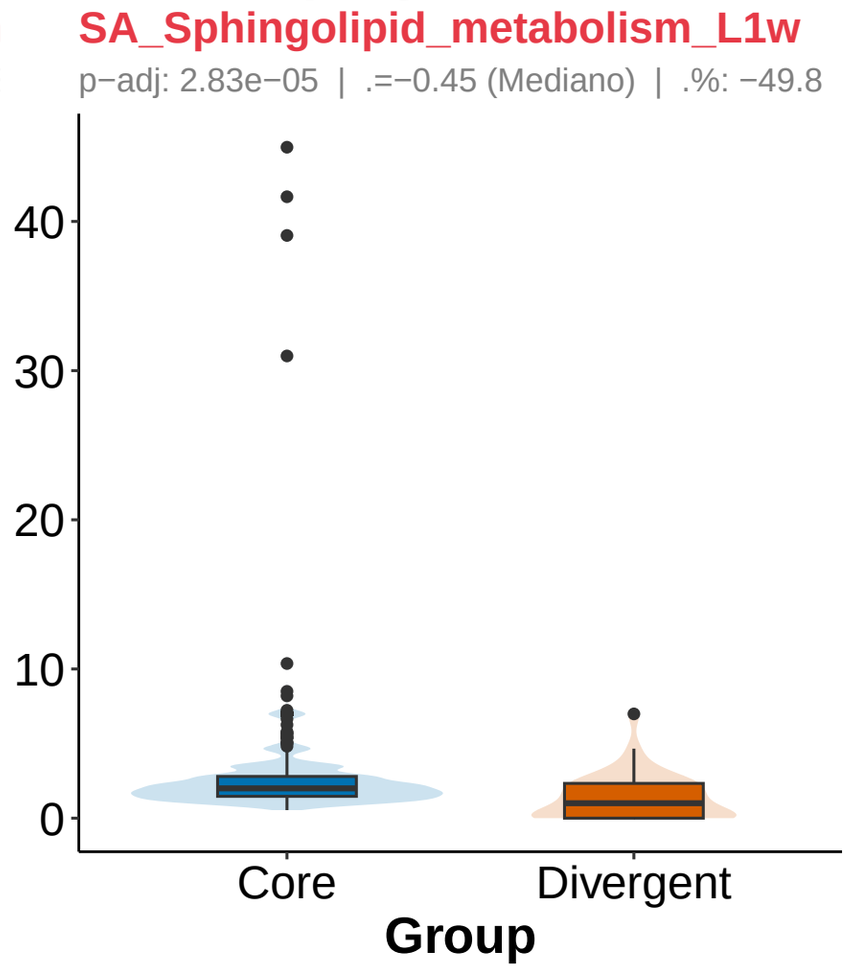

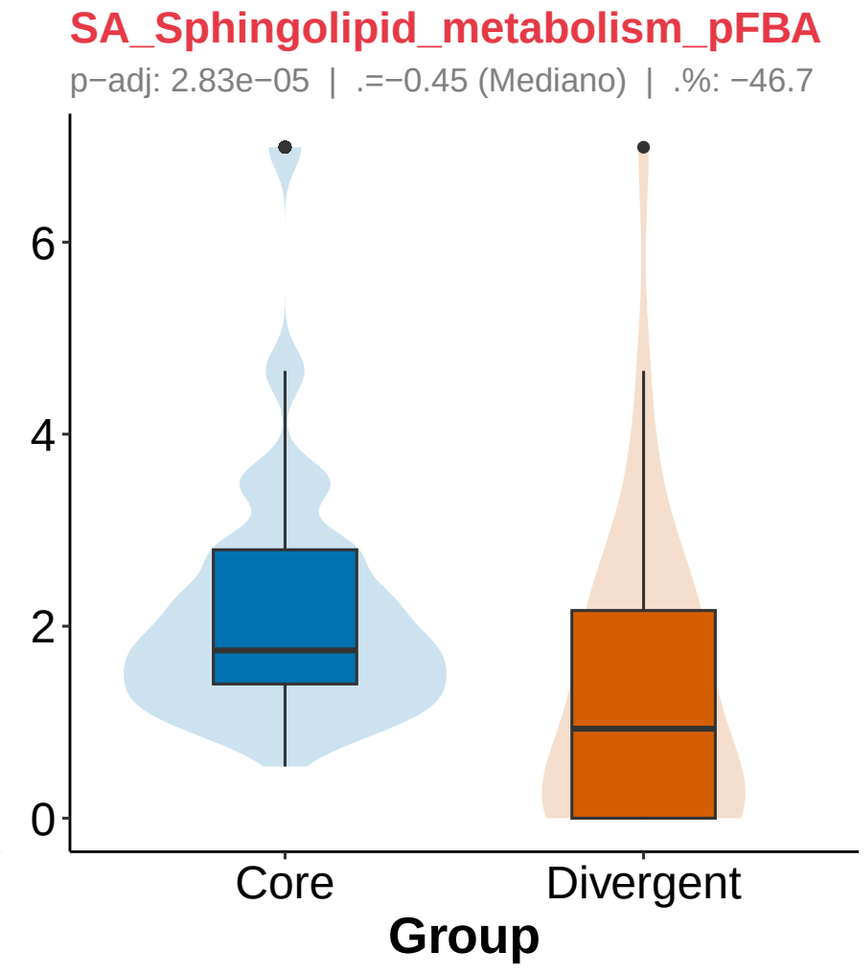

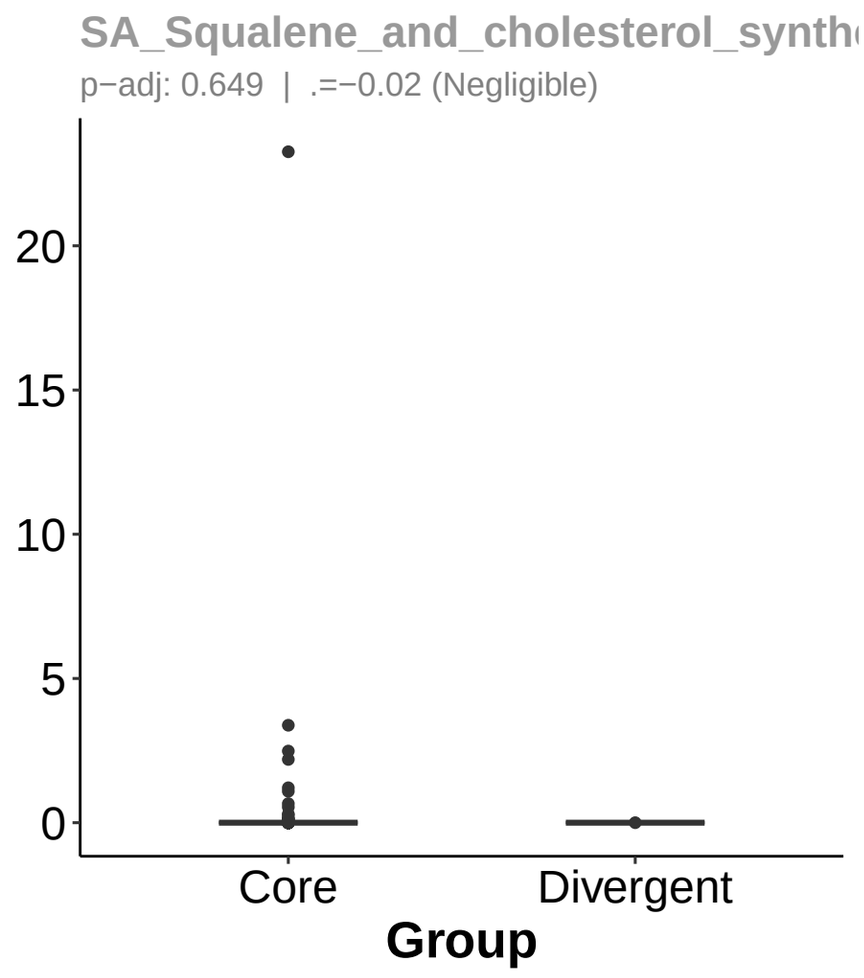

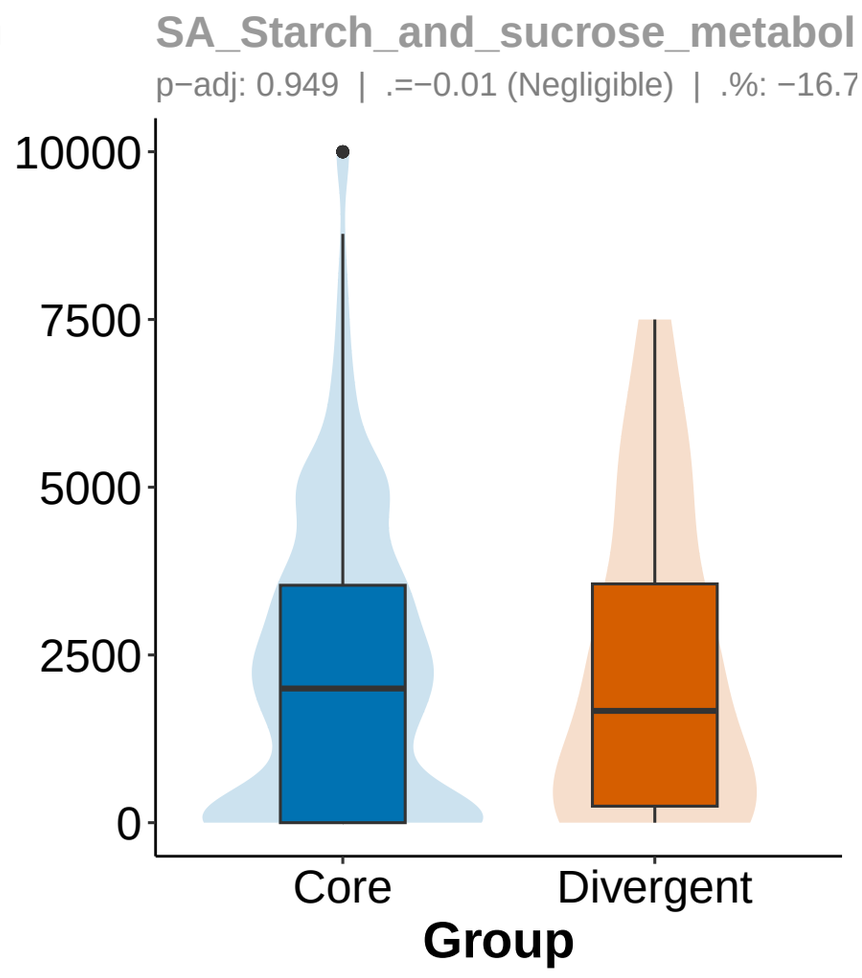

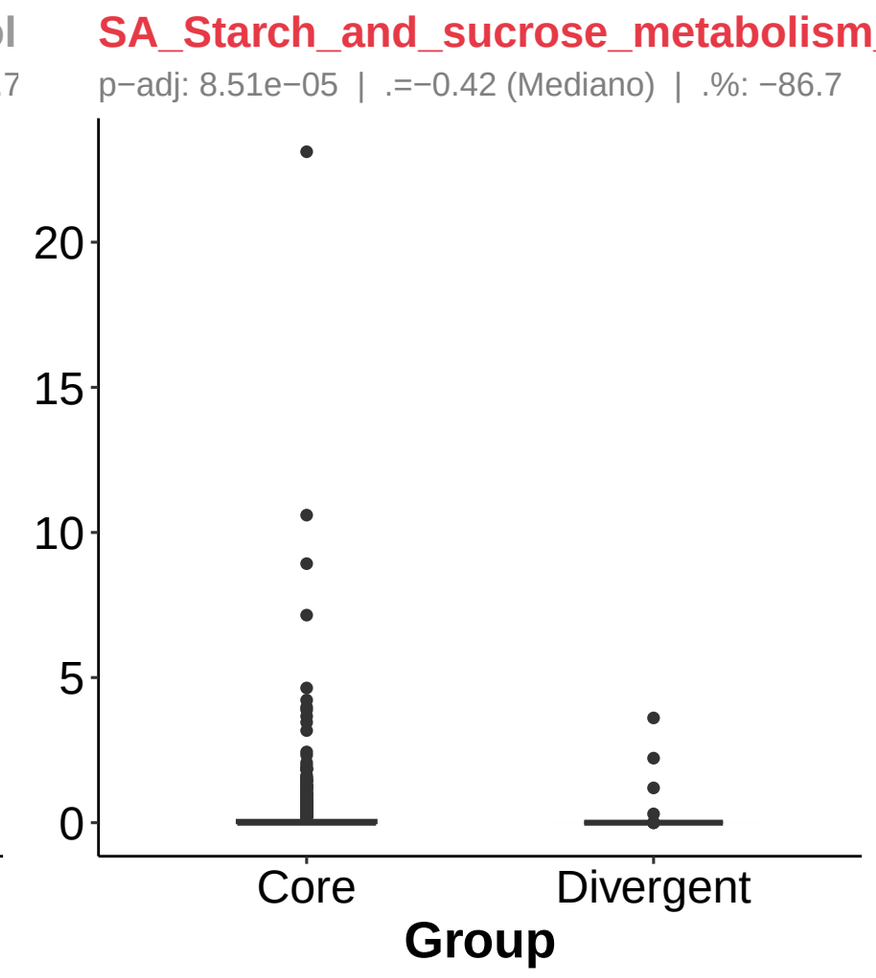

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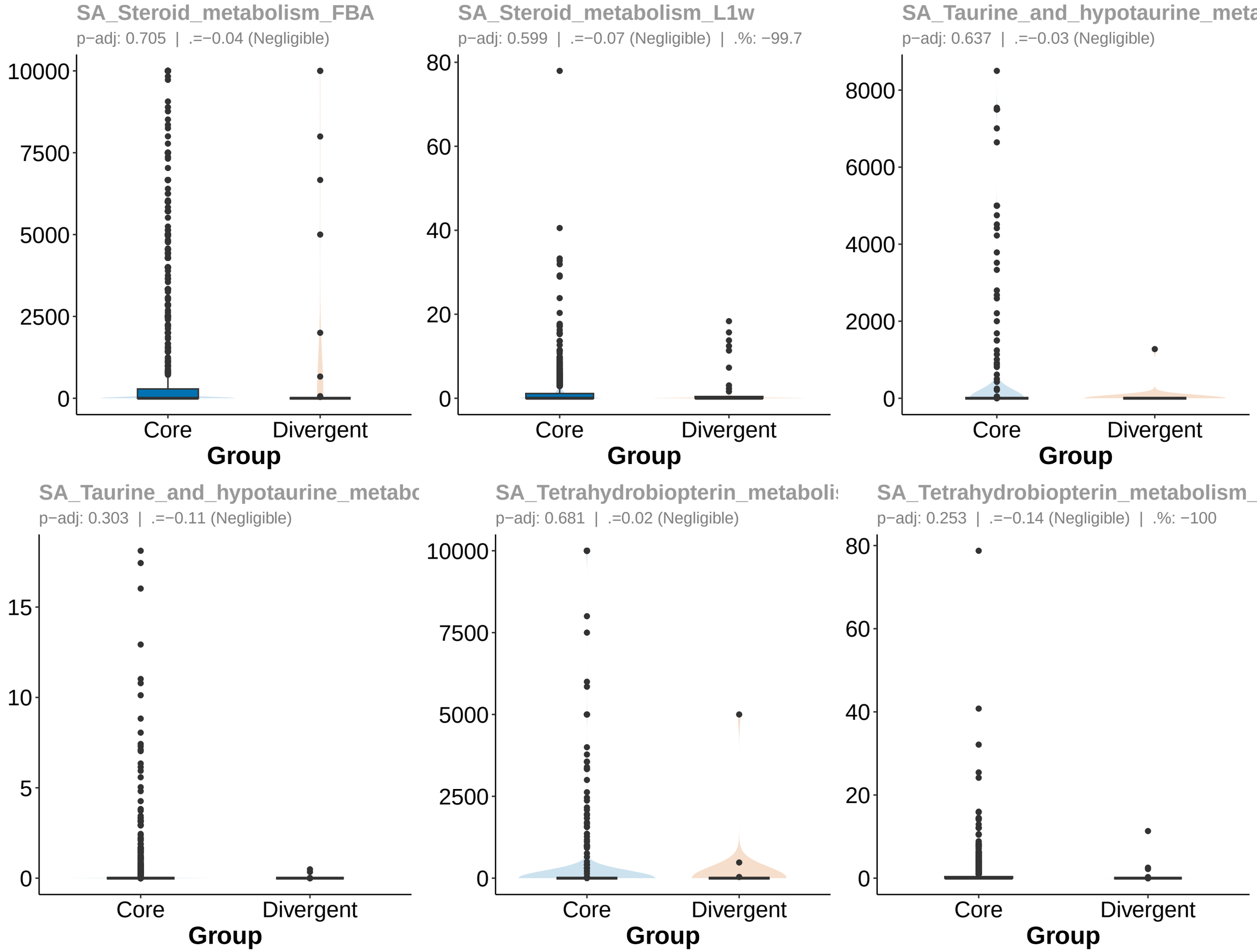

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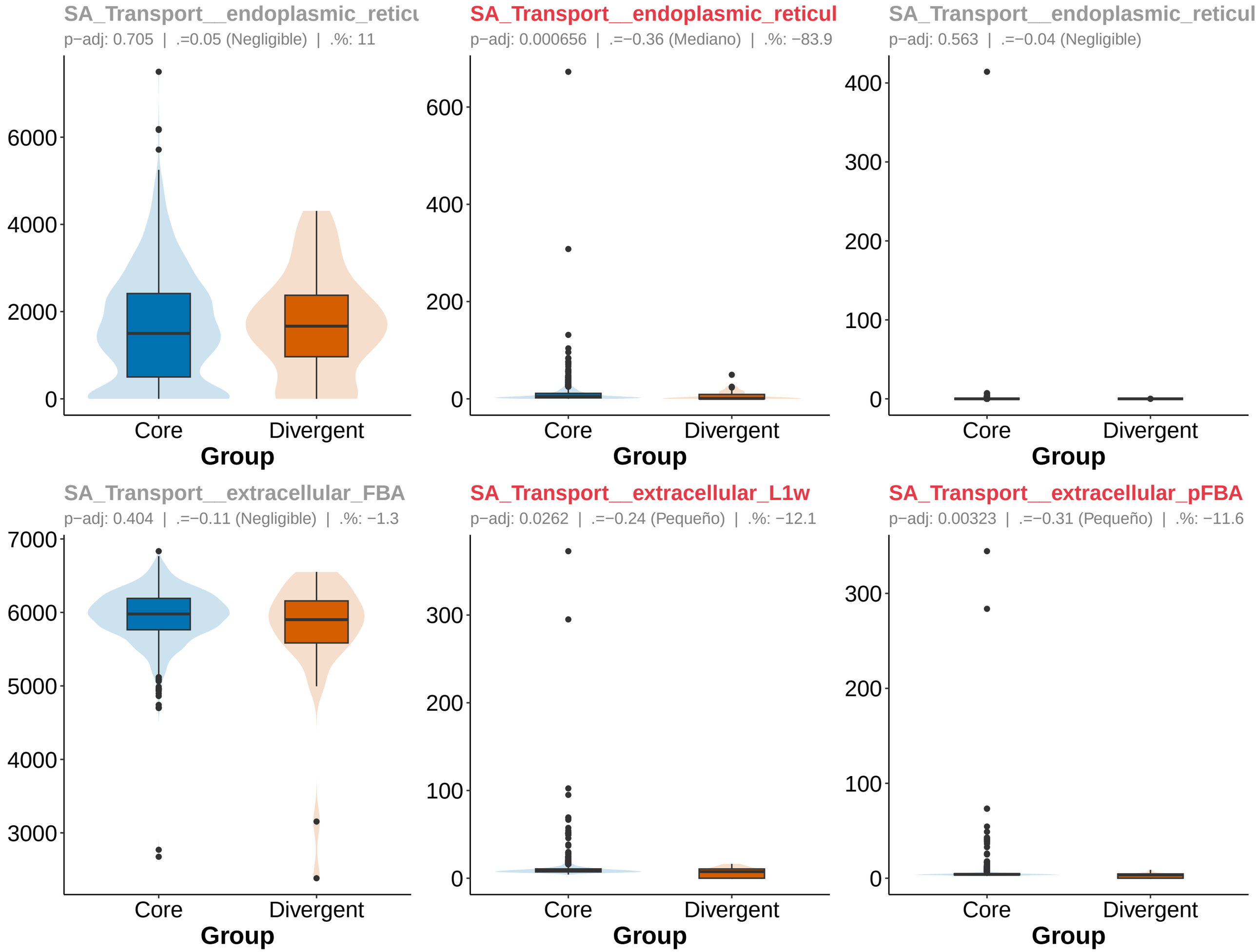

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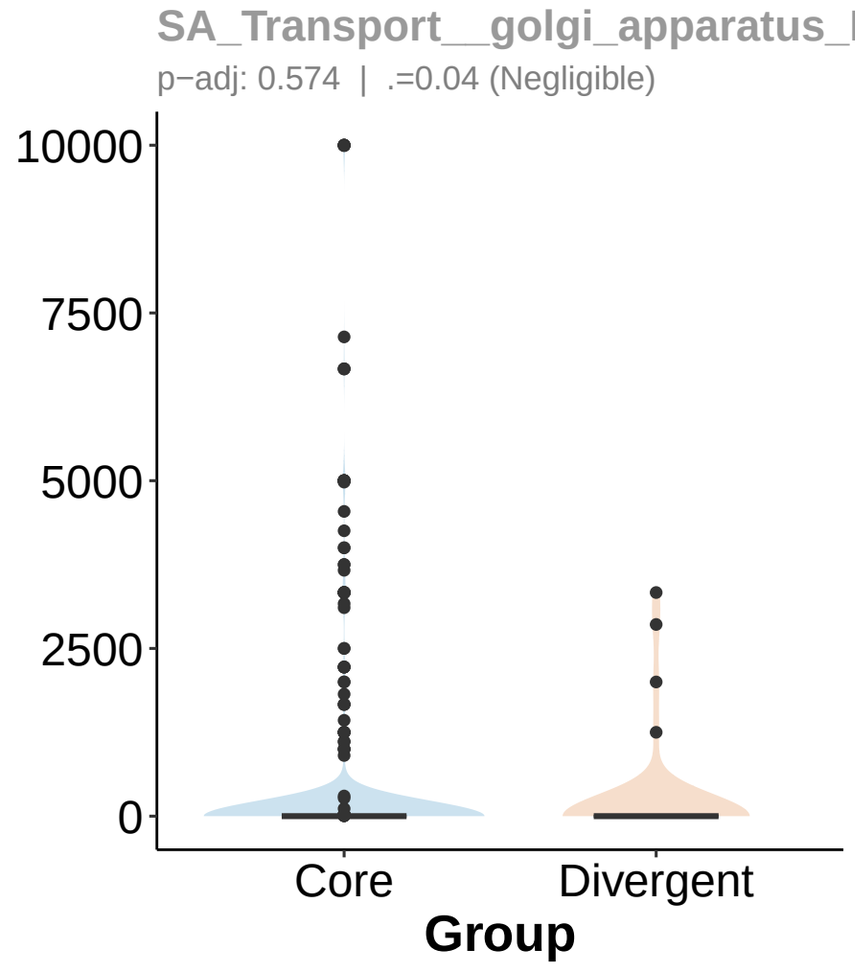
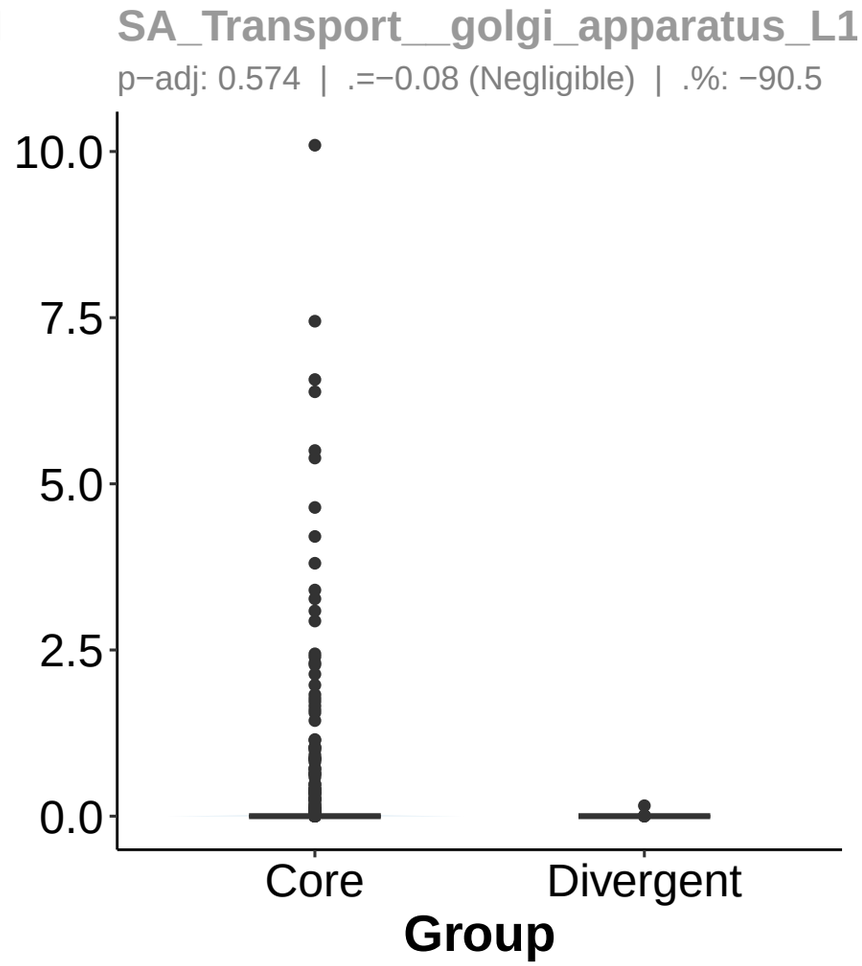
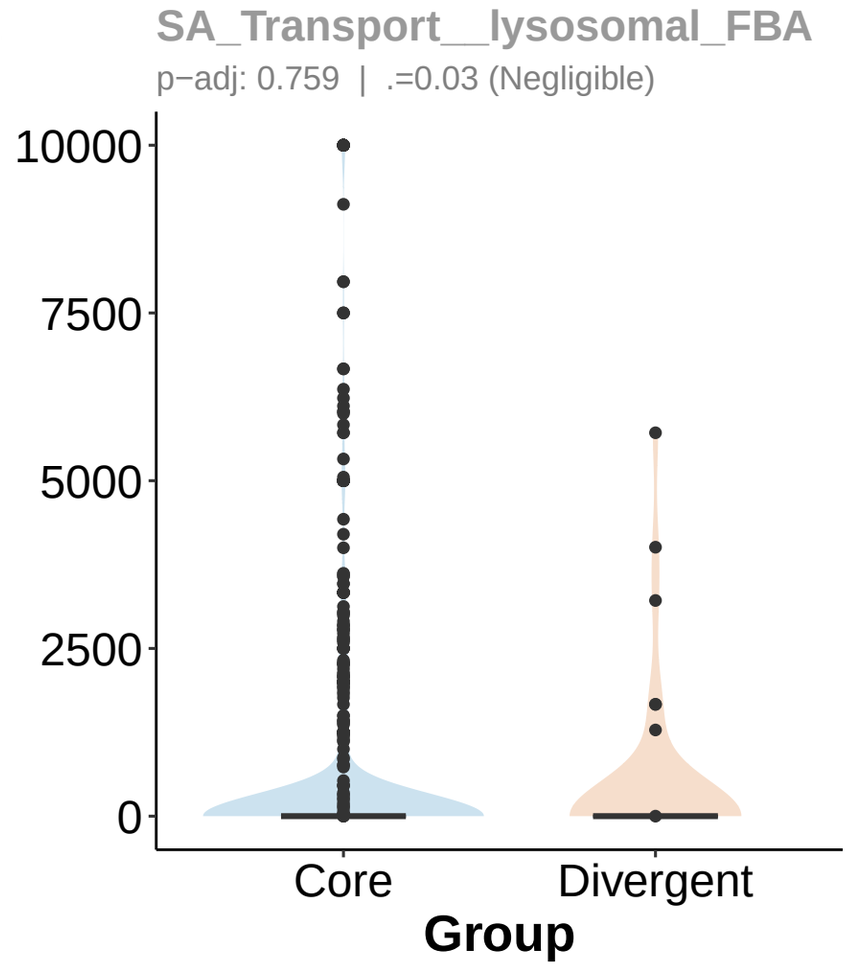
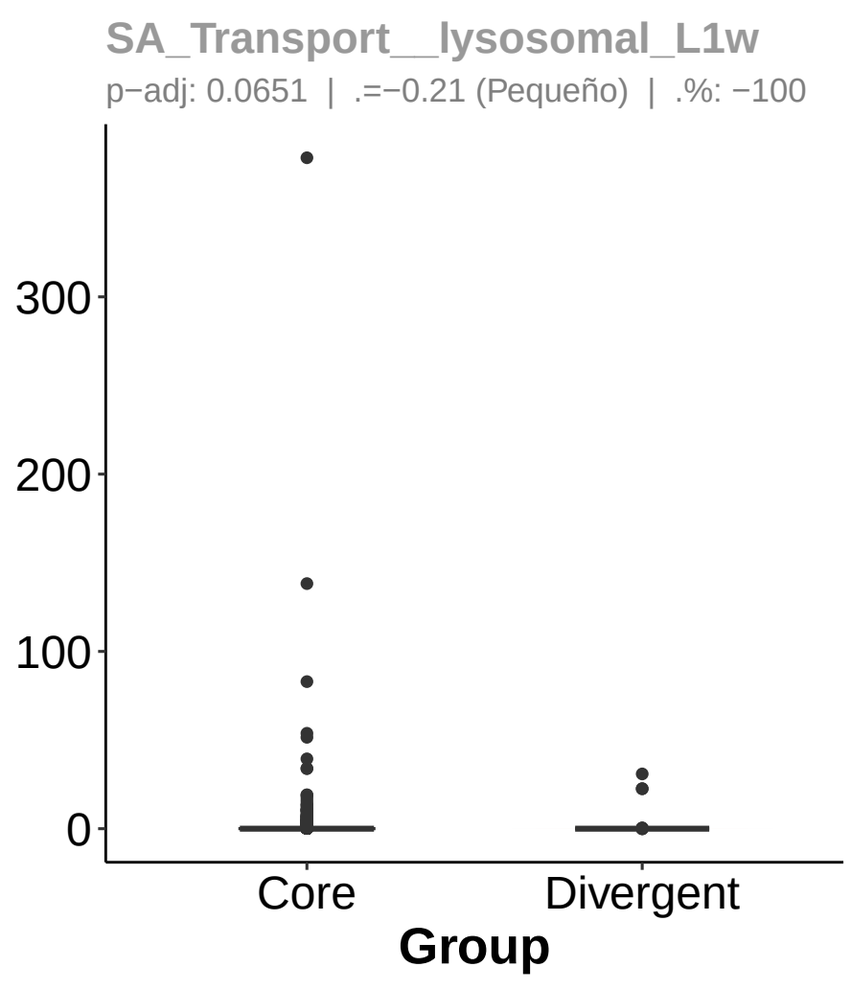
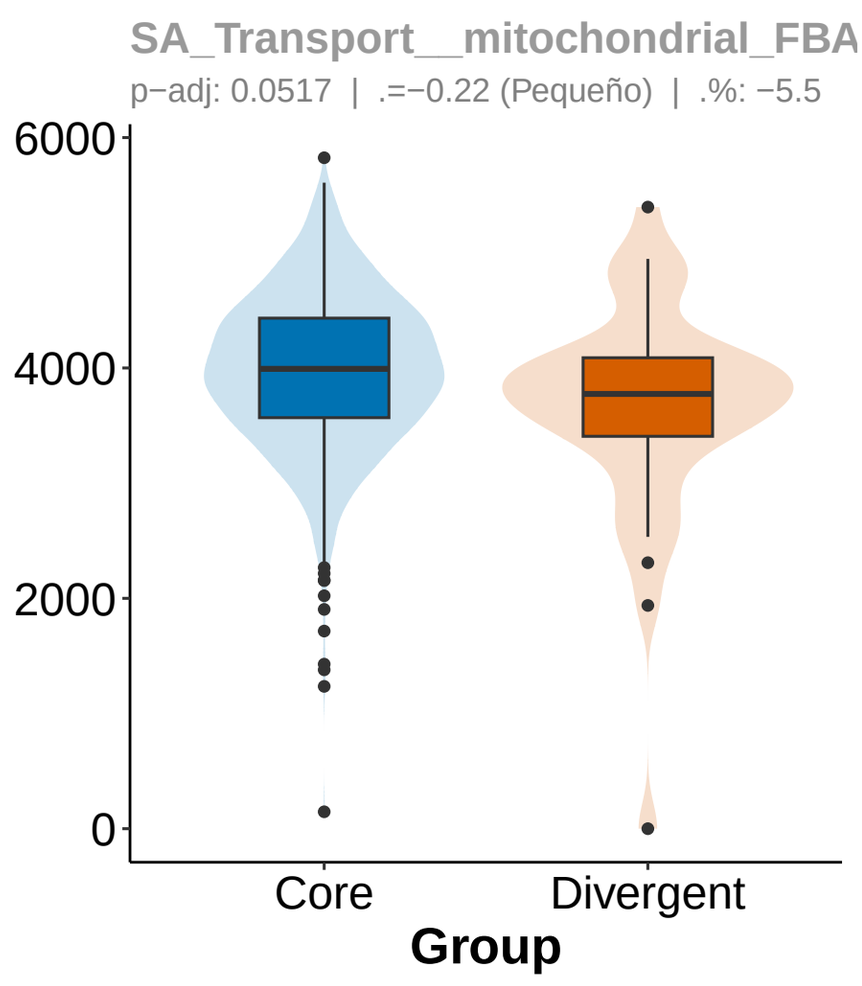
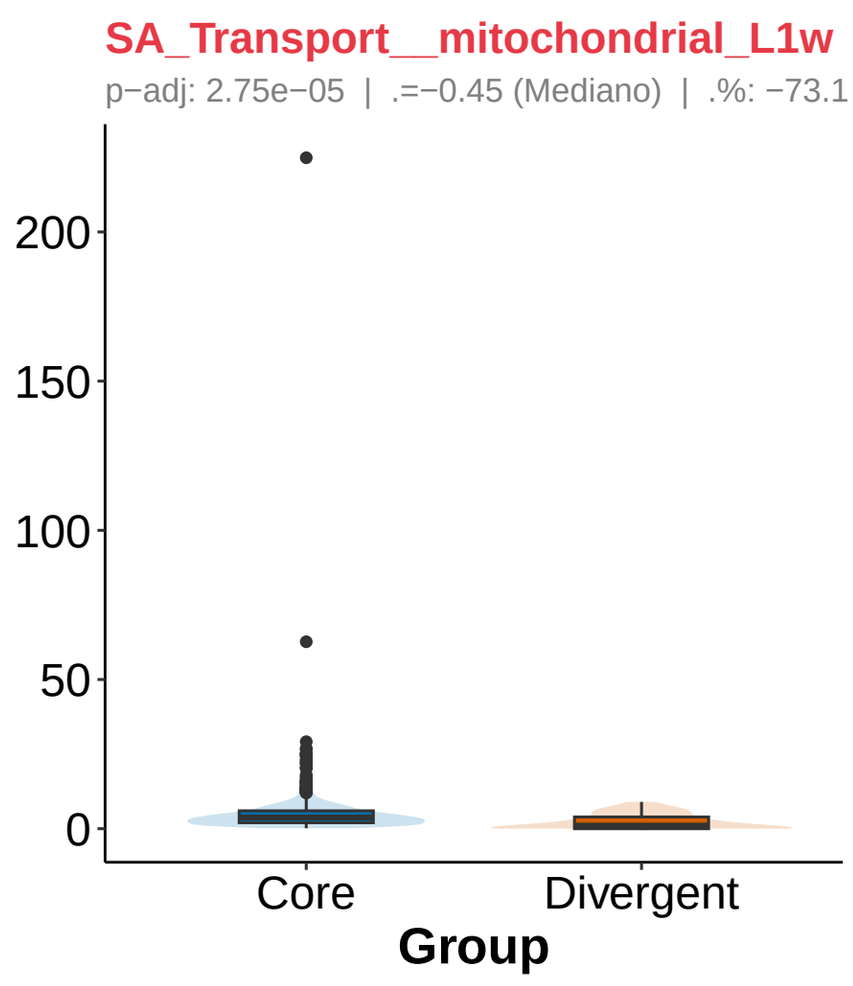




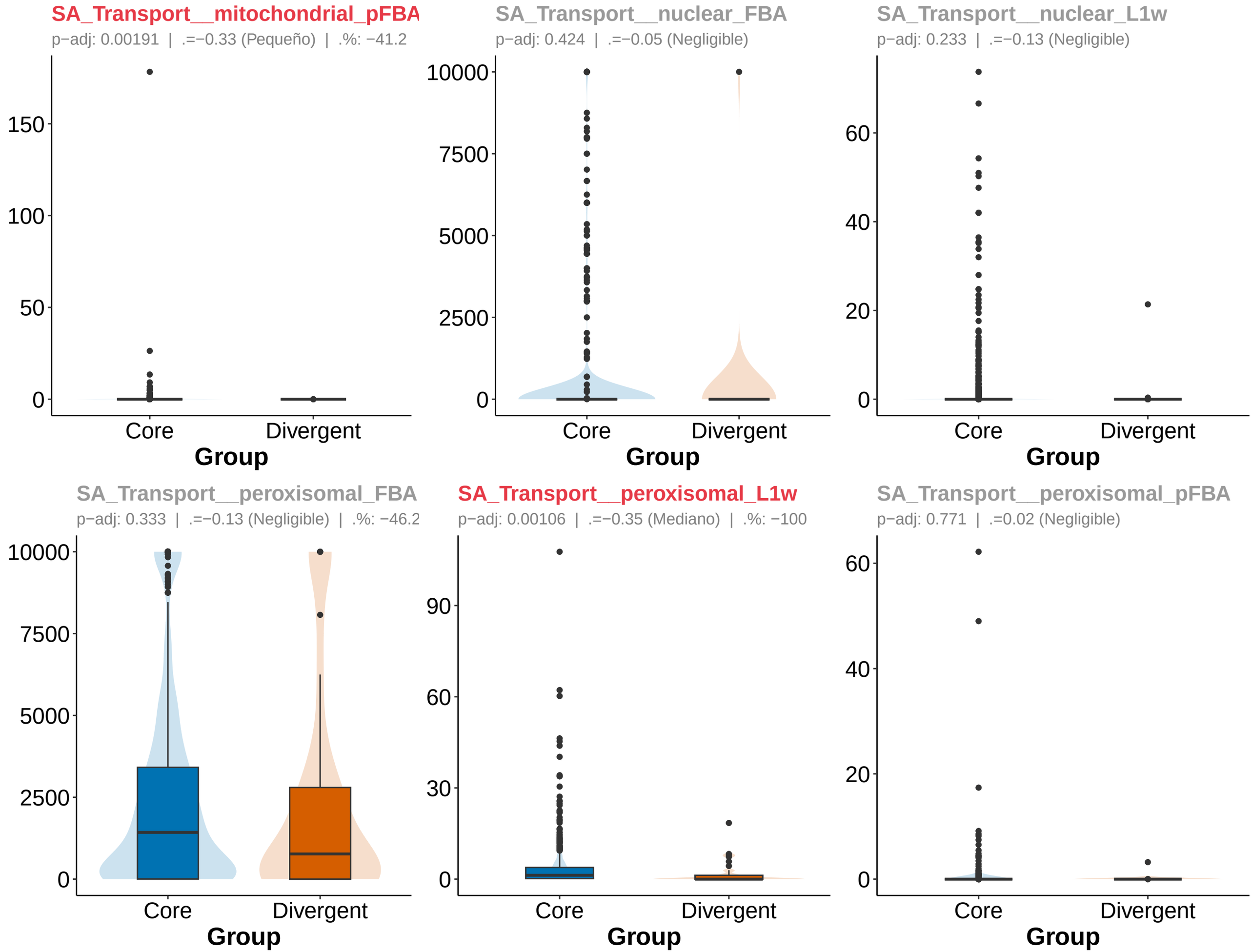




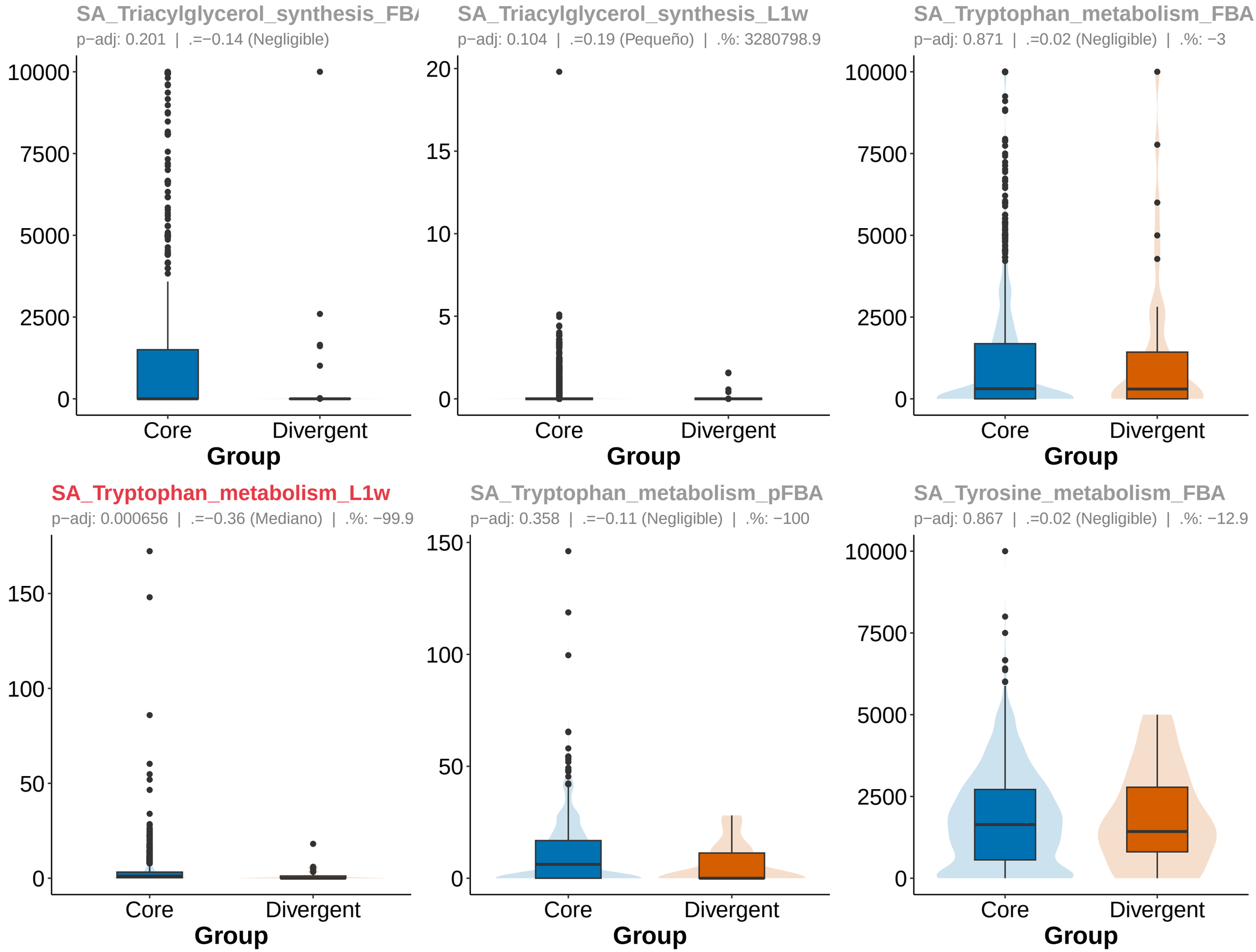




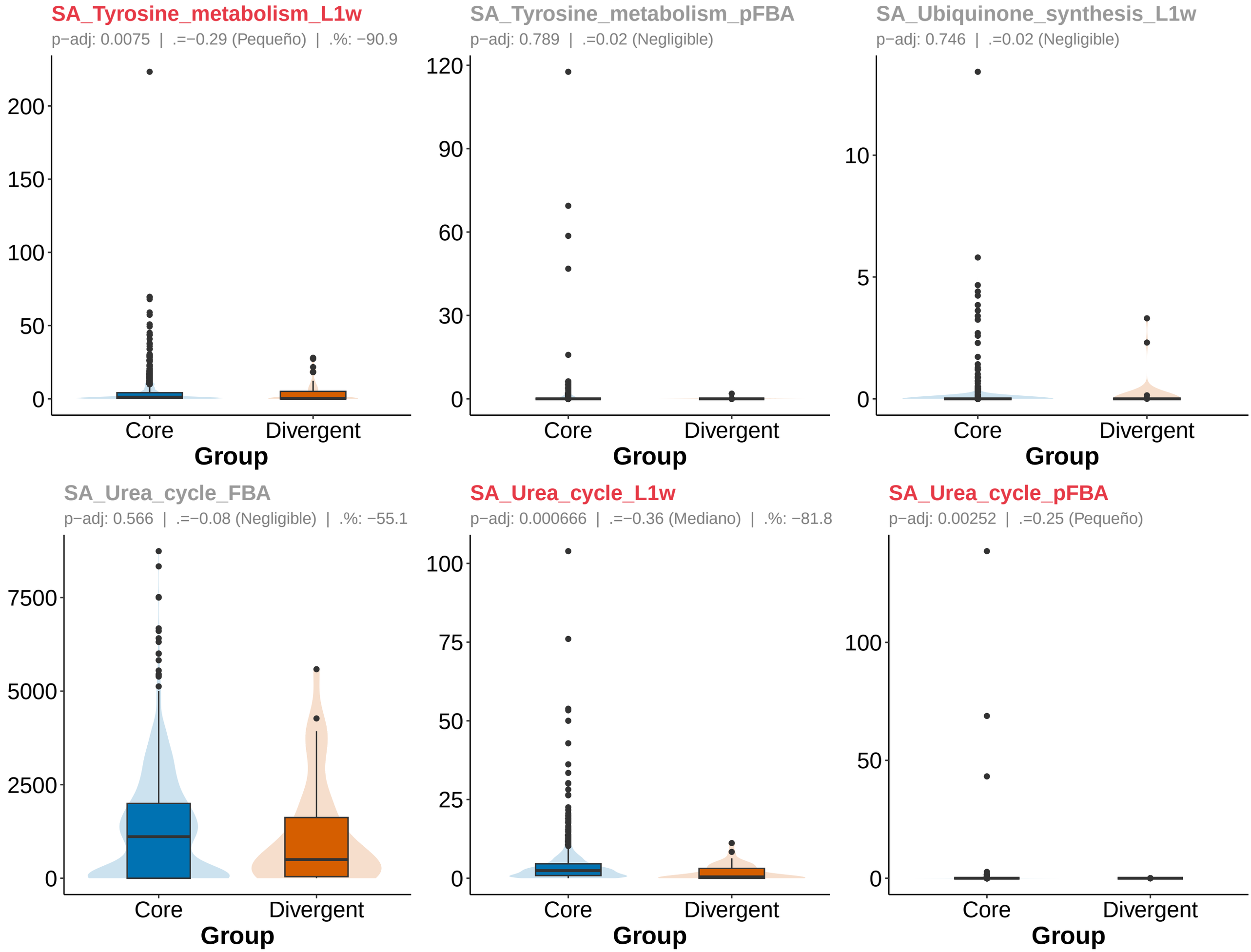
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



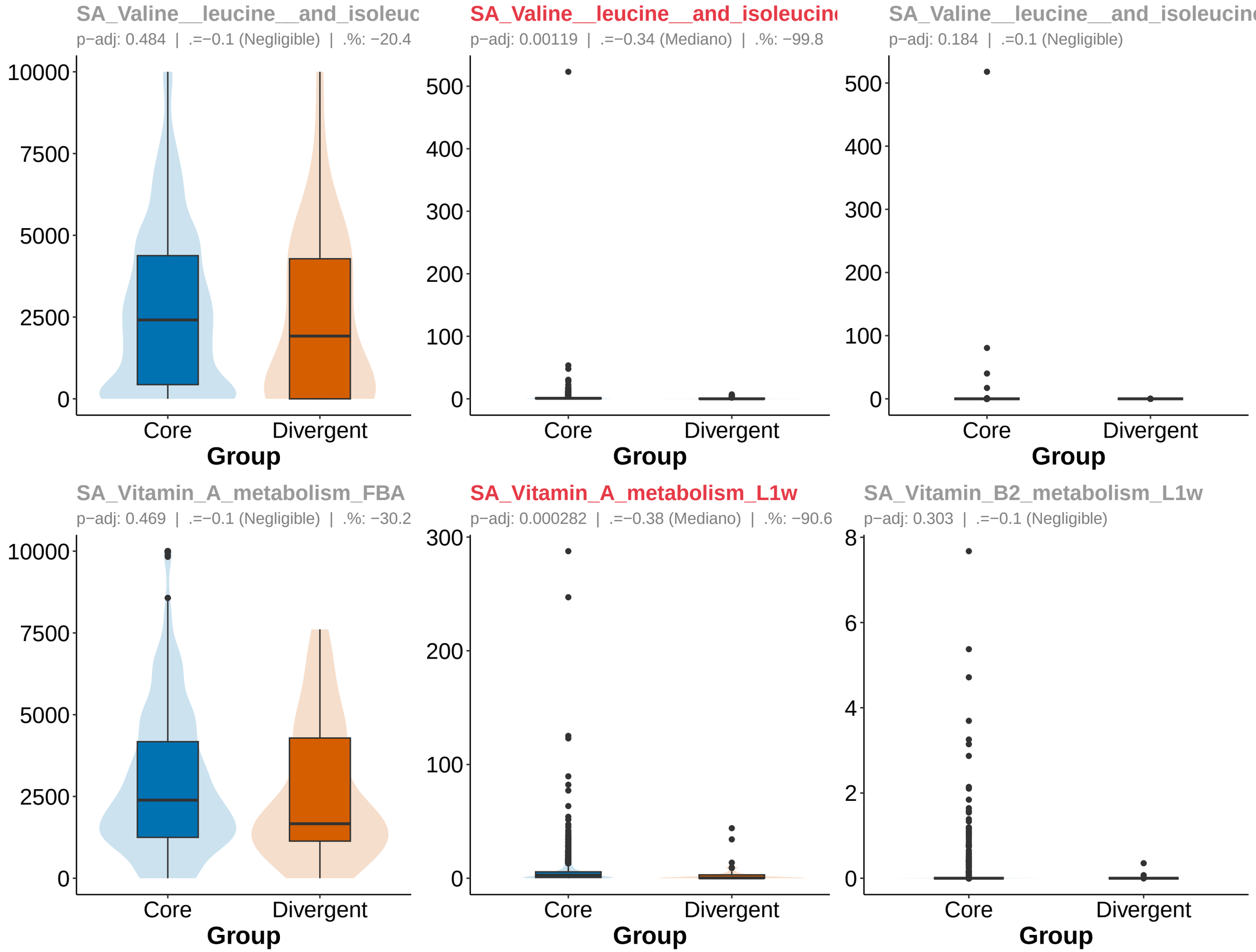
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



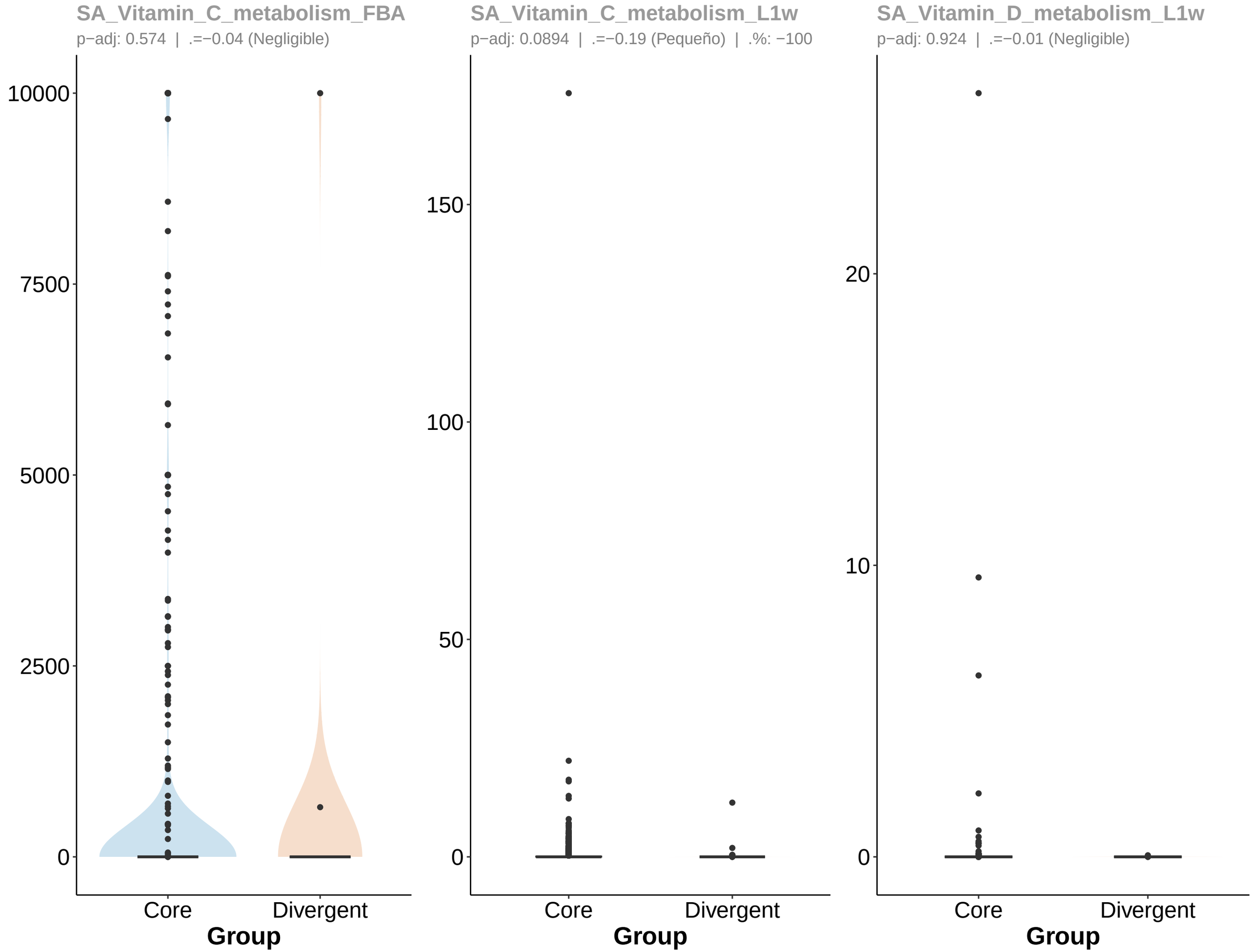
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



Metabolismo − . 5. Reprogramación Sistémica (SA_*)

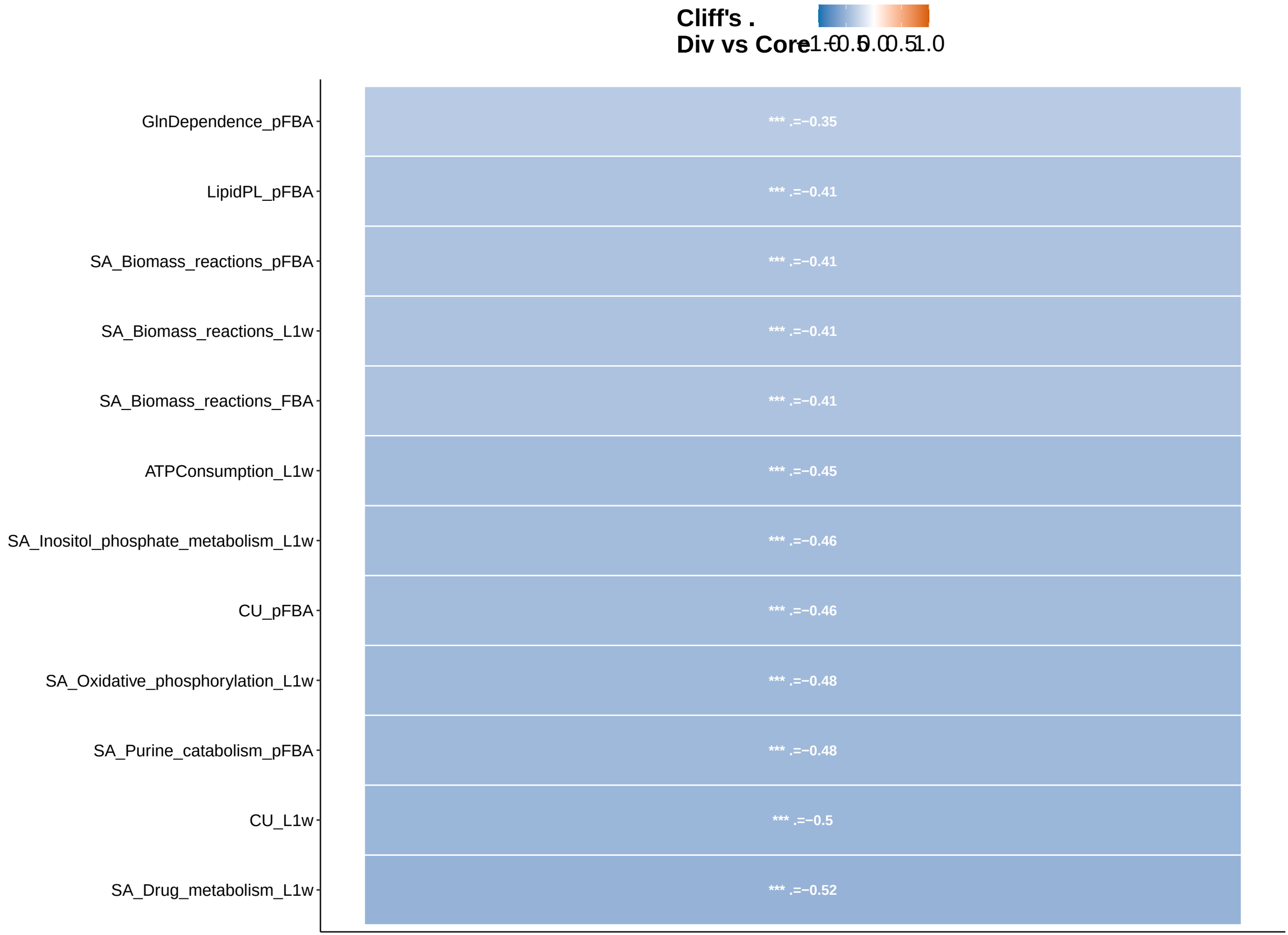


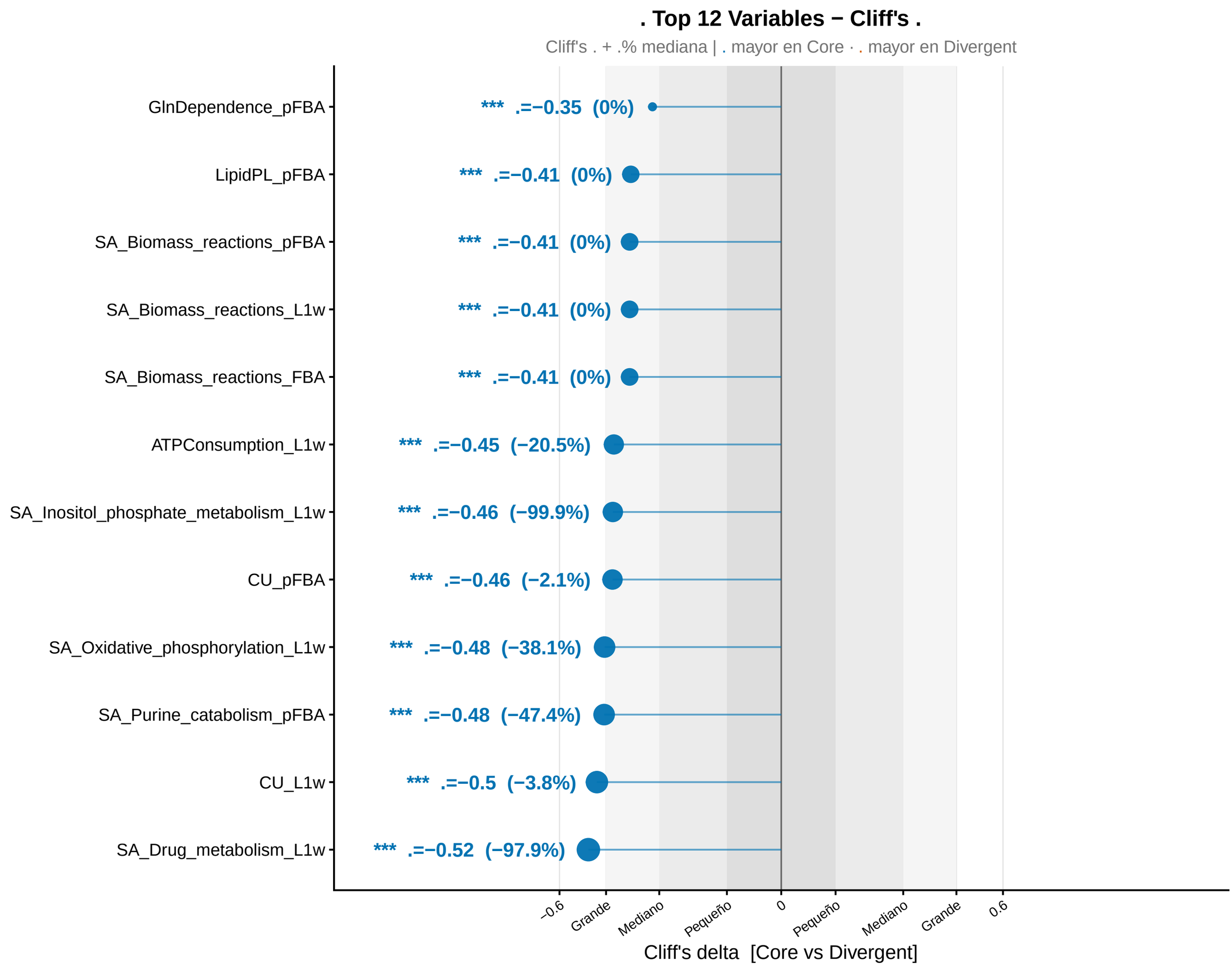
Metabolismo − . 5. Reprogramación Sistémica (SA_*)



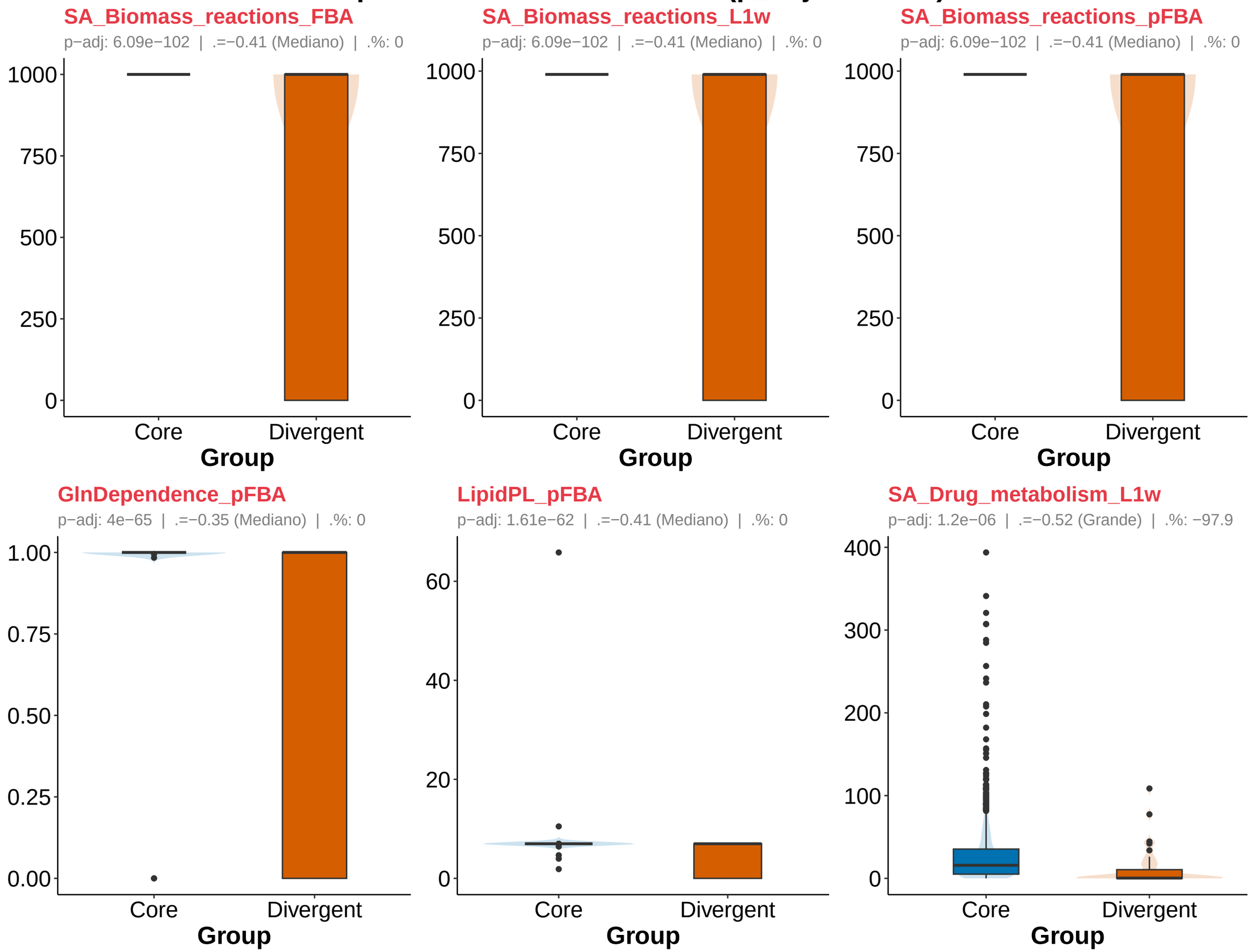
Cliff's .: Top 12 Variables

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

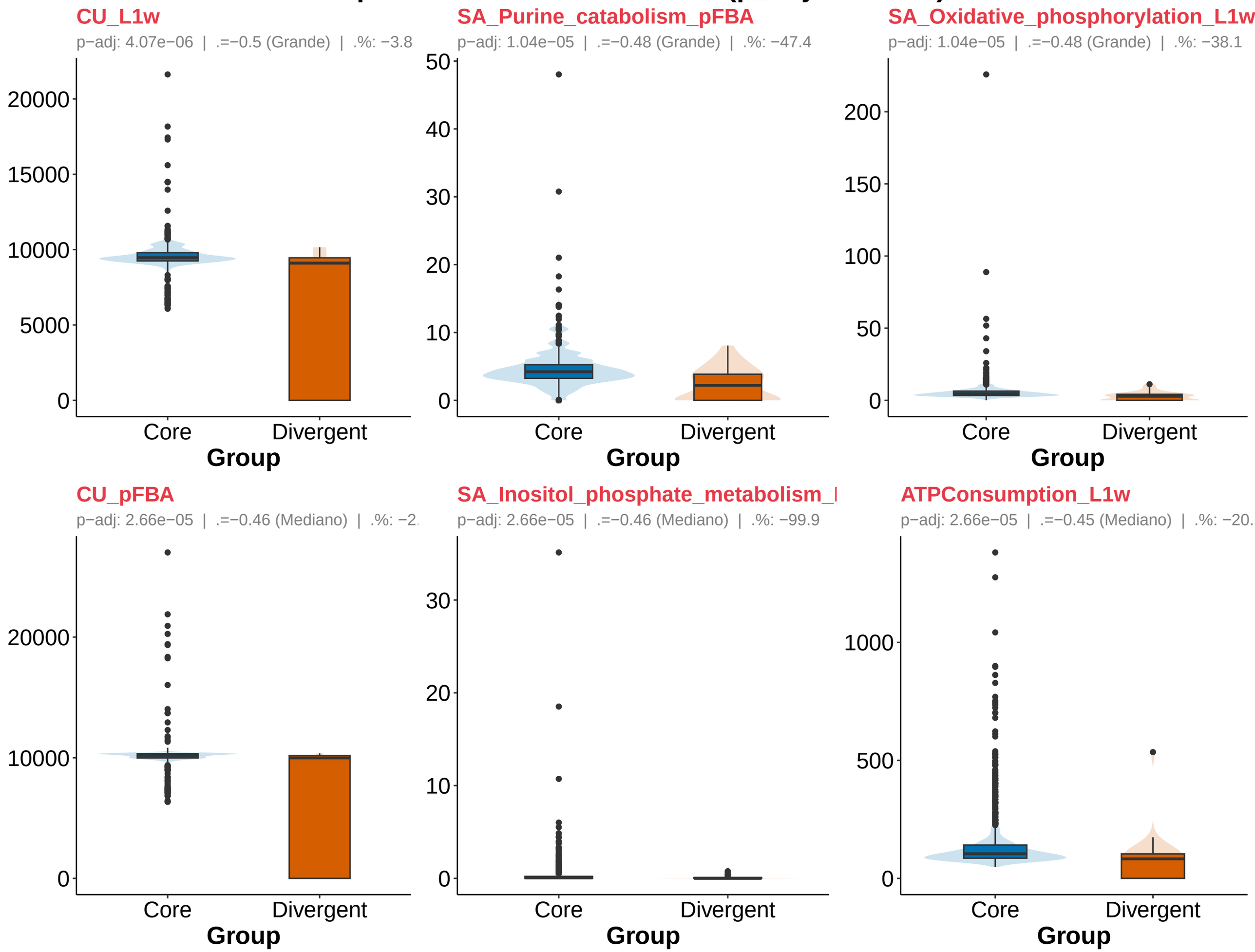



. Top 12 Variables − Cliff's .
Cliff's . + .% mediana | . mayor en Core · . mayor en Divergent
GlnDependence_pFBA
*** .=−0.35 (0%)
LipidPL_pFBA
*** .=−0.41 (0%)
SA_Biomass_reactions_pFBA
*** .=−0.41 (0%)
SA_Biomass_reactions_L1w
*** .=−0.41 (0%)
SA_Biomass_reactions_FBA
*** .=−0.41 (0%)
ATPConsumption_L1w
*** .=−0.45 (−20.5%)
SA_Inositol_phosphate_metabolism_L1w
*** .=−0.46 (−99.9%)
CU_pFBA
*** .=−0.46 (−2.1%)
SA_Oxidative_phosphorylation_L1w
*** .=−0.48 (−38.1%)
SA_Purine_catabolism_pFBA
*** .=−0.48 (−47.4%)
CU_L1w
*** .=−0.5 (−3.8%)
SA_Drug_metabolism_L1w
*** .=−0.52 (−97.9%)
−0.6
Grande
Mediano
Pequeño
0
Pequeño
Mediano
Grande
0.6
Cliff's delta [Core vs Divergent]

. Top 12 Variables Metabólicas (p_adj + Cliff's .)



. Top 12 Variables Metabólicas (p_adj + Cliff's .)

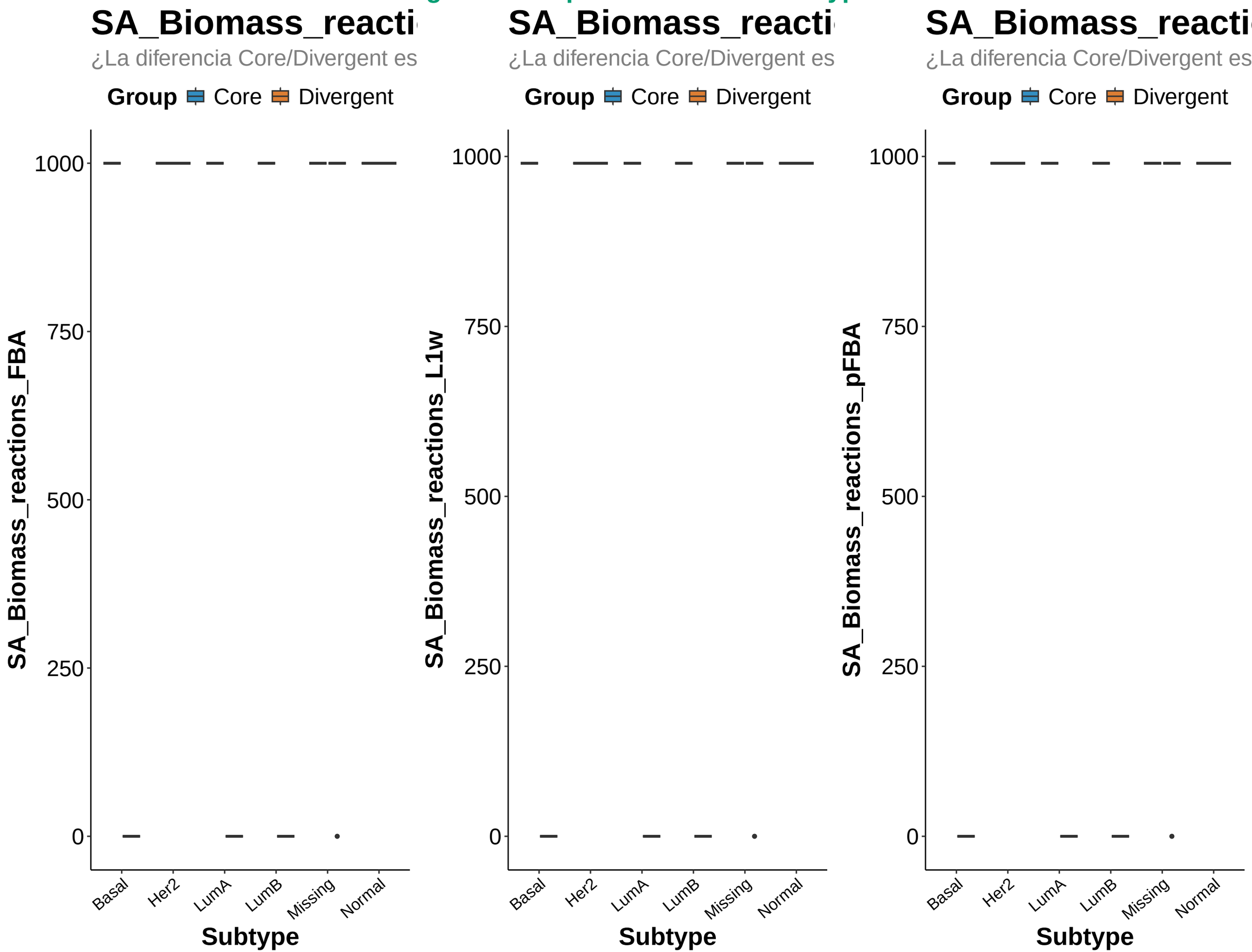


SECCIÓN III

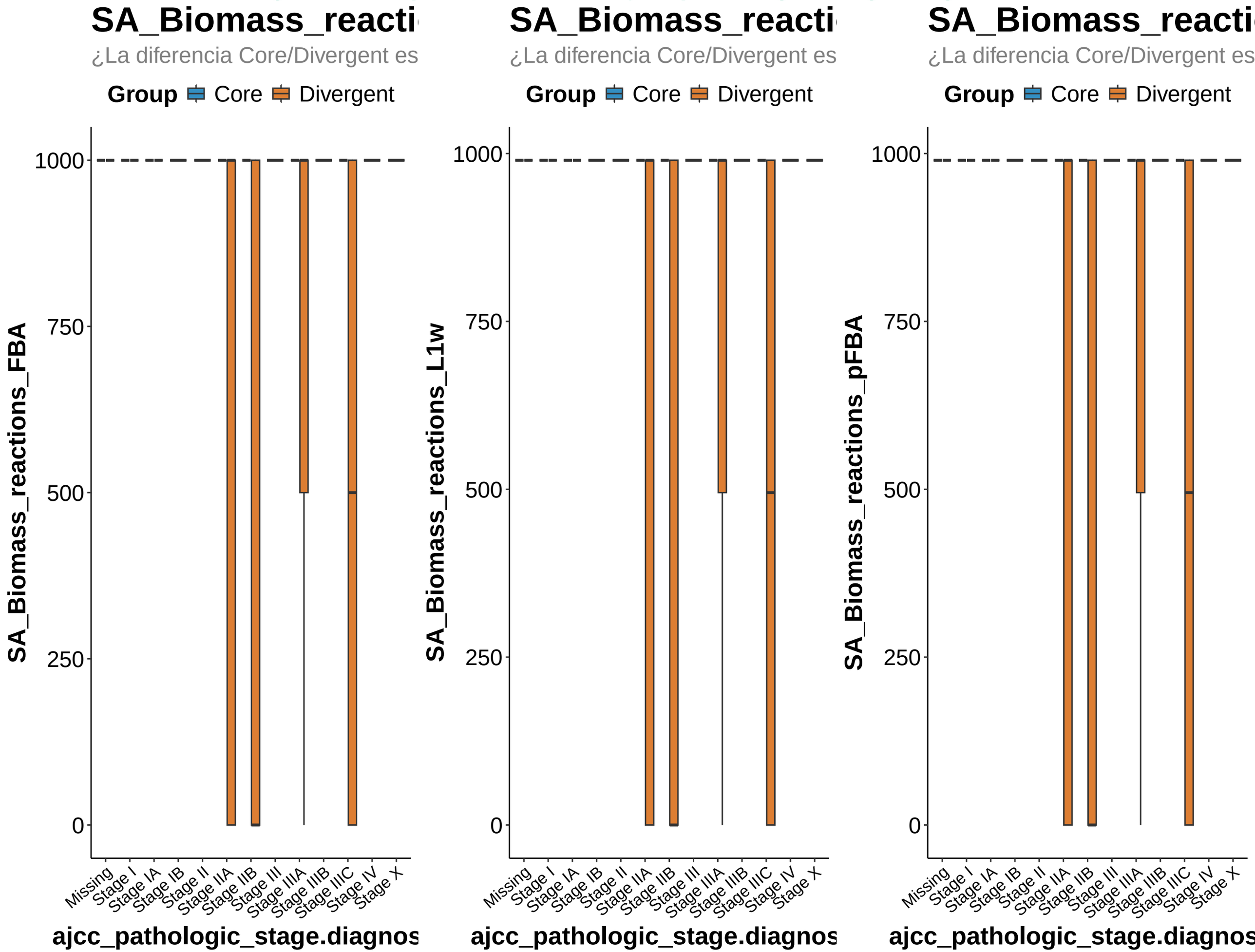
Análisis Integrado: Clínica <–> Metabolismo

¿Core vs Divergent es reflejo del subtipo? ¿O reprogramación metabólica independiente?

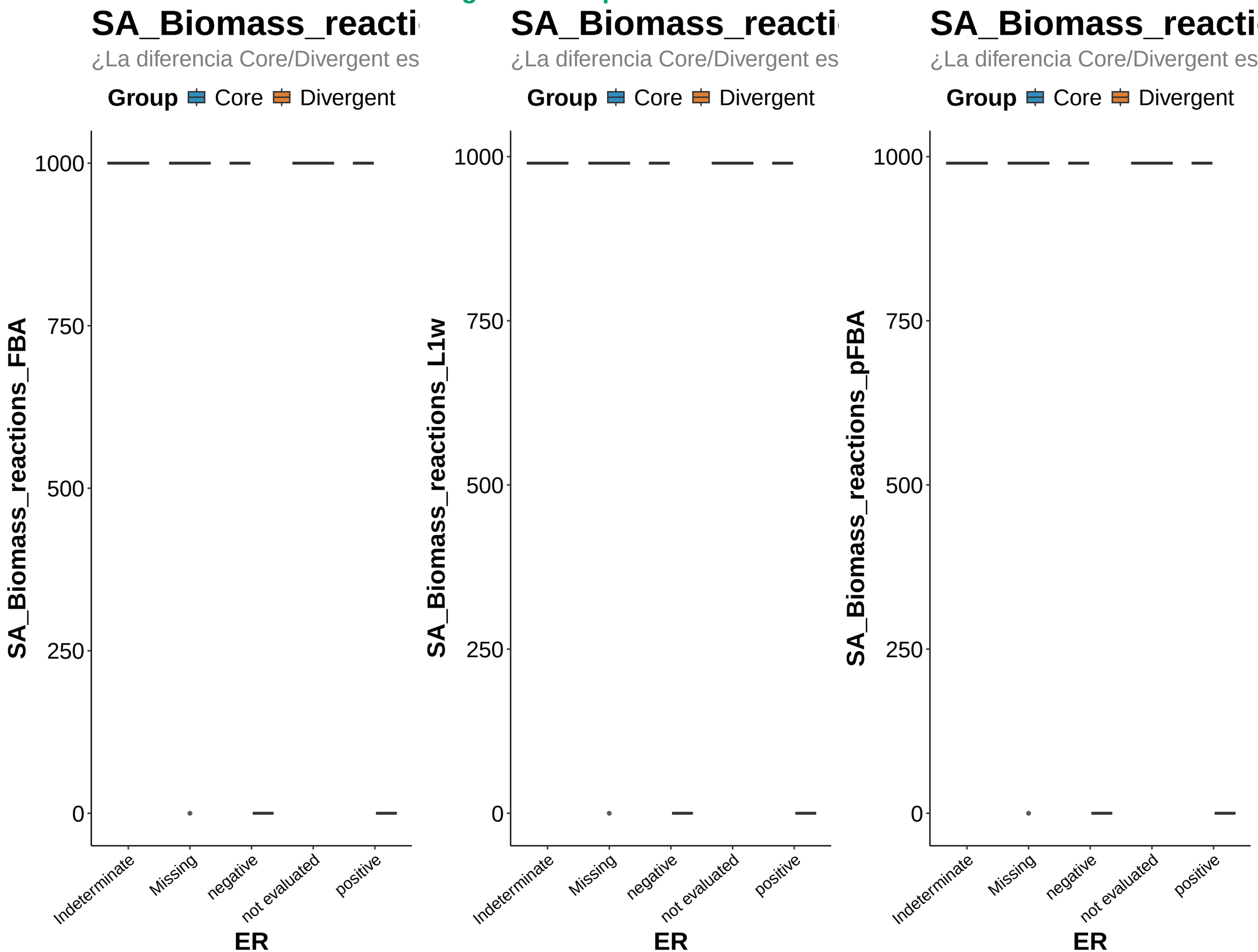
Integración: Top metabólicas × Subtype



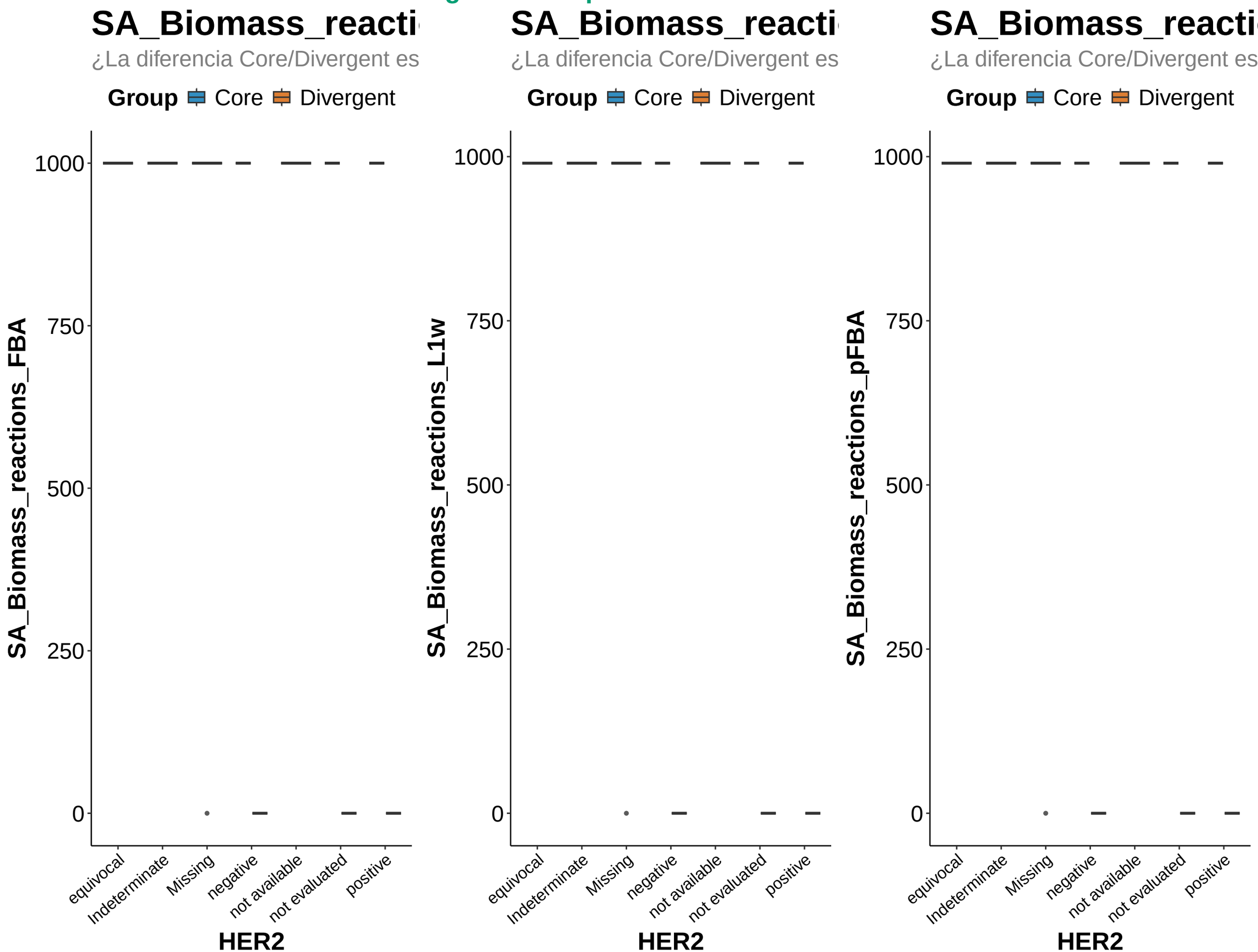
Integración: Top metabólicas × ajcc_pathologic_stage.diagnoses



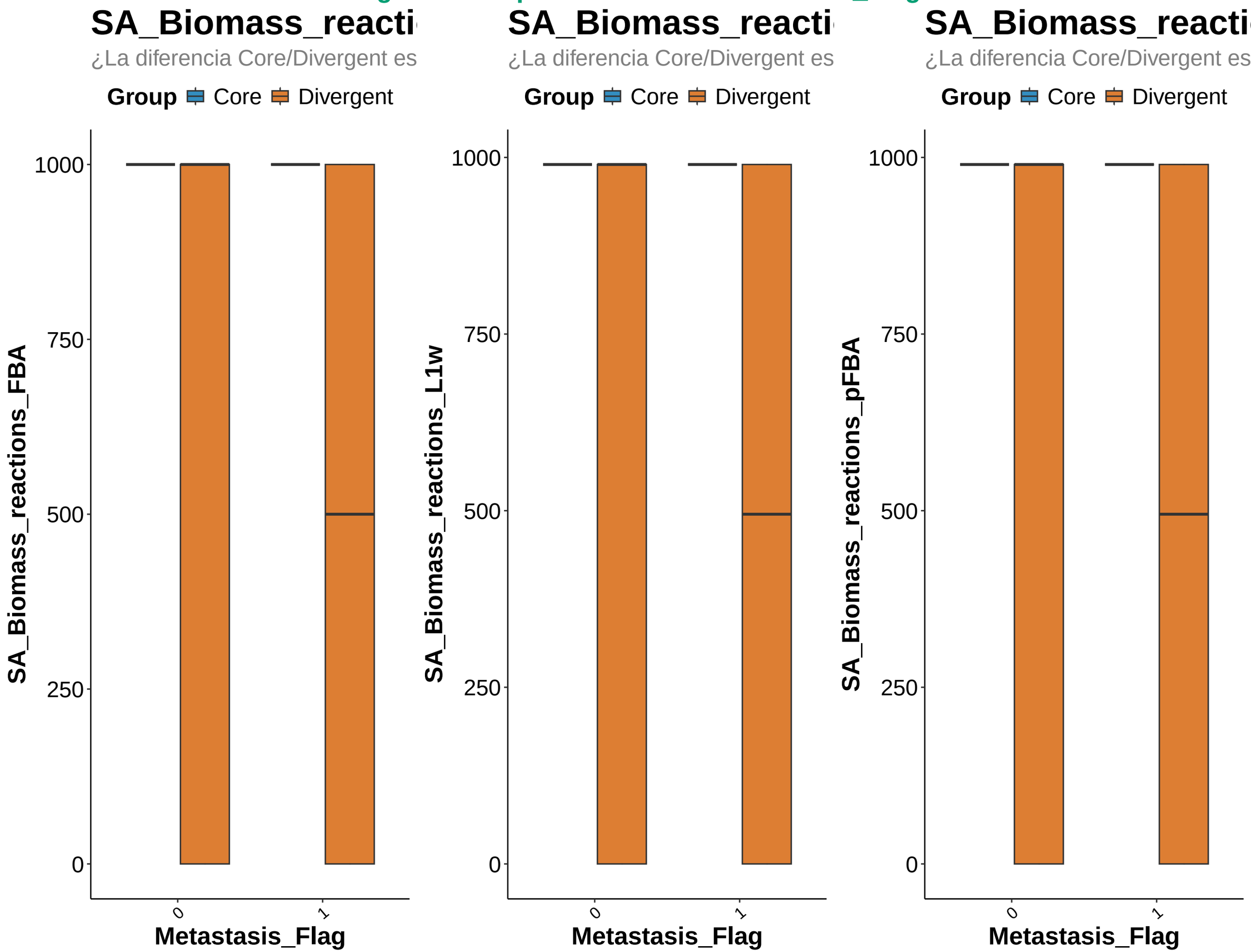
Integración: Top metabólicas × ER



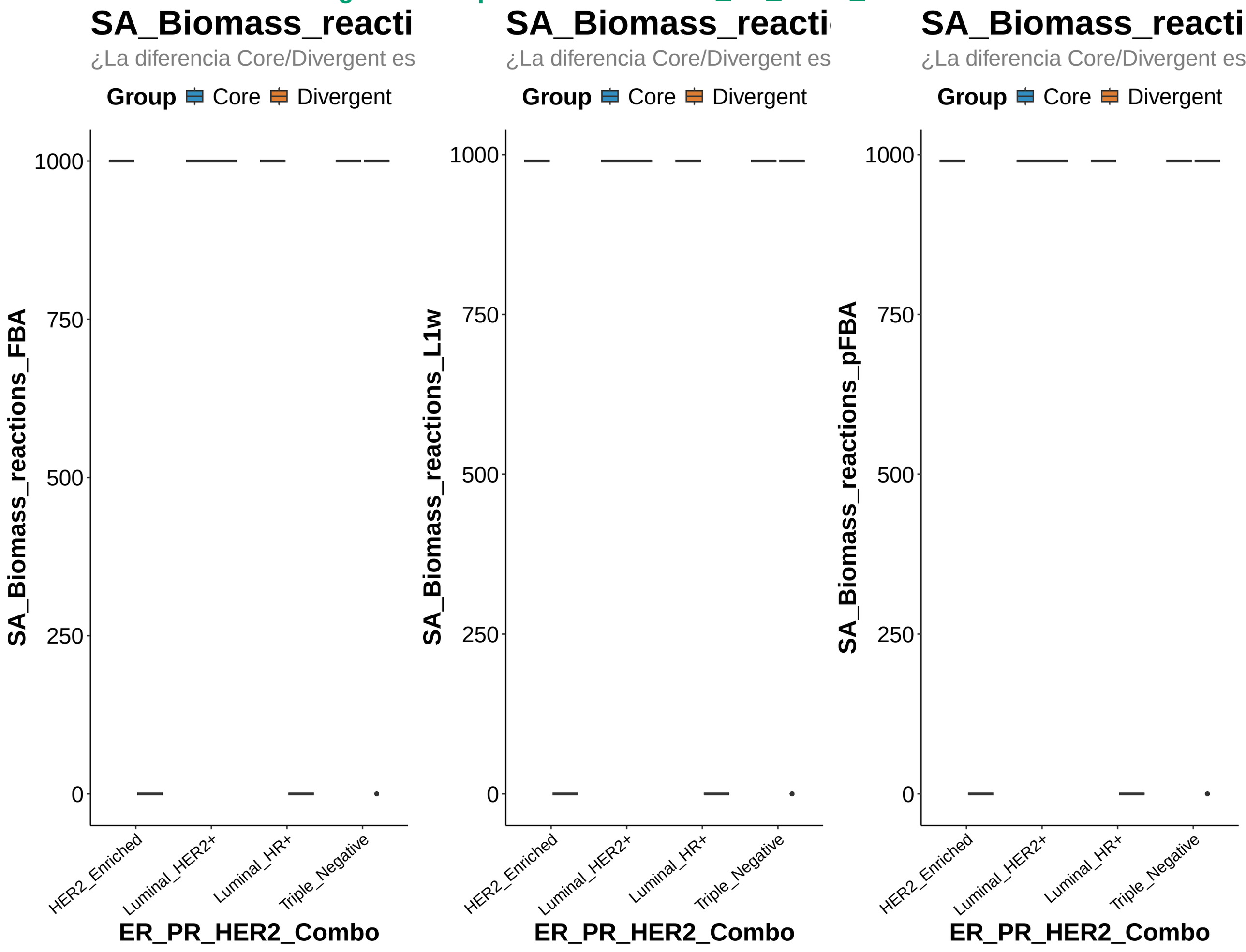
Integración: Top metabólicas × HER2



Integración: Top metabólicas × Metastasis_Flag

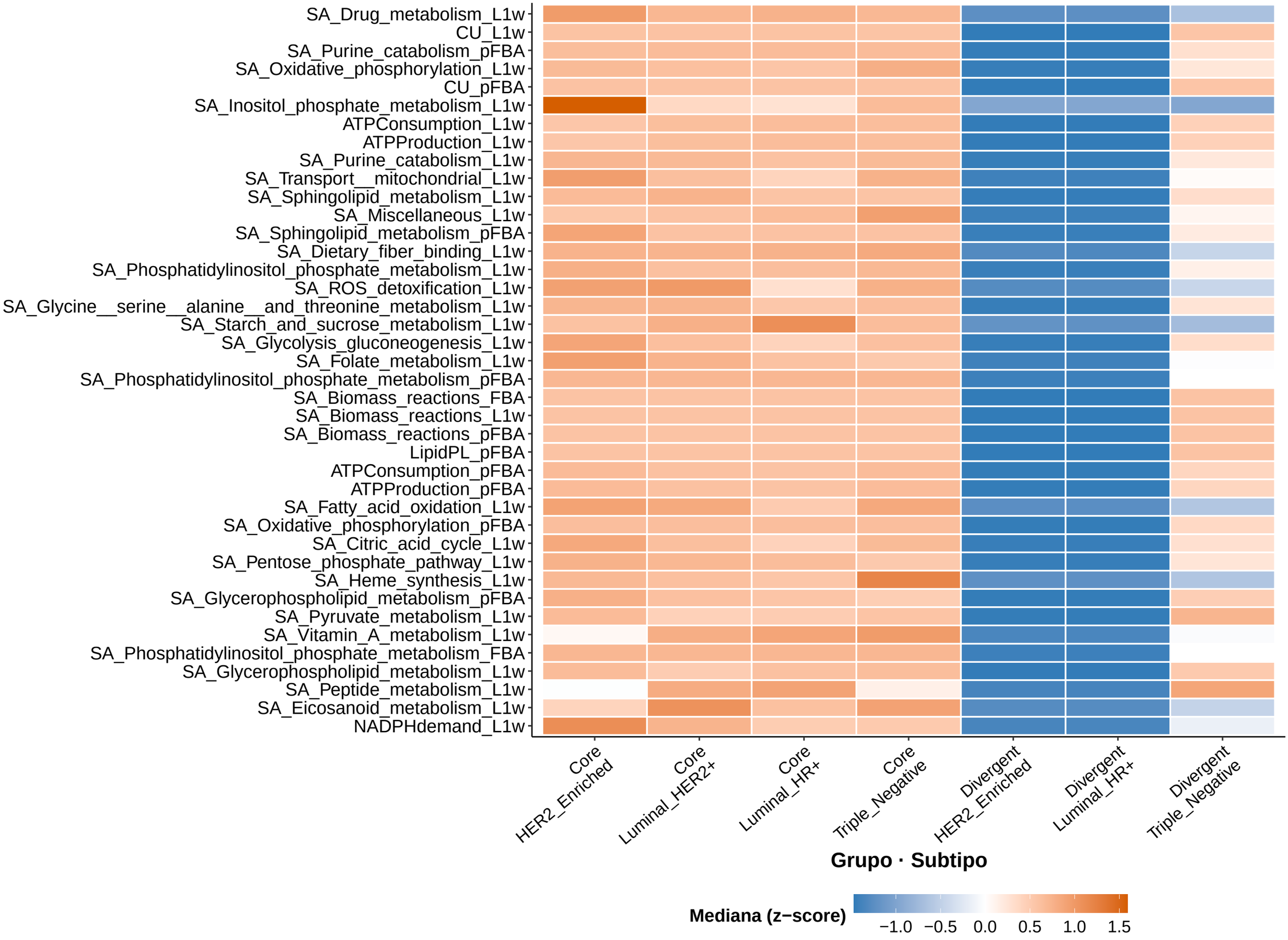


Integración: Top metabólicas × ER_PR_HER2_Combo



Firma Metabólica por Subtipo Molecular (ER_PR_HER2_Combo

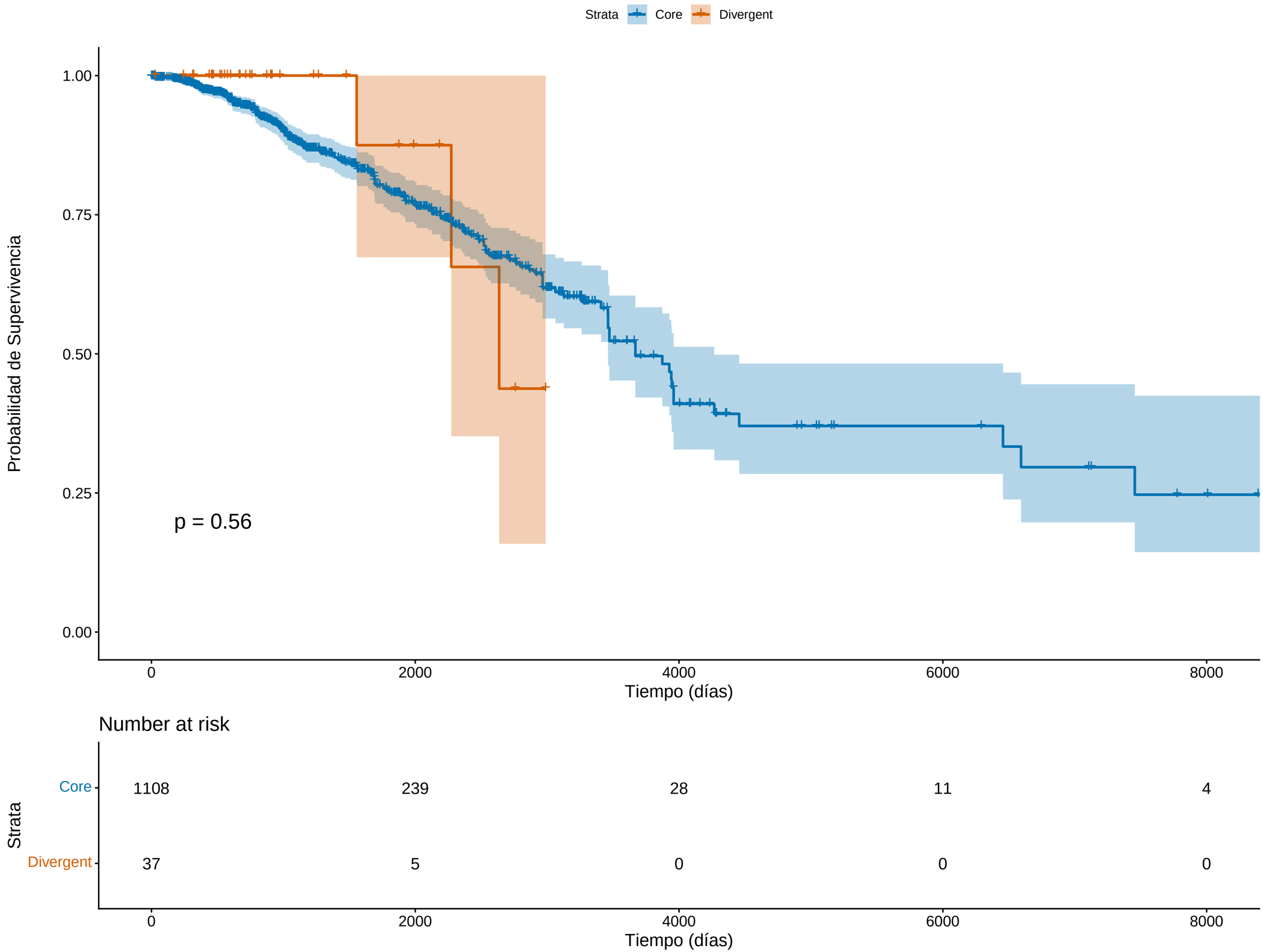
Variables ordenadas por Cliff's . | FDR < 0.05



SECCIÓN IV
Análisis de Supervivencia

¿El fenotipo metabólico Core/Divergent predice pronóstico?

Curva de Supervivencia Global: Core vs Divergent

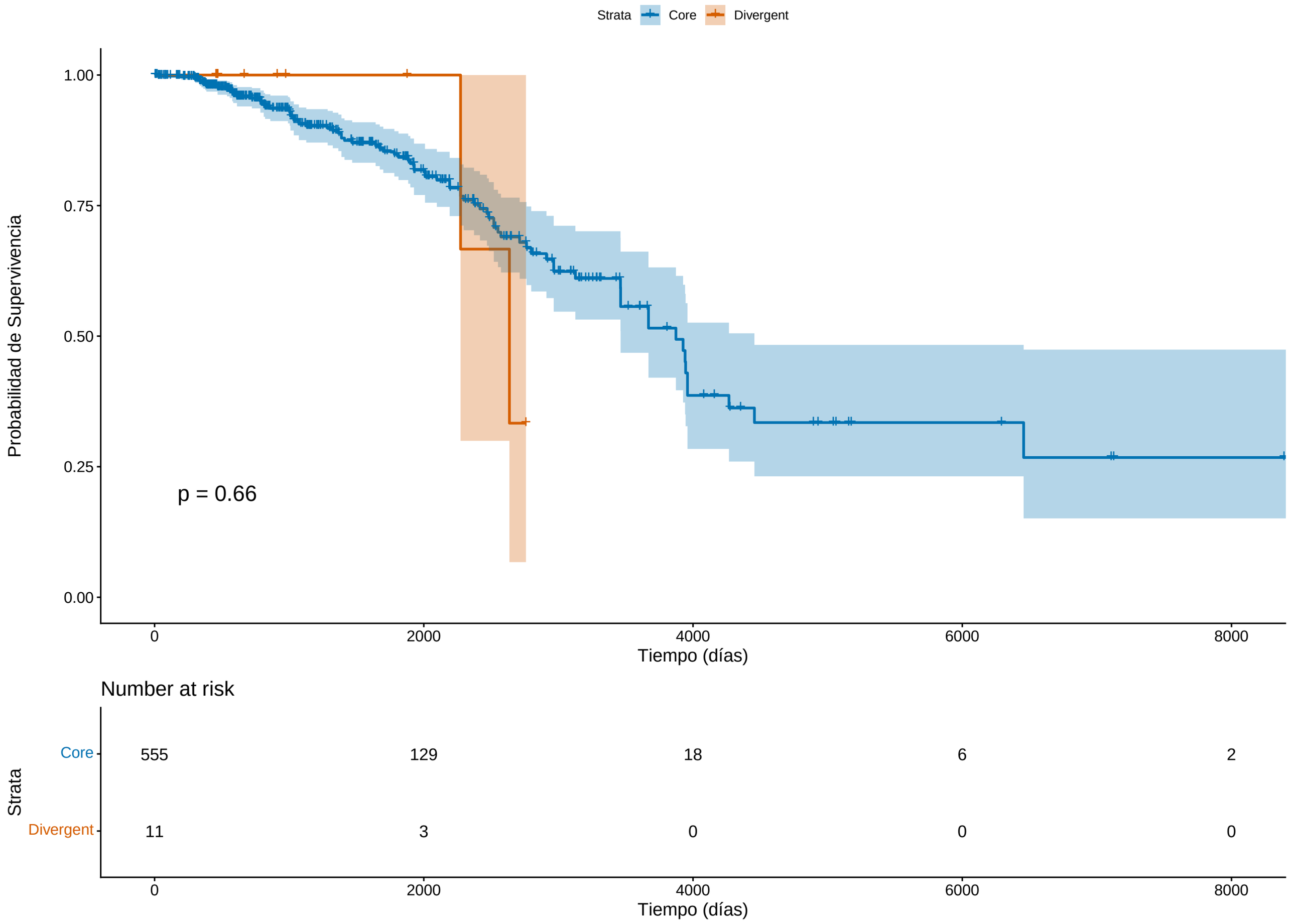


SECCIÓN IV−B

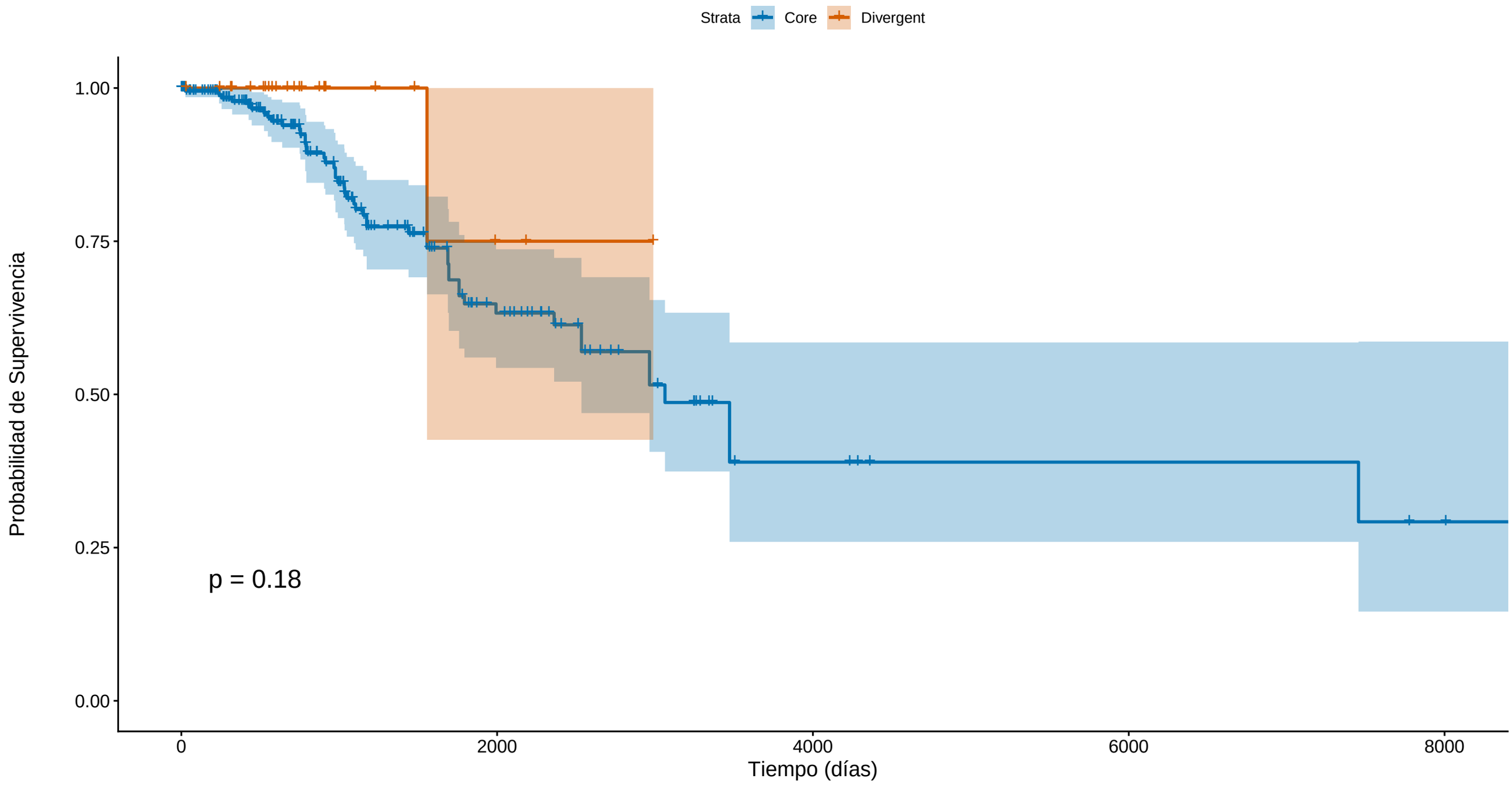
Supervivencia por Subtipo Molecular (ER_PR_HER2_Combo)

Cada subtipo: Core vs Divergent | mín. 5 pacientes por grupo

Supervivencia: Core vs Divergent
Subtipo: Luminal_HR+
Core n=555 | Divergent n=11



Supervivencia: Core vs Divergent
Subtipo: Triple_Negative
Core n=207 | Divergent n=25



Number at risk

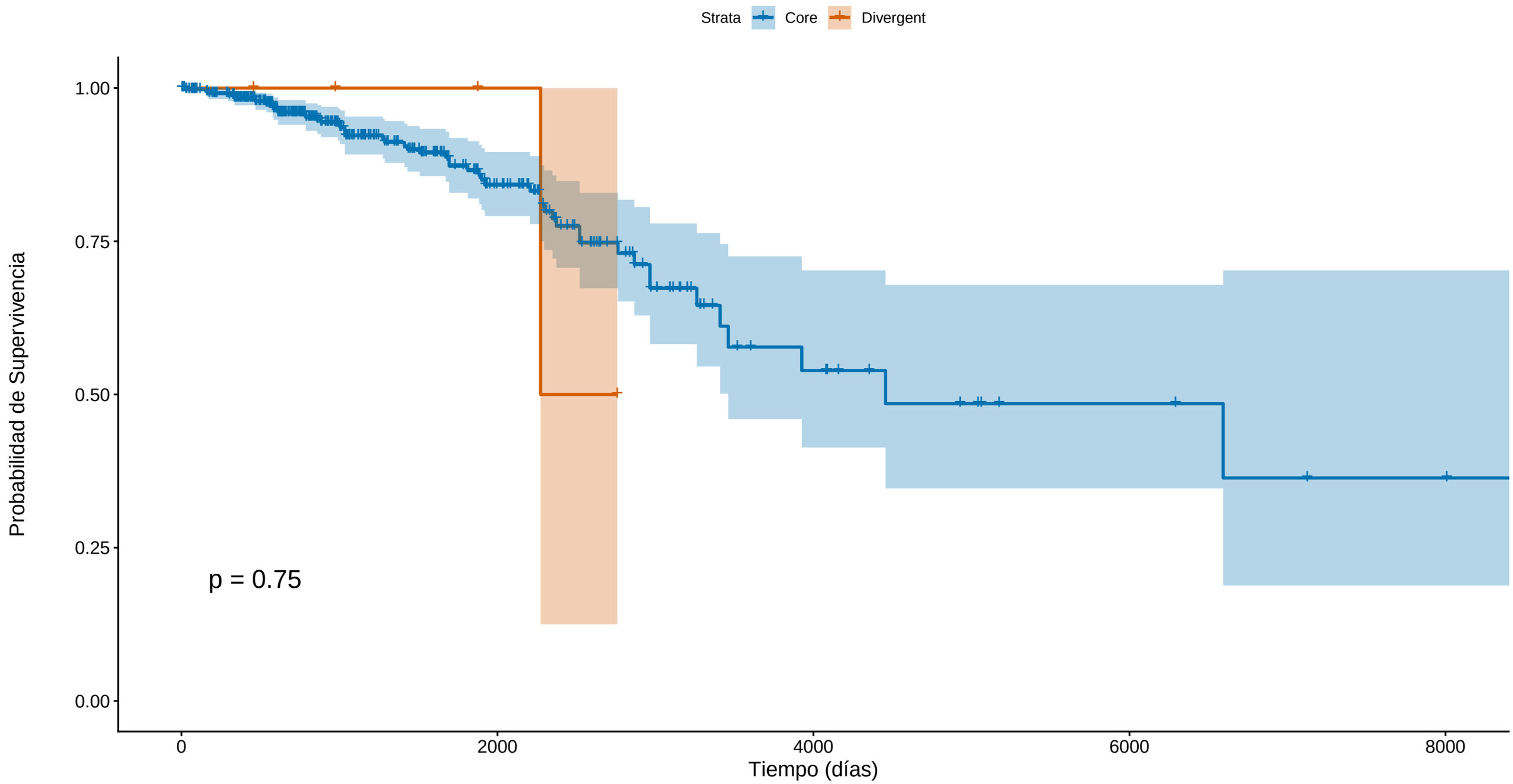
Strata	0	2000	4000	6000	8000
Core	207	42	7	4	2
Divergent	25	2	0	0	0

Tiempo (días)

SECCIÓN IV−C
Supervivencia por Subtipo Molecular (Subtype / PAM50)

Cada subtipo: Core vs Divergent | mín. 5 pacientes por grupo

Supervivencia: Core vs Divergent
Subtipo: LumA
Core n=467 | Divergent n=5

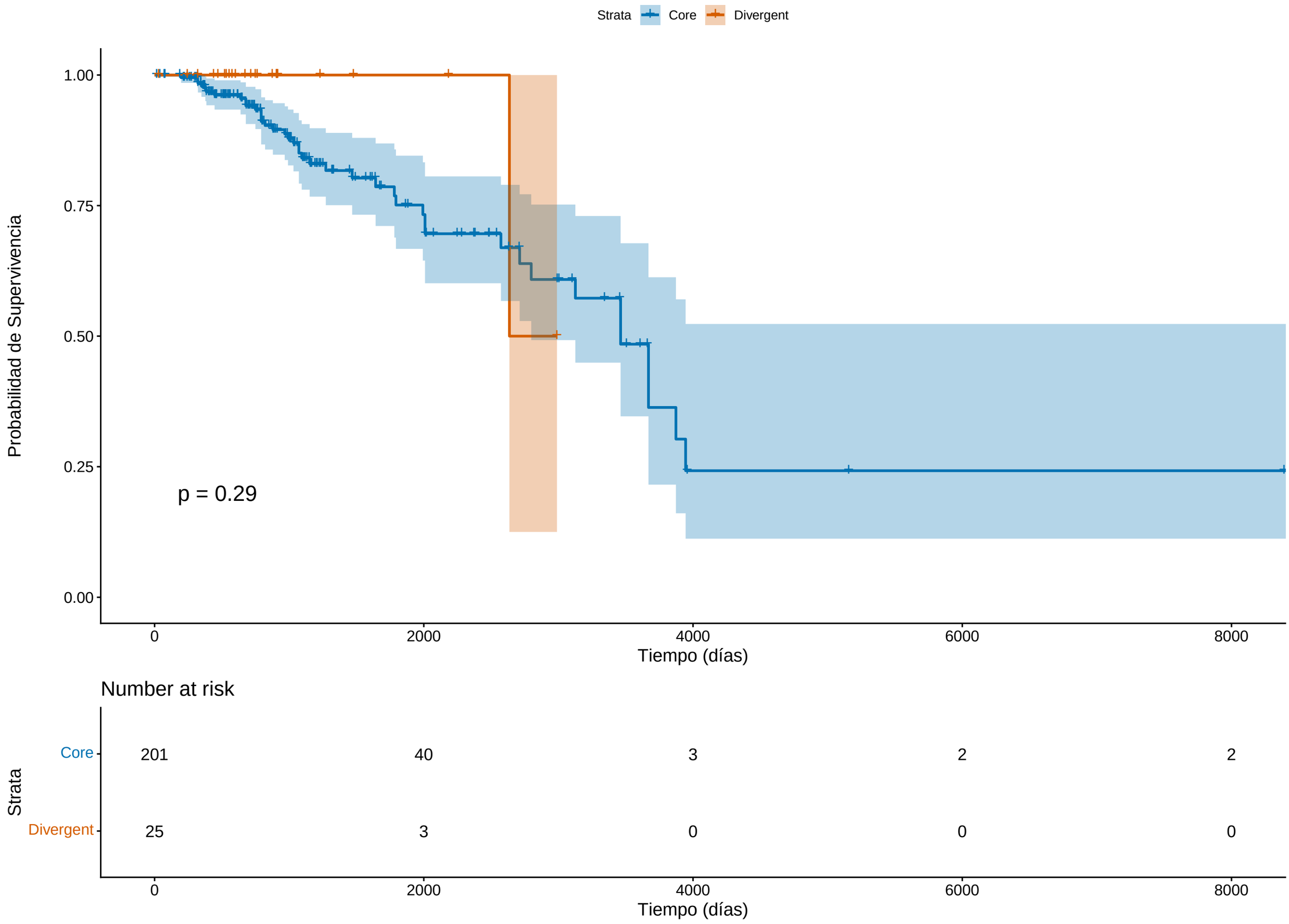


Number at risk

Strata	0	2000	4000	6000	8000
Core	467	100	14	5	2
Divergent	5	2	0	0	0

Tiempo (días)

Supervivencia: Core vs Divergent
Subtipo: Missing
Core n=201 | Divergent n=25



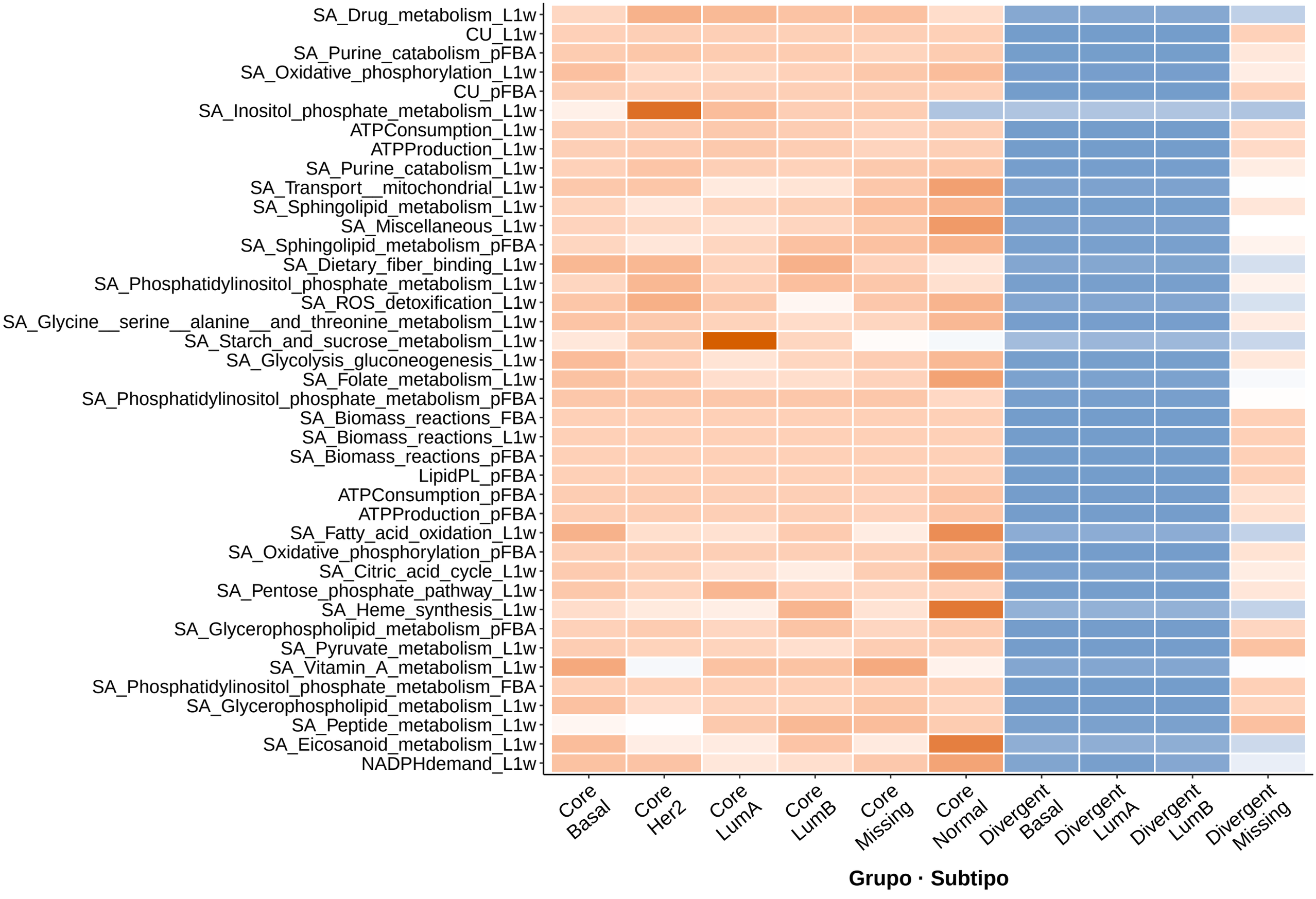
Resumen de Supervivencia

Core: N=1108 | Eventos=185 | Mediana OS (eventos)=1174 días | Mediana OS (todos)=943 días

Divergent: N= 37 | Eventos= 3 | Mediana OS (eventos)=2273 días | Mediana OS (todos)=762 días

N por Subtipo Molecular:

HER2_Enriched	Core	N= 72	Eventos=12
HER2_Enriched	Divergent	N= 1	Eventos= 0
Luminal_HER2+	Core	N=274	Eventos=35
Luminal_HR+	Core	N=555	Eventos=88
Luminal_HR+	Divergent	N= 11	Eventos= 2
Triple_Negative	Core	N=207	Eventos=50
Triple_Negative	Divergent	N= 25	Eventos= 1

Firma Metabólica por Subtipo Molecular − Top 40

Variables ordenadas por Cliff's . | FDR < 0.05